



Abound Academy



PMI PMP® EXAM-PREP

For 2023 Exams

**Over 650+ Realistic Questions with
explanations aligned with PMBOK
Guide, 6th Edition and 2021 ECO.**

BY ABOUND ACADEMY



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Amazon Edition

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Preface

The PMP certification is the gold-standard for Project Management professionals. With over 2 Million PMP professionals worldwide it has been important for PMI (Project Management Institute) which governs and certifies your PMP examination to make sure that only qualified professionals get certified. Unfortunately, this makes the PMP certification both costly and tough, nearly 30% of professionals who attempt the PMP examination are unable to pass on their 1st attempt.

If you are looking at ensuring your PMP certification you have come to the right place. This practice guide not only provides you access to over 300+ detailed Questions and Answers but also gives you an overview on the process of getting certified as a Project Management Professional (PMP) by the Project Management Institute (PMI).

The contents of this book include:

- Overview of PMP - Ensure you meet the criteria for experience and education
- Outline of Key players - PMP, PMI, Training Institutes & Prometric
- Revision - Definitions and concepts for all major PMP concepts
- Formulas and Calculation - Detailed explanation of formulae and sample calculation.
- Over 3 realistic practice tests as per the latest 2021 update

We truly enjoy assisting project managers pursue and achieve their credentials. Best of luck on your preparation and exam!

Team Abound

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Chapter 1: Introduction

1.1 About the Author

Abound Academy is a Professional Certification Provider Institution which provides content for major professional certification exams such as PMP®, Agile®, Disciplined Agile®, Scrum®, AWS®, Azure®, and many other such high-demand certifications. We offer our candidates with exam study materials like online courses, training books, realistic mock questions, and downloadable pdf for all the resources that are featured in our Academy. We help you to boost your professional career by providing a definitive way of getting you certified on your respective certification on your very 1st attempt.

As an academy, we have enabled more than 100,000 individuals with their certification requirements and delivered successful results for more than 50,000 students. Our mission is to act as a stimulant to bring a positive boost in career change for everyone. Our study material and exam simulators are made to help the professionals to get certified, and thus achieve their goals in their respective fields.

We believe that skills and their certification has the power to transform lives and the whole world. We are dedicated to providing best-in-industry training and mock tests that are delivered by highly experienced and competent industry experts. We thrive to work in partnership with communities over the boundaries. Our focus is to become the leading provider of high-quality online certification training to professionals over the boundaries.

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Chapter 2: About The Exam

Many PMP aspirants believe that the PMP exam is based on the PMBoK Guide. However, this is not the case. The PMP exam is not a test about or on the PMBoK Guide.

As the PMP exam moves to more person-based skills and agile approaches, these sources will increasingly not include the PMBOK Guide. With the structuring of the PMP Exam Content Outline according to persons, processes, and business domains, the PMI continues to underline the departure of PMBOK-focused topics.

Crucial for PMP aspirants is understanding and applying the concepts described in the areas, tasks, and capabilities listed in the PMP Exam Content Outline.

2.1 The Exam Focus

So far the domains of the PMP ECO corresponded to the 5 process groups from the PMBoK Guide (initiating, planning, implementing, etc.). The new examination will be based on three new areas:

People, Processes, and Business environment.

These new areas fit better with the PMI Talent Triangle® areas of Leadership, Technical Project Management, and Business and Strategic Work. Since project management takes place in a variety of industries, the Business Environment area only tests universal concepts and does not address the specifics of project types. The breakdown of questions between these areas is as follows:

- People: 42%.
- Process: 50%.
- Business environment: 8%.

2.2 The new PMP Exam Course Content

The task analysis (that PMI has performed in order to create the new PMP ECO) showed that many project managers use agile approaches or agile

concepts in hybrid lifecycles. To reflect this, the new study covers the entire value chain, including predictive, hybrid, and agile approaches.

The inclusion of agile concepts and the increased emphasis on the human aspects of projects represent a major change. Concepts such as servant leadership, conflict resolution, and retrospectives were previously the domain of the PMI-ACP® exam, but are now more frequently dealt with in the PMP® exam.

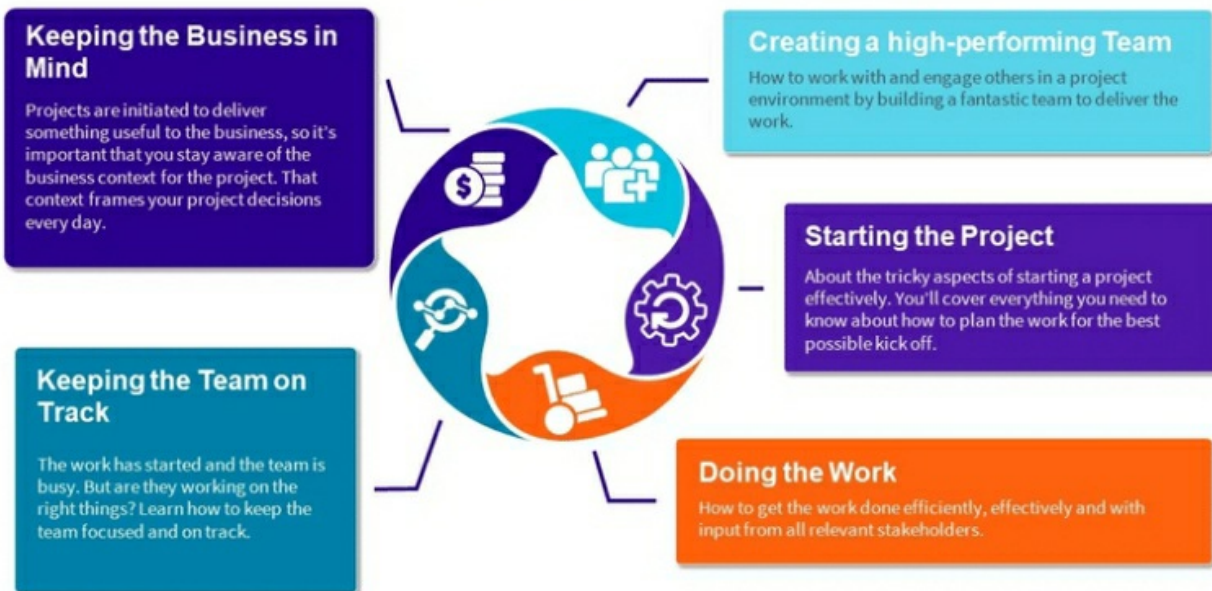
So far almost all PMP preparation courses and books are either based on the PMBoK Guide Knowledge Areas or on the 5 process groups (Initiate; Planning; ...) but with the new PMP Exam, PMI introduces a new "Flow" to the generalist approach and thus to all methods, be they predictive, hybrid or agile.

This new "Flow" includes the following 5 steps:

- building a high-performing team
- starting the project
- doing the work
- keeping the team on track
- keeping the business in mind

Each of these levels contains various Tasks and Objectives and is mapped to the corresponding area and tasks in the PMP Exam Content Outline.

The New PMP Flow in 2021!



2.3 Question Patterns

Another change recently announced by PMI is the introduction of some new question types. PMI will introduce question patterns that differ from the traditional multiple-choice format with four options and a correct answer.

New formats include drag and drop and clicking a graphics area. These new question types allow you to ask questions such as how to select a chart that best fits a described scenario, or determine which part of an image applies to a described situation.

Chapter 3: Revision Notes

Activity On Node (AoN)

The AoN diagram is a representation of the project network logic as well as the crucial activities scheduled. It showcases how each work package is planned as a sequence of related tasks. If you viewed what a network diagram looks like, you might have noticed the nodes and lines connecting them. These lines have arrows that indicate how these nodes are related.

Adaptive Life Cycle

An adaptive life cycle can also be termed agile because of its ability to respond to change. Another term used here is, therefore, change-driven lifecycle. An adaptive life cycle responds primarily to high-level changes in a project.

Affinity Diagram

An affinity diagram is a simple tool that is generally used in project management to sort and organize a tremendous amount of business ideas, which is written down after a couple of brainstorming among team players.

An affinity diagram is a crucial project management tool that helps in creating a mind map that can be used in sorting all unorganized and unsorted ideas in business.

Agile Life Cycle

An agile life cycle is basically the same thing as an adaptive one. These life cycles depend on changes in the project. Deliverables in such a lifecycle are realized after several iterations wherein a detailed scope has been defined and approved for each repetition.

Agile Manifesto

This is a form of proclamation that articulates the four fundamental values and twelve principles that its authors believe every software developer should use as a prime guide in their work.

These core values include; individuals and the interactions over processes and tools to be applied, customer collaboration over the contract negotiation, working software over comprehensive documentation, and responding to

change over following a given plan.

Agile Suitability Filter

These are tools that are put in place to help in assessing if an agile approach applied is fit for a given organization or the project of interest. As with any other subjective diagnostic tool, these are not fast and hard predictors of sustainability or instead project success.

In fact, suitability filters are typically used as conversation starters. They are used in acquiring more objective discussions about the general use of agile methods and in the identification of risks.

Agile Unified Process

This is basically a simplified form of the Rational Unified Process (RUP) which was developed by Scott Ambler. Agile Unified Process describes a simple and easy to understand approach in developing business application software with the use of agile techniques and concepts but still remain real to the RUP

Analogous Estimating

Analogous Estimating is a method used in estimating a selection of project parameters and measures of scale. It is usually used to determine the overall size of a particular parameter when information is derived from historical sources from similar past projects.

Analytical Estimating

This is a method of estimating project expenses for each activity. It considers the estimates for time duration tasks will require to be completed.

After the estimation, these figures are added together, computing the total project variables for cost and time. It is considered a very accurate means of calculating these figures via evaluation.

Anti-Pattern

An anti-pattern is a logical response that evaluates recurring problems which are usually risky and ineffective when it comes to business prosperity. More so, these problems are considered extremely counterproductive.

When it comes to project management, anti-pattern play a vital role in ensuring the limitation of previous patterns which are likely to affect business

output. An anti-pattern is considered a powerful tool when it comes to team brainstorming.

Backlog

A backlog is a list of prioritized tasks that a project team needs within the scope of work in a project management success. The backlog is driven by all factors from road mapping of the project to the most crucial part of the project.

In project management, Backlog plays a vital part by ensuring that all development is connected to the server and the owner of the project.

Behavior-Driven Development

Behavior-driven development also abbreviated as BDD is a methodology used in the documentation of user behavior, which majorly focuses on team behavior in the formulation of business strategies.

Behavior-driven development is a powerful tool when it comes to project management. It helps in understanding the team member's ethnicity in undertaking a given task. BDD also helps in detecting test-driven development among team players in project management.

Blending Of Approaches

The blending of approaches from its definition is a classification of different Agile aspects to create a single element in project management. In short, a mixture of strategies is a hybrid methodology that paces way to managers in trying different angles in business prosperity.

The blending of approaches is a method that helps to create more room for expansion in business growth. Engaging in blended approaches exposes the business to different conditions hence high chances of survival within the market.

Bottom-Up Estimating

Bottom-up estimating is an estimation model that focuses on project timeframe and its cost. The method focuses on the WBS' lower-level components and their estimates. Normally, work within an activity is broken down to realize a greater degree of detail and hence, confidence in the overall result.

Burndown Chart

This is a graphical representation of tasks to be completed. The chart looks at the variations considering an ideal burndown. It also bases this consideration on the work dedicated to the repetition planned. It can also be used in any project management as long as the project contains measurable improvement over time.

Burnup Chart

A burnup chart is a tool used to track the amount of work completed as well as show the total amount of work for a project or iteration. The tool consists of two lines on the same chart, whereby the project will be completed when the two lines meet. Hence, it allows you to visualize a more realistic completion date for the project.

Capability Maturity Model (CMM)

This is a way of assessing the maturity of a business procedure and its capabilities as well. Initially, the CMM model was created to determine the capacity of a software development process. Nowadays, however, it is used across a range of industries.

Like other maturity models, CMM allows organizations to assess themselves and compare themselves with external standards. Once done, it will offer recommendations where improvement is required.

Capacity Measures

Capacity measures are the process of evaluating, matching, and allocating the available resources with the work to be completed. Capacity measures are not only applicable to business but also projects and industries.

Resources involved in Capacity measures can be either tangible or intangible, depending on the project. Capacity measures are the backbone of goal achievement in project management. Resource allocation helps in estimating the time taken in completion of a task hence prosperity of a project.

Colocation

Colocation merely is placing all resources of business under a single location. Colocation can also be used to mean placing all of the active section of the team players in one physical location. This way, their ability to fulfill the project requirements are improved.

Colocation strategies include meeting rooms for the team as well as other amenities that improve communication within the team. All this is aimed at

improving their work-rate and efficiency towards the project.

Context Diagram

Context diagrams refer to visual tools which display the scope of business products and the entire system. The context diagram helps in showing the relationship between other arrangements with the business. The context diagram portrays the input to the business and the leading players involved and also the output in the system.

Project management needs a visual tool for a business to succeed. Context diagrams are the only tools a company can use in displaying all its progress, and so, Context diagrams are crucial for project management.

Contingency Plan

Sometimes, Plan A doesn't do well, and you would need a Plan B. Now Plan B, or whatever you consider secondary to the initial idea is what is called a contingency plan.

It is usually the second plan of action after the primary one has failed. Most times, project managers plan for failures and risks, and this allows them to be in a position to mitigate the same dangers. However, in the case of dangerous fruiting, then a contingency is applied to minimize the effects of the risks that occurred.

Control Chart

In every experiment, there is a control experiment and the test. The control is used to measure variables against the test. This is precisely the same thing that a control chart does.

A Control chart compares results from processes in a project with other methods and their results which were obtained before them. This is done to measure whether their operation is stable or it has a performance that is predictable. There are upper and lower limits that indicate the maximum and minimum values that will be accepted for the project.

Control limits are pre-determined from statistical methods. These are used to establish the project's natural capability for a certain process. If the deviation between the two results is inconsistent with the project expectations, then corrective measures are employed.

Cost Aggregation

Cost aggregation refers to the division of all costs for a financial account to the individual package. It follows the WBS parameters. Work package costs are estimated then combined to higher levels of the WBS and finally for the whole project.

Cost of Quality

Cost of Quality, also known as COQ refers to the entire expenses needed in bringing a set of products and services into a business. Value of Quality is evaluated by professional project managers who are assigned the project. Loss of quality is simply a combination of the cost of conformance and the cost of non-conformance.

Project management relies equally on the cost of quality in weighing business limits. Loss of quality tactics in project management helps in completion of activities at the required time.

Cost Performance Index (CPI)

A CPI compares the finances of a project versus its effectiveness as well as efficiency in production. Simply put, it showcases how each unit of cost has been spent on a given piece of work done. The CPI will show whether the project is performing well with the given budget or dismally.

Generally, a CPI will compare the earned value versus the actual cost.

Cumulative Flow Diagram

This is an analytical tool, which allows you to visualize your efforts and project progress. It visualizes how tasks mount up over time, as well as your distribution along the process stages. In most cases, the Cumulative Diagram Flow is fundamental when using Kanban as a custom project method. For efficiency purposes, the graph is built from different colored bands of tasks gathered in various columns.

Crashing

Crashing is a compression technique that is used in reducing a project schedule. Smashing technique is done through the addition of additional resources within a specific critical path. Every time a crash report is generated, a different essential way might be recorded hence changing the result of the entered project.

Critical Path Method (CPM)

The CPM provides a technique whereby functions for project activities are

written. The method allows a step-by-step analysis of critical and non-critical tasks. The program is used when planning for the critical, or rather, the shortest path meant to achieve an intended goal.

Decomposition

Decomposition is a technique used in business to decompose all your project strategies to expound on the most crucial parts in the project. The method divides the project scope and deliverables of the project into smaller units that are easily manageable.

Generally, decomposition presents the total work into work packages following certain laid-out criteria.

Delphi Technique

Project managers and stakeholders employ a range of estimation techniques to assess costs and other variables. One such estimation technique is the Delphi technique, which is typically founded on a consensus by the experts.

It is quite straightforward actually as the experts in the field come up with their own estimated figures. Afterward, they will have a sit-down and discuss their results and come up with one which everyone agrees with as the best estimate they can find.

Discrete Effort

This is work within a project that is associated directly with the elements of a work breakdown structure. It is a measurable parameter and forms part of the earned value management.

Dynamic Systems Development Method

Product development methods are varied depending on their form and dynamism. The dynamic systems development method is for one, an agile technique. Like other agile techniques, it conducts its development actions by a series of repetitions. The method also incorporates user-centered improvements.

This method operates a fixed cost and time model and makes use of the MoSCoW prioritization method when identifying requirements.

Earned Value Management

Earned value management is a method of measuring project performance. The procedure considers the project scope, time, and cost.

Earned value management integrates the scope starting point with that of the cost as well as the same for the schedule. By doing so it will be able to come up with the performance baseline.

Extreme Programming (XP)

Firstly, this is an agile method of software development. It emphasizes a high degree of responsiveness where customer demands are given precedence. Because of this, short development cycles and frequent releases are the norms.

Face-To-Face Pairing

Look-to-face pairing refers to project imply sittings. In every business, look-to-face pairing is a tactic used to ensure the healthy progress of a project within a company. Look-to-face pairing involves the physical availing of all team leaders and the employee on the same table presenting and discussing the progress of the project.

According to market analysis, face-to-face is a powerful strategy when it comes to business prosperity.

Fast-Tracking

Fast-tracking is a technique used in project management to accelerate activities. Typically, operations that were to be performed sequentially under an original schedule are performed simultaneously in parallel.

In other words, fast-tracking is applied when the project manager wishes to step-up the schedule; in situations of crisis or rather crunch time.

Feasibility Study

The economic sense of any project needs to be studied. A feasibility study ascertains the financial practicality of moving forward with such a project. Stakeholders would like to know how their money will be used to generate more money. Because in the end, it is all about profits.

A feasibility study gives these people the confidence to invest their money in a project without worrying about losses.

Feedback Loops

Feedback loops refer to the driving factors of the agile framework in agile methodology. Feedback loops act as an opportunity in increasing project productivity. Productivity encounters may be an individual or teamwork.

In project management, Feedback loops help the interaction and performance

of the entire project. Without feedback loops, then the project will not utilize its resources completely.

Fishbowl Windows

Fishbowl windows is an open dialog which is used in discussing projects among a large group within an organization. Fishbowl windows at times are also used for unconfereces for pictorial purposes.

In project management, Fishbowl is convenient in that all staff and personnel participate in project contribution. More so, all the participants are controlled and monitored by the project managers.

Fit For Use

In project management, the term fit for the purpose also known as fitness for use refers to the effectiveness of a design, system, or manufacturing method used in delivering goods and services that meets the customer's defined purpose, under the anticipated or specified operational conditions.

Fixed Priced Micro-Deliverables

Fixed price or fixed fee micro deliverables refer to the narrative description of a project's work requirement. For instance, it defines the project's budget, project-specific activities, timeline, and deliverables. Whereby, the vendor is committed to delivering everything that is written in the SOW within the allocated time frame, under the specified budget.

Funding Limit Reconciliations

In every project, cost budgeting is the critical factor to attaining the project's goal. Therefore, funding limit reconciliation is a business tool used in project management to control the flow of resources allocated for a given project.

Funding limit reconciliation helps in fund adjustments and fund limitation within a project, and therefore, funding limit reconciliation is a convenient tool for project management.

Graphical Evaluation and Review Technique (GERT)

GERT is a network analysis technique that makes use of the Monte Carlo simulation to bring about a probable approach to network logic and the formation of time estimates. It is an alternative to the PERT technique. However, it is not often used for complex systems.

Hybrid Life Cycle

The hybrid life cycle is the entire recording of all the phases involved in the project. A hybrid life cycle combines both predictive and adaptive life cycles. Predictive life cycles are those elements of the project that are well-known, whereas those elements that are still in the evolution process follow the adaptive life cycle.

Impact Mapping

This is a strategic planning technique that involves visual mapping for product and project development; it was developed by Gojko Adzic. It includes the collaborative efforts of the project manager and the technical staff in the creation of a visual mind map to answer the fundamental question of the goals, impacts, actors, and deliverables of the project. This visual map is often referred to as an impact map; hence, the term impact mapping.

I-shaped

This is a term that is designated to a certain kind of person. Normally, in any project, some people are well-versed in individual sections of the same. These guys will rarely lend their time and kill to do anything else. Such a person, in agile spheres, is called an I-shaped person.

The opposite is T-shaped. These guys lack the same depth but make up for it with their vast array of skills. They can do just about anything.

Iteration

This is a concept coined from iterative software development. It specifies a fixed time cycle for development work, which is usually a few weeks long. The life cycle of development consists of a few iterations or simply repetitions. The iterative development process prioritizes time over scope. Thus there are seldom any particular requirements to be achieved in an iteration.

Iterative and Incremental Development

Iterative and incremental development is simply a combination of both iterative and incremental development models. It was established as an alternative to the waterfall model.

The general idea was to develop a system through repeated cycles (iterations) and incrementally in small proportions. This way, the development team can learn from their mistakes and improve in future iterations.

Kanban Board

A Kanban board is a workflow visualization tool that allows project managers to optimize. Physical Kanban boards usually play the role of communicating progress, status, and other issues. The system is highly visual and thus allows teams to communicate more effectively on the work needed to be done and when it is due.

Kanban Method

This refers to a project management technique that pays attention to continual delivery while not overburdening the development team. Kanban is a Japanese word that loosely refers to a visual signal; as such, the Kanban method achieves its objectives by allowing managers to visualize their workflow for a given project.

Lead Time

The lead time is the amount of time a given activity can be brought forward concerning the activity that it is dependent upon.

Lean Approach

The lean approach is a means of process improvement essentially. In essence, organizations should be able to consider maximizing their product value and move the same towards their consumers.

Firstly, any products from the project process should conform to the client's requirements. This might mean lowered cost and at the same a higher quality of the product. To attain this, the project must embrace flexibility.

Linear Sequential Model

The linear sequential model moves through the phases of a project's life cycle both systematically and following a set sequence. Ordinarily, it is used for smaller projects with forthright requirements. This is because sequential development makes it harder to review the design based on testing and preliminary feedback.

Matrix Organization

A matrix organization can be defined as a diagram that is used to indicate the strongpoints of relationships between various factors, their causes and overall objectives for the entire spectrum. This will consider the rows and columns in which these items are found.

Minimum Viable Product (MVP)

This is a product aimed at the customer's short-term satisfaction. In essence, this methodology must be employed so that the project process can gain feedback from their product as early as possible. That is why the product may seem to satisfy the early needs and wants of customers.

MVP is a cost-effective way of gaining consumer insight into a product. This beats making a sample with an extensive list of features then releasing the same for your customers to sample.

Monte Carlo Simulation

Monte Carlo simulation refers to a computer-based technique that performs probabilistic forecasting of possible outcomes to aid in decision making. The simulation gives project managers a range of possible outcomes and the likelihood that each will occur.

Net Present Value (NPV)

The net present value is a discounted measure of project valuation. It compares the current value of an existing project against the current value of costs as a result of the project. The NPV can then be either positive or negative.

Nominal Group Technique

This is a problem-solving method whereby members of a group make their contributions to the cause. In essence, it involves brainstorming from several individuals whereby each person is encouraged to make their contribution to the problem.

Usually, team members are allowed to make their contribution towards the problem-solving effort. Afterward, a selection process ensues where a discussion on the various solutions goes on. The highly ranked solutions are then prioritized for selection.

Mobbing

Unlike most of the terms discussed here, mobbing refers to a rather dark part of the project process. It means bullying of an individual by a group. It usually occurs in the workplace and in this case, within the project team.

Network Diagram

This is simply a diagram that shows how different elements of the project are connected. The diagram indicates the hierarchy of various items. Additionally, the diagram ensures that you follow the items in the order in

which they depend on each other. This network helps others to understand the projects of different elements and how they are related.

Order of Magnitude Estimate

The order of magnitude estimate provides a vague but early idea of the time and money needed to complete a project. It makes the most use of historical data from previous projects to form estimates for similar projects.

Organizational Breakdown Structure (OBS)

This is a depiction of the organization of any project. It shows who heads different departments and who are their subordinates. The OBS considers the project's deliverables in accordance with the existing departments, teams, or units. The project work packages or activities are also included in the list.

Parametric Estimating

This is a technique for determining duration and cost. Basically, it will help determine the resources that are needed to fulfill any project activity. It is generally based on the use of historical data to establish the relationship between variables- for example, calculating the unit costs as well as the number of units required to complete the activity.

Paint Drip

This is not a project management term per se, but rather a form of artwork. It merely involves paint being poured onto a canvas. The colors can also be dripped onto the surface as well.

Phase Gate

This is an end-of-phase checkpoint where the project managers or instead leadership reviews the progress to decide whether to revisit the work done in the previous phase, end the project, or ultimately continue to the next stage.

PERT Chart

Also known as a Project Evaluation Review Technique, a PERT chart is used for the formation of tasks, the establishment of the same and their organization. The PERT chart is a diagram, and other diagrams can be formed inside it. These may include a precedence diagram and a network chart.

PERT Estimation

The Project Evaluation Review Technique estimation is used to compute the

time and expenses of a project. It is based on analyzing the worst-case situation that might occur for the activity in question.

Pessimistic Duration

This is an estimate of the most extended amount of time that is needed to complete a specific activity or task. Project managers consider this to plan for the worst possible scenario.

Precedence Diagramming Method (PDM)

It refers to the process of constructing a project schedule network diagram. It shows the order in which activities must be performed by using nodes to represent activities and arrows to indicate their dependencies; it, therefore, illustrates the logical relationship between project activities. Activities' relationships are represented graphically showing the sequence in which the activities are carried out.

Predictive Approaches

Such approaches help realize a predetermined end-result for a project process. Since different project outcomes may be predictable or otherwise, project managers must consider their strategies carefully.

A predictive approach is favored, where the result is known. Such an approach gives a specific and linear development structured plan that adheres to a strict time frame.

Product Backlog

This is essentially a decomposed scope of an adaptive project. The scope has been broken down into a set of requirements which are then called the product backlog.

RACI Chart

The acronym stands for responsible, accountable consult and inform. It is a matrix inside which all meaningful actions of the project are contained. The chart hosts the work to be done on the left column represented as activities whereas assigned resources are either grouped or indicated individually. This chart is when assigning roles and responsibilities in the case of internal or external resources.

Rational Unified Process (RUP)

This is a software development process. It is derived from Rational, a section of IBM that divides development into four sections. These sections center

around business modeling, design, analysis, and implementation. The four parts are specifically inception, elaboration, construction, and transition.

Remote Pairing

Today, a lot of work is done remotely. The bosses needn't have to be present physically. This is simply what remote pairing stands for. This is where managers will link up with the actual project team and provide guidelines for the project's processes. All this is done without them ever setting foot on the project premises.

Requirement Traceability Matrix

This is a means of tracking the project's requirements throughout the lifecycle of the said project. This way, the project will be able to deliver on its stipulated.

Resource Optimization Technique

Resource optimization can be used to change start and completion dates depending on the project duration or if a limiting resource is considered. This is a method that is used to merge the supply and demand of a resource meant for a project's needs.

Retrospective

The retrospective is a project meeting with the entire team meant to deliberate on certain items. These include the successful elements of the project phase. In case of failure, then what could be done to improve the same. Also included in the discussion are the factors and items to be incorporated and added to future projects as well as the organization's process assets.

Return on Investment (ROI)

Expected financial benefit due to a project expressed as a percentage of the total project investment. It is usually used to assess the overall profitability of a given project.

Risk Appetite

This is the amount of risk that an association is willing to stomach as they look forward to the gains. It is different to risk tolerance because tolerance is the performance variation an organization is prepared to accept.

Risk Mitigation

Before commencing any project, one needs to understand the risks that face

the project. Doing so can help prevent any damages that may occur because of such instances. During risk mitigation, action is taken to bring down the probability of a threat occurring. It is effective as compared to allowing the threat to occur then correcting the damage.

Rolling Wave Planning

Sometimes, deliverables or subcomponents may not be decomposed, especially the latter that is set to be accomplished in the future. Normally, a project management team will wait upon the agreement of deliverables and subcomponents so as to effectively develop a WBS. Such a technique is known as rolling wave planning.

Scope Creep

This is used to refer to the variations in project scope. These changes normally take place without an official change in the actual extent of the project taking place. The scope creep is always considered counterproductive because any changes that haven't been approved will have an impact on the cost. This impact is almost always on the negative side; hence, the negative reflection of scope creep.

Scrum Of Scrums (SoS)

Scrum of scrums is a scaling mechanism. In simple terms, it is used in scenarios with several scrum teams. This way, the mechanism gives those involved a way to collaborate in planning for the project at hand.

Doing so means that teams will be able to map out where conflicts occur and where integration is possible as a way of ensuring project success.

Serial Life Cycle

Unlike a long life cycle, the serial lifecycle suits routine-type projects. These are projects whose needs are intimately understood. If a project's model aims to fulfill a well-known problem and its requirements are precise, then the serial model suits it. It can also be referred to as the waterfall model.

Servant Leadership

To put it simply, servant leadership revolves around serving. It is different from traditional leadership whereby the latter focuses on the success of the organization mainly. A servant leader will often put his/her servant's needs forward instead of the company. This goes against satisfying the customer as the primary objective of any production process, but rather, satisfaction is

accorded to those involved.

It can and cannot negate the gains from production depending on where you stand. On the one hand, a motivated team performs well and produces quality products. On the other hand, too much emphasis on the team's satisfaction negates gains made from product quality.

Schedule Performance Index (SPI)

The ratio of planned value to the rate of earned value at a given point in time. It is meant to show whether or not the project is running according to schedule or not. If the SPI is less than one, then the project is behind schedule while if the SPI is more significant than one, the project is running ahead of schedule.

Siloed Organization

Any siloed organization is frowned upon mainly because of one shortcoming. That is the lack of information diffusion. A silo-mentality means that the organization in question does not want to share information with the other departments involved in the project process. This reduces efficiency aimed at improving production and the overall quality of the product.

Single-Loop Learning

It is also known as incremental learning. single-loop learning is the type of learning that takes place within an organization to fix a given problem. It is one of the most basic forms of learning within an organization and aims at solving organizational issues without altering the structure.

Smoke Testing

The term smoke test is often thrown around in the software development scene; it is also known as build verification testing. In simple terms, the test is applied to a given 'build' mostly software to verify whether all the essential features are functional. It is meant to ensure that the build is ready to be subjected to further testing; this is to reduce wastage from premature testing.

Stacey Complexity Model

Formulated by Ralph D Stacey, the Stacey complexity model categorizes tasks into significant categories; complex, simple, complicated, and anarchy. The model further goes on to explain the best approach to solve those tasks for each group.

Statement Of Work (SoW)

An explanation of work defines the method used in a specific project. Projects often involve several tasks whereby there are specific activities used to actualize these tasks. The SoW now defines these tasks. An SoW is often developed from the project scope starting point. It will only include the part of the project scope that shall be included with the related contract.

SWOT Analysis

One of the standout terms when it comes to project management is a SWOT analysis. If you are in project management, this is a critical piece of information that you must have. In full SWOT stands for Strength, Weakness, Opportunities, and Risks. These are essential factors project stakeholders consider first before commencing on the project. The SWOT analysis ensures that you will have all the prerequisite information about the project before you begin. This way, you will be adequately prepared to finalize the plan with ease.

Task Switching

Task switching refers to the executive function that involves the ability to shift one's attention between one task and another unconsciously. In project management, it is often seen as a drawback to productivity since your progress is hampered by the many things vying for one's attention.

Technical Debt

This is a concept often used in software development to mirror the implied cost of work as a result of choosing a limited solution over a long-term solution. Simply put it is the result of prioritizing a speedy delivery for a project over a perfectly executed project.

Test-Driven Development (TDD)

TDD is a software development practice that follows a particular cycle; first, a software developer writes an initial automated test scenario that defines the desired improvements to be made, the developer thus produces the minimum amount of code needed to pass the test, and a new code is developed with the acceptable standards.

Three-Point Estimation

A single-point estimation may not be considered accurate and hence a project team ought to consider uncertainties and risk estimates. The three-point

estimation techniques will aid in defining the estimated range for the project's timeframe.

Triple Constraint

Any project, however thoroughly planned, still faces some constraints. However, there are a couple of significant obstacles that almost always feature on any project. These limitations are grouped into four and are drawn in a triangle. The triangle represents the problems that always affect project successes. Nevertheless, these restraints are always balanced, and when one overcomes another, there will be an imbalance and a collapse in the projects.

Value Stream

A value stream refers to the entire set of steps from the start of your commodity creation up until the delivery of the final commodity to the customer. In project management, it is deemed most profitable to have a small and efficient value stream.

WBS (Work Breakdown Structure)

A WBS, or rather a Work Breakdown Structure is a game plan for any project implementation process. Typically, it is a diagrammatic representation of the steps that project personnel often undertake to manage a project until the end.

Every WBS will indicate the different levels of the project and their significance. Moreover, the objective of the said project is a primary agenda and is included in the first WBS level. Other secondary purposes follow the primary goal but with a lower rank.

These are among the standard terms concerning project management, which you will hear with project management officials and staff and their prospective teams. Understanding them is key to successful project implementation and finalization in the end.

Chapter 4: Formulas and Calculations

Range of / Expected Activity Duration (EAD)

The Range of Activity Duration, also known as the Expected Activity Duration, is the length of time that it is still expected for the project to take. The formula for calculating EAD is expressed as follows: **$EAD = O + P + (4L)/6$**

In this formula, the Expected Activity Duration is equivalent to one-sixth of the sum of the Optimistic Value plus the Pessimistic value plus four times the Most Likely Value. Let's look at an example.

Sample Calculation

A project has an Optimistic Value of 5 weeks, a Pessimistic Value of 20 weeks, and a Most-Likely value of 10 weeks.

Expected Activity Duration = Optimistic + Pessimistic + (4 Most-Likely) / 6

$$EAD = 5 + 20 + (4 \times 10) / 6$$

$$EAD = 25 + 40 / 6$$

$$EAD = 65 / 6$$

$$EAD = 10.83$$

The expected Activity Duration remaining for this project is 10.83 weeks.

Total Float

The Total Float is the total amount of time that a project may be delayed without delaying the end date of said project, or violating a schedule constraint/requirement.

It is expressed as follows:

$$TF = LS - ES$$

$$TF = LF - EF$$

As you can see, the Total Float may be calculated in one of two different ways. The Total Float is equivalent to the Late Start minus the Early Start.

The Total Float is also equivalent to the Late Finish minus the Early Finish. Let us look at an example.

Sample Calculation

A Project has an early start day of 2 and a late start day of 7.

Total Float = Late Start – Early Start

$$TF = LS - ES$$

$$TF = 7 - 2$$

$$TF = 5$$

So, in this case, the project may be delayed by a total of five days without delaying the end time of the project.

Communication Channel Formula

The Communication Channel is defined as the number of ways in which information can flow within the project. The Project Manager is always the center of the Communication Channel and, depending on the way that the project is structured, the information can flow in any direction.

The equation for the Communication Channel is: **$C = n(n - 1) / 2$**

In this equation, the Communication Channel equals half the total number of stakeholders multiplied by the number of stakeholders minus one. The variable 'n' is the placeholder for the number of stakeholders in the project. A stakeholder is defined as any team member, sponsor, project member, or any other person with a direct interest in the outcome of the project.

Sample Calculation

A project has 28 shareholders.

Communication Channel = Number of Stakeholders (Number of Stakeholders -1) /2

$$C = 28(28-1) / 2$$

$$C = 28(27) / 2$$

$$C = 756 / 2$$

$$C = 378$$

Therefore, in this particular case, there are 378 possible ways for information to move within this company.

Standard Deviation (SD)

The Standard Deviation is represented by the Greek letter sigma: σ . It is an analysis tool that measures the variation from the average.

SD is calculated using the following formula: $\sigma = (\text{Pessimistic} - \text{Optimistic}) / 6$

The Standard Deviation is equivalent to one sixth of the total of the Pessimistic Outlook minus the Optimistic Outlook.

Sample Calculation

A company has a Pessimistic Outlook of 20, and an Optimistic Outlook of 5.

$$\sigma = (\text{Pessimistic} - \text{Optimistic}) / 6$$

$$\sigma = (20 - 5) / 6$$

$$\sigma = 15 / 6$$

$$\sigma = 2.5$$

In SD, a low value means that the points are distributed close to the mean/average. A high value means that the points are distributed far from the average.

PERT Formula

PERT is an event-based, 3-point estimation technique – it uses the weighted average of the Pessimistic Duration, Optimistic Duration, and Most-Likely Duration to give an accurate estimate. Before we continue, let's define these three terms:

Optimistic – An estimate, taking into account all the known variables, of the shortest possible duration for the project

Pessimistic – An estimate, taking into account all the known variables, of the longest possible duration for the project

Most-Likely – An estimate, taking into account all the known variables, of the Most-Likely duration for the project

Beta can be calculated as follows: $\text{Beta} = (\text{P} + 4\text{M} + \text{O}) / 6$

Beta equals one-sixth of the sum of the Pessimistic value, the Optimistic Value, and the most likely value (multiply the Most Likely Value by 4 first).

Sample Calculation

A Project has an Optimistic Case of 12 weeks to finish, a Pessimistic Case of 32 weeks, and a Most-Likely Case of 18 weeks.

$$\text{Beta} = [32 + 4(18) + 12] / 6$$

$$\text{Beta} = (32 + 72 + 12) / 6$$

$$\text{Beta} = 116 / 6$$

$$\text{Beta} = 19.3$$

Therefore, the most likely result is that the project will take around 19.3 weeks to complete.

Risk Priority Number (RPN)

The Risk Priority Number is a numerical assessment of the risks attached to a process.

RPN is expressed as follows: **RPN = Detection × Occurrence × Severity**

Severity: a subjective expression of how badly the user will experience the effects of a failure. It is expressed on a scale of 1-10 – (1 = best / 10 = worst).

Occurrence: a subjective estimation of how likely an error is likely to occur. Again, it is expressed on a scale of 1-10 – (1 = best / 10 = worst).

Detection: a subjective estimation of how effective the controls are at detecting failures and the causes of failures before products/services reach the users. It is also expressed on a scale of 1-10 – (1 = best / 10 = worst).

A company releases a laptop with several potential software glitches. They won't be fatal to the laptop, but if they occur they will be frustrating to the user. We'll assign a severity rating of 5. The chances of the controls finding the failures or reasons for the failures is quite good, so we'll give it a Detection rating of 2. The chances of the glitches actually occurring after the product has reached the user is also quite low, so we will assign an Occurrence rating of 2. The Risk Performance Number will be determined as follows:

Sample Calculation

Risk Performance Number = Detection × Occurrence × Severity

$$\text{RPN} = 2 \times 2 \times 5$$

$$\text{RPN} = 20$$

Risk Priority Numbers have a scale of 1 – 1000.

A score of 1 is the absolute best risk score, with essentially no risk; and a score of 1000 is the absolute worst risk score, being riddled with risks.

This Project only has a Risk Score of 20, so the risks really aren't that bad. Remember that all the scales used for RPN are estimates, so RPNs can be mistaken and shouldn't be viewed as more than an estimate.

Return on Investment (ROI)

The Return on Investment tool is used to calculate the profit or loss gained by an investment – dependent on the amount that has been invested.

Return on Investment is expressed as a percentage, and helps to determine the viability of a project or investment.

It is expressed as: **$ROI = (\text{Net Profit} / \text{Cost of Investment}) \times 100$**

Sample Calculation

An investment in a Hotdog Stand has earned a profit of \$5000, and had an initial investment of \$500.

$\text{Return on Investment} = (\text{Net Profit} / \text{Cost of Investment}) \times 100$

$ROI = (0.1 / 5000) \times 100$

$ROI = 0.1 \times 100$

$ROI = 10$

In this case, you have an ROI of 10%, meaning there is a 10% return on your investment.

Payback Period

The Payback Period is the amount of time that it takes for an investment to recover or earn the cost of the initial investment. It is expressed by the following equation:

$\text{Payback Period} = \text{Initial Investment} / \text{Periodic Cash Flow}$

Let's look at an example.

Sample Calculation

A Hotdog Stand has an Initial Investment of \$500. It earns \$50 a day. Its Payback Period would be calculated as follows:

$\text{Payback Period} = \text{Initial Investment} / \text{Periodic Cash Flow}$

Payback Period = 500 / 50

Payback Period = 10 days

Benefit-Cost Ratio (BCR)

The Benefit-Cost Ratio is an indicator that is used to compare the overall value – expressed in monetary terms – of a project's costs as compared to its benefits. It expresses the project's overall value. BCR is expressed as follows: **BCR = Benefits/Costs**. The Benefit to Cost Ratio is calculated by dividing the Benefits by the Costs.

Sample Calculation

A project is currently spending \$1,000,000 per annum on necessary expenses and is earning a revenue of \$1,500,000.

Benefit-Cost Ratio = Benefits / Costs

BCR = 1,500,000 / 1,000,000

BCR = 1.5

With this indicator, the results may be read as follows:

BCR = 1 – the project will break even.

BCR < 1 – the project is failing, and not considered a good investment.

BCR > 1 – the project is doing well, and considered profitable.

In this case, your project will be considered as profitable.

Target Price

The Target Price is the sum of the Target Cost and the Target fee. If the Target Price is reached, the Buyer and Seller share the profits as per the cost-sharing document. If the Target Price is not reached, the profits are shared as per their profit-sharing agreement.

The Target Price is expressed as follows: **Target Price = Target Cost + Target fee**

Sample Calculation

A Project wishes to earn \$5 profit per item, and the item has a cost of \$20 to manufacture.

Target Price = Target Cost + Target fee

$$\$25 = \$20 \text{ cost} + \$5 \text{ fee}$$

Point of Total Assumption (PTA)

The Point of Total Assumption is a term related to Fixed Price Incentive Fee contracts. It is the point at which the profitability of the Seller decreases. The PTA is expressed as follows:

$$\text{PTA} = [(\text{Ceiling Price} - \text{Target Price}) / \text{Buyer's Share Ratio}] + \text{Target Cost}$$

Let's look at an example.

Sample Calculation

Target Cost: 2,000,000. Seller's Profit Target: 150,000

Target Price: 2,150,000 – The Target Price is equivalent to the target cost and the seller's profit target

Ceiling Price: 2,200,000 – the Buyer will not pay anything beyond this point

The Share Ratio (Buyer to Seller) is – in percentages – 70/30 on over-runs, and 50/50 on under-runs.

$$\text{Point of Total Assumption} = [(\text{Ceiling Price} - \text{Target Price}) / \text{Buyer's Share Ratio}] + \text{Target Cost}$$

$$\text{PTA} = ((2,200,000 - 2,150,000) / 0.70) + 2,000,000 = \$2,071,429$$

The Point of Total Assumption – where the transaction ceases to be profitable for the seller – for this transaction is \$2,071,429.

Earned Value (EV)

Earned value is a measure of the amount of work that has been performed – in terms of said work's authorized budget. Calculating Earned Value is a way of monitoring your project plan to check the project status – If you know the Earned Value of your project, you know whether or not adjustments are required, and by how much adjustments need to be made.

$$\text{EV} = \% \text{ complete (man-hours worked / total man-hours needed for project)} \times (\text{BAC})$$

The Earned value is equivalent to the number of man-hours worked divided by the total amount of man-hours needed for the project multiplied by the

project's completed percentage- Multiplying this total by the Budget at Completion will deliver the final result.

Sample Calculation

A company which needs to work a total of two hundred man-hours to complete their project has worked a total of fifty man-hours. The Budget at Completion is \$1000.

Earned Value = % Complete × Budget at Completion

Earned Value = (man-hours worked × 100 / total man-hours needed × 100) × Budget at Completion

EV = (50/200 × 100) × 1000 – Division first

EV = 25 × 1000

EV = \$25,000

In this case, the project has an Earned Value of \$25,000.

Net Present Value (NPV)

Net Present Value is a tool for calculating the current net-worth of a project. The formula for calculating NPV is represented as follows: **NPV = Value / (1 + r)^t**

In this equation, t is the predefined period of time, r is the rate of discount (typically the inflation rate), and Value is the Value of all the benefits.

Sample Calculation

A company invests \$20,000 in a project that will yield profits of \$100,000 in a timeframe of 2 years. The project has an inflation rate of 5%.

NPV = Value / (1 + r)^t

NPV = 100,000 / (1 + 0.05)²

NPV = 100,000 / (1.05)²

NPV = 100,000 / (1.054)

NPV = 94,876.6

The Net Present Value of the Project is \$94,876.6

Present Value (PV)

Present Value calculates the current worth of an investment that will produce

a lump sum at some point in the future. The Equation for Present Value is expressed as follows:

$$\text{Present Value} = \text{Future Value} / (1 + \text{interest rate})^{\text{Time Period}}$$

Sample Calculation

An Investment has a future value of 1,000. It is a 24-month investment and has a 2% annual interest rate.

$$\text{Present Value} = \text{Future Value} / (1 + \text{interest rate})^{\text{Time Period}}$$

$$\text{Present Value} = 1,000 / (1 + .02)^2$$

$$\text{Present Value} = 1,000 / 1.02^2$$

$$\text{Present Value} = 1000 / 1.0404$$

$$\text{Present Value} = 961.97$$

And, there you have it, your investment has a Present Value of, or is currently worth, \$961.97

Future Value (FV)

Future Value is the value of an investment at a point in the future based on the rate at which that investment is estimated to grow. The equation for Future Value is expressed as $FV = PV(1+i)^n$

$$\text{Future Value} = \text{Present Value} (1 + \text{Interest Rate})^{\text{timeframe of investment.}}$$

Sample Calculation

An Investment has a present value of \$1000. It is a 24-month investment and has an 8% annual interest rate.

$$\text{Future Value} = \text{Present Value} (1 + \text{Interest Rate})^{\text{timeframe of investment.}}$$

$$FV = 1000 (1 + .05)^2$$

$$FV = 1000 (1.05)^2$$

$$FV = 1000 (1.1025)$$

$$FV = 1102.5$$

In this case, your investment has a Future Value of \$1102.5 – after the 2 Year investment term, your investment will be worth \$1102.5.

Expected Monetary Value (EMV)

The Expected Monetary Value is a tool that is used to predict how much money will be made by making a specific decision. The formula for EMV is as follows: **EMV = Probability × Impact**

Let's look at an example.

You make a bet of \$10, with a friend, that you can pull a four out of a pack of cards on the first try without looking. Since there are 52 cards altogether, and 4 of each number, we can simplify the probability to be out of 13 – with each number counting as 1. This means that you now have a 1/13 probability to draw a 4 on the first try, and a 12/13 possibility to draw another number instead. How would we calculate the EMV?

Sample Calculation

First, calculate the EMV for a positive outcome.

EMV = Probability × Impact

$$EMV = 1/13 \times 10 = \$0.77$$

Then Calculate the EMV for a negative outcome

EMV = Probability × Impact

$$EMV = 12/13 \times 10$$

$$EMV = \$-9.23$$

Lastly, add the results of the positive and negative probabilities.

$$EMV = 0.77 - 9.23 = \$-8.46$$

In this case, the most likely result of your decision to make the bet is the loss of \$8.46 – in reality, we know that our friends would take all our money if we lose a bet, but that was a simple way of explaining how to calculate the EMV.

1st Formula for the EAC

This, the first equation for the EAC relies on the Budget at Completion and the Cost Price Index. The formula is as follows: **(EAC) = BAC / CPI** ie.. **Estimate at Completion = Budget at Completion / Cost Performance Index**. In this equation, the Estimate at Completion is equivalent to the Budget at Completion divided by the Cost Performance Index.

Let's look at an example.

In this particular example, we have a project with a Cost Performance Index of 1 and a Budget at Completion of 20,000. To calculate the Estimate at Completion, we will do the following equation:

Sample Calculation

Estimate at Completion = Budget at Completion / Cost Performance Index

$$EAC = 20,000/1$$

$$EAC = \$20,000$$

This project is estimated to have a cost of \$20,000 to complete all work.

2nd Formula for the EAC

The Second equation relies on the Actual Cost and the Bottom-up Estimate to Complete. The formula is as follows: **(EAC) = AC + Bottom-up ETC** ie..
Estimate at Completion = Actual cost + Bottom-up Estimate to Complete.

In this equation, the Estimate at Completion is equivalent to the Actual Cost plus the Bottom-up Estimate to Complete. Actual Cost is defined as the total of realized costs that have been incurred in order to fulfill the amount of work measured by the EV.

Sample Calculation

Let's look at another example – A company that has an actual cost of 10,000 and an ETC of 10,000.

Estimate at Completion = Actual cost + Bottom-up Estimate to Complete

$$EAC = 10,000 + 10,000$$

$$EAC = 20,000$$

This project is estimated to have a cost of \$20,000 to complete all work.

3rd Formula for the EAC

The third equation relies on the Actual Cost, the Budget at Completion, and the Earned Value. The formula is as follows: **(EAC) = AC + (BAC – EV)** ie..
Estimate at Completion = Actual Cost + (Budget at Completion –

Earned Value).

In this equation, the Estimate at Completion is equivalent to the Actual Cost plus the sum of the Budget at Completion minus the Earned Value.

Sample Calculation

For this example, our company will have a Budget at Completion of 20,000, an Actual Cost of 10,000, and an Earned Value of 10,000.

Estimate at Completion = Actual Cost + (Budget at Completion – Earned Value)

$$EAC = 10,000 + (20,000 - 10,000)$$

$$EAC = 10,000 + 10,000$$

$$EAC = \$20,000$$

This project is estimated to have a cost of \$20,000 to complete all work.

4th Formula for the EAC

This fourth, and last, formula for calculating Estimate at Completion relies on many more values.

It requires the Actual Cost, the Budget at Completion, the Earned Value, the Cost Price Index, and the Schedule Performance Index.

The formula is as follows: **(EAC) = AC + [(BAC – EV) / (CPI × SPI)]** ie..

Estimate at Completion = Actual Cost + [(Budget at Completion – Earned Value) / (Cost Performance Index × Schedule Performance Index)]

In this particular equation, the Estimate at Completion is a little bit more involved to arrive at. The first step is to subtract the Earned Value from the Budget at Completion, and to multiply the Cost Performance Index by the Schedule Performance Index. Divide the result of the first calculation by the result of the second calculation. The Estimate at completion is equivalent to the result of that final sum plus the Actual Cost.

It will be much simpler for us to look at an example. From our previous three examples, we know that our company has an Actual Cost of 10,000, a Budget at Completion of 20,000, an Earned Value of 10,000, and a Cost Performance Index of 1. We now only require the SPI, which is also 1.

Sample Calculation

Estimate at Completion = Actual Cost + [(Budget at Completion – Earned Value) / (Cost Performance Index × Schedule Performance Index)]

$$EAC = 10,000 + [(20,000 - 10,000) / (1 \times 1)]$$

$$EAC = 10,000 + (10,000 / 1)$$

$$EAC = 10,000 + 10,000$$

$$EAC = \$20,000$$

This project is estimated to have a cost of \$20,000 to complete all work.

Contract Formulae

Before we proceed with the Formulae to calculate the various contract types, we need to define some vital terms. These terms are cost and fee. In terms of contracts –

Cost is defined as the Expenditure of the Seller – whether Actual, Deemed, or Estimated.

A **fee** is defined as the Compensation earned, or deemed to be earned, by the Seller.

Cost Plus Percentage of Cost (CPPC)

In a CPPC Contract, there is no risk undertaken by the Seller. All expenses and risks are undertaken by the Buyer. In addition to this, the Seller also earns a percentage of the cost as a fee. The Contract is expressed as follows:

Contract = Cost plus a percentage of the cost as a fee.

For example, a European collector hires an expert to acquire a particular type of Persian Rug that the collector does not yet own. The Expert finds the Rug for sale for \$3,000. According to the CPPC contract between them, the Expert is entitled to 15% of the product cost as a profit. And so, the Collector (Buyer) pays the \$3000 for the rug, and an additional \$450 for the fee of the Expert (Seller). There is no risk to the Seller.

Cost Plus Fixed Fee (CPFF)

In the Case of a CPFF Contract, the buyer once again takes all the risks. However, because the contract is not based on a percentage of the product

cost, the fee is unchanging. There is a fixed, agreed-upon amount that gets paid to the Seller.

The contract is expressed as follows: **Contract = Cost plus fee of fixed amount.**

Let's use the same example as before.

A European collector hires an expert to acquire a peculiar type of Persian Rug that the collector does not yet own. The Expert finds the Rug for sale for \$3,000. According to the CPFF contract between them, the Expert is entitled to \$150 as a profit. And so, the Collector (Buyer) pays the \$3000 for the rug, and an additional \$150 for the fee of the Expert (Seller).

There is no risk to the Seller, but the Seller also cannot expect a change in their fee – it is fixed.

Cost Plus Award Fee (CPAF)

In the case of a CPAF contract, the risk is shared with the seller. The Buyer reimburses the Seller for the exact Product Price, plus an additional fee – which is dependent on the satisfaction of the buyer. This Contract is expressed as **Contract = Cost plus an award fee (some amount)**. Let us use the example of our collector for the third time.

A European collector hires an expert to acquire a particular type of Persian Rug that the collector does not yet own. The Expert finds the Rug for sale for \$3,000.

According to the CPAF contract between them, the Expert is entitled to be reimbursed for the product cost – as well as a profit that is dependent on the Collector's satisfaction.

And so, the Collector (Buyer) pays the \$3000 for the rug. If the Seller has found a particularly well-made, and well-preserved Persian Rug, he can expect a hefty tip – even as much as half of the product price. If, however, the Rug is Ragged and badly made the Seller cannot expect a large fee. Both parties share the risk.

Cost Plus Incentive Fee (CPIF)

In a CPIF Contract, the Buyer pays a small fee to the Seller, as an Incentive, and reimburses him for the Product Cost. This contract is expressed as Follows: **Contract= Cost Plus Incentive Fee**. Once more, we return to our collector as an example.

A European collector hires an expert to acquire a particular type of Persian Rug that the collector does not yet own. The Expert finds the Rug for sale for \$3,000. According to the CPIF contract between them, the Seller is only entitled to the reimbursement of the Product Cost, and a \$100 Incentive Fee. Both Parties carry risk.

4.1 Formulae for Indices

Cost Performance Index (CPI)

The Cost Performance Index is yet another tool to calculate cost efficiency – this time the efficiency of the budgeted resources – it is expressed as a ratio of EV to AC. The formula to calculate it is expressed as follows: **(CPI) = EV / AC** ie.. **Cost Performance Index = Earned Value / Actual Cost**.

In this Equation, the Cost Performance Index is equivalent to the Earned Value divided by the Actual Cost.

The Cost Performance Index has three possible results:

A Cost Performance Index of \$1 means that you are earning \$1 for every \$1 spent.

A Cost Performance Index of <1 you are earning less than \$1 for every \$1 you spend.

A Cost Performance Index of >1 means that you are earning more than \$1 for every \$1 you spend.

Sample Calculation

A Project has an Earned Value of \$15,000 and an Actual Cost of \$15,000.

Cost Performance Index = Earned Value / Actual Cost

$CPI = 15,000 / 15,000$

$CPI = 1$

This project has A Cost Performance Index of 1.

Schedule Performance Index (SPI)

The SPI is a measure of the schedule efficiency, and is expressed as a ratio of EV to PV.

SPI = EV/PV ie..Schedule Performance Index = Earned Value / Planned Value

The Schedule Performance Index is equivalent to the Earned Value divided by the Planned Value. The Planned Value is defined as the authorized budget that has been assigned to work that has been scheduled.

The Schedule Performance Index also has three possible results:

An SPI of 1 means that the project is exactly on par with the expected rate.

An SPI of <1 means that the project is behind the expected rate.

An SPI of >1 means that the project is ahead of the expected rate.

Sample Calculation

A Project has an Earned Value of \$15,000 and a Planned Value of \$15,000.

Schedule Performance Index = Earned Value / Planned Value

$SPI = 15,000 / 15,000$

$SPI = 1$

This project has a Schedule Performance Index of 1. To Complete Performance Index (TCPI). The To Complete Performance Index is a measure of the cost performance that the project has to achieve with the resources remaining to it in order to meet a specified goal.

This Index is expressed as a ratio – (Cost to finish work: Remaining budget). There are two different formulae for calculating the TCPI – depending on whether you have the BAC or the EAC.

How to calculate TCPI

Formula 1

The first formula to calculate the TCPI is as follows: **TCPI = (BAC – EV) / (EAC – AC).**

The To Complete Performance Index is equivalent to the (Budget at Completion minus Earned value) divided by the (Estimate at Completion minus Actual Cost).

A company has A Budget at completion of \$20,000, an Earned Value of \$10,000, an Estimate at Completion of \$25,000, and an Actual Cost of \$15,000. The TCPI would be calculated as follows:

Sample Calculation

To Complete Performance Index = (Budget at Completion – Earned Value) / (Estimate at Completion – Actual Cost)

$$\text{TCPI} = (20,000 - 10,000) / (25,000 - 15,000)$$

$$\text{TCPI} = 10,000 / 10,000$$

$$\text{TCPI} = 1$$

This project has a To Complete Performance Index of 1.

Formula 2

The second formula to calculate the TCPI is as follows: **TCPI = (BAC – EV) / (BAC – AC).**

The To Complete Performance Index is also equivalent to the (Budget at Completion minus Earned value) divided by the (Budget at Completion minus Actual Cost).

Using the same values as we did for TCPI formula 1, let's look at an example. In this case, the calculations will look like this:

Sample Calculation

To Complete Performance Index – (Budget at Completion – Earned Value) / (Budget at Completion – Actual Cost)

$$\text{TCPI} = (20,000 - 10,000) / (20,000 - 15,000)$$

$$\text{TCPI} = 10,000 / 5,000$$

$$\text{TCPI} = 2$$

This project has a To Complete Performance Index of 2.

Cost Variance (CV)

Cost Variance is defined as the amount of budgetary surplus, or budgetary deficit, at any given point in time – it is expressed as the difference in the values of the EV and AC. The formula for Cost Variance is as follows: **CV=EV-AC.**

In this equation, the Cost Variance is equivalent to the Earned Value minus the Actual Cost.

Sample Calculation

A project has an earned value of \$35, 000, and an actual cost of \$30,000.

Cost Variance = 35,000 – 30,000.

Cost Variance = \$5000

In this case, the project is under budget, meaning it is still well within an acceptable range. If the result is a negative value, then your project is over budget. If your result is a perfect Zero, then your project budget is precisely on par.

Schedule Variance (SV)

Schedule Variance is a measure of how far ahead, or behind, the project is of the planned delivery date at a given point in time – it is expressed as the difference between the Earned Value and Planned Value. The formula for calculating Schedule Performance is as follows: **SV=EV-PV**.

In this Formula, the Schedule Variance is equivalent to the Earned Value minus the Planned Value.

Sample Calculation

A project has an Earned Value of \$20,000 and a Projected Value of \$15,000.

Schedule Variance = Earned Value – Projected Value

Schedule Variance = 20,000 – 15,000

SV = \$5000

In this case, the project has a positive Schedule Variance of \$5000. Meaning it is \$5000 ahead of the projected value. If the result of the equation is negative, then it is behind the Projected Value, and if the result is zero then the project is on par.

Variance at Completion (VAC)

VAC is a projection of budgetary shortfall or budgetary surplus – it is expressed as the difference between the BAC and EAC. VAC is expressed as: **(VAC) = BAC – EAC**.

The Variance at Completion is equivalent to the Budget at Completion minus the Estimate at Completion.

Sample Calculation

A company has a Budget at Completion of \$30,000 and an Estimate at Completion of \$15,000.

Variance at Completion = Budget at Completion - Estimate at Completion

$$VAC = 30,000 - 15,000$$

$$VAC = \$15,000$$

Therefore, this Company has a Variance At Completion – in this case, a budgetary surplus – of \$15,000.

Estimate to Complete (ETC)

An ETC is another forecasting tool that predicts the costs required to complete the remaining work on a project. It is calculated using the following formula: **(ETC) = EAC – AC**. In this equation, the Estimate to Complete is equivalent to the Estimate at Completion minus the Actual Cost.

Sample Calculation

A project has an Estimate at Completion of \$15,000 and an Actual Cost of \$10,000.

Estimate to Complete = Estimate at Completion – Actual Cost

$$ETC = 15,000 - 10,000$$

$$ETC = \$5,000$$

The Estimate To Complete for this project is \$5,000.

Budget at Completion (BAC)

The BAC is defined as the sum of all the budgets that have been set in place to be used by the project. The Total Budget equals the Total activity cost estimates plus the Total contingency cost reserves.

Total Budget = Total Activity Cost + Total Contingency Cost Reserves

Sample Calculation

A company has a Total Activity Cost of \$10,000 and Total Contingency Cost Reserves of \$10,000.

Total Budget = Total Activity Cost + Total Contingency Cost Reserves

$$TB = 10,000 + 10,000$$

TB = \$20,000

This project has a Budget At Completion of \$20,000.

Chapter 5: PMP Practice Test 1 -Full Mock (180 Questions) - 2021

Question 1:

As a project manager for an IT organization, you are currently in the process of selecting a vendor to develop a new e-procurement software for your company. Several vendors are interested in bidding on the contract. During a bidder conference, one of the prospective suppliers asks about ownership of the intellectual property created from the project.

Which project document should have this information at this stage of the project?

- A. Seller proposals.
- B. Requirements documentation.
- C. Source selection criteria.
- D. Procurement contract.

Answer: B.

Explanation

Intellectual property rights are included among other non-technical requirements in the Requirements documentation (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 485.

Question 2:

During this stage of the project, the leader delegates. The team is highly creative with trust and openness. What stage of the Tuckman ladder your team has entered?

- A. Forming.
- B. Storming.
- C. Norming.
- D. Performing.

Answer: D.

Explanation

In the **Performing** stage, Leader delegates.

The Leadership style in each stage of the Tuckman ladder is as follows:-

Forming –Directive, **Storming** –Coach, **Norming**- Facilitator

Performing- Delegation, **Adjourning**- Reassure and communicate.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 338

Question 3:

During the planning phase of a software development project, two team members conflict in determining a risk associated with server performance. According to one, server performance is a high probability risk, and it requires further numerical analysis. According to another member, the server performance is not a risk considering modern days' architectural setup.

What should the project manager do to address this situation?

- A. Conduct Quantitative Risk analysis.
- B. Use Why-why diagram to make a constructive decision.
- C. Use a Probability and Impact matrix.
- D. Develop a Risk Response Plan.

Answer: C.

Explanation

Risk Probability assessment considers the likelihood that a specific risk will occur. Risk impact assessment evaluates the potential effect on project objectives such as cost, schedule, quality, and performance.

Option C is the best choice among all given options, which is also a useful tool for Perform Qualitative Risk Analysis process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 423,425.

Question 4:

Project stakeholders can be internal or external to the project. Which of the following are examples of Stakeholders external to the organization?

- A. Customer, Sponsor.
- B. Suppliers, Regulatory bodies.
- C. Project Manager, Team members.
- D. Competitors. PMO.

Answer: B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)-550.

Question 5:

In a Project review meeting, the Project Manager and the team are involved in analyzing Scope Verification and any deviation in terms of the agreed-upon scope. What should the project manager review to fetch this particular information?

- A. Scope baseline.
- B. Requirements traceability Matrix.
- C. Requirements Documentation.
- D. Issue Log.

Answer: C.

Explanation

The above scenario indicates that the Project Team is doing Scope control and are involved in the Control Scope process.

Option D is outright rejected as it's not the input for the Control scope process. All other options, A, B, C are input to the Control Scope process.

Option A -Scope baseline is not correct because it helps to do Plan Vs Actual analysis to determine if corrective or preventive actions are necessary.

Option B - Requirements Traceability Matrix is also incorrect because it helps to detect the impact of any deviation from the scope baseline on the project objectives.

Here, the Project Manager should review the Requirements Documentation to detect any deviation in the project's agreed-upon scope or the product. The Requirement documentation describes how individual requirements meet the business need (agreed upon scope) for the project.

So, among all options, option C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 169.

Question 6:

A project team is identifying and documenting the actions necessary to be performed to produce the project deliverables. Which of the following may be created as a result of this process?

- A. Work breakdown structure
- B. The Scope Baseline
- C. Requirement Traceability matrix
- D. A Milestone List

Answer – D.

Explanation

The Question suggests that the Team is engaged in the Define Activities process.

Activity list, Activity attributes, Milestone list, are primary outputs created during the Define Activities process.

Among all options given above, D is the correct one.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 183-186

Question 7:

Suseether is a Project Manager of a large IT infrastructure project that is running in multiple geographical locations. He awards a contract to a vendor for implementing 80 servers in Europe. Senior Management asks for the work status update and the vendor performance comparison report in the middle of execution.

Which of the following tools should the Project Manager refer to prepare the vendor performance comparison report?

- A. Trend Analysis.
- B. A Performance Review.
- C. Inspection.
- D. Audit.

Answer- B.

Explanation

A performance review is a technique used to measure, compare, and analyze the actual performance of work in progress on the project against the baseline.

The question suggests that the Project Manager has to perform the Control Procurements process and use the data analysis tool- Performance Reviews to prepare a vendor performance comparison report against the agreement. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 498.

Question 8:

A Senior Manager of the ABC company has disapproved a change request already reviewed and approved by the CCB. What document should the project manager review to know whether the Senior Manager has the authority to do the needful?

- A. Review of the Stakeholder Register.
- B. Review the RACI Chart.
- C. Review the work breakdown structure.
- D. Examine the change Management Plan.

Answer – D.

Explanation

The Change Management Plan provides the direction for managing the change control process and documents the roles and responsibilities of the Change Control Board (CCB). The change management plan also defines each committee member's role and level of authority within the group and committee rules and decision-making procedures. It also specifies the number of people needed to pass a vote, who has the power to break ties when voting on change requests, and who may have the ability to overrule the committee in some instances.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 116, 120.

Question 9:

An Agile coach explains to the team in a meeting about a technique that naturally helps to minimize the scope creep as it forces the team to work on the essential features first.

What technique did the Agile coach mention here?

- A. Scope Verification
- B. Three-point estimation
- C. Time-boxing
- D. Value analysis

Answer – C.

Explanation

Time-boxing helps to minimize scope-creep as it forces the teams to process essential features first, then other features when time permits.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 182.

Question 10:

In case of early contract termination, Buyer can terminate a contract when:

- A. The buyer can't make payment
- B. The seller failed to deliver the work promised.
- C. The market has changed, and there is no longer a need for the project.
- D. There are Termination clauses (Termination for Cause or Convenience) explicitly mentioned in the agreement.

Answer – D.

Explanation

A contract is generally terminated by mutual consent. However, the Buyer can take the call when it's explicitly mentioned in the Agreement.

Choices A and B are examples of Termination of Cause, and C is an example of termination of convenience.

Question 11:

The explicit application of Influencing skill of Project Manager is most applicable in: -

- A. Strong Matrix organization.

- B. Project-Oriented organization.
- C. Weak-Matrix organization.
- D. PMO Organization.

Answer – C.

Explanation

The authority of the PM is least in the Weak Matrix organization among all specified above. It's the proper utilization of influencing skills which the Project Manager requires for his success in the project as a Project Manager. Further, any matrix organization has two reporting bosses.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 350.

Question 12:

During execution, you have identified that the raw materials supplied to you are of lower grade than expected. What should the Project Manager do FIRST?

- A. Review Quality Management Plan.
- B. Update Project Document
- C. Create Workaround.
- D. Review the Requirement Documentation.

Answer – B.

Explanation

Issue log is the project document which should be updated first to register the issue.

Question 13:

The following contract clause does not allow for appeals in courts of law based on performance. What type of contract is mentioned in the given scenario?

- A. CPIF.
- B. FP-EPA.
- C. CPAF.
- D. FPIF.

Answer- C.

Explanation

An award fee is dependent on the satisfaction of the client and is subjectively evaluated. Award fee is not subjected to an appeal.

So, among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 472

Question 14:

Troy's project to build a new IT solution is in execution. A Change Request was submitted and approved to apply corrective action to the new production line element. The implementation of the Change Request requires technical knowledge, which no one in the project team has. What should Troy do as a Project Manager to resolve the current situation?

- A. Bring a qualified technical expert
- B. Escalate the issue to the sponsor.
- C. Defer the Change Request
- D. Document the issue to issue logs.

Answer - A.

Explanation

The Question focuses on resolving the issue rather than documenting the same. Recording in the issue log may be the first step the Project Manager should take, but it won't resolve the issue, as stated in the Question. If the Question was what the Project Manager should do first, then the answer could have been D.

So, A is the best option among all options provided.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 113-116.

Question 15:

A project manager has taken over a project which is under closure. The customer has accepted the deliverable, and the Project Manager is trying to determine who should sign off the project. What should the Project Manager review?

- A. Consult with Sponsor.
- B. Refer to the Project Charter.
- C. Ask Customers to sign.
- D. The Project Manager can sign himself.

Answer - B.

Explanation

In the Project Charter it is mentioned who will sign-off the project.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 124

Question 16:

As the project manager, Ryan identifies the possible departure of a critical team member who has the unique expertise required for the project. If that team member is promoted, this would be considered a risk _____.

- A. Category
- B. Event
- C. Trigger
- D. Quantification

Answer – C.

Explanation

A promotion would be the trigger to force the project team to execute the risk response plan. The correct answer is: Trigger

Question 17:

If you want multiple experts to identify project risks and wish to receive unbiased data, what is the most appropriate technique to use? Choose the Best answer.

- A. Interviewing.
- B. Delphi technique.
- C. Assumption analysis.
- D. Brainstorming.

Answer – A.

Explanation

Delphi Technique is not part of PMBOK6. so, it's not the best answer. Best Answer is interviewing, which can be done to an individual, as well as to a group. Further , Delphi technique is limited to technical risks, while interviewing addresses risks in detail and generates engagement of stakeholders.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 80,142, 282,409

Practice Standard for Project Risk Management, Project Management Institute Inc., 2009

Question 18:

Ryan is a Project Manager of a multi-million dollar project where team members are working from multiple countries. As a project manager, for handling multiculturalism, when should he especially consider cultural differences?

- A. When communication strategies are built based on the political environment of the organization.
- B. When the RACI chart gets finalized for assigning work to the available resources.
- C. When he is working in a co-located environment
- D. When he is considering recognition and rewards during team development

Answer: D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 341.

Question 19:

As an experienced Project Manager, you are planning out the communications methods to use as part of the stakeholder engagement. The most effective means for communicating and resolving issues with the stakeholders are:-

- A. Status reports.
- B. Email.
- C. Voice OEILS.
- D. Face-to-face meetings.

Answer – D.

Explanation

Face to face meetings are interactive, and hence are the most effective means for communicating and resolving issues with the stakeholders.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 371.

Question 20:

You are in the closing phase. You want to ensure you do not have any open risks. What should you review?

- A. Risk register
- B. Risk Report
- C. Issue log
- D. Lessons Learned Register

Answer – B.

Explanation

Risk Report should be reviewed during closing to check the existence of any risks which are still open.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 125

Question 21:

Arun is a project manager of a high-performance team. It recently came to his notice that the team uses an unofficial communication platform to communicate at the project level.

What should he do immediately?

- A. Review the Stakeholder Engagement Plan and ensure that the stakeholders are not affected by this unofficial platform.
- B. Review the Communication Management Plan and direct the team

- not to use this platform anymore.
- C. Immediately arrange communication training for the team and terminate the platform.
 - D. Ignore the situation as the project is running smoothly.

Answer – B.

Explanation

As per the scenario stated in the question, since the usage of an unofficial communication platform is not part of the Communication Management Plan, it should not be used. The Project Manager must prevent and protect the project level processes. The project manager should review the Communication Management Plan and direct the team to use the communication management plan's communication platform.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –377.

Question 22:

Lizan, the Project Manager, wishes to expand an existing list of sellers of her organization. Which of the following techniques should she consider?

- A. Bidder Conference.
- B. Make or buy analysis.
- C. Analytical techniques.
- D. Advertising.

Answer – D.

Explanation

Advertising is a good avenue to expand existing lists of sellers. Advertisements can be placed in general circulation publications in order to achieve this.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 487.

Question 23:

You are a Project Manager of the ABC project. Your team members - Anindyo, Tridib have successfully passed the PMP exam. You are satisfied and want to make sure that the success of the team members

gets recognized beyond the project level.

What document the Project Manager needs to update?

- A. Resource management plan.
- B. EEF.
- C. CPA
- D. Team performance assessment

Answer – B.

Explanation

Employee development plan records, skill assessments are recorded in EEF. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 344.

Question 24:

A software development project is using an adaptive development approach, which has six phases. At the end of the second phase, the project goes behind schedule. The Finish-to-start relationship between activities of the development and testing team is causing the delay. The testing team can only start after the development team's work gets completed. After discussion with experts, the development team decided to adopt a Test-driven development approach for some user stories, where testing and development can go in parallel.

Here adoption of Test-driven development is an example of:

- A. Resource leveling.
- B. Fast tracking.
- C. Crashing.
- D. Schedule optimization.

Answer – B.

Explanation

In Test-driven development, Test cases are prepared even before coding gets initiated. Here in this scenario, though you are using the Adaptive development approach, having Finish-to-Start dependencies in an adaptive environment makes little sense. Fast tracking is a schedule compression technique in which the activities or phases usually done in the sequence, are

performed in parallel for at least a portion of their duration.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s)- 215.

Question 25:

What is not true regarding the Identify Stakeholder process ?

- A. Communication management plans may act as valid input for identifying stakeholder processes only after the first iteration of the Identify stakeholder process.
- B. Change log and Requirement documentation are valid input for identifying stakeholder processes.
- C. Stakeholder identification processes can periodically happen throughout the project on a need basis.
- D. Post the first iteration of the stakeholder engagement process, a business document acts as a source of information about the project's stakeholders.

Answer : D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 511-515.

Question 26:

During the middle of a software development project that is using Agile-scrum methodology, the scrum master leaves the company. This event was documented as a risk during the initiation phase of the project. What corrective action needs to be taken care of in the situation stated above?

- A. Ask the Product owner to play a dual role.
- B. Ask the senior most team member to play the role of Scrum master.
- C. Examine the assumption log and take appropriate action.
- D. Review the project charter and work with the functional manager to resolve the issue.

Answer – D.

Explanation

The Project Charter also captures the overall project risks at a high level. Since the question states that the exit of the Scrum Master has already been documented as risk during Initiation, so it's obvious that, the project manager or any related responsible authority should consult the project charter, risk response plan developed to handle the stated risk, and then work with the functional manager to execute necessary steps involved in resolving the issue. Options A and B are not correct as role changes or implementation of dual roles are not examples of best practices. Option C is also ruled out as it won't provide any solution.

Among all options, Option D is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –81.

Question 27:

During the mid of execution, the customer comes up with new quality requirements that involve change in project schedule. Any change in schedule baseline can only be approved through which of the following processes?

- A. Validate Scope.
- B. Control Quality.
- C. Monitor and Control Project Work.
- D. Perform Integrated Change Control.

Answer – D.

Explanation

Any baseline related changes (as in this case schedule baseline) can only be approved through the Perform Integrated Change Control process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 223.

Question 28:

During the early stage of execution, a new Project Manager has joined the project. He wants to know; how the outcomes of uncertainties are documented, maintained, and communicated throughout the project. What should the project manager review to fetch this necessary information?

- A. Communication Management Plan.
- B. Stakeholder Engagement Plan.
- C. Risk register.
- D. Risk Management Plan.

Answer –D.

Explanation

Risk Management plan tells how the risks will be handled in the entire project life cycle.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 405

Question 29:

A Project is in execution. The Project Manager is trying to confirm all the information needs of the project and its stakeholders are met as it is planned. What should the Project Manager review to confirm the same?

- A. Stakeholder engagement Plan.
- B. Communication Management Plan.
- C. Both A and B.
- D. Project Scope Statement

Answer – C.

Explanation

Here in the question, the keyword is information not communication. The situation states that the Monitor Communication process is being carried out, where both Communication management plan and Stakeholder Engagement plan are valid input. Project Manager is trying to ensure whether information is floating and getting managed as per plan. Communication Management plan - contain plan for information creation, collection and distribution. Stakeholder Engagement plan - Helps to identify communication strategies required to engage stakeholders. As per the situation mentioned in the above question, both the Communication Management plan and Stakeholder Engagement plan need to be referred to come to a conclusion.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 390.

Question 30:

Auto-generated mail response from receivers' end in push communication states that the message is: -

- A. Communicated.
- B. Acknowledged.
- C. Confirmed.
- D. Clarified.

Answer – B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 373.

Question 31:

Your project team members have just started to work together after resolving several interpersonal conflicts. They are now adjusting their work habits and behaviors to support each other. Which stage of the Tuckman ladder has your team just entered?

- A. Norming.
- B. Performing.
- C. Forming.
- D. Storming.

Answer – A.

Explanation

During the norming stage, the project team members start to work together and adjust their work habits and behaviors to support each other.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 338

Question 32:

In the middle of an Agile software development project, a project stakeholder requests the Scrum Master for adding a new feature that will increase the project cost and schedule.

What should the Scrum Master do next?

- A. Ask the project stakeholder to raise a formal change request
- B. Conduct a meeting with the team to implement the change.
- C. Set up a meeting between the Project Stakeholder and the Product Owner.
- D. Review the situation with the Change Control Board.

Answer – C .

Explanation

The Product owner is responsible for guiding the direction of the product that includes features and functionality, cost, schedule. A and D are not correct as formal Change Control procedure that is followed in projects with predictive life cycle, is not followed in Agile projects. Agile team consists of Cross-functional team members who are responsible for carrying out the project work without any external dependencies. But the team cannot decide the product direction, so Option B is also not correct. Here, as per the scenario stated in the question, the best approach for the scrum master would be to set up a meeting between the project stakeholder and the product owner. The product owner is the person who is ultimately accountable for the end product that is built. The product owner generally works with customers and the teams to define and maintain the product direction.

So, among all options provided option C is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40,41

Question 33:

Ryan is a Project Manager who is currently engaged in planning Risk management activities for his project. Assessing overall project risk has been difficult for him as he is having trouble assigning cost impacts to identified risks due to a lack of high-quality numeric data.

What should Ryan do as a Project Manager?

- A. Do Quantitative analysis
- B. Evaluate using simulation techniques.
- C. Evaluate on a qualitative basis.
- D. Evaluate using a tomado diagram.

Answer- C.

Explanation

Post identification, Risk needs to be analyzed. Numerically analyzing Risk is costly, not possible in all projects. Qualitative analysis of Risk helps the project team know the probability and impact of Risk, further assisting the team in planning risk responses. A, B, D are examples of analyzing Risk quantitatively.

Among all options, C is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 421-427.

Question 34:

A German stakeholder joins in a multi-phase project. The sponsor has already informed the Project Manager that he is crucial for the project. After meeting with him, the project manager realizes that the new stakeholder should be involved in all status meetings and receive confidential information related to the project. The project manager adds him to the stakeholder register. What should he do next?

- A. Update the Communication Management Plan.
- B. Update the Stakeholder Engagement Plan.
- C. Update the Stakeholder Engagement Assessment Matrix.
- D. Update the Issue Log.

Answer – A.

Explanation

Communication Management Plan contains the plan for information creation, collection, and distribution. It also determines which stakeholder will receive confidential information.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -377.

Question 35:

During Planning while determining budget, the project manager wants to understand the financial success factors associated with the project. Which document should he refer to as a project manager to fetch this necessary information?

- A. Benefit Management Plan.

- B. Project Charter.
- C. Business Case.
- D. Cost estimates.

Answer – C.

Explanation

Business case is prepared by doing cost-benefit analysis with key stakeholders and contains information related to financial success factors associated with the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 251.

Question 36:

As a Project Manager, you have subcontracted some of the technically challenging work to a vendor. The vendor has delivered the subcontracted work as per the technical specifications mentioned in the contract. Still, you are not satisfied with the deliverables produced because they are not the exact ones you needed. What should you do next?

- A. Accept the deliverable.
- B. Proceed with claims administration.
- C. Withhold payment until the contested changes are addressed.
- D. Terminate the contract for causal

Answer – A.

Explanation

The question states that the vendor has completed the work as per the technical specifications mentioned in the contract. Therefore, the seller has met the acceptance criteria described in the terms and conditions of the agreement. Realistically, you would likely negotiate a new or amended contract with the seller to align the deliverable with your requirements better. However, the negotiation was not listed as an answer choice. Therefore, among all available options, accepting the deliverable is the best answer to the question asked.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 499.

Question 37:

During project execution, a Procurement software implementation process gets impacted due to a sudden change of the middleware technology version. The team implemented a risk response plan involving a costlier alternative solution. The project team determines that by applying this approach, contingency reserves, and management reserves, the project can come back to track without increasing the project budget. What should you do next?

- A. Submit a change request to update the cost baseline.
- B. Restart project execution with a new project team.
- C. Submit a change request to access the contingency and Management reserves.
- D. Ask the project sponsor to change Implementation strategy of the project.

Answer - A.

Explanation

The question suggests that a middleware version change has occurred during project execution, which triggered the Implement Risk Responses process. In this case, however, the scenario indicates that management reserves are necessary to restart project execution. Since Management reserves are not part of the cost baseline, an approved change request is required to transfer funds from management reserves into the cost baseline. Once the necessary portion of management reserves has been released and moved from management reserves to the cost baseline, the project manager can access the funds as part of the cost baseline and restart project execution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 254-255, 451.

Question 38:

During a risk review meeting, the project manager learns that a new company policy has been implemented, which prohibits overtime. This restriction in hours for the development team represents a unique constraint for the project. Where should this newly identified constraint be documented?

- A. The Constraint Control register.
- B. The Risk register.
- C. The Assumption Log.
- D. The Issue Log.

Answer - C.

Explanation

A risk review meeting is conducted as part of the Monitor Risks process. The risk register, assumption log, and issue log, get updated as a result of this process. The question describes a situation , results of which may need to be documented in some or all of these project documents.

The assumption log (sometimes referred to as the assumptions and constraints log) is a project document used to record all assumptions and constraints throughout the project life cycle.

So, among all options, option C seems to be the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 458.

Question 39:

Which of the following about the Cost Baseline is not true?

- A. Once baselines, Cost baseline can only be changed with only formal change control procedure even if Change is minor.
- B. Cost baseline includes work package Cost estimates as well contingency reserve.
- C. The cost baseline is the approved version of the time-phased project budget excluding any management reserves.
- D. Cost baseline includes cost of all control accounts as well as management reserve.

Answer: D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -264-265

Question 40:

Money spent during the project to avoid the failure is called:-

- A. Cost of Quality.
- B. Cost of Conformance.
- C. Cost of Non-Conformance.
- D. Fixed Cost

Answer: B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -283.

Question 41:

A new project manager joins the XGH project. He wants to review the project team directory as well as the roles and responsibilities of the team members. Which document should he review?

- A. WBS.
- B. RACI chart.
- C. Project Team assignments.
- D. Resource Calendar.

Answer - C.

Explanation

The Project Team assignments is a project document that records the team members and their roles and responsibilities for the project.

Choice A – WBS – Incorrect – WBS is the hierarchical decomposition of the total scope of work to be carried out by the project team. It doesn't have information on the Project Team directory.

Choice B – RACI chart – Incorrect – A RACI chart provides the level of involvement for each stakeholder associated with a task.

Choice C - Project Team assignments – Correct - The Project team assignments specify which resources are assigned to each activity. It's a project document that records the team members and their roles and responsibilities for the project. It also can include the project team directory.

Choice D – Resource calendar – Incorrect – A resource calendar contains resource availability information for the project work. It doesn't have information on the roles and responsibilities of the team members.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Question 42:

The following skill helps the project manager reduce tension and increase cooperation among Project team players in a highly stressful and demanding project environment.

What skill is stated in the given scenario?

- A. Compromise technique of conflict management.
- B. Emotional Intelligence.
- C. Political awareness.
- D. Management skill.

Answer - B.

Explanation

Successful Project Managers with good people skills always have great Emotional intelligence.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -349.

Question 43:

Ryan is a project manager of the ABX project. Some of the project deliverables have been subcontracted to the vendor CXA. Before signing the contract, the vendor asked whether they could get a part of the payment in advance before submitting the first set of deliverables. As a Project Manager, what should Ryan review to answer the vendor?

- A. Procurement Statement of Work.
- B. Agreement.
- C. CPA
- D. Project Charter.

Answer - A.

Explanation

Some may get confused between A and B. Payment schedule is recorded in Procurement SOW before the contract is signed.

After the contract gets signed, it's available in the Agreement. So, among all

options, A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 481

Question 44:

Your project has been terminated unsuccessfully before closure. Which of the following documents will be used to capture the termination reason?

- A. Lessons learned repository.
- B. Project Closure documents.
- C. Final Report
- D. Procurement File.

Answer – B.

Explanation

Project Closure documents capture the termination reason.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 128

Question 45:

You are a Project Manager of a complex and large automation project where you help the sponsor prepare a project charter. Only one of your vendor-SQZ from the preferred vendor list has the skill set to work on such projects. But you have come to know from your senior Management that the SQZ vendor you are looking for is not well suited for your organizational environment. The internal resources of the organization don't have the technical capability to do this project. What is the best course of action that you can take here as a Project Manager?

- A. Award the contract to the SQZ vendor who has the technical capability.
- B. Engage SW vendors into a Trial Engagement.
- C. Build bid documents and arrange bidder conferences with prospective sellers.
- D. Train your internal resources on the new technology.

Answer – B.

Explanation

Trial engagement is the modern trend of procurement management, where sometimes Buyer engages with Seller temporarily or on a trial basis before allotting them the big chunk of work. It helps Buyer to evaluate whether Seller is suited to Buyer's organizational culture.

In this scenario, among all options provided, B is the best solution. D is also a good option, but it may kill a lot of time.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 464.

Question 46:

Which of the following in relation to Project Charter is not true?

- A. The Project Charter establishes a partnership between performing and requesting organizations.
- B. The Project Charter is considered to be a contract.
- C. The Project Charter formally authorizes the existence of a project
- D. The Project Charter can contain key stakeholder list

Answer – B.

Explanation

A Project Charter cannot be considered as a contract because there is no consideration or any money promised or exchanged in its creation.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 77.

Question 47:

You are a Project Manager of a project where scope is well defined.

However, there is scarcity of resources in your organization, so you award a Fixed price contract to one of your preferred sellers.

What risk response strategy have you applied?

- A. Avoid.
- B. Share.
- C. Transference.
- D. Mitigate.

Answer – C.

Explanation

In this case Risk has been transferred to the third party.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) - 443

Question 48:

Ryan is a Project Manager of the TSE project of the BCD organization. The project involves work in a specific niche technology for a particular work package that is also part of the critical path. A contingency reserve of \$6000 got allotted to the project budget during planning, as there is only one available resource in the organization competent to work on the work package. Now, during the execution, the work package's work gets completed successfully, and the risk didn't occur.

What should the Project Manager do with the \$6000 contingency reserve?

- A. Utilize the budget for the celebration party for project success.
- B. Move the reserve amount to management reserve
- C. Keep it as a Contingency reserve for secondary risks.
- D. Remove the contingency reserve from the Project Budget

Answer – D.

Explanation

If identified risks do not occur, the unused contingency reserve may be removed from the project budget to free up resources for other projects or operations.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 202, 254, 439,443.

Question 49:

A team member worked extra 4 hours in the last 1 week to meet and exceed customer expectations. This is an example for:

- A. Scope Creep.
- B. Gold plating.
- C. Crashing.

D. Fast tracking.

Answer- B.

Explanation

This is another classic example of Gold plating. The main objective of the project team is to meet customer needs, not exceeding the same. Working 4 extra hours without a Change request is not recommended.

Question 50:

The design thinking approach of obtaining early feedback on requirements by providing a model of the expected product before actually building it is known as:

- A. Context diagram.
- B. SIPOC diagram.
- C. Prototype.
- D. Brain-writing.

Answer – C.

Explanation

Prototypes allow stakeholders to experiment with models of the final product or requirement.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 147.

Question 51:

Approved Change requests are implemented in which of the following processes: -

- A. Direct and Manage project work
- B. Control Quality
- C. Validate Scope
- D. Control Cost

Answer – A.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) 90,91.

Question 52:

You are preparing a report for an upcoming meeting with the senior management. The information should include the project risks that have the most potential impact on project outcomes.

What tool, the Project Manager, can use here to fetch this necessary information?

- A. Influence Diagram.
- B. Why-Why Diagram.
- C. Tornado Diagram.
- D. Pareto Chart

Answer- C.

Explanation

Tornado diagrams are a special type of Bar Chart, where the data categories are listed vertically instead of the standard horizontal presentation. This diagram is useful for deterministic sensitivity analysis – comparing the relative importance of variables.

Here, as per the scenario stated in the question, the project manager has to perform the Sensitivity analysis to determine the project risks having the most potential impact on project outcomes.

Tornado diagram is a popular tool for displaying the results of the Sensitivity analysis for the scenario stated above. So, among all options provided, option C is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 434.

Question 53:

Ryan is the project manager of a complex IoT device manufacturing project. By looking at its complexity, he invites a representative from various departments like Sales, Research and Development, Engineering, and Demand Generation to participate in the seller proposal evaluation process.

Which of the following tools and techniques the Project Manager used in the above-stated situation?

- A. Bench-marking.
- B. Market Research.
- C. Source selection analysis.
- D. Expert judgment

Answer – D.

Explanation

Expert judgment is based upon expertise in an application area, knowledge area, discipline, industry, etc., as appropriate for the activity being performed. Such expertise may be provided by any group or person with specialized education, knowledge, skill, experience, or training. The inclusion of representatives from various departments to evaluate seller proposals is an example of using expert judgment.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 487.

Question 54:

Multi-phase project will include kick-off meeting during: -

- A. At the start of the first phase.
- B. At the end of each phase.
- C. At the beginning of each phase.
- D. At the start of the planning phase.

Answer: C.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 86.

Question 55:

While performing project work, Ryan, the team player, notices that there is a potential new risk, which could impact the project's critical path. He informally discusses the same with the project manager, who advises him to do something immediately. What is the most probable suggestion the Project Manager gave to Ryan?

- A. Recommend the project be placed on hold for a day to avoid the

risk

- B. Document the risk in the risk register.
- C. Record the same in the issue log.
- D. Generate a risk report using quantitative and qualitative analysis of the identified risk to alert the team of the impacts.

Answer- B.

Explanation

The risk register serves as the central repository for planned risk responses and lists risk owners. In a project, any time a new risk emerges on a project, the first step to take is to immediately document it in the risk register. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 417.

Question 56:

Ryan is a project manager of the WSD project who has just completed planning. The head of the company wants Ryan to furnish a report that would include the procurement delivery method the project will use, type or nature of agreement considered for the project, and the procurement phases.

Which document should Ryan refer to give appropriate information to the BU head?

- A. Procurement Management Plan.
- B. Statement of Work
- C. Procurement Strategy.
- D. Bid Documents.

Answer – C.

Explanation

Procurement Strategy which is an output of the Plan Procurement Management process determines project delivery methods, type of legally binding agreement(s), and how the procurement will advance through the procurement phases.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 476, 481.

Question 57:

During the risk identification process, the project manager is trying to find out the individuals who will participate in the risk identification process and those available to play the role of risk owners.

What should the project manager review?

- A. Risk register
- B. Stakeholder Register.
- C. Communication Management Plan.
- D. Activity attributes.

Answer – B.

Explanation

Here the scenario suggests that the Project Manager is in the Risk Identification process. The Stakeholder Register, an input to Identify Risks process, contains the information of individuals who will participate in the risk identification process and those who are available to act as risk owners. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 413.

Question 58:

A project manager is in the process of developing the stakeholder register. He creates a list of potential stakeholders and decides to use a data-gathering technique to share a list of questions to the individual participants before the group creativity session. What technique the Project Manager applied in the given scenario?

- A. Brainstorming.
- B. Questionnaires and surveys.
- C. Brain writing.
- D. Stakeholder Mapping.

Answer - C.

Explanation

Data gathering techniques generally for the Identify Stakeholders process include questionnaires and surveys, brainstorming, and Brain writing. Brain writing is a refinement of brainstorming that allows individual participants to

consider the questions individually before the group creativity session. Therefore, the first step of Brain writing is providing the participants with the questions before the creativity session. This process will allow the participants to be better prepared, making the meeting more effective than a traditional brainstorming session. So among all options provided, C is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 511.

Question 59:

Legislation and regulatory are examples of : -

- A. Technical Risk.
- B. Management risk.
- C. Commercial risk
- D. External risk.

Answer: D.

Explanation

Very important topic for a real life exam.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 406.

Question 60:

In the middle of the project, the project manager realizes that the raw materials available in-house got exhausted, and he needs to purchase raw materials from outside. What analysis should the Project Manager do FIRST in the vendor selection process, considering the remaining project budget in mind?

- A. Alternative Analysis.
- B. Variance Analysis.
- C. Qualification Analysis.
- D. Cost-benefit Analysis.

Answer – D.

Explanation

Here the Project Manager is performing the Control resources process. For selecting the best supplier, the Project Manager should do Cost-benefit analysis to determine the corrective action in case of such project deviation. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 282,356.

Question 61:

Senior management has asked the Project Manager of the GHF project to provide a report containing information on an acceptable procurement timetable and assumptions planned for resource allocation. What should the Project manager review first to provide this necessary information?

- A. Communication Management Plan.
- B. Stakeholder Engagement Plan.
- C. Resource Management Plan.
- D. Project Management Plan.

Answer – D.

Explanation

The Project Manager should review the Procurement management plan to review timetables of key procurement activities and constraints and assumptions that can affect planned procurements.

Since this option is not provided, the Project Management plan is the best answer as the Procurement Management plan is a subsidiary plan for the Project management plan.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 466.

Question 62:

Good Interpersonal skills play a phenomenal role in successful stakeholder engagement.

All of the following are examples of interpersonal skills EXCEPT:-

- A. Resisting to change.
- B. Building trust.
- C. Resolving Conflict.
- D. Active listening.

Answer – A.

Explanation

Resisting to change is not a valid interpersonal skill.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 144-145, 153, 709.

Question 63:

You are the newly appointed project manager of the Qnx organization.

Currently, you are managing a project which has just moved to the planning stage. A PMP certified expert clearly explained to you that you must consider an Enterprise Environmental factor to determine how the work gets assigned to the people for any modern-day Project.

What should you consider to know how the work gets assigned to the people?

- A. RACI Matrix.
- B. Project Management Information System (PMIS).
- C. Resource Histogram.
- D. Work authorization system.

Answer – D.

Explanation

The work authorization system is a part of your company's Internal Enterprise Environmental Factors, and it's generally part of any change control system. It defines how work is assigned to people. If work needs to be approved by specific managers, the work authorization system will make sure that the right people are notified when a team member's work assignments change.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 38.

Question 64:

You are managing a project with an EV of \$15,000, PV of \$12,000 and AC of \$11,000.

How would you BEST describe this project?

- A. The project is ahead of schedule and within its budget

- B. The project is behind schedule and within its budget.
- C. The project is ahead of schedule and over its budget
- D. The project is behind schedule and over its budget

Answer – A.

Explanation

This is a calculation question that's asking you to use SPI and CPI to evaluate your project.

First calculate $SPI = EV / PV = \$15,000 / \$12,000 = 1.25$ – this means your project is ahead of schedule.

Then calculate $CPI = EV / AC = \$15,000 / \$11,000 = 1.36$ – so your project is within its budget or under budget.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 263,267.

Question 65:

Ryan is a Project Manager of a new RPA project. During the Risk review meeting, the team identifies a new type of technical risk and updates the necessary documents. Ryan decides to update the findings of the Risk level information in detail so that the risk identification process for future projects can include this new risk in their Risk Identification Checklist. What should get updated in the mentioned scenario?

Choose the best answer.

- A. Lessons Learned Register.
- B. Risk Report.
- C. CPA
- D. EEF.

Answer – C.

Explanation

The above scenario states that the team is engaged in the Monitor Risks process. Since the team has identified a new type of Technical risk, the Risk Breakdown Structure should be updated with the latest risk information so that the risk identification checklist for future projects can use the revised Risk Breakdown Structure (RBS). The Organizational Process Assets (OPA), which gets updated due to the Monitor risks process, include the Risk

breakdown structure.

Therefore, among all options, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 458.

Question 66:

Throughout the project, the stakeholders have been raising various problems, gaps, inconsistencies, and conflicts. Which of the following should be used to capture such information?

- A. Change Log.
- B. Risk Register.
- C. Issue Log.
- D. Stakeholder Register.

Answer – C.

Explanation

The problems, gaps, inconsistencies, and conflicts are collectively known as issues. Issues are recorded in the issue log.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 96.

Question 67:

Ryan is a Project Manager of a company that prepares highly specialized scientific devices. He wants to prevent any costly mistakes in association with frequently performed tasks.

What preventive action is best suited for the above-stated scenario?

- A. Encourage the project team to use a Quality checklist.
- B. Ensure proper training for each team member on a periodic basis.
- C. Make sure that the Quality Management Plan is clear and concise.
- D. Immediately identifying this as risk and recording his concern in the risk register.

Answer – A.

Explanation

A Quality checklist is a structured tool used to verify that a set of required

steps has been performed. It's a valid data gathering tool used in the Manage Quality process. Quality checklists are also used to incorporate the acceptance criteria included in the scope baseline.

Here, in this scenario, among all options, option A seems to be the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 292, 718.

Question 68:

The project team is doing risk analysis and assessment on those identified risks which can positively or negatively affect an organization's strategic goals. Which characteristics of risk assessment are mentioned in the above scenario?

- A. Propinquity
- B. Data Quality.
- C. Strategic Impact
- D. Quantitative

Answer – C.

Explanation

The project team may consider other characteristics of risk in addition to probability and impact when prioritizing individual project risks for further analysis. In assessment of other risk parameters, Strategic impact is assessed to analyze risks that affect strategic goals of the organization.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 423, 424.

Question 69:

In your project, SPI is 0.82, CPI 1.3. You analyzed the baseline and decided to return the schedule on track to meet the customer plan. You will: -

- A. Check tasks which can be removed from the baseline.
- B. Checking with team tasks can be automated and save time.
- C. Check the option of introducing over time in the project.
- D. Check with your sponsor.

Answer – C.

Explanation

Since the CPI is over one and SPI is less than one, it means that the project is currently under budget, but is behind schedule. So, among all possible options given in the above, C seems to be the best option. you need to check the option of crashing the project. Over time is an example of crashing. so among all options provided , C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 215.

Question 70:

In a smooth going project, if CPI is 1.4 and SPI is .97 then expected value of TCPI value should be :-

- A. Greater than 1.
- B. Less than 1.
- C. Equal to 1.
- D. Equal to current CPI value.

Answer – B.

Explanation

In the above scenario, Project looks in good condition from a project health point of view. It's under budget and slightly behind schedule. Since CPI is greater than 1, TCPI value should be less than 1, as the project is in good shape from the Cost point of view.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 263,267.

Question 71:

The Change Control Board has approved a Change request involving scope change. But the sole technical resource planned to work on the CR has suddenly gone for two weeks' vacation. If the work start date gets delayed, then the upcoming Project Milestone will be impacted. In a Project team meeting, a team member said there is an alternate way of addressing the current situation, and the related plan is available. What should the Project Manager do?

- A. Inform the current situation to the stakeholder which is most concerned with the Change Request.
- B. Have a meeting with the sponsor for buying out extra 2 weeks' time in the project.
- C. Create a Change Request for an additional resource.
- D. Review the Risk Register with the team and implement the risk response as stated by the team member.

Answer: D.

Explanation

The situation looks like a case of Known-Unknown risk, and a Risk response plan is already available. So, reviewing the risk register and implementing the Risk Response plan, as the team member stated to the Project Manager, is the best course of action in the current scenario.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 449- 452.

Question 72:

What will be affected if any angle of the iron triangle is not kept in balance?

- A. Cost
- B. Quality.
- C. Scope.
- D. Schedule.

Answer: B.

Explanation

The change to any angle of the iron triangle may impact the quality. A, C, D are incorrect because they are three angles or sides of the iron triangle. They must be kept in balance or else quality will suffer.

Question 73:

A salience model is a combination of which of the following stakeholder characteristics?

- A. Power, Interest and Involvement

- B. Influence, Impact and Interest.
- C. Power, influence, and impact.
- D. Urgency, Power, and Legitimacy.

Answer – D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 513.

Question 74:

A Project Manager has experienced greater than anticipated delays in his current project. To keep the project on schedule, the project manager will need to access the contingency reserves allocated for this risk. What should the project manager do NEXT?

- A. Submit a change request to access the contingency reserves.
- B. Submit a change request to update the schedule baseline.
- C. Access the contingency reserves as needed to meet the project objectives.
- D. Take permission from the Executive membership team.

Answer – C.

Explanation

The Project Manager manages the schedule and cost reserve and can access the contingency reserves as necessary to meet the project objectives. Since, Contingency reserves are the estimated duration within the schedule baseline, there is no need to follow formal change control procedure for accessing the contingency reserve.

So, options A and B are incorrect. Since PM manages the Schedule/cost baseline and the schedule and cost reserve, option D is not required.

Among all options, option C is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) 202,255, 439,443,702.

Question 75:

The project you are leading is behind schedule. A team member suggests a change that may help to bring the project back on track. You agree

**with the team member and issue a request for corrective action.
This request would be an outcome of which of the following processes?**

- A. Develop Project Management Plan.
- B. Develop Project Charter.
- C. Perform Integrated Change Control.
- D. Monitor and Control Project Work.

Answer – D.

Explanation

The scenario describes a project that falls behind schedule. Such a determination could have been a result of comparing the actual project performance with the schedule baseline. Reviewing the differences between planned and actual project performance is known as variance analysis. When issues are discovered during the Monitor and Control Project Work process, change requests are submitted to rectify the problem. Therefore, in this scenario, a change request for corrective action is raised as a part of the Monitor and Control Project Work process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 105-108.

Question 76:

The project team is currently working with an externally hired expert. The expert determines a change in some standards, procedures, policies currently followed by the team. As per the above situation, which of the following processes is getting performed?

- A. Plan Quality Management
- B. Control Quality.
- C. Manage Quality.
- D. Manage Team.

Answer- C.

Explanation

The situation states that the Project Team is involved in the Manage Quality Process, and the audit process is going on. In this case, an external auditor found the need to change some standards, policies, procedures of the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 294.

Question 77:

During execution a new Project Manager joins the ABC project of TRG company. As he is new to the project, he wants to review the meeting plans, workflow authorization sequence, and glossary of common terminology being followed in the project.

What should be reviewed to get this information ?

- A. Communications Management Plan.
- B. Work Breakdown Structure.
- C. Team Charter.
- D. Resource Management Plan.

Answer – A.

Explanation

The Question implies that the Project Manager should review the Communications Management Plan. The Communications Management plan records the glossary of common terminology, workflows with possible sequence of authorization, meeting plans, etc.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 377.

Question 78:

A project is in execution. A key stakeholder raises a change request for adding a new functionality. The Project Manager decides to do impact analysis before passing the change request to the Change Control board. Which of the following documents would help him assess the impact of the change on the project scope?

- A. Requirement Documentation.
- B. Risk register.
- C. Requirements Traceability Matrix.
- D. Change Management Plan.

Answer - C.

Explanation

The question states that the Project Manager is involved in the Perform Integrated Change Control process. He or she should review the Requirements Traceability Matrix to assess the impact of the change on the project scope.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 116.

Question 79:

Arun is a Project manager of the MMX project. In the recent past, one of his stakeholders has turned neutral from supportive. After having a face-to-face meeting with the stakeholder, he understands that the stakeholder is not receiving the status update of the change requests raised by him on time.

Which document should the Project Manager review first to resolve this issue?

- A. Change Log.
- B. Issue Log.
- C. Assumption Log.
- D. Stakeholder Register.

Answer – A.

Explanation

Change requests and their statuses are documented in change logs and communicated to the appropriate stakeholder.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 525.

Question 80:

Arun is a non-technical Project Manager. His project's two highly skilled resources - Syed and Sandeep, conflict with the best technical solution to meet a project requirement. However, it is not impacting the project schedule. Who is initially responsible for the conflict resolution?

- A. The project sponsor is initially responsible for the conflict resolution.
- B. The project manager is initially responsible for the conflict

- resolution.
- C. Both team members are initially responsible for the conflict resolution.
 - D. Either team members are responsible for the conflict resolution.

Answer – C.

Explanation

Initially both the team members are responsible for the conflict resolution. If the conflict escalates, the project manager should help facilitate a satisfactory resolution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 348.

Question 81:

The iterative process of increasing the level of detail in a project management plan as greater amounts of information and more accurate estimates become available is termed as :-

- A. Rolling wave planning.
- B. Progressive Elaboration.
- C. Planning package.
- D. Agile Release Planning.

Answer: B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -715.

Question 82:

A Change request for performing a corrective action has been approved by the Change Control Board. What is the NEXT step?

- A. Implement the change.
- B. Update the Change Log.
- C. Update the Project Management Plan.
- D. Analyze the impact of the change.

Answer - B.

Explanation

First step after CCB approval is to record the Change in the Change log. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 120.

Question 83:

Ryan is a Project Manager who has just implemented an approved Change Request that involved crashing with two new resources. This requires all of the following to be updated EXCEPT:-

- A. Scope baseline.
- B. Project schedule.
- C. Project costs
- D. Risk register.

Answer – A.

Explanation

The point of crashing is to accelerate the schedule without reducing the project scope. The change proposed in this scenario would not incur adjustment to scope. So the scope baseline is the exception in the mentioned scenario.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 215.

Question 84:

A new Project Manager joins the FDC Project. He wants to review the meeting and communication guidelines, which also states the Project Team's expected behavior in all the Stakeholder engagement meetings. What should the Project Manager review to fetch this necessary information?

- A. Resource Management Plan.
- B. Team Charter.
- C. Project Charter.
- D. Communication Management Plan.

Answer – B.

Explanation

Here the question is talking about Ground Rules. All Ground rules are documented in Team Charter which talks about Team Value, Meeting guidelines, Team agreements, etc.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)-319, 528.

Question 85:

Ryan is a servant leader for an agile project which has too much product complexity. What should he suggest the team do to address this paint point?

- A. Encourage the team to apply the Agile Principle “Simplicity, the art of maximizing the amount of work not done”.
- B. Ask the team to apply a waterfall approach to reduce the complexity.
- C. Ask the team to follow the team charter.
- D. Advise the team to eliminate the complex user stories.

Answer – A.

Explanation

As per the scenario stated in the question, the servant leader should encourage the team to think of the simplest possible way to alleviate the issue and apply the Agile Principle of “Simplicity, the art of maximizing the amount of work not done.” These may undoubtedly help to reduce the product level complexity.

Here, among all options, option A is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 59.

Question 86:

Ryan is managing a road construction project. Due to unseasonable weather conditions, the team productivity might be above or below target. This is an example of: -

- A. Ambiguity risk
- B. Variability risk

- C. Mitigated risk
- D. Opportunity risk.

Answer – B.

Explanation

Examples of variability risks include: productivity may be above or below target, the number of errors found during testing may be higher or lower than expected, or unseasonal weather conditions may occur during the construction phase.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 398.

Question 87:

You need to procure a highly specialized chemical solution for a research project. There is only one vendor available that provides the materials you need. This scenario is an example of what kind of market condition?

- A. Constraint
- B. Single source.
- C. Sole source.
- D. Oligopoly.

Answer – C.

Explanation

Sole source is the best choice because it describes the marketplace condition in which only one vendor can provide the goods or services your project requires. A constraint is not a valid market condition. B, single source, describes the marketplace condition in which there are multiple vendors that can provide the goods or services your project demands, but you prefer to work with just one in particular. D, an oligopoly, is a market condition in which the actions of one vendor affect the actions of the other vendors.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 473.

Question 88:

In a Sprint review meeting, the client formally accepted the work and congratulated the Project Team. The next day, the project manager sets

up a meeting with the client to make arrangements for the final payment. What should the project manager do first when meeting the client as part of project closure?

- A. Obtain final acceptance of the deliverable.
- B. Settle procurement claims.
- C. Prepare a Final report.
- D. Obtain formal acceptance of the deliverable.

Answer - A.

Explanation

The question implies the project is in the Close Project or Phase process. When closing a project, it is necessary to obtain final acceptance of all project deliverables in order to adequately complete and close the project. By obtaining final acceptance, the project manager is essentially 'confirming' the delivery and formal acceptance of deliverables by the client. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 123.

Question 89:

CCB has approved a change request to add a feature. The project manager holds an approved Change request review meeting with the project team to ensure the deliverables getting implemented are correct and valid. In which process the project team is currently in?

- A. Direct and Manage Project work
- B. Control Quality.
- C. Validate Scope.
- D. Control Scope.

Answer – B.

Explanation

Inspections carried out as part of the Control Quality process may uncover defects or areas of noncompliance with project requirements, which, in turn, may generate change requests. It is the responsibility of the project team to ensure that all approved change requests are getting implemented in correct fashion as per stakeholder need. Approved Change request review ensures

Approved Change request implementation and successful completion.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 306.

Question 90:

While analyzing the risks in a project, a Project Manager updated the risk register with the risk urgency assessment ratings. In which process the Project Manager is currently in?

- A. Identify Risks.
- B. Monitor Risks.
- C. Perform Qualitative Risk Analysis.
- D. Plan Risk Management

Answer – C.

Explanation

Risk urgency information is updated to the Risk register during the Perform Qualitative Risks Analysis process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) 427.

Question 91:

The Client team has accepted the deliverables of a year-long project in the final Iteration review meeting. It is one month late, but the Client team and the sponsor are happy with the project outcome. The project manager is glad to be done with the deliverables but is concerned about the cost and schedule overruns. He decides to re-run the initial cost-benefit analysis using the actual project costs.

Where should the Project Manager review to obtain the initial cost-benefit analysis?

- A. The Project Management Plan.
- B. The Project Charter.
- C. The Business Case.
- D. The Benefits Management Plan.

Answer- C.

Explanation

The business case has the complete cost-benefit analysis for the project. It can provide the Project Manager with all the initial information used to justify the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 125.

Question 92:

Ryan is the new Project manager of a large project where team members are working from multiple countries. The project team and sponsor are currently engaged in a meeting to discuss a portion of the work. Ryan tells everyone that the work will commence from Canada, and the documentation is available for the type of effort involved.

What did the Project Manager review to address the information stated by him in the meeting?

- A. Communication Management Plan.
- B. WBS Dictionary.
- C. Activity Attributes.
- D. Stakeholder Engagement Plan.

Answer – C.

Explanation

Activity Attributes can be used to identify the place where work has to be performed, the project calendar the activity is assigned to, and the type of efforts involved.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -186.

Question 93:

In a meeting with an IT Project Manager, the Delivery Director asks for information on seller invoices paid for the project. What should the Project Manager review?

- A. Work Performance Data.
- B. Work Performance Information.
- C. Procurement Management Plan.
- D. Procurement SOW.

Answer - A.

Explanation

Choice A - Work performance data - Correct- It can also include information on the seller invoices that have been paid.

Choice B - Procurement SOW - Incorrect - It contains information in relation to payment schedule.

Choice C - Procurement Management plan - Incorrect - Tells how payment will be done.

Choice D - Work Performance Information - Incorrect - It gives Plan Vs Actual information on seller work.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 496.

Question 94:

During execution, some key stakeholder advises the Project Manager to add two experts from the Engineering team in a few upcoming project activities. What should the project manager do FIRST?

- A. Update Stakeholder register by adding two new experts from the Engineering team.
- B. Capture the concern of key stakeholders in Lessons Learned document
- C. Record the key stakeholder concerns in the Issue Log.
- D. Discuss the suggestion of the key stakeholders with the Sponsor.

Answer – C.

Explanation

All stakeholder related concerns raised during execution should be recorded in the issue log first.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 525.

Question 95:

Your company is sponsoring an exhibition cricket match for raising funds for unprivileged children. However, your company cannot afford to pay the \$5,00000 award if some batsman hits six sixes in an over. So, the company decides to take an insurance policy in the event someone

does achieve it. What risk response strategy has been applied here?

- A. Sharing.
- B. Mitigation.
- C. Transference.
- D. Avoidance.

Answer – C.

Explanation

The use of insurance to shift the negative impact of a risk—in this case, the payment of \$5,00000 —is an example of risk transference.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443.

Question 96:

Your company has accepted a Robotic Process Automation project. Due to a lack of in-house skills and experience in all technologies needed for the project, you decide to go for a Joint Venture with another company that has industry level experienced experts. Here Joint Venture is an example of: -

- A. Contract
- B. Risk sharing.
- C. Risk transfer.
- D. Trial Engagement

Answer – B.

Explanation

Share is one of the Risk response strategies for Opportunities. It involves transferring ownership of an opportunity to a third party so that it shares some of the benefits if the opportunity occurs. Risk-sharing often involves payment of a risk premium to the party taking on the opportunity.

A Joint Venture is an example of Sharing.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 444.

Question 97:

Tridib is a Project Manager of a highly technical Space Research project of ABC company. The micromanaging and task-oriented nature of Tridib helped the first three phases of the Project complete on time and budget. The team knowledge base also improved as they got the chance to learn new technology solutions. The Project Manager also rewarded the project expediter and technical lead for their excellent performance. Still, in a board meeting, several senior management members raised concern about Tridib's Project Management skills as four skilled team members of the Project resigned after the last project release. What is wrong with this situation?

- A. The Project Manager lacks management skills.
- B. Project Manager lacks Leadership skill.
- C. Project Manager failed to keep good relationship with higher management
- D. Nothing. 4 key team members chose better career opportunities.

Answer- B.

Explanation

The following scenario suggests Tridib is a task-oriented Project Manager, but he needs to improve his Leadership skill. Micromanagement is a kind of autocratic leadership style, and PMI forbids its usage. Further, the Zero-sum-reward system's application, by rewarding only the Project Expediter and Technical lead, may have caused dissatisfaction in the mind of other contributors.

Of all the above options, B looks the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 64.

Question 98:

A server manufacturing company receives product performance complaints about their latest launched product. Which of the following tools and techniques should the team use to determine whether the product needs any modification?

- A. Scatter Diagram.
- B. Control Chart.

- C. Histogram.
- D. Affinity Diagram.

Answer – B.

Explanation

Control charts are used to determine whether or not a process is stable or has predictable performance. By using a control chart – the data representation tool, the project team can verify whether the product has performance related issues and whether any modifications are required.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 304.

Question 99:

The Global delivery head of an IT services company meets a Project Manager and asks him to immediately start a business critical project that needs to be completed as quickly as possible. He also confirmed that he is allocating a figure of \$50,000 as an initial project fund.

What should the project manager do first?

- A. Perform high level estimation to confirm whether the allocated budget is enough for the project.
- B. Perform project schedule analysis to come up with an estimated end date.
- C. Gather measurable project objectives and high level requirements to create the project charter.
- D. Ask the Global delivery head to take the Go/No-go approval from the CEO.

Answer – C.

Explanation

The first step in project initiation is to formally authorize the existence of the project. In the given scenario, the project manager has to understand the measurable project objectives, high level requirements, project success criteria and come up with a project charter and get it approved by the Global Delivery Head (who is the project sponsor) in this case.

Option D is not correct as the question states that the Global Delivery Head is ready for fund allocation. So, in most probable scenarios, Go/No-Decision

has already been taken.

Options A and B can only be taken care of during planning, so both the options are ruled out.

Among all provided options, option C is the best answer as any project work can begin after the project is formally authorized.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 75,81.

Question 100:

As a Project Manager of a Telecom equipment manufacturing project, you are quickly required to accumulate information from many people located across different countries of Asia and Europe. What is the best tool the Project Manager can use in this scenario?

- A. Brainstorming.
- B. Questionnaires and surveys.
- C. Benchmarking.
- D. Brain writing.

Answer – B.

Explanation

Questionnaires and surveys are a written set of questions designed to quickly accumulate information from a large number of respondents. This tool is extremely useful when quick turnaround is needed, and when respondents are geographically dispersed.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 143.

Question 101:

The Project Manager has to furnish the project progress report to the senior management, including the latest customer inspection results for his project. What should the Project Manager review?

- A. Requirement Traceability Matrix.
- B. Work Performance Data.
- C. Work Performance Information.
- D. Trend Analysis report

Answer – C.

Explanation

The question states that the Project Manager has to furnish the results of the Validate scope process. Work Performance Information, which is output for the Validate Scope process, includes information about project progress, such as which deliverables have been accepted and are not accepted and reasons behind acceptance and non-acceptance decisions.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 166.

Question 102:

Ryan is a Project Manager of an IT development project. The Project has a mixed bag of Internal and External human resources. Senior Management has asked for a report that would reflect work assignment and roles and responsibilities of human resources working in the project. What should Ryan review to fetch this required information?

- A. WBS.
- B. Communication Management Plan.
- C. Resource Requirements.
- D. RACI Chart

Answer- D.

Explanation

RACI is a common type of RAM that uses responsible, accountable, consult and inform statuses to define the involvement of stakeholders in project activities. It shows the project resources assigned to each work package. A RACI chart acts as an useful tool to ensure clear assignment of roles and responsibilities when there is a team consisting of Internal and external resources.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 317.

Question 103:

Several experts worldwide have gathered to build an Implementation Strategy for making data protection software for New York Stock Exchange.

What can guide them with information about legal and regulatory requirements and constraints to be taken care of while developing a document that will formally authorize the project's existence?

- A. Historical information in relation to data protection.
- B. Benefit Management Plan.
- C. Quality Policies of the NYSE
- D. Enterprise Environmental Factors (EEF).

Answer – D.

Explanation

Here the question states that several experts have gathered to build implementation strategies. In this case, EEF needs to be reviewed to take necessary guidance on legal and regulatory requirements which can impact the upcoming project in future. Further, EEF is an input for Develop Project Charter Process, and currently Develop Project Charter Process is getting performed.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 78.

Question 104:

You are a Project Manager of a Chemical Engineer project which has just been initiated. The project involves some specialized critical tasks with a considerable amount of pure risk for which no internal skills are available. There is only one vendor in the current market to handle such tasks. The sponsor has told you that the vendor is no more interested in your company as they had few payments related disputes in the past, which were ultimately sorted in court.

What should the Project Manager do to overcome this difficult situation?

- A. Delay the project and provide training to internal resources.
- B. Set up a meeting with the sole source vendor and try to negotiate by using your collaborative management skill.
- C. Advise sponsors to terminate the project as it's too risky.
- D. Award the contract to another vendor and advise them to make necessary arrangements.

Answer – B.

Explanation

The situation here states that the Project cannot be successful without the involvement of the sole source supplier. So, the best solution is to negotiate with the supplier with the exchange of some open dialogues and reach a win-win consensus. Negotiation is a powerful tool which can bring back trust and harmony among buyer and Sole source sellers.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 341, 347, 474.

Question 105:

A software development project is on the verge of closure. The project manager is now carrying out all of the activities associated with project closure.

Which of the following activities should the project manager and team perform as a part of the process?

- A. Update all the Project documents and mark them as the final version.
- B. Update Enterprise Environmental factors.
- C. Ask Customer for a letter recommending the contract can be closed.
- D. Update Organizational Process Asset to include the final version of Pre-approved sellers list seller performance evaluation documentation.

Answer – A.

Explanation

During Closing, all Project documents may be updated and marked as final versions as a result of Project Closure. Question here asks about tasks associated with Closing activities. C and D are both associated Control Procurements processes, and B is not a valid choice. Some may get confused between A and D, but we need to remember that OPA's are updated for both Control Procurements and Close Project or Phase processes. So, among all options provided A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 127.

Question 106:

All CCB members have gathered in one meeting room to assess a requested change submitted by a key stakeholder of High power and high interest. They are using a technique to decide whether the Change Request can be accepted or rejected, or need further clarification to be finally accepted. What tool has been used in the above given scenario?

- A. Cost —benefit analysis.
- B. Alternative Analysis.
- C. Status meeting.
- D. Performance Analysis.

Answer – B.

Explanation

The Question states that the CCB members are performing an Integrated Change Control process. As a part of the process, Alternative analysis is a tool that's being used in the PICC process to assess the change requests and decide which are accepted, denied, or need further clarification to be finally accepted.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 119.

Question 107:

The sponsor, enterprise architect and few technology experts are engaged in a meeting to discuss a document that describes how and when the benefits of the project will be delivered and what mechanism should be in place to measure those benefits.

Which of the following is not an example of key elements of the document mentioned in this scenario?

- A. Strategic alignment
- B. Risk and Assumption.
- C. Benefit Owner.
- D. Quality Report.

Answer – D.

Explanation

The question states that the sponsor, architect and a few technology experts are discussing the Benefit Management Plan. The key elements of the Benefit Management plan are Target benefits, Strategic alignment, Risks, Assumptions, Benefit owner, metrics, and Timeframe of realizing benefits. So among the options provided, all A, B, C are key elements of the Benefit Management Plan, and D is the wrong option. Quality Report is a project document and it doesn't fall under key elements of the Benefit Management Plan.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 33.

Question 108:

An experienced Project Manager explains to a Junior Project Manager the importance of a particular data-gathering technique mainly used in the Initiation and Planning process group. Such a technique is used to identify a list of ideas in a short time and primarily comprises two parts: Idea generation and Analysis. What technique has been referred to in this scenario?

- A. Brainstorming.
- B. Benchmarking.
- C. Focus Groups.
- D. Questionnaires and Surveys.

Answer – A.

Explanation

Here in this scenario, the experienced manager is talking about the Brainstorming technique which is mainly used in processes like Develop Project charter, identify risks, collect requirements, Identify Stakeholders, Plan Quality Management. Such technique is used to identify a list of creative ideas in a short period of time and mainly comprises two parts: Idea generation and Analysis.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 80, 142, 414,511.

Question 109:

A Project Manager is conducting a kick-off meeting in absence of a sponsor. In the meeting, some stakeholders started raising concerns on a

few objectives of the Business case which they feel needs to be changed. The Project Manager recorded the stakeholder concern in the Minutes of Meeting document, and moved to the next topic by stating that he will come back after discussing the concern with the sponsor. What conflict management technique has been applied over here?

- A. Accommodate.
- B. Compromise.
- C. Withdraw
- D. Force.

Answer – C.

Explanation

In this situation, the modification in the business case will need to be addressed by the project sponsor. The Project Manager has applied an Avoid conflict resolution technique here.

Withdraw/avoid is a conflict resolution technique which may include postponing the issue to be better prepared or to be resolved by others. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -29, 348-349.

Question 110:

As a Project Manager, you are finalizing all open claims and confirming the formal acceptance of the seller's work. Which process are you in?

- A. Control Procurement.
- B. Close Project or Phase.
- C. Validate Scope.
- D. Monitor Communication.

Answer – B.

Explanation

The situation states that the Project Manager is doing Closing Activities. The work related to finalization of open claims or confirmation of Formal acceptance of sellers' work happens as a part of closing.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 123.

Question 111:

Ryan is strategic leader of the CRAC company. He is heading a department that standardizes project related governance processes and acts as liaison between the balanced scorecard and the program, projects, and portfolios of the organization. He has a high degree of authority and the project managers of all business critical projects are reporting to him.

What kind of Organization Structure type Ryan is leading?

- A. Ryan is heading a strong matrix organization.
- B. Ryan is heading a PMO of type controlling.
- C. Ryan is acting as the COO of the CRAC company.
- D. Ryan is heading a PMO of type directive.

Answer – D.

Explanation

The Question states that Ryan is leading the Project Management Office of the CRAC company. There are three types of PMO: - Supportive, Directive, Controlling. According to the situation stated in the question, Ryan is heading PMO of type Directive. Directive PMO's control and manage the project and Project Managers directly report to the PMO.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 48.

Question 112:

During the closing, the Project team uses a technique to analyze the interrelationships between different project variables that contributed to the project outcomes to improve performance on future projects. Which of the following tools can be used to show the results of such analysis?

- A. Scatter diagram.
- B. Fishbone diagram.
- C. Flowchart
- D. Affinity diagram.

Answer – A.

Explanation

The question states that the Project team is involved in regression analysis. Scatter diagram shows relationship among two variables of distributed value range. So, among all options, Option A becomes the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 126, 293.

Question 113:

A new Project Manager joins the Software development project during the middle of execution. Before meeting all key stakeholders, he wants to prepare a report that compares the current and desired level of engagement of stakeholders involved in the project.

What should he review to fetch this necessary information?

- A. Stakeholder Register.
- B. Stakeholder Engagement assessment matrix.
- C. Stakeholder Engagement plan.
- D. Communication Management Plan.

Answer – B.

Explanation

As per the question, the Project Manager should consult the Stakeholder Engagement assessment matrix to compare the Stakeholder's current and desired level of engagement. Some may get confused with the Stakeholder Engagement Plan, but the Stakeholder Engagement Plan documents "how the adjustments to stakeholders' level of engagement will be achieved".

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 521.

Question 114:

Which of the following is not true in relation to Critical Path?

- A. Critical paths may have zero or negative float
- B. It's the longest path that has the highest total float
- C. There can be more than one Critical Paths in a project
- D. It's used to estimate the minimum project duration.

Answer – B

Explanation

A, C, D are true statements. B is false. The critical path is the sequence of activities that represents the longest path that has the least total float (Usually zero) through a project, which determines the shortest possible Project duration.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 210.

Question 115:

The recent sprint review meeting has detected a lot of Quality problems in the project. The Project team is currently involved in the evaluation of the causes of poor quality. What process is getting performed here?

- A. The recent sprint review meeting has detected a lot of Quality problems in the project.
- B. Direct and manage project work.
- C. Manage Quality.
- D. Monitor risks.

Answer – C.

Explanation

Manage Quality process increases the probability of meeting quality objectives by identifying the ineffective processes and the causes of poor quality.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – .288.

Question 116:

The functional manager asks the project manager how the project will be impacted if two key resources are removed from the project. To come up with a correct response project manager is required to do: -

- A. Resource Leveling.
- B. Schedule Compression.
- C. Fast Tracking.
- D. What —If analysis.

Answer: D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 213.

Question 117:

Ryan is a Project Manager of a manufacturing project. Recently, a thunderstorm has destroyed 50 percent of the raw materials immediately required for the project. Risk response actions are owned by a preferred vendor who has other competing demands in the current situation. What should the project manager do to overcome this situation?

- A. Look for an alternative solution.
- B. Adjust the Project Schedule.
- C. Apply influencing skills to encourage the preferred vendor to take necessary actions immediately.
- D. Talk to Sponsor and request him to add more funds in the management reserve.

Answer – C.

Explanation

As per the given scenario, the project manager has to initiate the Implement Risk Responses process. Some risk response plans may be owned by people outside the project team, and the current scenario also depicts the similar scenario. In this case, the project manager has to apply his influencing skill to encourage the nominated risk owner (in this case the preferred vendor who needs to supply raw materials) to take necessary actions immediately.

Further, Influencing, an example of Interpersonal and Team skills is a valid Tools and techniques used in the Implement Risk Response process. So, among all options provided, C is the best answer. One can think of options A, B, D, only after option C fails.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 451.

Question 118:

Ryan is a junior Project Manager who has just joined the ABC Organization. He is having a meeting with a Senior Manager to understand the importance of OPA. Which of the following is not an example of OPA?

- A. Pre-Approved Supplier list.
- B. Project Files from previous projects.
- C. Lessons Learned Register.
- D. Reporting Methods.

Answer: C.

Explanation

C is the correct answer. A, B, D are examples of OPA. C is an example of Project Document.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 39-41, 559.

Question 119:

Which of the following regarding Manage Project Knowledge is True?

- A. Managing project knowledge involves just documenting it so that it can be shared.
- B. Managing project knowledge involves just obtaining lessons learned at the end of the project in order to use it in future projects.
- C. Co-location improves sharing of tacit knowledge among individuals.
- D. Beliefs, experience, know-how are examples of explicit knowledge.

Answer- C.

Explanation

Tacit knowledge remains in the mind of individual experts and normally shared through conversations and interactions between people. Co-location helps team members to work closely with one another and give chances for face-face interactions which helps in gaining tacit knowledge. Only Explicit knowledge can be documented, so option A is not correct. Lessons learned are generally captured throughout the project, so option B is incorrect. Belief, know-how, experience are examples of tacit knowledge, so option D is also incorrect. Therefore, among all options, Option C is the right answer.

Refer to the PMI website for more details.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 100.

Question 120:

Ryan is a Project Manager of the VCR project, which is currently under planning. Now, he and the project team are sequencing activities and have just begun to identify dependencies.

What can the project team use to get started as quickly as possible?

- A. Enterprise environmental factors.
- B. Organizational process assets.
- C. Milestone list
- D. Scheduling tool.

Answer – B.

Explanation

Organizational process assets include historical documents from past projects, along with templates. Templates, in particular, can help expedite the preparation of network diagrams.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 189.

Question 121:

A senior project manager is doing knowledge-sharing to a newly appointed Junior Project Manager on Scope Control and Scope changes related to Change Request management. He is recommending updating the lessons learned register always while doing the scope control from the very beginning.

What is the reason behind the Senior Manager's suggestion?

- A. Lessons learned earlier in the project can be applied in later phases in the project to improve Scope Control.
- B. Lessons learned define how product scope will be controlled.
- C. Lessons learned give performance information over a period of time.
- D. Lessons learned give comparison information of Plan Vs Actual.

Answer- A.

Explanation

Lessons learned register is used as input as well output for the Control scope

process. As an output, lessons learned should be updated with the actions chosen for scope control. This information can be used in later phases to improve the Scope control and verification process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 169,171.

Question 122:

The project scope statement consists of multiple components. All of the following are included within the Project Scope Statement EXCEPT for which one?

- A. Product Scope Description.
- B. Deliverables.
- C. Project Requirements.
- D. Project Exclusions.

Answer – C.

Explanation

The project scope statement includes the product scope description, deliverables, acceptance criteria, and project exclusions. The project requirements are considered to be part of the requirements documentation.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 155.

Question 123:

At the end of the project, all of the following are equal EXCEPT for : -

- A. BAC.
- B. PMB.
- C. PV.
- D. EAC.

Answer: D.

Explanation

BAC is the Budget at Completion, which is the summation of PV for the project, which is sometimes referred to as the Performance Measurement Baseline (PMB). These are set at the beginning of the project, and all

represent the total authorized budget for the project, which, if changed, changes them all. Estimates at Completion (EAC), however, will be greater or less than the planned budget depending on whether the project comes in over or under budget.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 263-267.

Question 124:

Ryan is a Project Manager of a software development project. He is working with the project team to prepare the Project Management plan. The project requires at least three resources who have specialized knowledge on a particular niche technology.

What should the Project Manager review to find out whether such resources are available internally?

- A. Organizational Process Assets.
- B. Enterprise Environmental Factors.
- C. WBS Dictionary.
- D. Project Team assignments.

Answer: B.

Explanation

Resource availability and employee capability are examples of Internal EEF. The project team should review the Internal EEF to fetch information about resources and knowledge available internally within the organization.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 38.

Question 125:

During the development of a software development project's project management plan, the project management team decides to build a working environment where a self-organizing team can do their work without any centralized control.

What needs to be specially taken into consideration while doing resource planning?

- A. Acquire industry standard Subject Matter Expert from Market
- B. Appoint a technical architect for the role of a project Manager.

- C. Ensure that the project scope is defined in detail.
- D. Acquire generalized specialists.

Answer – D.

Explanation

As per the situation stated in the question, the project management team has decided to adopt an agile development approach for project work execution. Agile project teams are generally self-organizing and cross-functional. Cross-functional teams consist of team members with all the skills necessary to produce a working product. In these teams, the team functions independently without centralized control. The project manager may not be called a project manager in these teams. Instead, the project manager may adopt a Laissez-faire leadership style, allowing the team to establish their own goals. In resource planning, successful self-organizing teams consist of generalized specialists rather than subject matter experts. These generalized specialists then continuously adapt to the changing environment. In the resource acquisition process, only those people should be hired who can welcome changes any time in the project and even request constructive feedback. So, among all options Option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –311.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 42.

Question 126:

**Projects are initiated by an entity : -
(Choose the Best Answer)**

- A. External to the Project
- B. Internal to the Project
- C. CEO of the Buyer Organization.
- D. PMO.

Answer – A.

Explanation

Projects are initiated by someone external to the project, such as Sponsor, Program or Project Management office, or any other authorized representative.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 77.

Question 127:

The Project team is struggling in the requirements gathering phase of a complex project undertaken to meet modern-day business needs.

Stakeholders from the Buyers' side who want to help are finding it very difficult to articulate their requirements. The Project Manager decides to send a team of experts to the Buyer's worksite to experience the day to day work to uncover the hidden needs.

The Interpersonal technique that the Project Manager is trying to apply over here is popularly known as: -

- A. Interviewing.
- B. Job Shadowing.
- C. Reverse Shadowing.
- D. Facilitation.

Answer – B.

Explanation

Job Shadowing provides a direct way of viewing individuals in their environment and how they perform their jobs.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 145.

Question 128:

The project team has chosen an adaptive product development cycle.

Which of the following leadership styles will you have to adopt to increase the project's success probabilities? (Choose the best answer)

- A. Transformational.
- B. Laissez-faire.
- C. Servant Leader.
- D. Charismatic

Answer- B.

Explanation

Laissez-faire is an example of Hands-off style where project managers allow team members to make their own decisions and establish their own goals. A few may get confused with the option C - Servant Leader. One should remember, Servant Leader is a role, Servant Leadership is a leadership style. In an adaptive environment, both Laissez-faire and Servant Leadership styles are followed on a need basis.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 65.

Question 129:

You are a Project Manager of XYZ company. You are currently performing Stakeholder Analysis for an upcoming project to align expectations and gain support for the project. The initial list of Key Stakeholders is first time identified as a part of which process?

- A. Develop Project Charter.
- B. Stakeholder Register.
- C. Business Case.
- D. Stakeholder Engagement plan.

Answer – A.

Explanation

The question here asks about the process that fetches the initial list of key stakeholders.

Choice A - Develop Project Charter - Correct - Project Charter, the main output of Develop Project Charter process contains key stakeholder list and high level classification of roles and responsibilities of everyone involved in the project.

Choice B - Stakeholder Register - Incorrect - It is an example of a Project Document that records all Stakeholder information.

Choice C - Business Case - Incorrect - It is a Business Document which is input for the Develop Project Charter process. Business case justifies why the project was undertaken.

Choice D - Stakeholder Engagement plan - Incorrect - It is an output for the Plan Stakeholder Engagement Process. It provides an actionable plan to interact effectively with the stakeholders.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) – 81, 509.

Question 130:

The project team has religiously planned all the risk responses for all Identified risks. Still there are some risks which still remain because they are deliberately accepted. How should the project manager handle these risks, if they occur?

- A. Use Contingency reserve to manage these risks.
- B. Use Management Reserve to manage these risks.
- C. Submit a Change request to change the baseline.
- D. Create a workaround to manage these risks.

Answer – A.

Explanation

The question talks about the Residual risks. It's a Known-Unknown risk. Contingency reserve is used to manage these risks and such risks are handled by the Project Manager.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 448.

Question 131:

A new Project Manager joins the project to replace a Project Manager terminated for poor performance and poor stakeholder relationships. As a newly appointed Project Manager, he would like to understand how to interact with all the critical stakeholders involved in the project. Which of the following would be best for him to review?

- A. Stakeholder Register.
- B. Stakeholder Engagement Plan.
- C. RACI
- D. Project Charter.

Answer – B.

Explanation

The Stakeholder Engagement Plan guides how the various stakeholders can be best involved in the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 525.

Question 132:

The following analysis recommends preventive actions if necessary by warning the Project Manager ahead of time that there may be a problem later in the schedule if the established circumstances persist. What is the data analysis tool referenced here?

- A. Variance Analysis.
- B. Alternative Analysis.
- C. Trend Analysis.
- D. Schedule forecasts.

Answer – C.

Explanation

Trend analysis is used to forecast future performance based on past results.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 111.

Question 133:

The Project team is preparing a report with the help of a tool to illustrate those risks which have been categorized by using more than two parameters. The tool that can be used in this scenario is an example of: -

- A. Bubble Chart.
- B. Control Chart.
- C. Pie Chart.
- D. Probability and Impact Matrix.

Answer – A.

Explanation

When risks are categorized using more than two parameters, the Probability and Impact matrix cannot be used and other graphical representations are required, such as the Bubble chart.

Bubble charts help to display three-dimensional data. First, the risks are analyzed by an evaluation of these dimensions, then the criticality of the risks

are evaluated based on the results from the analyses, and finally the results are presented in a bubble diagram.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 425- 426.

Question 134:

Which of the following estimating techniques takes risk and uncertainty into consideration?

- A. Analogous estimating.
- B. Three-point estimating.
- C. Parametric estimating.
- D. Bottom-up estimating.

Answer – B.

Explanation

Three-point estimate helps us to define an approximate range of an activity's duration.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 244,245, 724.

Question 135:

As a newly joined Project Manager you are currently reviewing the estimation of processing speed of an application prepared by the old Project Manager. The Old project manager applied Three-point estimation and recorded the following information: The PERT estimate is 9 seconds. The best-case estimate is 4 seconds and the worst-case estimate is 18 seconds. Since Most-likely duration estimate is not provided in the document, you are required to calculate this in order to complete the estimation process of the activity. What is the Most-Likely duration estimate, as per the given data?

- A. 4 Seconds.
- B. 11 Seconds.
- C. 8 Seconds.
- D. 18 Seconds.

Answer: C.

Explanation

$PERT = (Pessimistic + 4 \text{ times Most Likely} + Optimistic) / 6$
 $9 = (4 + 18 + 4M) / 6$
 $54 = 22 + 4M$
 $4M = 54 - 22 = 32$
 $M = 8$

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 244, 245, 724.

Question 136:

The Project manager and the team are working on a Fixed Price Project. The project is currently under planning. Team has proactively gathered requirements from all possible sources and came up with a very detailed Work Breakdown Structure (WBS).

What estimation technique is most suitable in the current scenario?

- A. Analogous estimating.
- B. Three-point estimating.
- C. Parametric estimating.
- D. Bottom-up estimating.

Answer - D.

Explanation

Bottom-up estimating is recommended when detailed scope is available. Here, as per scenario stated in the question, the team has successfully gathered requirements from all possible sources and came with the detailed Work Breakdown Structure.

So, among all given options, Option D becomes the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 244, 245.

Question 137:

As a Project Manager, you are updating the Stakeholder Register to include three Specialists who have just joined your project temporarily to implement a Business-critical task that involves Pure Risk.

How will the newly appointed specialists be classified in the Stakeholder Register based on their direction of influence in the project?

- A. 3 Specialists will be marked as Outward in the Stakeholder Register.
- B. 3 Specialists will be marked as Sideward in the Stakeholder

- Register.
- C. 3 Specialists will be marked as Downward in the Stakeholder Register.
 - D. 3 Specialists will be marked as Influential in the Power/Interest grid.

Answer – C.

Explanation

The question implies that the three specialists are hired temporarily to complete important project work. It suggests that their Direction of Influence in the project is Downward as they can influence the project with their specialized knowledge in a temporary capacity.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 513.

Question 138:

An unidentified risk occurs after the project team implements a risk response plan. Which document should be updated FIRST to record this information?

- A. Risk Register.
- B. Risk Report.
- C. Issue Log.
- D. Lessons Learned Register.

Answer - C.

Explanation

Whenever a risk occurs, it becomes an issue. All Issues raised during execution are recorded in the Issue log first. People may get confused with the statement “after implementation of the risk response plan,” some may answer option A.

Secondary Risk occurs after the Risk Response plan gets implemented, but it’s a Known-Unknown risk.

According to the question, an unidentified risk occurs after the project team implements a risk response plan; it never stated the outcome of the process had raised the Risk. The Issue Log is an output of the Implement risk responses process that records all issues identified during the process.

Unidentified Risk that occurs is considered an issue, and we are required to do a workaround for the same.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 452.

Question 139:

A Project Team is having a meeting with all identified stakeholders to state the objectives and goal of the project and gain commitment from stakeholders on the project objectives. In such meetings who talks about the Strategic direction of the project?

- A. Project Manager.
- B. Project Sponsor.
- C. Project management Team.
- D. CEO of the Buyer company.

Answer – B.

Explanation

In this scenario Kick-off meetings are getting conducted. It's the responsibility of the Sponsor to state the strategic direction of the project in the Kick-off meeting.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 86.

Question 140:

Troy is a Project Manager of a complex automation project which is currently under planning. In a meeting with the Project team to determine initial Stakeholder communication strategies, Troy sees that the Team has done well in stakeholder analysis and grouped stakeholders based on Power-Interest-Attitude. What does this signify?

- A. Team has prepared Power/Interest gnd to map identified stakeholders in a project
- B. Team has prepared a Stakeholder Cube to classify stakeholders on the basis of Power, Interest and Attitude.
- C. Team has used a Data analysis tool to classify stakeholders according to their direction of influence.
- D. Team has prepared a Salience model to classify Stakeholders

based on Power, Interest Attitude.

Answer – B.

Explanation

The Question states that the Team has prepared the Stakeholder Cube. The Power-Interest grid has two axes, and if you add the 3rd axis, you'll get a Cube. Here Attitude is the third axis added to the Power-Interest grid. Stakeholder Cube is the refinement of the grid model, which combines the grid elements into a three-dimensional model that can help the Project Team develop Stakeholder communication strategies. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 513.

Question 141:

An aviation company decides to build a new website to meet its growing strategic needs. The company chooses to award a contract to a well-known IT services company for building their new website. The process of implementing a legal agreement for delivery followed by awarding a contract happens as a part of which of the following processes?

- A. Control Procurements.
- B. Develop Project Charter.
- C. Acquire Resources.
- D. Conduct Procurements.

Answer – D.

Explanation

Contract Procurements is the process of obtaining seller responses, selecting a seller, and awarding a contract. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 482.

Question 142:

Ryan is a Project Manager of a Health care project. During planning, the project team determines that the products' testing is heavily dependent on the on time delivery of the IoT devices by one of the

suppliers. The following situation is an example of: -

- A. Internal Dependencies.
- B. Discretionary Dependencies.
- C. Hard logic Dependencies.
- D. External Dependencies.

Answer – D

Explanation

External dependencies involve a relationship between project activities (Testing the product in this case) and non-project activities (delivery of the IOT devices from an external supplier as in this question) . These dependencies are usually outside of the project team's control.

Here as per the situation stated in the question, the project team has to depend on the on time delivery of the IOT devices in order to perform product level testing . Such a situation is outside the control of the team. so, among all options, option D is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 192.

Question 143:

During the closing, a Project Manager is trying to determine what to do with unused physical resources that are no longer required for the project. What should the Project Manager do next?

- A. Review the Project Management Plan.
- B. Review the Procurement Management Plan.
- C. Review the Requirement Documentation.
- D. Review the issue log.

Answer – A .

Explanation

Here, in this case, Resource Management Plan, the Project Management Plan's subsidiary plan needs to be referred to know how Project resources can be released. Since the Resource Management plan is not there in provided options, reviewing the Project Management plan is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 121,318.

Question 144:

Ryan is a Project Manager of a large IT automation project of a geographically dispersed team. Which of the following is not a challenge to build an effective team?

- A. Cultural differences.
- B. Time zone differences.
- C. Doing work from Home offices.
- D. Security and compliance issues.

Answer: C.

Explanation

The availability of communication technology like email, audio/video conferencing, social media, web-based meetings has made the existence of virtual teams or geographically dispersed teams feasible. In the Virtual team model, working from home offices are generally allowed.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 333.

Question 145:

You are a Project Manager of an IT project that involves technologies which are very new in the market. During risk response planning, keeping a number of ambiguity risks in mind, you and team divide the project into phases and decide to adapt an approach of building a prototype of the actual product and take constructive feedback from stakeholders before the start of execution of the first phase. Here in this scenario prototyping is an example of: -

(Choose the best answer)

- A. Risk Exploitation strategy for handling opportunities.
- B. Risk Mitigation strategy for handling threats.
- C. Quality Control Strategy.
- D. Rolling wave planning.

Answer – B

Explanation

Mitigation may involve prototype development for risk reduction. Ambiguity risks are addressed through exploration and experimentation, of which Prototyping is a great example. Here the Project team is trying to transform the ambiguity risk into Known-Unknown by building a prototype.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443.

Question 146:

An External reviewer informs the Project Manager that the documented results of the Control quality activities are not captured in the right format.

What should the Project Manager review to know the prescribed format of capturing Control Quality results?

- A. Quality Report.
- B. CPA
- C. Configuration Management Plan.
- D. Quality Management Plan.

Answer: D.

Explanation

The Question implies that an Auditor (External reviewer) has informed the Project Manager that data has not been captured in the correct format in the Quality Control Measurements document. Quality control measurements are documented results of Control Quality activities. Here in this case the Project Manager should review the Quality Management plan to find out the correct format to be used.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 305.

Question 147:

The members of the Project Team are involved in developing actionable plans to interact effectively with stakeholders. What can assist the project team in the selection of the optimal stakeholder engagement strategy mix?

- A. Risk Management Plan.

- B. Risk Register.
- C. Stakeholder Register
- D. Assumption log.

Answer – A.

Explanation

The question states that the project team is involved in Stakeholder engagement planning. Here in the question "optimal stakeholder engagement strategy is the key word".

Stakeholder register provides a list of project stakeholders including additional classification data and other information.

Risk register contains the identified risks of the project and usually links them to the specific stakeholders e.g. risk owners.

Assumption log provides information about assumptions and constraints which may be linked to a specific stakeholder.

Risk management plan may contain risk thresholds or risk attitudes (risk averse/risk neutral/risk taker) that can assist the project team in the selection of the optimal stakeholder engagement strategy mix.

So, among all options provided, choice A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 518, 407.

Question 148:

A project that involved multiple numbers of vendors is on the verge of closure. There have been significant changes in contractors and suppliers during the project's life cycle to meet the growing project needs.

Where should these changes be reflected?

- A. Stakeholder register.
- B. Procurement SOW.
- C. Resource Requirements.
- D. Make or Buy decisions register.

Answer – A.

Explanation

As the project work progresses through the execution phase, the contractors and suppliers may change. These changes are reflected in the stakeholder

register.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 501.

Question 149:

As a Project Manager, you are worried about consistent changes that the Project has gone through. When the Project is almost 80% complete, you decide to do scope verification to ensure that things are under control. What needs to be reviewed to find variance in scope?

- A. Requirement Traceability Matrix
- B. Scope baseline.
- C. Requirement Documentation.
- D. Lessons Learned Register.

Answer – B.

Explanation

Here in this scenario, as the question states, the project manager is involved in the control Scope process, and the scope baseline can be used to find any variances in scope. During the Control Scope process, the scope baseline is compared to actual results to determine if a change, corrective action, or preventive action is necessary.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 169.

Question 150:

During the start of execution, a Project Manager has planned a face-to-face meeting with a key stakeholder who has missed the kick-off meeting. What is the most valuable information that needs to be conveyed and explained to the stakeholder during the meeting? Choose the best option.

- A. Project Scope.
- B. Project Schedule.
- C. Project Risks and assumptions.
- D. Project Goal.

Answer- D.

Explanation

The Question implies that the Project manager wants to ensure that the stakeholder clearly understands the Project Goal as well as how his contribution will enhance project Success.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 524.

Question 151:

During the middle of the execution, management decides to close the contract of a seller who is not performing well. Contract closure is part of which of the following processes?

- A. Control Procurements.
- B. Conduct Procurements.
- C. Validate Scope.
- D. Close Project or Phase.

Answer- A.

Explanation

Even if a contract is terminated before completion, procurement needs to be closed out, which is done as a part of the Control Procurements process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 492, 499.

Question 152:

A newly joined Project Manager who has joined the project during execution finds that the Project Management Plan is not aligned with the Business Case. What should the Project Manager do next?

- A. Raise a Change Request to update the Business Case.
- B. Report to Sponsor regarding the gap between Business Case and Project Management Plan.
- C. Escalate the issue to his Senior Management
- D. Update his findings in the Risk Register.

Answer – B.

Explanation

It's the Sponsor who is accountable for the preparation and modification of the Business Case, the document that states objectives and the reason for Project Initiation. If the Project Management Plan is not aligned with the Business Case, it's the duty of the Project Manager to get in touch with the Sponsor, who can take the final call regarding the situation mentioned above. Among all options provided, B looks the best answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -30,77-78, 125,251.

Question 153:

You are the new Servant Leader for your company's latest software development project running in Agile. Which of the following defines your role?

- A. Your primary job is to ensure the Project's success.
- B. Your main role is to ensure the processes and help the project team to remove impediments.
- C. Your primary responsibility is to manage the Minimum Viable Product (MVP).
- D. Your main role is stakeholder management

Answer - B.

Explanation

The question here is referring to the Scrum Master, who acts as Servant Leader of the team. His job in the project is ensuring the process, not the project. The main focus of Servant Leadership is understanding and addressing the development team members' needs to enable the highest possible team performance.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101,154.

Question 154:

During the early stage of execution, a new project manager joins the manufacturing project where the environment is counterproductive. In his first meeting with the team, he noticed that team members are quarreling, and people do not listen to one another's point of view. What does the project manager do next to overcome this situation?

- A. Declare reward for early project completion.
- B. Avoid the situation, as it is so common in the storming phase.
- C. Refer to the Team charter, and update the same with the team.
- D. Update Communication Management Plan.

Answer – C.

Explanation

The Team Charter establishes clear expectations regarding acceptable behavior by project team members. Early commitment to clear guidelines decreases misunderstandings and increases productivity. Team charter works best when the team contributes to it.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 320.

Question 155:

Two weeks before the start of execution, the Project Manager and the team had just completed the initial risk response planning and updated the risk register with agreed upon risk response strategies, contingency plans, risk triggers and other necessary information.

What should the project manager do next?

- A. Gain risk response plan acceptance from the sponsor.
- B. Implement the agreed upon risk responses.
- C. Update the project management plan.
- D. Schedule a Risk review meeting in order to confirm the risk response plan.

Answer – C.

Explanation

As per the question, the project is in the pre-execution stage and the project team has just completed the plan risk responses process.

Choice B is wrong as the project is yet to move to execution.

Choice D is ruled out as Risk review happens as a part of the Monitor Risks process.

Option A is outright rejected.

The Project Management Plan gets updated as a part of the plan risk responses process and strategies adopted for preventing threat and availing

opportunities may result in Project Management Plan update in most possible scenarios. The Schedule Management Plan, Cost Management Plan, Quality Management Plan, Procurement Management Plan, Procurement Management Plan ,along with all the baseline documents may get updated , as a result of the plan risk responses process.

So, among all options, option C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 447.

Question 156:

The Project team is currently engaged in sequencing activities of a particular work package which is part of the project. The values are: Task A will take 5 days to complete. Task B duration is 3 days and follows Task A. Task C has 4 days and follows task B. Task D will take 4 days and follows Task A, Task B, Task F. Task E has 3 days and follows Task C and D. Task F starts along with Task A and has duration of 4 days. There is a Finish-To-Start relationship that exists between all predecessors and successors and Task E is the last task to be performed. What is the Critical path?

- A. A-B-D-E.
- B. A-B-C-E.
- C. Both A and B.
- D. F-D-E

Answer – C.

Explanation

A Project can have more than one critical path. Here, the shortest time the work package can be completed is 15 days and both the series of tasks A-B-D-E and A-B-C-E are in critical path.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 188-190.

Question 157:

Ryan is a Project Manager of Qnx company, which has just attained an advanced level Quality certification, as a result of which threshold of quality metrics needs modification for his project, which is currently in execution. Some deliverables are already ready and passed the user

acceptance test. The sponsor and stakeholders responsible for managing changes have approved the change request regarding change in quality metrics.

What should the Project Manager do next?

- A. Apply quality metrics to all project deliverables, other than those who are ready.
- B. Apply change to all deliverables in the project.
- C. Evaluate the change with other constraints in the project
- D. Discuss the issue with the sponsor and take everyone's buy in.

Answer – B.

Explanation

The situation states that the change request for updating the quality metrics has been approved by the Sponsor and the Change Control Board.

Since the new Quality requirement is mandatory, the change should be applied to all deliverables, even to the finished ones.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 90,95.

Question 158:

The Project Management team has decided to award the work associated with three complex work packages to a preferred vendor.

What should the Project Manager review to know when the resources are required to be available for the project and when they should be acquired?

- A. Project schedule.
- B. Resource calendars.
- C. Resource requirements.
- D. Stakeholder register.

Answer - A

Explanation

The situation states that the Acquire Resources process has been performed and a decision has been taken to hire a vendor to work on 3 work packages. Resource Calendar determines resources needed for the project are actually

available for the project or not, so B is incorrect.

Resource Requirements identify which resources are required to be acquired, so the choice C is not the right option.

Stakeholder registration in this scenario may reveal stakeholders' needs or expectations for specific resources to be used on the project, so the choice D is also incorrect.

The Project Schedule shows the activities and their planned start and end dates to help determine when the resources need to be available and acquired. So, among all options available, A is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 331.

Question 159:

During the intermediate stage of the project, a new project manager joins. The project is running behind schedule. After reviewing all documents with the Project Team, the Project Manager realizes that a few uncontrolled changes have happened in the project due to sudden changes in regulatory requirements.

The uncontrolled expansion to project scope without adjustments to cost, schedule, resources referred to as: -

- A. Gold plating.
- B. Rework
- C. Scope creep.
- D. Cost of Non Conformance.

Answer – C.

Explanation

The Uncontrolled expansion of Project or Product scope without adjustment to time, cost, and resources is referred to as Scope creep. In project level, Scope creep can happen due incomplete Project Scope Statement, ineffective Change Control system, reasons external to organizations; e.g., market conditions, regulatory requirements, etc.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 154, 168, 722.

Question 160:

The Scrum team is engaged in a meeting to ensure the backlog is ready

for the next sprint. What kind of meeting is mentioned in the given scenario?

- A. Sprint Review meeting.
- B. Sprint Retrospective meeting.
- C. Project milestone meeting.
- D. Backlog refinement meeting.

Answer – D.

Explanation

Often, Scrum Teams come together once per Sprint, or once per week to have their "Refinement Meeting". The Product Owner shares what Product Backlog items (PBI) need to be refined and the whole team discusses them.

Option – A – Sprint Review meeting – Incorrect – The purpose of the Sprint Review is to inspect the outcome of the Sprint and determine future adaptations. It's a working session where the scrum team and stakeholders review what was accomplished in the sprint and what has been changed in their environment.

Option – B - Sprint Retrospective meeting – Incorrect – The purpose of the Sprint Retrospective is to plan ways to increase quality and effectiveness. Such meetings help the team identify improvement actions by gathering feedback on how they worked in the ongoing sprint. It's the last meeting of the sprint.

Option – C - Project milestone meeting – Incorrect - It's a made up term.

Option – D - Backlog refinement meeting – Correct – The Product Owner and the team prepare some stories for the upcoming sprint or the iteration. Generally, the discussion happens on the top items of the Product Backlog. Such a meeting occurs in the middle or near the end of an iteration. The purpose of this meeting is to ensure the backlog is ready for the next sprint. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 52.

Question 161:

Your project uses a hybrid framework for its new product development project. The team has organized the work into small sprints and uses a board to display its work and monitor progress. What framework the project used in the above-stated scenario?

- A. Scrum.
- B. Scrumban.
- C. Kanban.
- D. Paired programming.

Answer – B.

Explanation

Scrumban is a project management framework that combines essential features of two popular agile methodologies: Scrum and Kanban. The Scrumban framework merges the structure and predictable routines of Scrum with Kanban's flexibility to make teams more agile, efficient, and productive. In Scrumban, the work is organized into small Sprints, and the Kanban board is used to visualize and monitor the work.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 108.

Question 162:

Your team has adapted the Feature-Driven development approach to meet a specific need of a large software development project.

Which of the following are primary roles on a Feature-Driven development project?

Select any two.

- A. Project Manager.
- B. Scrum Master.
- C. Domain Expert.
- D. Product Owner.

Answers – A, C

Explanation

There are six primary roles on a Feature-Driven Development project: -

- I. Project Manager.
- II. Chief architect.
- III. Development manager.
- IV. Chief programmer
- V. Class owner.
- VI. Domain expert.

Question 163:

Two teams are not getting along well and always creating conflicts. The team lead approached the Project Manager for help. What should the project manager do first to manage the conflict?

- A. Analyze the context of the conflict.
- B. Interpret the source and stage of the conflict.
- C. Evaluate the appropriate Conflict resolution solution.
- D. Analyze both the team member's influence in the project.

Answer – B.

Explanation

The Project Manager should first interpret the source and stage of the conflict. That will guide him to analyze the context of the conflict which will further assist him to decide an appropriate conflict resolution approach.

Task 1 of the People domain states the same. It reveals 3 enablers for the task statement Manage Conflict.

Manage Conflict - Enablers

- Interpret the source and stage of the conflict.
- Analyze the context of the conflict.
- Evaluate/recommend/reconcile the appropriate conflict resolution solution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 348

PMI PMP Examination Content Outline – June 2019 – Domain 1 – Task 1

Question 164:

The successful use of Scrum depends on people becoming more proficient in living five values. Which of the following are part of Scrum Values?

Choose any two.

- A. Integrity.
- B. Respect.
- C. Team work
- D. Openness.

Answers– B, D.

Explanation

The five Scrum Values include: - Commitment, Focus, Openness, Respect, and Courage. These values give direction to the Scrum team with regard to their work, actions, and behavior.

scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page – 4.

Question 165:

Two days after a successful Iteration review session for a Software development project, a stakeholder complains that the software is not user friendly and end-users find it challenging to use.

What technique can the team employ to prevent this in the future?

- A. Conduct Smoke tests before every release.
- B. Observe stakeholders interacting with a preliminary version of the software.
- C. Adapt DevOps at work and plan more frequent releases.
- D. Define organizational usability standards.

Answer – B.

Explanation

Usability testing is an important way that teams can test their software to make sure it is easy to use, and agile teams conduct frequent reviews by testing the software and incorporating the improvements back into the deliverables. A very common way to perform usability testing is to observe users while they interact with early versions of the software.

Question 166:

In a software maintenance project, the team pulls features from the backlog to start their work. Each feature sometimes takes a different amount of time to finish.

What type of delivery method does this describe?

- A. Iteration-based agile.
- B. Swarming.
- C. Scrum.

D. Flow-based agile.

Answer – D.

Explanation

In flow-based agile, the team creates a board of their flow and WIP (work in progress) limits. The visualization of the work and the WIP limits manage the scope of work for the team. Sometimes, flow-based agile is more helpful because the team can manage interruptions to the project-based work.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 24,25.

Question 167:

Two team members are not going well among each other and always creating conflicts. The Project Manager decides to break the team and reassign team members. What conflict resolution strategy he employed?

- A. Withdrawal.
- B. Forcing.
- C. Collaborating.
- D. Smoothing.

Answer – B.

Explanation

Here the Project Manager used his positional power to resolve the ongoing conflict. So, the best answer is B.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 349.

Question 168:

Match the following

SL no	Name	Description
1	Anti-Pattern	a. Improves collaboration between development and operations stuff.
2	Continuous Delivery	b. The practice used in software development to frequently incorporate new and changed code into their project code repository.
3	Continuous Integration	c. A known flawed pattern of work that is not advisable.
4.	DevOps	d. The practice of delivering feature increments immediately to customers through the use of automation technology.

- A. 1-d, 2-d, 3-a, 4-b.
- B. 1-c. 2- d, 3-a, 4-b.
- C. 1-c, 2-d, 3-b, 4-a.
- D. 1-a, 2-d, 3-b, 4 -c.

Answer - C.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 150, 151.

Question 169:

The Scrum Master is accountable for the Scrum team's effectiveness. How does the Scrum Master serve the Scrum Team? Select any two.

- A. Coaching the team members in self-management and cross functionality.
- B. Helping in effective Product Backlog management
- C. Determining corred Life Cycle approach for the team.
- D. Helping the Scrum Team focus on creating high-value Increments that meet the Definition of Done.

Answer – A, D.

Explanation

The Scrum Master ensures the process, not the project. Option C is automatically ruled out. The Product Owner maintains the Product Backlog. Therefore, choice B is also not correct.

The options A and D are correct answers.

The Scrum Master serves the team in several ways, including:

- i. Coaching the team members in self-management and cross-functionality.
- ii. Helping the Scrum Team focus on creating high-value increments that meet the Definition of Done.
- iii. Causing the removal of impediments to the Scrum Team's progress.
- iv. Ensuring that all Scrum events take place and are positive, productive, and kept with the timebox.

scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page – 6.

Question 170:

The Scrum Master comes to know that the Team is not happy because the new product owner has directed them to complete 70 story points for the next sprint.

What should the Scrum Master do?

- A. Schedule a meeting with the team to find a way to achieve the Sprint goal directed by the Product Owner.
- B. Advise team to add 2 more developers in the team.
- C. Schedule a meeting with the Product Owner and explain to him that the team is empowered to make Sprint Goal decisions. No one should interfere in it OIT
- D. Escalate the situation to the Senior Management.

Answer – C.

Explanation

The Scrum Master is responsible for removing any barriers in the Scrum Team. The development team has the best insight into task execution. The team is empowered to make their own decisions and commitments. They decide the Sprint Goal, which includes how many story points they will take for an Iteration. The Product Owner can help the team reprioritize the Product backlog and decide the sprint goal but cannot direct them to take up tasks

with 70 story points.

The Scrum Master should coach the roles and responsibilities of the Scrum Team to the Product Owner and encourage him to allow the team to set their Sprint Goal.

Question 171:

Ryan is the newly joined Project Manager for an IT services project. A Team member meets him and conveys that he is not happy with his previous appraisal.

What should the Project Manager do?

- A. Review the Key Performance Indicators set for the team member.
- B. Determine appropriate feedback approach.
- C. Assure the team member that you will address this next time.
- D. Determine training needs for the team member.

Answer – A.

Explanation

The Project Manager should review the Key Performance Indicators set for the team member and determine whether the appropriate decision was considered on the previous occasion.

Task 3 in People Domain of PMI PMP Examination Content Outline states, “Appraise Team member performance against Key Performance Indicators.” This supports the selection of choice A as the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 345.

PMP Examination Content Outline – June 2019 – Domain 1 – Task 3.

Question 172:

In a typical large agile project, who works with the stakeholders and the teams to define the product direction?

- A. Project Manager.
- B. Sponsor.
- C. Team facilitator.
- D. Product Owner.

Answer – D.

Explanation

The Product Owner works with stakeholders, customers, and the teams to define the product direction. Typically, the Product Owners have a business background and bring a deep Subject Matter expertise to the decisions.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.

Question 173:

Sprints are the heartbeat of Scrum, where ideas are turned into value. When a Sprint is considered complete?

- A. The fixed time allocated to the Sprint runs out
- B. All the Product Backlog achieves the Definition of Done.
- C. Soon after the end of the Sprint review meeting.
- D. Team completes all tasks of the Sprint backlog.

Answer – A.

Explanation

The Sprints describe a time boxed iteration in a Scrum. They are fixed length events of one month or less. They generally start with a Sprint Planning meeting and end with the Sprint Retrospective meeting.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

Question 174:

Match the following

SL no	Name	Description
1	Functional requirements.	a. Reasons why a project has been undertaken.
2	Project requirements	b. Describe the behavior of the product.
3	Quality requirements.	c. Describe the actions, processes, or other conditions the project needs to meet.
4.	Business requirements.	d. Capture any condition or criteria needed to validate the successful completion of a project deliverable.

- A. 1-b, 2-c, 3-d, 4-a.
- B. 1-b, 2-a, 3-d, 4-c.
- C. 1-d, 2-a, 3-c, 4-b.
- D. 1-c, 2-b, 3-d, 4-a.

Answer – A.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 148.

Question 175:

A critical success factor for Agile teams is strong Product ownership. Which of the following are the responsibilities of a typical Product Owner?

Select any two.

- A. Managing the Sprint Backlog.
- B. Coaching the team members in self-management and cross-functionality.
- C. Managing the Product Backlog.
- D. Developing and explicitly communicating the Product Goal.

Answers – C, D.

Explanation

The Product Owner is accountable for effective Product Backlog management. The Product Owner works with stakeholders, customers, and the teams to define the product direction.

Choice A - Managing the Sprint Backlog – Incorrect - Responsibility of the Development team.

Choice B - Coaching the team members in self-management and cross-functionality – Incorrect - Responsibility of the Scrum Master.

Choice C - Managing the Product Backlog – Correct - The Product Owner is accountable for effective Product Backlog management.

Choice D - Developing and explicitly communicating the Product Goal – Correct - The Product Owner works with stakeholders, customers, and the teams to define the product direction.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.

Question 176:
Match the following

SL no	Name	Description
1	Daily Scrum	a. One of the four values of the Agile Manifesto.
2	Increment	b. An Agile Principle.
3	Responding to change Over following a plan.	c. Scrum event or ceremony.
4.	Business people and developers must work together daily throughout the project.	d. Scrum artifact.

- A. 1-a, 2-c, 3-b, 4-d.
- B. 1-c, 2-d, 3-b, 4-a.
- C. 1-c, 2-d, 3-a, 4.b.
- D. 1-d, 2-a, 3-b, 4-c.

Answer – C

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 8,9, 101.

Question 177:

In an Agile Software development project, the team discovered a new risk in the middle of a sprint. Where would the team record the newly discovered risk?

- A. Risk Register.
- B. Issue Log.
- C. Scrum Backlog.
- D. Product Backlog.

Answer - D.

Explanation

The Product Backlog is the ordered list of what is needed to improve the product. In Agile-Scrum, all user-centric requirements, risks, issues, and everything about the project gets recorded in the Product Backlog.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 52, 153.

Question 178:

During the middle of the fourth Sprint for eight Sprint releases, the development team realized that they would not meet the release goal. What should the team do next?

- A. Add more developers to the team.
- B. Propose a new release date.
- C. Schedule a meeting with the Product Owner to reprioritize the backlog.
- D. Consult with the Project Manager for improving the team velocity.

Answer – C.

Explanation

In general, in agile projects, the cost and the schedule are fixed, while the scope is flexible. Therefore, the options A and B are ruled out. Option D also makes little sense here. So, meeting with the Product Owner to re-prioritize the backlog, is the best answer among all the choices provided.

Question 179:

Match the following.

SL no	Tools and Technique	Process its used
1	Control chart.	a. Manage Quality.
2	Cost of Quality.	b. Control Quality.
3	Audit	c. Plan Quality Management .

- A. 1-c, 2-b, 3-a.
- B. 1-b, 2-c, 3-a.

- C. 1-b,.2-a, 3-c.
- D. 1-a, 2-c, 3-b.

Answer - B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 277, 288, 298.

Question 180:

**You are the sponsor of a project that is following the Scrum framework. Which of the following are roles for the Scrum framework?
Select any two**

- A. Product Owner.
- B. Project Manager.
- C. Subject Matter Expert
- D. Scrum Master.

Answers: A, D.

Explanation

The Scrum team consists of a Product Owner, Development Team, and Scrum Master.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

Chapter 6: PMP Practice Test 2 - Full Mock (180 Questions) - 2021

Question 1:

Quality is an example of:-

- A. Technical Risk.
- B. External Risk
- C. Communication Risk
- D. Commercial Risk

Answer - A.

Explanation

Quality is the degree to which a set of inherent characteristics fulfills requirements. Quality demand changes are always subjective to technical risk.

Question 2:

Arun is a Project Manager of a large project which has just moved to the Planning stage. The project sponsor is anxious to prepone the Project kick-off and start the execution as soon as possible. He also suggests the Project Manager skip the communications management plan for now and prepare the same during execution on a need basis.

How should the project manager respond to the sponsor to support the completion of the communications management plan?

- A. The communications management plan is required as a part of the project management plan.
- B. The communications management plan is required for estimation and work assignment analysis.
- C. The communications management plan assists stakeholder engagement
- D. The communications management plan is required for the preparation of project report

Answer - C .

Explanation

The Communications Management Plan provides the documented approach to effectively and efficiently engage with the stakeholders by presenting relevant information promptly. Therefore, of the choices provided, option C is the best response, the project manager can give to the project sponsor to support the completion of the communications management plan before moving to execution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 377.

Question 3:

A program manager uses a diagram to prepare a snapshot of the status of various projects occurring in the organization that he is managing. Which diagram or chart is the Program manager using?

- A. Milestone Chart
- B. Bum down Chart
- C. Histogram.
- D. Pie-Chart.

Answer - A.

Explanation

A milestone is used to represent a group of activities or significant events or commitments in the project. A milestone chart shows a group of milestones in an organized way similar to a Gantt chart with one milestone per line vertically with a description on the left and the milestone located horizontally along a time scale showing when it occurs.

Question 4:

Ryan is the project manager of a complex AI automation project who is currently engaged in a Risk review and analysis meeting with his core technical team. The main agenda of the meeting is to convert those risks that may arise due to market conditions and the future developments in regulatory frameworks into the Known-Unknown risks and develop strategic response plans whenever applicable.

What type of risk is being discussed in the situation stated above?

- A. Variability risk
- B. Ambiguity risk
- C. Stochastic Uncertainty.
- D. Emergent Risk

Answer - B.

Explanation

Ambiguity risk, also known as “epistemic uncertainty.” The Greek word episteme means knowledge since they describe uncertainties arising from lack of knowledge or understanding. Areas of the project where imperfect knowledge might affect our ability to achieve project objectives include Elements of the requirement or technical solution, Use of new technology, Market conditions, Competitor capability or intentions, etc. Ambiguity risks are addressed through exploration and experimentation, seeking to define the scope and boundaries of those areas where we have a deficit of knowledge or understanding. The aim is to transform ambiguity risks into “known-Unknowns.”

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 398.

Question 5:

The Project - XFG is running smoothly with a team of Generalizing specialists. The control of the product planning and delivery has been delegated to the team. The Project Manager is trying to build a collaborative decision-making environment while ensuring that the team can respond to changes.

What development approach is being followed in the XFG project?

- A. Predictive.
- B. Iterative.
- C. Traditional.
- D. Agile.

Answer - D.

Explanation

Agile projects run iteratively and incrementally. Here, the team is responsible for planning and delivery. The question suggests that the Project is following

the Agile approach, where the team has the power of controlling the planning and execution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 74.

Question 6:

The project team identified a new risk after an approved Change Request on scope change got implemented. Where should the new risk be documented?

- A. Change Log.
- B. Risk Register.
- C. Issue Log.
- D. Assumption Log.

Answer - B.

Explanation

The question implies that the project team is in risk monitoring activity. The new Know-Unknown risk should be recorded in the risk register.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 458.

Question 7:

The customer is highly satisfied with the outcome of the Digital Banking project delivered by the seller. The customer and the sponsor congratulated the Project Manager of the performing organization and handed over a letter stating the Contract completion.

According to the situation mentioned above, what process the project team has just concluded?

- A. Close Project/Phase.
- B. Control Procurements.
- C. Closed Procurements.
- D. Validate Scope.

Answer -B.

Explanation

The Question indicates the completion of the Control Procurements process, where the customer has provided the formal written notice of the Contract Completion to the seller.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 499.

Question 8:

After formal acceptance of deliverables by the customer, the Project Manager and the team has prepared a report that contains a summary of project performance from the scope, quality, cost, schedule point of view. The report also includes an overview of the project's risks or issues and how they were addressed. After reviewing the report with the Sponsor, the Project Manager decides to share the same with all Stakeholders who are authorized to receive the report.

What report has been prepared by the Project Manager and team?

- A. Work Performance Report.
- B. Project Closure Report.
- C. Final Report
- D. Lessons Learned Register.

Answer – C.

Explanation

The Question states that the project team is doing Project Closure activities, and they have prepared the Final Report to measure project success and stakeholder satisfaction.

A few get confused with the Option - A (Work Performance Reports), which is not correct. Work Performance Reports are the physical or electronic representation of work performance information intended to generate decisions, actions, or awareness. Examples include Progress report, Status report, etc. And above all, the Work Performance Report is not an input or output for closing.

Among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 127,128.

Question 9:

In a project that involves an adaptive Life Cycle, two processes are

repeated for each iteration to confirm scope verification. What are the names of the processes as mentioned in the above scenario?

- A. Validate Scope and Control Scope.
- B. Control Quality and Validate Scope.
- C. Define Scope and Validate Scope.
- D. Direct and Manage Project Work and Validate Scope.

Answer- A.

Explanation

In an adaptive environment two processes (Validate Scope and Control Scope) are repeated for each iteration to confirm Scope Validation. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 131.

Question 10:

Your company has signed a contract with a vendor-FGT for the supply of critical raw materials. Senior Management has asked the Project Manager to furnish the information concerning the legal jurisdiction and the currency to be used for payment to the FGT vendor. Which document should the Project Manager review to fetch this necessary information?

- A. Procurement Management Plan.
- B. Bid Documents.
- C. Procurement SOW.
- D. Source Selection Criteria.

Answer - A.

Explanation

Legal jurisdiction and currency that will be used for payment are recorded in the Procurement Management plan. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 481.

Question 11:

The ZRAC company is ready to launch its new IOT device in the

market. The CEO of the company would announce the new product launch in the company's upcoming press release. Here press release is an example of: -

- A. Pull Communication.
- B. Push Communication.
- C. Interactive Communication.
- D. Small Group Communication.

Answer - B.

Explanation

Press releases are an example of Push Communication.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 374.

Question 12:

A technique that subdivides the project scope and project deliverables into smaller, more manageable components, until the project work associated with accomplishing the project scope is defined in sufficient detail to support executing, monitoring, and controlling the work is also known as: -

- A. Decomposition.
- B. Rolling-wave technique.
- C. Level of Effort.
- D. WBS.

Answer - A.

Explanation

Question is asking about a technique that will breakdown project scope in to more manageable parts, i.e. Decomposition. Decomposition helps to create WBS.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 158.

Question 13:

During Planning, the Project team is struggling in the requirement

gathering process. The project has a significant number of cross-functional requirements in association with security, performance, and disaster recovery that are not finalized yet due to stakeholder differences.

Which of the following is usually the best technique to discover and resolve these kinds of issues?

- A. Questionnaire.
- B. Interviewing.
- C. Facilitated Workshop.
- D. Focus Group.

Answer- C.

Explanation

Facilitated workshops can be used to quickly define and confirm the Cross-Functional requirements and reconcile Stakeholder differences.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 145.

Question 14:

The Change Control Board has approved the addition of two resources in a Software development project. The Project Manager promptly updates the approval status in the Change log. What should be the next course of action?

- A. Initiate the Resource on boarding process.
- B. Update the Project Management plan and the Project Documents.
- C. Ask Sponsor to add more funds in the Management reserve.
- D. Evaluate the CCB decision with the Project Team.

Answer - B.

Explanation

The question implies that a Change request has been approved and the Change log is updated. Next step will be to update the Project Management Plan, Project Documents before initiating the Resource hiring process. C is also a good option, but the question never stated whether the sponsor is part of CCB or not. If Sponsor's presence is not explicitly mentioned in the

question, we should assume Sponsor is also part of CCB.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 115.

Question 15:

The Project Manager, sponsor, and key stakeholders are having a brainstorming session to discuss the success measures of the ongoing Invoice automation project. They are reviewing a document that describes how and when the project's benefits will be delivered and provide guidelines to measure the tangible and intangible benefits. What document is referred to in the above-stated scenario?

- A. Business Case.
- B. Benefits Management Plan.
- C. Cost Management Plan.
- D. Project Charter.

Answer - B.

Explanation

The Benefits Management Plan is a Business document that is created before the initiation of the project. It states how and when the benefits of the project will be delivered.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 33.

Question 16:

The Project Management team has decided to adopt the Agile Scrum methodology for its upcoming software development project. In Agile, which of the following meetings is not a timeboxed event?

- A. Product Backlog refinement meeting.
- B. Sprint Review meeting.
- C. Sprint Retrospective meeting.
- D. Daily Scrum.

Answer – A.

Explanation

In Scrum, all the ceremonies are time-boxed. The events - Sprint Review, Daily Scrum, Sprint Retrospective, cannot be extended. Product Backlog Refinement is one such meeting that is not time-boxed; instead is a continuous process and can be conducted whenever the team wants. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 52, 53, 101.

Question 17:

A year-long project got scrapped during the middle of execution. The project expenditure at the time of termination was \$65000. Here the amount \$65000 is an example of: -

- A. Direct Cost
- B. Indirect Cost.
- C. Sunk Cost
- D. Opportunity Cost.

Answer - C.

Explanation

Sunk Cost means the cost that is incurred cannot be recovered. In this scenario , the amount \$65000 is an example of Sunk Cost.

Question 18:

As a Project Manager, you are doing workload and work assignment analysis of your project team. What tool can be used in this scenario? Choose the best Answer.

- A. RACL.
- B. WBS.
- C. Resource Histogram.
- D. Project Organization chart.

Answer - A.

Explanation

A responsibility assignment matrix is also known as a RACI matrix. It shows project resources assigned to each work package. At a high level it defines the responsibilities of the project team.

A RACI analysis is useful for:

- Workload Analysis: when used against individuals or departments, the overloads can be quickly identified.
- Re-organization: to ensure that key functions and processes are not overlooked.
- Employee Turnover: newcomers can quickly identify their roles and responsibilities.
- Work Assignment: allows duties to be redistributed effectively between groups and individuals.
- Project Management: allows for flexibility in matrix management situations.
- Conflict Resolution: provides a forum for discussion and resolving inter-departmental conflict.

Here as per the scenario stated in the question, a lot gets confused between RACI and Resource Histogram, as the question talks about workload analysis along with work assignment analysis.

In short , RACI helps in workload and work assignment analysis , whereas resource histogram helps in resource allocation planning and revising , and also workload analysis in certain situations.

R (Responsible) of RACI covers the workload analysis part, whereas R (Responsible), A (Accountable), C (Consulted), I (Informed) covers work assignment analysis. RACI stands for :

Responsible: person who performs an activity or does the work. This helps in quick work load analysis.

Accountable: person who is accountable for the correct and thorough completion of the task. This must be one person and is often the project manager.

Consulted: person that needs feedback and contributes to the activity.

Informed: person that needs to know of the decision or action

Resource Histogram is generally a bar chart that helps mainly in resource allocation planning. It shows the number of resources needed per time period; it also illustrates where there is a spike in the need for resources. In certain situations, you can use it to do workload analysis but you cannot do work assignment analysis using Resource histogram.

Among all the options given , A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 317.

Question 19:

During planning, the Project Manager is trying to determine the type and quantity of resources required for the work packages decomposed to scheduled activities. What should the project manager review to get this information?

- A. Project Schedule.
- B. Resource Calendars.
- C. Stakeholder Register.
- D. Resource Requirements.

Answer - D.

Explanation

As per the scenario stated in the question, the project manager is involved in the Develop Schedule process of which Resource Requirements is a valid input .

Resource Requirements identify the types and quantities of resources required for each work package or activity in a work package.

Option A - Project Schedule is the output of the develop schedule process , so it's not the correct option.

Options C -Stakeholder Register is not an input for Develop schedule process, so it's not the correct choice.

Option B- Resource Calendar, though a valid input for Develop Schedule process, provides information in relation to Resource Availability, so it's also not the right choice.

So, among all options Option D is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 208,325.

Question 20:

What is not true regarding Work packages?

- A. Work packages can be decomposed into smaller components called Activities.
- B. The work package is the work defined at the lowest level of the

- WBS for which cost and duration can be estimated and managed.
- C. Each work package can be associated with more than one Control account.
 - D. The level of detail for work packages will vary with the size and complexity of the project.

Answer - C.

Explanation

A, B, D are correct statements. C is false. A control account can have more than one work package. But each work package is associated with only one Control Account.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 157-160.

Question 21:

The project team is currently involved in the Determine Budget process. For establishing an authorized Cost baseline, the Project Manager and the team want to know whether indirect costs are to be included in the Project budget or not.

What should the team review to get this necessary information?

- A. Basis of Estimates.
- B. Cost Estimates.
- C. Cost Management Plan.
- D. Requirement Documentation.

Answer - A.

Explanation

As per scenario stated in the question, the project team is currently involved in the Determining Budget process and the project team wants to know whether indirect costs are to be included in the project budget.

Options A, B, C are valid inputs for the process. Since, Requirement Documentation is not a valid input for the Determine Budget process, option D is ruled out.

The cost management plan describes how the project costs will be structured into the project budget, so C is also not correct. Many can get confused with the option B -Cost Estimates. But cost estimates provide work package level

estimates and additional information or any supporting details in relation to cost estimates are recorded in Basis of Estimates. Basis of estimates specify any basic assumptions dealing with the inclusion or exclusion of indirect or other costs in the project budget.

Therefore, among all options, option A is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –250.

Question 22:

All of the following are part of Scope baseline except:-

- A. Scope Management Plan.
- B. Project Scope Statement
- C. WBS.
- D. WBS dictionary.

Answer- A.

Explanation

Scope Management plan is not a part of scope baseline.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 161.

Question 23:

Ryan is a Project Manager of the GHD project that has just completed planning. In a meeting, the sponsor told him the most likely exit of 3 supportive stakeholders very soon. Ryan promptly adds the information to the risk register. For preparing a risk response plan, he has to find out at what phase of the Project Life Cycle; the influence of these stakeholders is mostly required.

What should the Project Manager review to fetch this required information?

- A. Stakeholder Register.
- B. Stakeholder Engagement Plan.
- C. Communication Management Plan.
- D. Project Charter.

Answer – A.

Explanation

Stakeholder Register contains stakeholder details, expectations, roles and contribution and their influence in the project. It also states at what phase of the Project Life cycle; they will have maximum influence or impact on the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –514.

Question 24:

Arun is a Project Manager of an automation project which has just moved to execution. Two key stakeholders were supportive initially, playing a neutral role for the past few days. What tool needs to be used to analyze the current situation and improve these two stakeholders' engagement levels?

- A. Stakeholder Engagement Assessment Matrix.
- B. Mind mapping.
- C. Root Cause Analysis.
- D. Influence Diagram.

Answer - C.

Explanation

The question is a little bit tricky. The question implies you need to find out the reason; why two key stakeholders have turned neutral from supportive. Option A is wrong because it just implies the Current vs desired level of engagement of the stakeholder.

The options B and D are outright rejected. They can be easily ruled out.

Among all options, C is the best answer, because RCA identifies underlying reasons for the level of support of project stakeholders in order to select an appropriate strategy to improve their level of engagement.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –521.

Question 25:

Sandeep is a Project Manager of a highly technical middle-ware migration project. The project is in execution. The sponsor and some key

stakeholders are not happy with the performance of the delivery team. After meeting the team, he realized that the team members are struggling with the latest PMIS software, which is affecting the on-time task update.

What should he do first to resolve this problem?

- A. Capture the Lessons learned in the Lessons Learned Register.
- B. Modify the Project Team assignments and Organization Process Assets.
- C. Review the Resource Calendar to find a slot for training.
- D. Raise a Change Request for updating the training requirements in the Resource Management Plan.

Answer- C.

Explanation

Here, as per the scenario stated in the question, the Project Manager has to resolve the problem. The team needs to be trained in the latest PMIS software to fill the knowledge gap. Since the project is in execution, the Project Manager has to review the Resource Calendar to find a training slot. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –339.

Question 26:

Ryan is a project manager of a software development project which is running behind schedule with the CPI - 1.3. The project manager decides to bring the project back on track by applying the crashing technique. What will happen to the project?

- A. Costs and risks will increase.
- B. Risks will increase.
- C. Activities will be performed in parallel.
- D. The scope will be reduced.

Answer – A.

Explanation

Crashing a project may increase both costs and risks.

The options C and D are ruled out. Option C is applicable for fast tracking

and crashing never reduces project scope.

Option B is partially true, but not the best answer. Crashing may increase risk, but more than that it adds cost.

Crashing is basically a schedule compression technique that involves adding or adjusting resources in order to compress the schedule while maintaining the original project scope. It mainly works for activities on the critical path where an additional resource will shorten the duration. Crashing may increase cost and risk.

Here, among all options, option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 215.

Question 27:

You are a Project Manager of an invoice processing software application. The project is in execution, and in the last project review meeting, the customer has raised concern about the processing speed of the invoices. As per the latest Customer need, 50 invoices must get processed every 10 seconds. As a Project Manager, you are using a tool to detect how often 50 invoices fail to get processed every 10 seconds to improve the end product quality. What tool the Project Manager used over here?

- A. Checklists.
- B. Check sheets.
- C. Histogram.
- D. Control Chart

Answer- B.

Explanation

A Check sheet is a type of checklist used to keep track of data, such as Quality problems uncovered during the inspection, and document how often a particular defect occurs. The question implies that the Project Manager is doing the Control Quality process and using a check sheet as a tool to detect how many times the invoice processing speed is below the expected value. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)-302.

Question 28:

Your project uses a structured tool in both the Manage Quality and the Control Quality processes. It helps to verify the sequential execution of the steps to be performed.

What tool has been mentioned above?

- A. Checklists.
- B. Check sheets.
- C. Histogram.
- D. Control Chart

Answer - A.

Explanation

The Checklists help in managing and controlling quality activities in a structured manner. It's used in both the Control Quality and the Manage Quality processes.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 292,302.

Question 29:

The following process is concerned with the correctness of the project deliverables. The process ensures project outputs are complete, correct, and meet customer expectations.

What process has been referred to here?

- A. Manage Quality.
- B. Control Scope.
- C. Control Quality.
- D. Validate Scope.

Answer- C.

Explanation

Control quality is generally performed before Validate scope to ensure correctness of the project deliverables.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 298.

Question 30:

The Project Team is using a quality tool to understand and estimate the Cost of Quality of a process by using the workflow branching logic and associated relative frequencies.

What tool has been referred to in the above scenario?

- A. Flowchart.
- B. Control Chart
- C. Histogram.
- D. Mind Mapping.

Answer- A.

Explanation

Flowchart, one of the Seven QC tools used in all Quality management processes. It is a useful tool for the determination of the Cost of Quality. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 284.

Question 31:

A startup company is building an innovative software solution to meet the modern-day Healthcare need. The first working version of the software is ready. The company's CEO has reached some prospective customers and given them free access to the software to collect and analyze their valuable feedback to gain their confidence in the product. What the CEO intended to demonstrate to the customers?

- A. MMF of the product.
- B. A complete product demonstration of the software solution.
- C. Company capability in the healthcare domain.
- D. MVP of the product to collect the maximum amount of validated learning about customers with the least amount of effort.

Answer – D.

Explanation

Here as per the question, the team has developed the MVP (Minimum Viable Product). The CEO has shared the MVP with some prospective customers to collect the maximum amount of validated learning about customers with the least amount of effort. The primary benefit of an MVP is that you can

understand your customers' interest in your product without fully developing the product. On the other hand, learning gained from customer feedback greatly helps in final product development.

Question 32:

Which one of the following is not true regarding the Control Resources process?

- A. The Control Resources process ensures that resources are used and released according to the plan and project needs.
- B. The Control Resources process monitors the resource expenditure.
- C. Control Resource Process should be performed continuously in all project phases and throughout the project life cycle.
- D. The Control Resources process is concerned with physical resources as well as human resources.

Answer – D.

Explanation

The Control Resources process is concerned with physical resources such as equipment, materials, facilities, and infrastructure. Team members are addressed in the Manage team process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 353.

Question 33:

Which of the following estimating techniques use a statistical relationship between historical data and other variables to calculate cost or duration estimate for project work?

- A. Bottom up estimating.
- B. Parametric estimating.
- C. Three-Point estimating.
- D. Analogous estimating.

Answer- B .

Explanation

Parametric estimating uses a statistical relationship between historical data

and other variables to calculate an estimate for activity parameters, such as cost, budget and duration.(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 200,201,244.

Question 34:

Which of the following processes help to ensure that stakeholders clearly understand the project goals, objectives, benefits and risks for the project, as well as how their contribution will enhance project success?

- A. Manage Communication.
- B. Manage Team.
- C. Manage Stakeholder Engagement
- D. Monitor Stakeholder Engagement

Answer – C.

Explanation

Manage Stakeholder Engagement processes foster appropriate stakeholder involvement.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 524.

Question 35:

Ryan is a Project Manager of an Invoice processing project. He has identified 8 important stakeholders for the project. The number of communication channels on this project are:-

- A. 9.
- B. 77
- C. 36.
- D. 28

Answer - C.

Explanation

Total number of stakeholders in the project -: $8+1$ (Project Manager).

No of Communication channels = $N(N-1)/2 = 9*8/2=36$.

Question 36:

The Project team applied the crashing technique to bring the project

back on track. The Project Manager is sharing the latest project information in a meeting with all key stakeholders, who looked worried at the start of the session. He uses the Stoplight chart, Earned value graph to display the current state and the future forecast of the project. Which process is getting performed in the scenario stated above?

- A. Manage Stakeholder Engagement
- B. Manage Communication.
- C. Monitor Communication.
- D. Control Schedule.

Answer – B.

Explanation

The situation states that the Project Manager is sharing the Work Performance Reports in a meeting with stakeholders in the form of a presentation. This reflects that the Project Manager is currently involved in the Manage Communication process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 379 -384.

Question 37:

A Decision Tree Analysis is a graphic representation of various alternative solutions that are available to solve a problem.

Decision tree analysis can be done as a part of which of the following processes?

- A. Perform Qualitative Risk Analysis.
- B. Perform Quantitative Risk Analysis.
- C. Monitor and Control Project Work.
- D. Identify Risks.

Answer- B.

Explanation

Decision tree analysis is a data analysis tool used in the Perform Quantitative Risk Analysis process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 435.

Question 38:

Troy is the Project Manager of the BNM Project. Ten high storage capacity servers need to be procured. The accounting department has informed him that he needs a unilateral contract since it falls under commodity procurement. Which among the following contract types has been suggested by the Accounting department?

- A. An Internal SOW.
- B. A Fixed Price Contract
- C. A Purchase Order.
- D. A Letter Contract.

Answer- C.

Explanation

Purchase Orders are normally Unilateral (signed by one party). Usually used for simple commodity procurement.

PMP Exam Prep 9th edition 2018 – Rita Mulcahy, Page(s) – 512.

Question 39:

Ryan is the new Product Owner of the Software development project. His key responsibilities include: -

- A. Managing the Scrum Team.
- B. Managing the Product Backlog.
- C. Managing the Project.
- D. Coaching Agile philosophy to the team.

Answer - B.

Explanation

The Product Owner works with stakeholders, customers, and the teams to define the product direction. He or She is accountable for managing the Product Backlog.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.
[scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf](https://www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf) – Page – 6.

Question 40:

A Project Manager is doing Closing activities of a project. Which of the following activities will not be performed as part of this process?

- A. Formal Acceptance of project deliverables.
- B. Final Acceptance of project deliverables.
- C. Release of Physical and human resources.
- D. Archival of Project Information.

Answer A.

Explanation

Option A is done as part of Validate Scope. B, C, D are examples for Closing activities.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 121-128.

Question 41:

Changes are evident in all projects. Ideally ,Projects should have a structured Change Management Plan. When will you develop the Change Management Plan?

- A. During the Perform Integrated Change Control process.
- B. During Monitor and Control Project Work Process.
- C. Along with Developing the Project Charter Process.
- D. As a Part of Developing Project Management Plan Process.

Answer – D.

Explanation

The Change Management Plan is a component of the project management plan and is developed as part of the Develop Project Management Plan process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 88.

Question 42:

During execution, a new stakeholder with High Power and High Interest joins your project. He has also ensured his support for the project. You have promptly updated the Stakeholder Register and marked him as

**Upward in the Direction of Influence chart.
What should the project manager do next?**

- A. Submit a Change Request to update the Stakeholder Register.
- B. Submit a Change Request to update the Project Management Plan.
- C. Update the Stakeholder Engagement Assessment Matrix.
- D. Introduce the new stakeholder to your team.

Answer – B.

Explanation

Here the scenario states that the Identify Stakeholder process is being carried out and a new stakeholder with High Power and Interest has joined your project and corresponding Stakeholder Register has been updated. So, option A is out of Question. Option C is a good option, but the best time to update the Matrix is when you have updated the Project Management Plan, and there is a clear vision of the current level of engagement of stakeholders. Further, Stakeholder Engagement Assessment Matrix is an example of Data Representation technique that gets updated during planning Stakeholder engagement. For a new stakeholder, the desired level is set during planning, then its current level gets modified any time during the life cycle of the project. And as per the question the Stakeholder support is already ensured. Project Management Plan components that may be updated as a result of the addition of the new Stakeholder (Identify Stakeholder Process) described by the scenario may include the Requirements Management Plan, Communications Management plan, Risk Management plan, and Stakeholder Engagement Plan. With new stakeholders being introduced into the project, it is likely that each of these components will need to be revised. For revision of each of these control documents, a formal Change Request is required. So, among all options provided B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 515.

Question 43:

The Change Control Board has approved a change request for updating the project schedule. The above event occurs as a part of which of the following processes?

- A. Perform Integrated Change Control.
- B. Validate Scope.
- C. Control Quality.
- D. Monitor and Control Project Work.

Answer - A.

Explanation

Perform Integrated Change Control is the process of reviewing all Change Requests; approving changes and managing changes to deliverables, Project Documents, and the Project Management Plan; and communicating the decisions.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 113.

Question 44:

Arun is a Project Manager of a complex Cloud Integration project. The project is in the planning stage and needs three DevOps expert consultants, which is not available internally. After reviewing the Preferred Supplier List with the Procurement Manager, Arun decides to hold a bidder conference with three prospective sellers.

What type of contract is most suitable in the scenario stated above?

- A. Fixed Price with economic price adjustments (FP-EPA).
- B. Cost Plus Incentive Fee (CPIF).
- C. Time and Material (T&M).
- D. Cost Plus Award Fee (CPAF).

Answer- C.

Explanation

The question implies that the Project Manager wants to hire Technology experts. So, the T&M contract will be most suitable in the scenario stated above. The Time and Material contracts are a hybrid type of contractual agreement with aspects of both cost-reimbursable and fixed-price contracts.

When to use the Time and Material Contract: -

- a. Staff augmentation
- b. Acquisition of experts
- c. SOW is not available in the current state or cannot be prescribed quickly.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 472.

Question 45:

Ryan is a Project Manager of a website development project, which is managed using the Cost Plus Incentive Fee(CPIF) contract. The Project budget is \$88,000. Halfway through the project, at the point where it was planned that half the budget would have been spent, only \$41,000 has been spent, and 45% of the work has been completed. What is the project's Schedule Performance Index(SPI)?

- A. 1.15.
- B. 0.82.
- C. 0.86.
- D. 0.90.

Answer - D.

Explanation

$$SPI = EV / PV = 45\% / 50\% = 90\% = 0.90.$$

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 263.

Question 46:

Ryan is a Project Manager of a Software development project. The project would be running using the adaptive development approach. He wants to use a tool to determine the high-level summary timeline of the release schedule and the number of iterations in a release. What is the best tool the Project Manager can use in the scenario mentioned above? Choose the best answer.

- A. Critical Path Method.
- B. Agile Release Planning.
- C. Schedule Compression.
- D. Simulation.

Answer- B.

Explanation

Agile Release Planning provides a high-level summary timeline of the release schedule as well as the number of iterations in a release.

Characteristics: -

Helps the Product Owner and the Team to decide: -

- a. The percentage of work needs to be completed for each iteration.
- b. Duration that may take to have a releasable product based on business goals, dependencies, and impediments.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 216.

Question 47:

The project is getting released soon. The Project Manager is about to start the Closing Activities. The Sponsor advised him to share some confidential project information with all key stakeholders before commencing the project closure activities. What should Project Manager do to know the name of the person or people responsible for authorizing the release of confidential information?

- A. The Project Manager can authorize himself, as he is the Project Manager.
- B. The Project Manager can ask Sponsor to do the needful, as he has initiated the project.
- C. Review Project Charter to know the name of the responsible person.
- D. Review the Communication Management Plan to know the name of the responsible stakeholders who are authorized to release and receive the confidential information.

Answer – D.

Explanation

The Communication Management Plan states the name of the person responsible for authorizing the release of confidential information. The Project Manager should review the Communication Management plan to fetch this necessary information.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 377.

Question 48:

You are a Project Manager of a Software development project. Two of your team members are working in a work package – "Sales Module" that has two activities – Develop Sales Module, Test Sales Module. Both the activities have Finish-To-Start (FS) relationships with zero Lead time. What nature of dependencies exist between the activities Develop Sales Module and the Test Sales Module?

- A. Mandatory Dependency.
- B. External Dependencies.
- C. Discretionary Dependencies.
- D. Internal Mandatory dependencies.

Answer- D.

Explanation

The Question implies that both the activities are involved in the Finish-To-Start relationship. The activity Test Sales Module can only start once the Develop Sales Module activity gets finished. Therefore, mandatory dependencies exist among these two activities. Now since both the activities are project activities and there is no external influence, their relationship is also Internal in nature. So, considering all options, D is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 190,191.

Question 49:

The Project Manager is not happy with the cost estimates submitted by four preferred vendors. They are different from what he actually expected. He decides to have a separate meeting with all of them. What should he review before going for the meetings?

- A. Source Selection Criteria.
- B. Make or Buy Decisions.
- C. Procurement SOW.
- D. Vendor Bid Analysis.

Answer – C.

Explanation

The Project Manager should review the Procurement SOW before going for

the meetings. Significant differences in estimation can be an indication that the Procurement SOW was deficient or ambiguous or that the preferred sellers misunderstood the SOW.

The Procurement SOW clearly states the process of determination of the Independent cost estimate . If we analyze the question, we can assume that the project manager is unhappy because the cost estimates submitted by 4 different vendors are not matching with the Independent cost estimate.

Among all given options, reviewing the Procurement SOW before the meetings is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 479.

Question 50:

You are appointed as the full-time manager for a software integration development project. You are reporting to the Practice Head of the Integration track, who will manage the project budget.

What type of organization structure are you part of?

- A. Organic.
- B. Strong —Matrix.
- C. Hybrid.
- D. Functional.

Answer – C.

Explanation

The role of Project Manager is mostly part time in Organic and function org structure. So, A and D are not correct. In the Strong-Matrix organization, the project manager is required full time, but at the same time it's generally the project manager who takes care of the budget, so B is also not correct. The roles are mixed in hybrid organization, and the above scenario is an ideal example of Hybrid organization structure. Here Practice Head is playing the role of Functional Manager who is taking care of the project budget. So among all choices provided C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 47.

Question 51:

Emergent risks can only be recognized after they have occurred. What is

the best way of tackling such risks?

- A. Developing Project Risk Management Plan.
- B. Developing Project Resilience.
- C. By following Integrated risk management techniques.
- D. By building sufficient Management reserves.

Answer – B.

Explanation

Emergent risks can be tackled through developing project resilience. This requires each project to have the right level of budget and schedule contingencies, flexible project processes, empowered project team and frequent review of early warning signs.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 399.

Question 52:

Ryan is a Project Manager of a manufacturing project. Hurricane alerts may delay the shipments of the main components of the project that would affect the project schedule.

What should the Project Manager do next?

- A. Implement a workaround.
- B. Evaluate another seller.
- C. Review the risk management plan.
- D. Review and update the contingency plan.

Answer – D.

Explanation

In this situation, the risk has triggered, it has not yet occurred. So option A is ruled out. Option B is also ruled out here, as in this scenario the risk has triggered natural calamity. Option C is close but the question states that Project schedule will get impacted, if such risk occurs, so in this kind of scenario, the review of Risk Management Plan may not be the ideal choice. By considering elimination rules, D becomes the best answer because the question has keywords like “hurricane alert”. Further, the project manager has to do a comprehensive identification considering a broad range of sources

of risk to ensure that as many uncertainties as possible that might affect project objectives have been identified, as well as the Identify Risks process should take input from a range of project stakeholders to ensure that all perspectives are represented and considered. By this way the Project Manager can identify all risks that also include this hurricane risk and accept actively and assign contingency plans with trigger conditions. The plan can be updated because all risk processes are iterative. So among all options provided, D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 445.

Question 53:

During the mid of the project, the project manager is trying to assess whether the project-level communications distributed to the stakeholders across different countries are aligned to the Project Management Plan. What should the Project Manager review in this scenario?

- A. Communication Management Plan.
- B. Stakeholder Engagement Plan.
- C. Project Communications.
- D. Stakeholder Engagement Assessment Matrix

Answer - C.

Explanation

The Question states that the Project Manager is involved in the Monitor Stakeholder Engagement process. He needs to assess whether the distribution of Project Communications is in line to the Project Management Plan (i.e. Communication Management Plan and Stakeholder Engagement Plan in this scenario). Though A and C are input to the Monitor Stakeholder Engagement process, it cannot give all information. Option D is ruled out because Stakeholder Engagement Assessment Matrix is a tool for assessing Stakeholder engagement. Project Communications(Output of Manage Communications process) is a valid input for Monitor Stakeholder Engagement process. Project communications include all project level communications that have been distributed to all stakeholders as defined in the Communication Management Plan and Stakeholder Engagement Plan. So, considering all options, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 532.

Question 54:

The customer has formally approved the result of the last sprint review meeting. The project has just moved to the Closing phase. The Project Manager is trying to determine the final dates of accomplishments so far. What should the Project Manager review to derive the required information?

- A. Milestone list.
- B. Project Communications.
- C. Issue log.
- D. Lessons Learned Register.

Answer – A.

Explanation

The question states that the project manager is trying to determine the final dates on which the project milestones have been accomplished so far. For this, the Milestone list needs to be reviewed which is also a valid input for the Closing process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 124,186.

Question 55:

Working software is the primary measure of progress. This principle is religiously followed by which of the following life cycle approaches?

- A. Predictive.
- B. Agile.
- C. Iterative.
- D. Incremental.

Answer – B.

Explanation

Working Software is the primary measure of progress and is one of the twelve Agile Principles.

Question 56:

The Business Case documents the business need and cost-benefit analysis that justifies the need for the project. During the project closure phase, the accuracy of the Business case gets recorded in which of the following documents?

- A. Business Case.
- B. Project or Phase Closure Documents.
- C. Lessons Learned Register.
- D. Quality Report.

Answer – C.

Explanation

Lessons Learned register is updated as key output for Close Project or Phase process. It includes final information on phase or project closure. The accuracy of the business case is also documented in the Lessons Learned Register as a part of the process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 127.

Question 57:

Ryan is a Project Manager of a Software development project that involves work on DevOps and RPA technology. Since internal resources are scarce, the Project Management team has decided to outsource the work to a suitable vendor.

What tool can the Project Manager use to fetch the information on specific seller capabilities?

- A. Market research.
- B. Advertising.
- C. Source selection analysis.
- D. Make-or-buy analysis.

Answer – A.

Explanation

As the scenario states, the Project Management team is involved in planning. Market research, also a tool for the Plan Procurement Management process, includes the examination of industry and specific seller capabilities.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 473.

Question 58:

You are the owner of a company whose organizational structure type is Organic. Your company has been awarded a small project with a limited budget. Which of the following processes can be overlooked in such a project?

- A. Perform Qualitative Risk Analysis.
- B. Perform Quantitative Risk Analysis.
- C. Identify Risks.
- D. Plan Risk Management.

Answer – B.

Explanation

Availability of budget and time is a key factor that determines the need for the Perform Quantitative Risk Analysis process. A small project with a limited budget may decide to do away with this process if the project management team decides that quantitative statements about risk and impacts are not needed.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 429.

Question 59:

Which of the following is not true regarding the Cost Estimates?

- A. Cost estimates include quantitative assessments of probable costs required to complete project work, as well as contingency amounts to account for identified risks.
- B. Cost estimates can be presented in summary form or in detail.
- C. Indirect costs can be part of cost estimates if they are included in the project estimate.
- D. Cost estimates don't include the management reserve.

Answer - D.

Explanation

Cost estimates, which is the main output for the Estimate Costs process generally include management reserves to cover unplanned work.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 246.

Question 60:

Midway through the project, the project management team decides to reassess some risks which they think are no longer valid. What process is getting performed in the above-stated scenario?

- A. Perform Qualitative Risk Analysis.
- B. Perform Quantitative Risk Analysis.
- C. Identify Risks.
- D. Monitor Risks.

Answer – D.

Explanation

The Risk reassessment is generally performed in the Monitor Risks process. The risk reassessment meeting should be scheduled regularly to maintain a healthy Risk management process.

The following list of activities may happen as part of the Risk reassessment meeting:-

- a. Identification of the new risks.
- b. Evaluation of the current risks and the risk management processes.
- c. Closure of the outdated risks.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 457.

(The Standard for Risk Management in Portfolios, Programs, and Projects) - Project Management Institute Inc., 2019, Page(s) – 149.

Question 61:

Ryan is a new project manager of a software development project who has joined during the execution. He is trying to determine whether the modification of any of the project documents are subject to the change request. What the Project Manager needs to review to fetch this

information?

- A. Review Change Management Plan.
- B. Refer to the Scope Management Plan.
- C. Review the Configuration Management Plan.
- D. Examine Requirement Traceability Matrix.

Answer – C.

Explanation

A change request must be submitted to update any baseline or Project Management Plans. The standards outlined in the Configuration Management plan determine whether any of the Project documents are subject to the change control procedures.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 41, 701.

Question 62:

Arun is the delivery director of the new business development program. He has assigned you as the Project Manager and also given other staff for your project. You want to know more about the project, but all you have received so far was your department manager's email with little detail about the project.

To learn more about the project, which of the following options would be your best course of action in this situation?

- A. Interview the delivery director of the new business development
- B. Hold a brainstorming session with your team.
- C. Identify all the stakeholders who will be impacted by the project.
- D. Visit the project management office (PMO) to gather organizational process assets.

Answer – A.

Explanation

Interviews, an example of a data-gathering technique of the Develop Project Charter process, can be used to obtain information on high-level requirements, assumptions or constraints, approval criteria, and other information from stakeholders by talking to them directly. Although the

question has never stated that you have already started the development of the project charter, in this situation, since you know nearly nothing about the project, to better understand the essence of the project, interviewing the director of new business development is the best course of action among all the given choices.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 75.

Question 63:

In a status meeting of a project which is in execution, the sponsor introduced a new stakeholder. The new person requests two scope level changes in the project.

What should the Project Manager do first?

- A. Raise a Change Request to change the scope baseline.
- B. Update the Stakeholder Register.
- C. Document the stakeholder concern in the Issue Log.
- D. Update the CCB team on changes advised by the new stakeholder.

Answer – B.

Explanation

Whenever a new member joins the project, the first thing the project manager should do is to add him in the Stakeholder Register, thereby formally introducing him in the project stakeholder community. Post introduction, all his requests can be analyzed or taken care of.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 505.

Question 64:

Ryan is a Project Manager of a construction project that is a part of the Cost reimbursable contract. In a meeting with the seller, the seller asked him to share the payment for the cost of items prepared for the project that got destroyed in a recent thunderstorm. What should the Project Manager do next in this situation?

Choose the best answer.

- A. Ask the seller to bear the entire cost as they are accountable for the loss.

- B. Review the Force majeure clause in the contract
- C. Review the Force majeure and the Risk of loss clause in the contract
- D. Agree with the seller's proposal

Answer – C.

Explanation

Option A and Option D are not true, as the Project Manager has to review the contract to make a decision. In case of materialistic loss due natural disaster or fire, for which the Force majeure clause of the contract should be consulted. The clause states that in case of “act of nature” situations such as fire, storm, neither parties are judged faulty. In general, it is resolved by the seller receiving an extension of time on the project. But in this question, the seller is asking the buyer to share the cost for loss. So, in this scenario, the Project Manager needs to review the "Risk of loss" clause too along with the Force majeure clause. Risk of loss clause allocates the risk between the parties to a contract in the event of loss of goods and services during performance of the contract. So, among all options provided, option C is the best answer.

Rita Mulcahy 9th edition, RMC Publications, Inc.,2018, Page -531.

Question 65:

The Project team is investigating the underlying reasons for poor stakeholder management in their project. They are performing the RCA to find out the root cause of the problem by breaking the identified problem statement into discrete branches. What tool should be used to represent the problem diagrammatically?

- A. Root Cause Analysis.
- B. Why-Why diagrams.
- C. Flowchart.
- D. Histogram.

Answer – B.

Explanation

The scenario states that the project team is doing RCA for identifying the reasons in relation to poor stakeholder engagement. The results of the RCA needs can be represented by using a data representation tool.

RCA is a data analysis technique. On a few occasions as stated in the above scenario, findings of the RCA can be represented by using Fishbone diagrams, also known as Why-Why diagrams or Cause and Effect diagrams. This type of diagram breaks down the causes of the problem statement into discrete branches, helping to identify the main or root cause of the problem. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 293 ,533.

Question 66:

Ryan is the new quality auditor of a software development project. He wants to review the results from the previous audits before conducting the formal audit meeting. What should he review to know the results of the previous audits?

- A. Quality report.
- B. Test and Evaluation documents.
- C. Issue log.
- D. CPA

Answer – D.

Explanation

The situation states that the new auditor wants to review the results of the previous audits before conducting the formal audit meeting. The Quality audit happens in the Manage Quality process. The OPA, which is also an input for the Manage Quality process, stores the results from previous audits. So, among all options, option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 291.

Question 67:

A CEO of a paint manufacturing company involves his senior leadership team to prepare a business case for authorizing a project to establish guidelines for handling toxic materials.

This business case originated as a result of : -

- A. Organization needs.
- B. Legal requirement
- C. Ecological impacts.

D. Social need.

Answer – B.

Explanation

The establishment of proper guidelines periodically for handling toxic materials is a mandatory legal need which every manufacturer needs to follow.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 78.

Question 68:

A new risk gets discovered during the closing phase of the project. What should the Project Manager do first?

- A. Update Risk Register.
- B. Update Issue Log.
- C. Ignore the risk, as the customer has already approved the formal deliverables.
- D. Update Risk Management Plan.

Answer – A.

Explanation

A risk can be discovered at any point of time in the project, and all identified risks can be recorded to Risk Register throughout the life cycle of the project, as the process is iterative. B is not correct, because in this situation Risk has been identified, risk has not occurred. C is incorrect, as per PMI ethics it's the responsibility of the project manager to address such a situation at any point in the project Life cycle. D is incorrect, as the Risk management plan has no role here. Here, as per the question, new risk gets discovered during closing, so in this case, the Risk Register should be updated, with information in relation to new risk. Therefore, among all options provided A is the best answer.

Question 69:

You are the newly joined Project Manager of an automobile manufacturing company. Senior Management has involved you in the preparation of the Project Charter for their prestigious in-house fuel

efficient car project. What should you do to know the marketplace conditions and current industry standards?

- A. Examine the Historical information and lessons learned in the repository.
- B. Review the Business Case.
- C. Review the Enterprise Environmental Factors(EEF) external to the organization.
- D. Review the Enterprise Environmental Factors(EEF) internal to the organization.

Answer – C.

Explanation

The question states that the newly joined Project Manager has been involved in the preparation of the project charter of the company's upcoming internal project. Here, all options are input to the Develop Project Charter process. But the question asked "for the review" of Market conditions and industry standards, which are examples of the External Enterprise Environmental factors, because they are outside the control of the Project Team. So among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 39, 78.

Question 70:

The Work Breakdown Structure(WBS) plays a vital role in creating a scope baseline for project planning, execution, and control. The scope baseline serves as an input to many of the project processes. Which of the following is not true?

- A. The Scope baseline serves as an input for Quality Planning.
- B. The Scope baseline serves as an input to Cost budgeting.
- C. The Scope baseline serves as an input to Risk identification
- D. The Scope baseline serves as an input for Planning Cost Management

Answer – D.

Explanation

Scope baseline is an input for Quality planning (Plan Quality Management Process), Cost budgeting (Determine Budget process), risk identification (Identify Risk process). So, options A, B, C are not the right answers. Whereas Scope baseline is not an input for Planning Cost Management, and developing initial Cost Management Plan. It is later referred for cost estimation which is a part of Estimate Cost process and cost budgeting, part of Determine Budget process.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 235,248,277,409.

Question 71:

The Project Team has successfully implemented the required risk responses, which has brought the project back in good shape. The project manager asks the team to update all the information concerning challenges faced, which went well for implementing the risk responses into a document for future reference. What document the Project Manager tried to mention in the scenario stated above?

- A. Risk Register.
- B. Risk Report.
- C. Lessons Learned Register.
- D. Issue Log.

Answer – C.

Explanation

Here, as per the scenario stated in the question, the Project Manager is asking to update all experiences of the recently completed Risk Response implementation process into the Lessons Learned Register document.

All 4 provided options stated above may get updated as a part of the Implement Risk Responses process.

Option D -Issue Log is not the correct answer as the Issue log records the Issues encountered during the risk implementation process.

The Risk register or the Risk report may get updated to capture any changes to the agreed-upon risk responses as a result of the risk response implementation. So the options A and B are also not correct.

The Lessons Learned Register is a living project document that gets continuously updated throughout the life cycle of a project. All experiences

of the Risk response implementation process are recorded in the Lessons Learned register for future reference.

This includes:-

- a. Challenges faced.
- b. The approaches that worked well during risk implementation.

All contents of the Lessons Learned Register are moved to OPA during the Project or Phase closure.

So, among all the provided options, option C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 452.

Question 72:

Ryan is the Project Manager of the Qnx organization. He has been awarded an internal project for building a new payroll processing system for his company. The main project objectives include the automation of a few of the current processes done manually as of today. What tool should the Project Manager use to determine the financial success factors for the project objectives?

- A. Flowchart
- B. Affinity Diagrams
- C. Cost-benefit analysis
- D. Cost of quality

Answer – C.

Explanation

The Cost-Benefit analysis is a financial analysis tool used to estimate the strengths and weaknesses of the alternatives, as automation in this case, in order to determine the best alternatives in terms of the benefits provided. It helps the Project Manager to determine if planned quality activities are cost-effective.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 282.

Question 73:

Conflict is an inevitable consequence of organizational interactions. Which of the following practices can reduce the amount of conflict within a team?

Choose the Best answer.

- A. Reward and recognition.
- B. Team ground rules.
- C. Always involving a third party to resolve a conflict.
- D. Managing team by applying autocratic leadership style.

Answer - B.

Explanation

Team ground rules mean the expected behavior for project team members among one another during the course of the project. Setting ground rules, group norms can definitely reduce conflict, as a team needs to follow specific guidelines.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 348.

Question 74:

The following process may confirm the risk owners for individual project risks and is performed regularly throughout the life cycle of the project, as defined in the Risk Management plan. Sometimes, it is conducted in Agile projects before the start of each iteration.

What process has been referred to in the scenario mentioned above?

- A. Perform Qualitative Risk analysis.
- B. Identify Risks.
- C. Plan Risk Responses.
- D. Collect Requirements.

Answer – A.

Explanation

Risk owners for individual project risks may be nominated in the Identify risks process and confirmed in the Perform Qualitative Risk analysis process. The process is performed throughout the life cycle of the project, as defined in the Risk Management Plan. Often in projects with adaptive development approaches, it's being conducted before the iteration begins.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 411, 421.

Question 75:

During Phase Gate, the project performance and progress are compared to all of the following documents except:

- A. Business Case.
- B. Procurement Documentation.
- C. Project charter.
- D. Project Management Plan.

Answer – B.

Explanation

Phase gate is held at the end of a phase. During Phase Gate, the project performance and progress are compared to project and business documents that include: - Business Case, Benefit Management Plan, Project Charter, Project Management Plan.

So, among all the options provided, Procurement Documentation is the correct choice. Project performance and progress are not compared to the procurement documents.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 21.

Question 76:

Ryan is a newly joined Project Manager of a large, multi-phase software development project. In his first status review meeting, some key stakeholders raised concern on the status of the change requests raised by them, as most of their requests are either deferred or in a rejected state. What should the Project Manager do next?

- A. Consult with the members of the Change Control Board.
- B. Review the Change Management Plan.
- C. Review the Change Log.
- D. Check current CPI and SPI of the Project.

Answer – C.

Explanation

The disposition of all the Change Requests are recorded in Change Log as a project document update. Since the details of all approved, deferred, and

rejected change requests are stored in the change log, the Project Manager should review the change log first. Option A and Option D are also good options, but those should be considered after reviewing the Change log. So, among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 120.

Question 77:

Ryan is a Project Manager of a large multi-year project that is being executed using an agile development approach. The Project Team used the bottom-up estimation technique for project work related to all near term releases. A list of the planning packages has been prepared for all future releases. What tool or technique the project team has used in the given scenario?

Choose the Best Answer.

- A. Decomposition.
- B. Expert judgment
- C. Agile Release Planning.
- D. Rolling Wave Planning.

Answer – D.

Explanation

As per the above scenario detailed planning is available for project work that need to be executed in the near term. This can be assumed because the Project team used Bottom-up estimation technique, which can only be done when work breakdown structure components are decomposed into more detail. Question also states for all future releases only planning packages are planned (Hint: The statement in the question “work breakdown structure components above the work package level with known work content but without detailed schedule activities” signifies the definition of planning packages). This confirms the Project has used Rolling Wave Planning technique. As per definition, Rolling wave planning is an iterative planning process which will allow near-term activities to be planned in detail while leaving activities further out planned at a higher level. So among all options provided D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) – 160,185, 721.

Question 78:

Ryan is a Project Manager of an IT project. He is currently preparing a report for senior management that would show the resource requirements by time period and applied schedule reserves. What should the Project Manager review to fetch the required information?

- A. Schedule Data.
- B. Project Schedule.
- C. Project Calendars.
- D. Resource Requirements.

Answer – A.

Explanation

The Schedule data for the project schedule model is the collection of information for describing and controlling the schedule. It contains resource requirements by time periods, often displayed in the form of a resource histogram. It also contains alternative schedules like Best-Case or Worst-Case scenarios. As per the question stated above, the best thing the project manager can do is reviewing the Schedule Data.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 220.

Question 79:

An automobile manufacturer initiated a project to build more fuel-efficient cars in response to gasoline shortages. Project Initiation is complete, and several experts have gathered to identify the areas required to be taken care of while building the best fuel-efficient car. They are comparing the specifications of popular branded fuel-efficient cars available in the market with their planned product. What tools and techniques have been used in this scenario?

- A. Brainstorming.
- B. Benchmarking.
- C. Focus Groups.
- D. Facilitation.

Answer – B.

Explanation

As per given scenario, the experts are involved in Benchmarking while doing the planning. Benchmarking is a data gathering tool used in quality planning and requirement gathering. It involves comparing actual or planned products, processes, and practices to those of comparable organizations to identify best practices, generate ideas for improvement, and provide a basis for measuring performance.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 143.

Question 80:

Ryan is a Project Manager of a Telecom equipment manufacturing project. During a project performance review with the team, he notices the following: (a) Activity A, on the critical path, is delayed by 4 days. (b) Activity B, not on the critical path, is delayed by 9 days. (c) Activity C, on the critical path is delayed by 2 days (d) Activity D, not on the critical path, is delayed by 6 days. In what sequence should the Project Manager and the team prioritize their efforts in addressing these delays?

- A. Activity A, Activity B, Activity D, Activity C.
- B. Activity B, Activity D, Activity A, Activity C.
- C. Activity A, Activity C, Activity B, Activity D.
- D. Activity B, Activity A, Activity D, Activity C.

Explanation

Answer – C.

Tasks on Critical path should be addressed first, else it may impact the project schedule. Here, of all choices provided, C is the most reasonable choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 210.

Question 81:

As a Project Manager, you are trying to determine the project deliverables which are subject to quality review. What needs to be reviewed here?

- A. Requirements Management Plan.
- B. Scope Management Plan.
- C. Risk Management Plan.
- D. Project Management Plan.

Answer – D.

Explanation

The Quality Management Plan, the component of Project Management Plan determines Project deliverables and processes subject to quality review. Among all the given choices, since the option Quality Management Plan is not there, Selection of option D i.e. Project Management Plan is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 286.

*** Expect these kinds of questions in exams.***

Question 82:

The Project Team is currently involved in reviewing the historical information from similar past projects that would assist them in developing parametric or analogous estimates.

What technique has the team applied in the given scenario?

- A. Historical information review.
- B. Expert judgment.
- C. Data Analysis.
- D. OPA

Answer – A.

Explanation

Some may get confused with OPA, but OPA is an input, not tools or techniques. Historical information review, a technique used in Budget determination can assist in developing parametric or analogous estimates. It may include project characteristics(parameters) to develop mathematical models to predict total project costs.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 253.

Question 83:

You are a Project Manager of a software development project. During the early stage of execution, it came to your notice that the training is required for the technical team to build a quality product. Here training is an example of: -

- A. Appraisal Costs.
- B. Prevention Costs.
- C. Internal Failure Costs.
- D. Sunk Costs.

Answer – B.

Explanation

Sunk cost, means Cost that is incurred cannot be recovered, So D is not correct.

Option C is an example of Cost of Non-conformance, but the question talks about Cost of Conformance, so C is also incorrect.

Both the options A and B are examples of Cost of Conformance, but the question states that “the training is required for the technical team to build a quality product”, but appraisal costs help to assess the quality, So, A is also not correct.

The Prevention costs are one of the classic examples for the Cost of Conformance and it helps to build a quality product. The examples for Prevention costs include, training, document processes, equipment, Time to do it right. So, among all options provided, B is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 283.

Question 84:

All baseline documents are approved. The Project management team is engaged in reviewing those documents that guide the selection of internal and external resources needed to complete the project work. Which process is currently getting performed in the mentioned scenario?

- A. Plan Procurement Management
- B. Plan Resource Management.
- C. Acquire Resources.

D. Develop Project Management Plan.

Answer – C.

Explanation

The scenario states that the Acquire Resources process is getting performed. The Project management team is engaged in reviewing the Resource and the Procurement Management Plan so that resources can be boarded to complete the project work. Both the Resource Management Plan and the Procurement Management plans are input to the Acquire Resources process, which is being reviewed in the given scenario.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 328- 330.

Question 85:

In a Research and Development project, several experts have gathered in a focus group session to model the key relationships and dependencies to determine the areas of maximum uncertainty on overall project objectives. A risk diagramming technique is being used in the given scenario and a S-curve has been produced as an output. Which tool has been used in the given scenario?

- A. Sensitivity Analysis.
- B. SWOT Analysis.
- C. Simulation.
- D. Influence Diagrams.

Answer – D.

Explanation

The situation states that the experts have gathered to determine the areas of maximum uncertainty on over all project objectives, which means, Perform Quantitative Risk Analysis is being performed here. Influence diagram is an example of Risk Diagramming technique that's been used as a tool in the Perform Quantitative Risk Analysis process to address similar situations as stated above. Influence diagrams are graphical aids to decision making under uncertainty. It represents a project or situations within a project as a set of entities, outcomes, and influences, together with relationships and effects between them.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 436.

Question 86:

The work of a particular work package is essential for the Project Manager since the buyer company's CEO will review the verified deliverables of the same. If he gets satisfied, the company can earn more business in similar application areas. There is one experienced resource available from next week. Since a week's delay will not impact the project schedule, the project manager adjusts the start and the end date of the work package based on the experienced worker's availability. Which of the following risk response strategies the Project Manager applied in the given scenario?

- A. Exploit
- B. Enhance.
- C. Active Acceptance.
- D. Avoid.

Answer - A.

Explanation

The exploit strategy may be selected for high priority opportunities where the organization wants to ensure that the opportunity gets realized. As per the scenario, moving the work package to a date when the more experienced resource is available, thereby increasing the probability of occurrence to 100 percent, is an example of the Exploit strategy.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 444.

Question 87:

Concerning the performance of a recently manufactured deliverable that will go for customer inspection next week, the control chart has determined two data points outside the upper specification limit but within the upper control limit. The rest of the data points are within control and specification limits. What should the Project Manager do next?

- A. Present the deliverable for customer inspection, as the process is in

control.

- B. Do not present the deliverable for customer inspection, as the process is out of control.
- C. Do not present the deliverable for customer inspection, as the process is outside limits specified by the customer.
- D. Present the deliverable for customer inspection, let the customer decide.

Answer – C.

Explanation

Control limits represent the performing organizations standards for quality, while specification limits represent customer's expectations or the contractual requirements for the performance or the quality of the project. In general, most of the projects keep Control limits lower than customer specified limits for meeting customer needs. As per the question, the process is stable as per quality standard of the performing organization. But it's not acceptable from a customer point of view. So among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 304.

Question 88:

The Project Team is involved in developing an actionable plan to interact effectively with the stakeholders. What can guide them with the information regarding the roles and responsibilities of the team members and other stakeholders listed in the stakeholder register?

- A. Stakeholder Register.
- B. Project Charter.
- C. Communication Management Plan.
- D. Resource Management Plan.

Answer – D.

Explanation

As per the question, the project team is involved in the Plan Stakeholder Engagement process, and all 4 options are input to the process. But, it's the Resource Management Plan, which may contain information regarding roles

and responsibilities of the team members and other stakeholders listed in the stakeholder register.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 518.

Question 89:

A Project Manager appoints an expert for defect analysis of the deliverables in the current sprint. He conducted the process of translating the Quality Management plan into executable quality activities and then used a tool to show the number of defects per deliverable and the number of times each process became non-compliant. What tool did he use to show a graphical representation of numerical data?

- A. Histogram.
- B. Flowcharts.
- C. Matrix diagrams.
- D. Affinity diagrams.

Answer – A.

Explanation

The scenario suggests that the Project Manager has involved an external expert for defect analysis of deliverables in the current sprint. The expert conducted the Manage Quality process and used the Histogram tool for showing the number of defects per deliverable.

The histogram is the tool used in the Manage Quality process for graphical representation of numeric data. It can show the number of defects per deliverable and the number of times each process became non-compliant.

So among all options, option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 293, 304.

Question 90:

In a project that had several conflicts, most of the time, the Project Manager used his positional power to resolve most of the conflicts, as quick resolutions were required on most occasions. What conflict management technique has been used in the given scenario?

- A. Withdrawal.
- B. Forcing.
- C. Smoothing.
- D. Problem Solving.

Answer – B.

Explanation

As per the scenario stated above, the project manager applied a Forcing approach to resolve conflicts in a quick frame of time. Forcing is the conflict management technique that involves pushing one's viewpoint at the expense of others and mostly offers only win-lose solutions. In general, positional power is used to apply such strategies. So, among all options provided B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 349.

Question 91:

During planning, the project team is using a technique to assess the communication styles and identify the preferred communication method, format and the content for planned communication activities. Which process is being carried out in the given scenario?

- A. Plan Stakeholder Engagement
- B. Identify Stakeholders
- C. Plan Communications Management
- D. Manage Stakeholder Engagement

Answer – C

Explanation

As per the stated scenario, the Project Team, here, is applying Communications styles assessment technique which is a common Interpersonal and Team Skills technique used during planning communications. A, B, D are incorrect, because the Communications Styles Assessment techniques are not used in these processes.

So, among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 375.

Question 92:

Ryan is a Project Manager of an IT project which is in execution. All of a sudden, a significant change in organizational structure happens in the buyer organization. Which of the following is least likely to change as a result of this?

- A. Stakeholder Register.
- B. Stakeholder Engagement Plan.
- C. Resource Management Plan.
- D. Scope Management Plan.

Answer – D.

Explanation

In case of changes to organization structures, the stakeholder register, and the Project Management Plan (e.g. Stakeholder Engagement, Communication Management Plan, etc.) needs to be mandatorily updated. So, options A, and B are incorrect. Option C is also incorrect because the Resource Management Plan may contain information regarding roles and responsibilities of the team and other stakeholders listed in the Stakeholder Register, so this may also change.

Option D - Scope Management Plan describes how the project scope will be defined, developed, monitored, controlled, and validated.

Since the project is already in execution, most likely the Scope Management Plan would not change.

So, among all options, option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 518.

Question 93:

You are a Project Manager of a construction project. The work associated with the 3 work packages has been allocated to a preferred vendor. During the mid of the project, a group of experts appointed by you are conducting the structured review of the work being performed by the preferred vendor. Which of the following processes is getting executed in the given scenario?

- A. Control Procurements.

- B. Conduct Procurements.
- C. Validate Scope.
- D. Manage Quality.

Answer – A.

Explanation

As per the given scenario, the experts are engaged in Procurement Inspection activities. The Inspection is a valid tool & technique for the Control Procurements process. Inspection is the structured review of the work being done by the contractor. So, among all options provided, A is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –498.

Question 94:

**As a Project Manager, you are reviewing the plans for rewards and recognition. Which of the following factors play a pivotal role in determining the rewards and recognitions?
Choose the best answer.**

- A. Political Climate.
- B. Cultural differences.
- C. Size and complexity of the Project
- D. Nature of contractual agreement

Answer – B.

Explanation

The question is tricky; all four options may act as factors for determination of recognition and rewards in different situations. But among all, the cultural difference is the best choice. Reward will only be effective only if they satisfy the need that is valued by the individual.

The factors associated with Cultural differences play a pivotal role in determining recognition and rewards.

Among all options, option B is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 341.

Question 95:

Ryan is a Project Manager of a manufacturing project. The project budget is \$50,000. The project is 3 months into its 6 month schedule. It was anticipated that \$25,000 worth of the project would be completed by now. So far, \$24,000 has been spent and the value of the completed work at this point is \$22,500. What is the project's VAC?

- A. \$5,555.
- B. \$3,191.50.
- C. - \$3,191.50.
- D. - \$5,555.

Answer – C.

Explanation

Here, BAC = \$50,000, EV= \$22,500, AC= \$24,000

$CPI = EV/AC = \$22,500 / \$24,000 = 0.94$

$EAC = BAC/CPI = \$50,000 / 0.94 = \$53,191.48$

$VAC = BAC - EAC = \$50,000 - \$53,191.48 = -\$3,191.48$ (negative value)

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 262 -267.

Question 96:

The Bid documents are used to solicit proposals from prospective sellers. Which of the following processes create bid documents?

- A. Conduct Procurements.
- B. Control Procurements.
- C. Develop Project Charter.
- D. Plan Procurement Management

Answer – D.

Explanation

The Bid documents are one of the key outputs for the Plan Procurement Management process. They generally include: - RFI, RFQ, RFP.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 466, 477.

Question 97:

A newly joined Project Manager, who has just joined during execution wants to have information in relation to the glossary of common terminologies being used in the project. What should he review to know this information?

- A. Requirement Documentation.
- B. Communication Management Plan.
- C. Stakeholder register.
- D. Stakeholder Engagement Plan.

Answer - B.

Explanation

Glossary of all common terminologies in relation to the project are recorded in the Communication Management Plan.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 377.

Question 98:

Your company has been awarded a multi-billion dollars' worth dam construction project that will run for 4 years. Such projects are also termed as: -

- A. Program.
- B. Large Project.
- C. Megaproject.
- D. Waterfall Project.

Answer – C.

Explanation

Waterfall is a development approach in the field of project management, so option D is not correct. Programs are not large projects, so A is also incorrect. A very large project may be referred to as a megaproject. As per guidelines, Megaprojects cost US\$1 billion or more, affect 1 million or more people, and run for years. So, among all options provided C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 11 , 463.

Question 99:

Ryan is a Project Manager of a software development project which is currently in execution. A Change Request for adding a new feature has been raised by a key stakeholder. What should the Project Manager do NEXT?

- A. Ask the Team to implement the change.
- B. Analyze the impact of change with the Team on other Project Constraints.
- C. Send the Change Request to the CCB.
- D. Reject the Change Request as base lines are already created.

Answer – B.

Explanation

As the scenario suggests, the Project Manager and Team should analyze the impact of change on other constraints (Schedule, Cost, Quality, Risk, etc.) first, in order to assess the reasonableness of the requested change before sharing the same to the CCB.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 120.

Question 100:

An Agile team is doing POC of a new technology solution to gather feedback from the relevant stakeholders. How should the team accomplish this task?

- A. Create a detailed presentation and showcase the same to the customer.
- B. Develop a persona.
- C. Build a prototype.
- D. Propose a new contractual agreement with the customer.

Answer - C.

Explanation

A prototype can be used to demonstrate the product increment and gather feedback from the stakeholders.

Question 101:

In a Software development project, Daily Scrum is consistently taking longer than expected, affecting the team's day-to-day work plan. Stakeholders are discussing topics outside the meeting agenda, which is causing the delay. What should the Scrum Master do?

- A. Advice team to set up two daily stand-up meetings a day.
- B. Ask the team to leave the meeting after 15 minutes.
- C. The Scrum Master should help the Stakeholders understand why the 15-minute time-box should be respected and guide them find ways to do it as required.
- D. Ask the Product Owner to take strict actions against the stakeholders.

Answer – C.

Explanation

The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide. He/ She helps in removing barriers between stakeholders and the Scrum Team. The Scrum Master is also responsible for removing impediments to the scrum team's progress.

As per the scenario stated in the above question, Choice C is the best answer. The Scrum Master should help the Stakeholders understand why the 15-minute time-box should be respected and guide them find ways to do it as required.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41. [scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf](https://www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf) – Page – 6.

Question 102:

Match the following.

SL no	Approach	Goal
1	Predictive	a. Speed.
2	Incremental	b. Correctness of the solution.
3	Iterative	c. Customer value via frequent deliveries and feedback.
4.	Agile	d. Manage Cost.

- A. 1-d, 2-c, 3-a, 4-b.
- B. 1-d, 2-a, 3-b, 4-c.
- C. 1-c, 2-a, 3-b, 4-d.
- D. 1-b, 2-a, 3-c, 4-d.

Answer – B.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 18.

Question 103:

In a Backlog refinement meeting for a Software development project, the Product Owner presents a new story to the team that needs thorough research and analysis. What may be the best advice the Product Owner can give the team?

- A. The Advice team adds the stories in the Sprint backlog.
- B. Advise the team to do a why-why analysis for the n story.
- C. Advise team to complete the story in the next sprint
- D. Advise the team to spike the features of the new story for further understanding.

Answer – D.

Explanation

The Spikes are helpful when the team needs to learn some critical technical or functional element.

Here, as per the scenario stated in the question, since the new story requires thorough research and analysis, the best approach for the Product Owner will be to request the team to spike the story's features and come up with a detailed analysis report.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 53,55.

Question 104:

The latest audit report suggests Quality improvements for some of the Business-critical processes of a software development project. Which of the following are examples of quality improvement tools used to evaluate and analyze opportunities for Quality Improvement? Select any two.

- A. Design For X.
- B. Plan-do-check-act.
- C. Six Sigma.
- D. Check sheets.

Answer - B, C.

Explanation

Plan-do-check-act and Six Sigma are two of the most common quality improvement tools used to analyze and evaluate opportunities for improvement.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 296.

Question 105:

Ryan is the Project Manager of a Software maintenance project that is following Flow-based agile. What are the examples for measurements used in the Flow-based agile? Select any two.

- A. Cycle time.
- B. Velocity.
- C. Lead time.
- D. Spike.

Answer – A, C.

Explanation

Lead time, cycle time, response time are examples of measurements used in the Flow-based agile.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 64.

Question 106:

In the middle of a Software development project, the product owner leaves the company. The senior management suggests the Scrum Master for playing a dual role. What should the Scrum Master do?

- A. Start playing the dual role.
- B. Reject the proposal of the Senior Management.
- C. Discuss with the team and suggest a team member's name for the role of the Product owner.
- D. Advise the Senior Management to hire a new Product Owner as it's a specialized role.

Answer – D.

Explanation

The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide. He/She helps in scrum implementation within the organization. Here, as per the scenario stated in the question, the Scrum Master should coach the Senior Management, every role in a Scrum team is unique, having one person for both roles creates a conflict of interests. The best approach would be to look for a new Product Owner, as a dedicated person is required for the position.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.
[scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf](https://www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf) – Page – 6.

Question 107:

Ryan is a Project Manager of an IT Project that is using Agile-Scrum. A stakeholder requests a design change in one of the work packages. Who should assess the change?

- A. CCB.
- B. Product Owner.
- C. Project Manager.

D. The Development Team.

Answer – D.

Explanation

One of the 12 Agile Principles states that - " The best architectures, requirements, and designs emerge from self-organizing teams "

The Development team has the best insight on task execution. They should evaluate and work on the change.

Option A is not correct, as it's not applicable for Agile projects. The Project Manager, if present, acts like a Servant Leader. His job is ensuring the process, not the project. Therefore, option B is also incorrect.

Since the team is responsible for creating the architecture, they should assess the change. The team should also consult the Product owner to help them define the direction.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9,41.

Question 108:

Match the following

SL no	Name	Description
1	Empiricism	a. Work must be visible to both the doer and the receiver.
2	Transparency	b. Scrum Value.
3	Lean thinking	c. Knowledge comes from experience and decisions are made based on observation.
4.	Commitment	d. Reduce wastes and focus on essentials.

A. 1-c, 2-a, 3-d, 4-b.

B. 1-c, 2-d, 3-b, 4-a.

C. 1-d, 2-a, 3-c, 4-b.

D. 1-b, 2-c, 3-b, 4-d.

Answer - A

Explanation

scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page – 3,4.

Question 109:

Ryan is the project manager of a multiphase project. Phase 1 has come to an end. What is the most important stuff to ensure before beginning the next Phase?

- A. Ensure Lessons Learned are captured.
- B. Confirm the formal acceptance of the deliverables.
- C. Ensure Work Performance report is prepared for phase 1.
- D. Verify the resource availability.

Answer – B.

Explanation

All four choices are essential. But, a phase must be formally closed and accepted before the next phase starts. So, option B is the most important one that the project team should consider among all choices.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 165.

Question 110:

The Sprint Review meeting helps to inspect the outcome of the Sprint and determine future adoptions. Who participates in the Sprint Review meeting?

- A. Product Owner, Development Team.
- B. Development Team, Scrum Master.
- C. Scrum Team, Relevant Stakeholders.
- D. Project Manager, Product Owner, Scrum Master.

Answer – C.

Explanation

During the Sprint Review meeting, the Scrum Team (Product Owner, Development Team, Scrum Master) and Stakeholders review what was

accomplished in the Sprint and what has changed their environment.

So, among all options, Option C is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

[scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf](https://www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf) – Page – 9.

Question 111:

Match the following.

SL no	Conflict Management Technique	Description
1	Compromise.	a. Can result in win-win situation.
2	Force.	b. Occasionally results in a lose-lose situation.
3	Smooth.	c. Empathizing areas of agreement rather than difference.
4.	Collaborate.	d. Pushing one's view point at the expense of others.

A. 1-b, 2-c, 3-c, 4-a.

B. 1-c, 2-d, 3-b, 4-a.

C. 1-d, 2-b, 3-c, 4a.

D. 1-b, 2-d, 3-a, 4-c.

Answer – A.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 349.

Question 112:

Ryan is the new Project Manager of a Software development project. In his project, the team follows an Agile methodology where the release occurs continuously or whenever a shippable product gets created. What agile methodology the above-stated project is using?

A. Scrum.

- B. Kanban.
- C. Battlefield Agility.
- D. Large Scale Scrum.

Answer: B.

Explanation

The Kanban allows for a continuous flow of work and value to the customer. In Kanban, there are no fixed length sprints. Instead, the team pulls tasks from a prioritized backlog of things that need to be done. Releases occur frequently or whenever a shippable product gets created.

Choices A and D are not correct because, in Scrum or Large Scale Scrum, the product gets released on a particular cadence, determined by the Sprint's length.

Choice C is also incorrect because the Battlefield Agility doesn't focus on continuous delivery. Among all choices, Choice B is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 103, 104.

Question 113:

A junior project manager asks a senior Project Manager about documents created during execution and is used to evaluate the achievements of the quality objectives stated in the Quality Management Plan. They may include a dedicated checklist and detailed Requirement Traceability matrices as a part of the document. What is the document the junior Project Manager referring to?

- A. Test and Evaluation document.
- B. Check sheets.
- C. Quality Reports.
- D. Requirement Documentation.

Answer: A.

Explanation

Test and Evaluation documents describe the activities used to determine if the product meets the quality objectives stated in the quality management plan.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 296

Question 114:

In the middle of a Sprint, the Scrum team knows that additional code changes need to be in place for the Sprint backlog's work items due to new government regulations. The team advises the Product Owner to extend the Sprint length to accomplish the updated sprint goal. What should the Product Owner do?

- A. Increase the Sprint length and advise the team to update the Sprint backlog.
- B. Update the Sprint backlog and direct the team to finish the work as a part of the current sprint.
- C. Cancel the sprint and work with the team to reprioritize the product backlog.
- D. Ask the team to cancel the Sprint.

Answer – C.

Explanation

The Sprints are fixed length events of one month or less to create consistency. Choice A is incorrect as it is not a recommended option.

The Product Owner cannot update the Sprint Backlog; the Development team is empowered to revise the Sprint backlog. So, Choice B is also wrong.

The team cannot cancel a sprint. A product owner can cancel a sprint. So, Choice D is incorrect.

Among all options, choice C is the best answer. The Product Owner should cancel the Sprint and work with the team to reprioritize the Product backlog, so that team can start working on the change as quickly as possible.

scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page – 7,8.

Question 115:

Kanban is a popular framework used to implement agile and DevOps software development. Which of the following are core properties of Kanban?

Select any two.

- A. Limit Work in Progress.
- B. Work is done within time-boxed sprints, generally on 2-4 weeks.

- C. Product is released on a particular cadence, determined by the Sprint's length.
- D. Visualize the workflow.

Answer – A, D.

Explanation

The core properties of the Kanban method include:

Visualize the workflow, Limit Work in Progress, Manage flow, Make process policies explicit, Implement feedback loops, Improve collaboratively.

The choices B and C are properties for Scrum Methodology. So, they are incorrect.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 104.

Question 116:

An Agile coach advises the team to share the Project progress for a Mission-Critical project with everyone to enable information sharing across the company. What should the Scrum team do?

- A. Share Project Progress communication to all stakeholders via email.
- B. Invite the Project Managers of other teams in the Iteration review meeting.
- C. Share the Project progress communication with the HR team and ask them to share with all relevant stakeholders.
- D. Use the Information radiator and place it in a prominent location of the office.

Answer – D.

Explanation

An information radiator is a comprehensive display of key team information, openly shared with all team members and other stakeholders and continuously updated. It is something similar to a big visual chart placed in a prominent location. It includes information such as the task board, the count of automated tests, the team velocity, incident reports, continuous integration status, the state of work packages, or the progress of the team.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) –

152.

Question 117:

On a project following the predictive approach, the change control board (CCB) is conducting a meeting to evaluate a change request to increase the scope of the project. After reviewing the change request, the CCB must now decide whether to approve the request. They utilize a decision matrix to evaluate the requested change according to a set of predefined criteria.

What decision-making technique is the CCB using?

- A. Voting.
- B. Autocratic decision making.
- C. Multi-criteria decision analysis.
- D. Nominal group technique.

Answer- C.

Explanation

Multi-criteria decision analysis utilizes a decision matrix to provide a systematic analytical approach to evaluate requested changes according to a set of predefined criteria.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 119.

Question 118:

Match the following

SL no	Name (RBS)	Example
1	Technical Risk.	a. Suppliers and Vendors.
2	Management Risk.	b. Competition.
3	External Risk.	c. Estimates , assumptions, and constraints
4.	Commercial Risk.	d. Resourcing and Communication.

- A. 1-c, 2-d, 3-a, 4-b.
- B. 1-c, 2-d, 3-b, 4-a.
- C. 1-d, 2-c, 3-a, 4-b.
- D. 1-d, 2-a, 3-b, 4-c

Answer - B

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 406.

Question 119:

An agile coach suggests the Project Management Team select a methodology that would support frequent releases and osmotic communication for the company's upcoming Software development project. What methodology is best suited for frequent releases and osmotic communication?

- A. Scrum.
- B. Crystal.
- C. Meta Scrum.
- D. Kanban.

Answer – C.

Explanation

The common properties for the Crystal methodology include frequent delivery and osmotic communication.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 107.

Question 120:

Which one of the following is not true about the decision-making technique called Plurality?

- A. A decision that is reached based on the decision of the largest block in a group.
- B. The method is generally used when the number of options nominated is more than two.

- C. Plurality is a popular voting technique.
- D. A decision that is reached with the support from more than 50% of the members of the group.

Answer – D.

Explanation

Plurality or relative majority is a decision that is reached whereby the largest block in a group decides, even if the majority is not achieved. A, B, C are characteristics of Plurality whereas D is the characteristic of Majority. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 144.

Question 121:

You are a Project Manager of a Software development project of a Matrix Organization. The Change Management Plan defines no role for the sponsor in the Change Control Board. Currently, CCB has approved a Change Request that involves the addition of two resources. What should the Project Manager do NEXT?

- A. Evaluate the impact of the change on other constraints.
- B. Update Project Management Plan.
- C. Seek Sponsor authorization.
- D. Implement the approved Change Request

Answer- C.

Explanation

The Sponsor authorization is mandatory for approval of a Change Request. Only approved changes should be incorporated into a revised baseline. It's the sponsor who is accountable for adding funds to the project. Among all options provided, C is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 117.

Question 122:

In Release planning, the product backlog items are grouped into multiple more minor releases. Who plans a release?

- A. Scrum Team.
- B. Scrum Master, Product Owner.
- C. Scrum Master, Development Team, Stakeholders.
- D. Product Owner, Stakeholders.

Answer – A.

Explanation

Entire Scrum Team (Scrum Master, Product Owner, and the Development Team) participate in release planning to derive a product roadmap.

Question 123:

During execution, two change control board members have a conflict on the outcome of a change request. After the sponsor and the Project Manager's intervention, the team applied a multi-criteria decision analysis technique to reach a decision that both the team members happily accepted. Here adoption of multi-criteria decision analysis is an example for which of the following conflict resolution techniques?

- A. Forcing.
- B. Problem solving.
- C. Reconciliation.
- D. Smoothing.

Answer: B.

Explanation

Here, the Sponsor and Project Manager have adopted a collaborative/problem-solving conflict resolution technique to reach a win-win situation. So, among all options, option B is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 349.

Question 124:

Conflict is evident in project management. At what team formation and development phase is conflict likely to be highest?

- A. Forming.
- B. Norming.

- C. Storming.
- D. Performing.

Answer – C.

Explanation

The level of conflict is the highest in the Storming phase. At this phase, the team begins to address the project work. If team members are not open to different ideas and perspectives, the environment can become counterproductive.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 338.

Question 125:

The Team Charter is a document that establishes operating guidelines for the team. It includes: Select any two.

- A. Team values.
- B. Contractual agreement
- C. Meeting guidelines.
- D. Recognition plan.

Answer – A, C.

Explanation

The Team Charter generally includes.

- a. Team values.
- b. Communication guidelines.
- c. Decision making criteria and process.
- d. Meeting guidelines.
- e. Conflict-resolution process.
- f. Team agreements.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 319.

Question 126:

Your project has a resource working in niche technology. The Functional manager approaches you to release the resource for working on a business critical assignment of another project. What should you do

before having further discussion and negotiation with the functional manager?

- A. Review the Project Calendar.
- B. Check for non-billable resources who can complete assignments assigned to the resource to be released.
- C. Review the RACI to know the current assignment of the resource.
- D. Immediately escalate the situation to the higher management.

Answer: C.

Explanation

Here in this scenario, the project manager should review the work assigned to the resource and his current responsibilities in the project. For this RACI chart needs to be reviewed. After reviewing the RACI chart, he can discuss and negotiate with the Functional Manager. Here among all options, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 317.

Question 127:

Ryan is a new project manager of a large IT infrastructure project, which has 120 team members. All team members are working on a building taken for lease as a part of the Project direct cost. In his first meeting with the sponsor and key stakeholders, he knows that the project must be extended for another year to meet additional customer needs. The approved change request is already in place for the same. What should Ryan review first to know the building's lease expiration date where the project team members are operating?

- A. Review the Project Schedule.
- B. Review the Resource Calendar.
- C. Have a meeting with the previous Project Manager.
- D. Review the Contractual Agreement of the building.

Answer: B.

Explanation

Resource Calendar specifies when and how long identified physical or Team

resources will be available during the project. Here the building under the lease is a physical resource for the project. Based on this, the Project Manager can decide whether the building's lease should be renewed or not. D is a good option, but in many organizations, the Procurement Administrator handles such situations, mostly where corporate procurement strategy is centralized. The question never stated whether the organization's procurement strategy is centralized or decentralized. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 323.

Question 128:

For a very large project with a large team size, what Agile methodology or framework is recommended? Select any two.

- A. Scrum.
- B. Disciplined Agile (DA).
- C. Large Scrum Scrum (LeSS).
- D. Kanban.

Answer – B, C

Explanation

For a very large project with large team size, one should consider scaled agile or lean frameworks such as Disciplined Agile (DA), Large Scale Scrum (LeSS), SAFe, etc.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 121.

Question 129:

Arun is the Project Manager of the RTG project, and he is part of the Matrix organization. During planning, he understands the need for extra human resources. Whom should the Project Manager Approach First for the additional resources?

- A. Project Sponsor.
- B. Functional Manager.
- C. Program Manager.
- D. Procurement Administrator.

Answer – B.

Explanation

In a matrix organization, a Functional Manager acts as the resource manager. So the right answer is B.

Question 130:

On a project, the change control board (CCB) is conducting a meeting to evaluate a change request to increase the project's scope. After reviewing the change request, the CCB must now decide whether to approve the request. They utilize a decision matrix to evaluate the requested change according to a set of predefined criteria. What decision-making technique is the CCB using?

- A. Voting.
- B. Autocratic decision making.
- C. Multi-criteria decision analysis.
- D. Nominal group technique.

Answer- C.

Explanation

Multi-criteria decision analysis utilizes a decision matrix to provide a systematic analytical approach to evaluate requested changes according to a set of predefined criteria.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 119.

Question 131:

A project sponsor asks the project manager how well the project is doing in terms of budget. The project manager reviews the Work Performance Data and informs the sponsor that the project is doing well. What is wrong with the project manager's approach?

- A. Nothing is wrong with the project manager's approach.
- B. The project manager should have reviewed work performance information.
- C. The project manager should have reviewed the cost baseline.
- D. The project manager should have consulted the accounting

department.

Answer – B.

Explanation

To determine how well the project is doing in terms of budget, the project manager should have reviewed the work performance information rather than work performance data. Work performance data by itself should not be used in the decision-making process as it has only out-of-context meaning. Work performance information, on the other hand, is correlated and contextualized, and provides a foundation for project decisions.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) -109

Question 132:

The Sponsor and some relevant stakeholders are doing the need assessment to evaluate the feasibility of the latest strategic need of the organization by considering all related assumptions and constraints. What can act as valid input for this effort?

- A. Lessons learned from previous projects.
- B. Business Case.
- C. Project Charter.
- D. Benefit Management Plan.

Answer – A.

Explanation

Need assessment often precedes the business case. It involves understanding the business goals and objectives, issues and opportunities and recommending proposals to address them. The results of it may be summarized in the business case document.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 30.

Question 133:

The following document provides the framework for the Schedule Management Plan allowing for consistency with the estimates and the resulting schedules. What document is referred to here?

- A. Work Breakdown Structure.
- B. Network Diagram.
- C. Risk Breakdown Structure.
- D. Stakeholder Register.

Answer – A.

Explanation

Work Breakdown Structure(WBS) provides the framework for the schedule management plan
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 182

Question 134:

An agile team of a software development project has been rewarded with a higher hourly billing rate because they have completed the work earlier than the contractual deadline. What is the contract type followed in the Agile team's project?

- A. Time and Materials with Not To Exceed clause.
- B. Fixed Price with economic price adjustments.
- C. Cost reimbursable.
- D. Graduated Time and Materials.

Answer – D.

Explanation

Graduated Time and Materials is a shared financial risk approach. Suppliers are rewarded with a higher hourly rate when the delivery is earlier than the contracted deadline. Conversely, there will be a rate reduction for late delivery.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 78.

Question 135:

As a project manager, you lead a research and development project, and you always follow all project management best practices. You have just completed project initiation, and you need to begin the planning phase of your project. What should you do next?

- A. Create the Project Management Plan.
- B. Develop the Stakeholder register and the Stakeholder Engagement Plan.
- C. Get the project charter approved.
- D. Assign Resources to the project.

Answer – A.

Explanation

The question states that you, as the project manager, have completed project initiation and are about to start the planning phase of your project. The project management plan gets created during the Develop Project Management Plan process, part of the Planning Process Group.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 82.

Question 136:

Match the following

SL no	Stakeholder's direction of Influence	Description
1	Upward.	a. Representatives outside the project team.
2	Outward.	b. Specialists contributing knowledge or skills in a temporary capacity.
3	Downward.	c. Senior management of the performing or customer organization.

- A. 1-b, 2-c, 3-a.
- B. 1-c, 2-b, 3-a.
- C. 1-c, 2-a, 3-b.
- D. 1-b, 2-c, 3-a.

Answer – C.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) - 513.

Question 137:

Naveen is the Program Manager of an IT services company. He is reviewing the agile metrics report for two of his IT projects. He finds that Team A has a Velocity of 75, and Team B has a 50. Which team has the higher throughput?

- A. Team A. as higher the velocity, higher is the throughput
- B. Team B, as lower the velocity, higher is the throughput.
- C. Both Team A and Team B, as every project is unique.
- D. Velocity is not the right metric to compare throughput across teams.

Answer – D.

Explanation

Each team's velocity is unique. If team A has a velocity of 50 and team B has a velocity of 75, it doesn't mean that team B has higher throughput. Since each team's estimation culture is unique, their velocity will be as well. Resist the temptation to compare velocity across teams. Measure the level of effort and output of work based on each team's unique interpretation of story points. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 64.

Question 138:

A project is running behind schedule. Earned value analysis shows the project needs a twenty percent faster run rate than the current state. In a meeting, senior management suggested adding five more people to an activity currently assigned to one person. The project manager disagrees, as he feels noting such an increase will not produce the desired increase in speed.

The approach of the senior management can lead to: -

- A. An increase in the overall SPI of the project.
- B. The law of diminishing returns.
- C. Anti-pattern.
- D. Improvement in Project Performance.

Answer – B.

Explanation

The Senior Management's approach can lead to a Law of diminishing returns. According to it, when one factor (e.g., resource) used to determine the effort required to produce a unit of work is increased while all other factors remain fixed, a point will eventually be reached at which additions of that one factor start to yield progressively smaller or diminishing increases in output.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 197.

Question 139:

The Project Manager recommends the team use a scope model that will visually depict the product scope by showing a business system. What tool the project manager suggested?

- A. Context diagram.
- B. Prototype.
- C. JAD.
- D. Requirement Traceability Matrix

Answer – A.

Explanation

The context diagram is used to establish the context and boundaries of the system to be modeled.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 146.

Question 140:

The Kanban method allows a continuous flow of work and value to the customer. Which of the following conditions may be best suited for using the Kanban method?

- A. Projects where the product needs to be released on a particular cadence.
- B. Projects focusing on continuous delivery.
- C. Projects where variability in the workload exists.
- D. Projects where work is done within time-boxed sprints.

Answer – B, C

Explanation

The Kanban method may be best used where there is a need of the following conditions:

- a. Flexibility – Not bound by time boxes.
- b. Focus on continuous delivery – Focus on workflow.
- c. Increased productivity and quality – Productivity and quality are increased by limiting work in progress,
- d. Increased efficiency – Checking all possible tasks and removing non-value adding activities.
- e. Team member focus – Limited work in progress allows the team to focus on the current work.
- f. Variability in the workload – Work arrives in an unpredictable way.
- g. Reduction of waste – Transparency makes waste visible so it can be removed.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 103-104.

Question 141:

Ryan is a Project Manager of a multi-phase RPA project. While reviewing the working software of phase 4, he found out that the product developed has less functionality than was planned for the same phase during the design stage. Which of the following tools and techniques the Project Manager used in the given scenario?

- A. Variance analysis.
- B. Trend analysis.
- C. Reserve analysis.
- D. Technical Performance analysis.

Answer – D.

Explanation

Technical performance analysis compares technical accomplishments during project execution to the schedule of technical achievements. Deviations, such as less functionality than planned at a key milestone, can help to forecast the degree of success in achieving the project scope.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) – 456.

Question 142:

Project reporting is one of the most valuable tools for the project teams and the stakeholders involved in the project. Project reporting happens as a part of which of the following processes?

- A. Monitor and Control Project Work
- B. Manage Communications.
- C. Monitor Communications.
- D. Conduct Procurements.

Answer. B.

Explanation

The project reporting helps to track the current progress of the project. It's the act of collecting and distributing project information to the relevant stakeholders. It's a valid tool for the Manage Communication process, and the project reporting happens during execution as a part of the Manage Communication process.

A few may get confused with option A - Monitor and Control Project Work, as the Work Performance Reports are output for the process. But one should remember that the Work Performance Reports are developed as part of the Monitor and Control Project Work process, but are distributed or presented during the Manage communication process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 385.

Question 143:

A new Agile PMO joins an IT organization. He wants to understand whether the organization's four agile teams are continually improving their process and productivity for their respective ongoing projects. What should the PMO do?

- A. Review the Burn-down charts of each project.
- B. Set up separate meetings with each Agile team and ensure regular risk review meetings are conducted in the project.
- C. Set up separate meetings with each Agile team and ensure retrospectives are getting conducted after the Working Software

- demo for each iteration.
- D. Review the RAID logs for each project

Answer: C.

Explanation

Retrospectives help the team learn from its previous work on the product and its process. Holding regular retrospectives allows an Agile team to improve their process and productivity continually.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 50-51.

Question 144:

In a retrospective meeting, an Agile coach suggested using a tool to display work remaining, work completed, and work in progress for work related to user stories for the current release.

What tool does the agile coach recommend?

- A. Burndown chart.
- B. Burnup Chart.
- C. Control Chart.
- D. Cumulative Flow Diagram.

Answer – D.

Explanation

Cumulative Flow Diagrams are valuable tools for tracking and forecasting the delivery of value.

Choice – A - Burndown Chart – Incorrect – Shows work remaining.

Choice – B - Burnup Chart – Incorrect – Shows work completed.

Choice – C - Control Chart – Incorrect – It is a graphic display of process data over time and against established control limits.

Choice – D - Cumulative Flow Diagram – Correctly – Shows work remaining, completed, work that is in progress. Cumulative Flow Diagrams help us gain insight into project issues, cycle times, and likely completion dates.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 70.

Question 145:

Match the following.

SL no	Name	Description
1	Storyboarding.	a. Consolidates ideas created through individual brainstorming sessions into a single map .
2	Mind mapping.	b. A prototyping technique.
3	Quality function Deployment.	c. An example of a scope model.
4.	Context diagram.	d. A facilitation technique.

- A. 1-d, 2-a, 3-b, 4-c.
- B. 1-b, 2-a, 3-d, 4-c.
- C. 1-c, 2-a, 3-d, 4-b.
- D. 1-c, 2-b, 3-d, 4-a.

Answer – B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 144 -147.

Question 146:

A user story is the smallest unit of work in an agile framework. Which of the following are characteristics of good User Stories?

Select any two.

- A. UX design.
- B. Single loop learning.
- C. Negotiable.
- D. Small.

Answer: C,D

Explanation

The characteristics of a good User Story: -

I – Independent.

N- Negotiable.

V- Valuable.

E- Estimable.

S- Small.

T- Testable.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 155.

Question 147:

In the middle of a Software development project, a new Product Owner joins. She wants to review the work completed so far in the current release and scope changes during the release cycle. What should the product owner review?

- A. Burndown Chart.
- B. Burnup Chart.
- C. Milestone Chart.
- D. Sprint review Chart.

Answer – B.

Explanation

The Burnup charts track the work that has been completed. It can also show the changes in project scope, making the impact of those changes visible.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 63,150.

Question 148:

Ryan is a Project Manager of a Software development project. During planning, after an exhaustive brainstorming session with some key stakeholders, he decides to document the commonality and differences in understanding of the product scope into a diagram using a tool. What tool the Project Manager used in the given scenario?

- A. Mind mapping.
- B. Affinity diagram.
- C. Matrix diagram.
- D. Decision tree diagram.

Answer – A.

Explanation

The question states that the “Collect Requirements” process is getting performed. So, options C and D are incorrect. Option B is also wrong as the Affinity diagram helps in the grouping of ideas. It never states commonality and differences of opinion among stakeholders.

Mind mapping consolidates ideas created through individual brainstorming sessions into a single map to reflect commonality and differences in understanding and generate new ideas.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 144.

Question 149:

In the middle of a sprint for a Software development project, a stakeholder presents a new story to the Product Owner. The Product Owner wants to show the overall story concept to the team. In which of the following meetings he will discuss the new story with the team?

- A. Iteration Planning meeting.
- B. Iteration Review meeting.
- C. Backlog refinement meeting.
- D. Iteration retrospective meeting.

Answer – C.

Explanation

In iteration based agile, the product owner often works with the team to prepare some stories for the upcoming iterations during the middle of the iteration.

Here, as per the question, a new story has arrived in the middle of a sprint. The Product Owner presents the overall story concepts to the team in the Backlog refinement meeting. The team discusses and refines it into as many stories as required.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 52-53.

Question 150:

A newly formed Scrum Development team is looking for templates for

User stories, test cases. Whom should they approach?

- A. Agile PMO.
- B. Scrum Master of another project.
- C. Product Owner.
- D. Quality Control team.

Answer: A.

Explanation

The Scrum Team should approach the Agile PMO responsible for developing and implementing standards by sharing templates for user stories, test cases, Cumulative flow diagrams, etc. Agile PMO also provides training on Agile tools and concepts.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 82.

Question 151:

Match the following

SL no	Situation	Best course of action.
1	An Unidentified risk occurs in the middle of a project.	a. Update the issue log and implement the fallback plan .
2	A new risk is discovered during project execution.	b. Update the Contingency plan.
3	An identified risk gets triggered.	c. Update the risk register.
4.	An identified risk occurs, and the primary response strategy is not effective.	d. Update the issue log and plan workaround.

- A. 1-d, 2-b, 3-d, 4-c.
- B. 1-c, 2-a, 3-b, 4-d.
- C. 1-a, 2-c, 3-d, 4-b.
- D. 1-d, 2-c, 3-b, 4-a.

Answer - D.

Explanation

Risk Register - Whenever a new risk gets discovered, update the Risk register.

Issue log - Whenever a risk occurs (Identified or Unidentified), update the Issue log.

Contingency Plan - Activated once pre-accepted risk is triggered. It is a Primary Risk Response Strategy.

Fallback Plan - Implemented when the Primary risk response strategy is not effective. Can also be called a rollback plan on certain occasions.

Question 152:

As a Project Manager, you are helping the sponsor develop a document that formally authorizes the project's existence. To build an implementation strategy, you are trying to review High-level strategic and operational assumptions and constraints. What should the project manager review?

- A. Project Charter.
- B. Project Scope Statement.
- C. Feasibility analysis report.
- D. Business Case.

Answer – D.

Explanation

High-level strategic and operational assumptions and constraints are normally identified in the Business Case before the project gets initiated and will flow into the Project Charter. The Assumptions Log is used to record all assumptions and constraints throughout the project life cycle.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 81.

Question 153:

Ryan is the Project Manager of a Mission-critical software development project that is in planning. In a meeting with the team, he finds out that there are significant gaps between the current and desired engagement levels for two most critical stakeholders. What should the Project Manager do next?

- A. Escalate the situation to the Sponsor.

- B. Review and update the Benefits Management Plan.
- C. Schedule a brainstorming session to evaluate the engagement needs for stakeholders.
- D. Report the issue to the PMO.

Answer – C.

Explanation

The Project Manager cannot update the Business documents (Business Case, Benefits Management Plan), so choice B is incorrect.

The Project Manager is accountable for Stakeholder Engagement. So, escalating the situation to the Sponsor is not a great approach. Therefore, option A is also not the right choice.

The question never mentioned the type of PMO, and in general, PMO is not responsible for Stakeholder Engagement. So, Choice D is also not correct.

The Project Manager is responsible for Stakeholder collaboration. The best approach here is scheduling a Brainstorming meeting with the team to evaluate the engagement needs for the two stakeholders to optimize alignment between stakeholder needs, expectations, and project objectives. In this way, the engagement level gaps can be filled .

Here, among all options, option C is the best answer.

PMI PMP Examination Content Outline – June 2019 – Domain 1 – Task 9.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 521-522.

Question 154:

Match the following.

SL no	Agile Manifesto-Four values		
1	Individuals and Interactions	OVER	a. following a plan .
2	Working Software	OVER	b. contract negotiation.
3	Customer collaboration	OVER	c. Processes and tools.
4.	Responding to change	OVER	d. Comprehensive documentation.

- A. 1-a, 2-c, 3-d, 4-b.
- B. 1-c, 2-d, 3-b, 4-a.
- C. 1-d, 2-a, 3-b, 4-c.
- D. 1-c, 2-d, 3-a, 4-b.

Answer – B.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 8.

Question 155:

The CPI and SPI are below 1 for a project which is in execution. After reviewing all related risk response plans, the Project Manager vows to bring the project back on track.

What should the project manager do?

- A. Crashing.
- B. Fast tracking.
- C. Resource Smoothing.
- D. Resource Leveling.

Answer – B.

Explanation

Crashing impacts the project cost directly, and since CPI is low, option A is incorrect. Choice C and D are resource optimization techniques. Resource

smoothing won't change the critical path, so the application of Resource smoothing would not positively impact the project schedule. Resource leveling gets used when limits on resource availability are paramount, so it may not be the right option. Therefore, both choices C and D are not correct.

Fast tracking is a schedule compression technique in which activities or phases are performed sequentially, done in parallel for at least a portion of their duration. It's a cost-effective technique that helps to reduce the duration of the critical path. Further, the question also states that the project manager has decided to bring the project on track after reviewing planned risk responses, so he should be well aware of the risks that may arise due to fast tracking.

So among all options Option B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 215.

Question 156:

DSDM is focused on eight principles. Which of the following are principles that guide the use of the DSDM framework? Select any two.

- A. Focus on the Business need.
- B. Focus on high-value activities.
- C. Situationally specific
- D. Communicate continuously and clearly.

Answer – A, D.

Explanation

Eight Principles that guide the use of the DSDM framework include: -

1. Focus on the Business need.
2. Deliver on time.
3. Collaborate.
4. Never compromise Quality.
5. Build incrementally from firm foundations.
6. Develop iteratively.
7. Communicate continuously and clearly.
8. Demonstrate control (use appropriate techniques).

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 110.

Question 157:

In a Backlog refinement meeting, a team member requests to add a new feature in the current product release that may dramatically improve an application's performance. Other team members are concerned about prioritizing the mandatory features first.

What should the Product Owner do?

- A. Extend the duration of the next two sprints and add everything for the current release.
- B. Ask the team to implement only the mandatory features.
- C. Discuss the situation with the Scrum Master and the Sponsor, and ask them to take a final call.
- D. Take a decision based on the business value of the mandatory features and the recommended feature.

Answer – D.

Explanation

The Product Owner should rank the work based on its business value. He should determine the ROI of each feature independently to conclude.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.

Question 158:

Ryan is a Project Manager of a Software development project. He is scheduling a meeting with his team to discuss a problem that key stakeholders described in detail in a separate meeting. What would be helpful for the upcoming meeting?

- A. Inviting key stakeholders in the upcoming meeting.
- B. Discussing the problem raised by the key stakeholders with the sponsor.
- C. Reviewing the Minutes of Meeting (MOM) captured in his meeting with the key stakeholders.
- D. Sharing questionnaires to the team in relation to the problem before the meeting.

Answer – C.

Explanation

Meeting minutes (MOM) are formal communication artifacts used to summarize the discussion points for a meeting. Meeting minutes may include a list of attendees, the purpose for the meeting, the original agenda, the events that occurred during the meeting, materials presented such as slides or reports, outstanding questions or concerns, and action items discussed along with relevant votes, milestones, assignments of responsibility, and deadlines. If the project team can get some exposure to a formal report of what caused the stakeholders to be concerned and the context of the discussion, it may help with brainstorming ideas for the solution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 361, 381,386.

Question 159:

An agile coach is doing a knowledge-sharing session with all the Project Managers of the Qnx company. He is using PowerPoint presentations with images, graphs, bulleted words for crystal clear knowledge transfer. What type of knowledge is the agile coach sharing?

- A. Explicit knowledge.
- B. Tacit knowledge.
- C. Tangible knowledge.
- D. Technical knowledge.

Answer – A.

Explanation

Here, the Agile coach is sharing explicit knowledge with the team. The knowledge that can be codified using images, numbers, and words is called Explicit knowledge.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 100.

Question 160:

In the middle of a software development project following agile-scrum, a stakeholder comes up with six new performance requirements. How can the scrum team ensure that the performance requirements are addressed in the solution? Select any two.

- A. Add a performance tester for performance testing.

- B. Add six requirements to the definition of Done.
- C. Add six requirements as items in the product backlog.
- D. Hire an external auditor for performance testing.

Answer: B,C.

Explanation

Performance like security, scalability and maintainability is a non-functional feature. Non-functional features are normally added to the definition of “Done” because they apply to all functional features. For some non-functional features, such as performance, it’s possible to create non-technical, independent items in the Product Backlog. This is usually done when we have to make an improvement in the previously done features.

Question 161:

In the middle of a Software development project following Agile-Scrum, a new Product Owner joins. After meeting the team, he discovers that the Product Backlog items are ordered inefficiently. What should he do?

- A. Apply Planning poker technique to prioritize the product backlog.
- B. Prioritize the Product Backlog by CD3.
- C. Ask the team to adjust the Product Backlog.
- D. Review and update the Business Case.

Answer – B.

Explanation

Planning Poker is used for estimating User Stories. It is not an ideal technique for ordering the Product Backlog items. So, choice A is not correct. The Product Owner is accountable for ordering the Product Backlog items, whereas the team manages the sprint backlog and the task execution. Therefore, choice C is not the recommended answer. The Product Owner cannot update the Business Case, so the choice is D is outright rejected.

The value model, CDT, can prioritize the Product Backlog. Cost of Delay Divided by Duration is a prioritization method that maximizes the value delivered in a given period. The CD3 uses the "Weighted Shortest Job First" queuing method to rank items in the Product Backlog based on business value.

It's impossible to compare radically different stories because their values are expressed in incomparable measure units. To get around this problem, the product owner can convert everything to money using the cost of delay. And then can divide the cost of delay by duration to calculate the CD3. The higher the CD3 score, the higher the story should be in the backlog.

So, here, among all options, option B is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 59.

Question 162:

The condition or capability that is required to be present in product, service, or result to satisfy an agreement is termed as -

- A. Deliverable.
- B. Requirement.
- C. Product scope.
- D. Process.

Answer - B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 131.

Question 163:

You are an owner of a small Start-up company. You have been awarded a fixed price project by the Government of Oman, which you are required to complete in 3 months. The project charter is approved. Key people affected by the success and failure of the project have been identified. After working with the team, you have decided to execute the project by using the agile approach. What should you do?

- A. Work with team and build an Implementation strategy
- B. Complete Stakeholder analysis.
- C. Conduct Kick-off meetings.
- D. Perform Qualitative Risk analysis.

Answer – C.

Explanation

The situation states that a 3-month project involving a Fixed-Price Contract was awarded to a small company. In this case, we can assume that project scope is also well defined since price and timeline are both fixed, and it's the characteristics of selecting a fixed price contract. And this is not an example of a large project. If the project is small, the kickoff meeting occurs shortly after the initiation process (in the planning process group). The meeting includes all team members, as the same team is involved with the planning and execution. Many aspirants go for option A, but we should always remember that implementation strategy is determined before the project gets approved in most cases.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 86.

Question 164:

Arun is the Delivery head for an IT organization. He wants to initiate the usage of an adaptive delivery approach slowly in the company. For more than ten years, the company is successfully using the predictive model, and the majority of the critical stakeholders are resistant to change. Currently, the company has acquired a million-dollar project with medium uncertainties.

What should be Arun's approach here?

- A. Maintain the waterfall approach.
- B. Discuss the situation with the CEO and try to influence him to shift to an adaptive model.
- C. Adapt Scrum, and direct the PMO team to initiate a training plan.
- D. Adapt a Hybrid model.

Answer – D.

Explanation

Switching to a new technique overnight is not easy. For this reason, planning a gradual transition is a more pragmatic approach.

Teams resistant to change adapt better to a gradual transition where both predictive and adaptive delivery models are used, called the Hybrid model.

The question says that the degree of uncertainty is medium for the new project that the company has recently acquired. Adapting a hybrid technique can be ideal in this kind of situation. If the company is successful with a

hybrid approach, more complex projects can be tested in the future.
Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 30.

Question 165:

At the Sprint Review meeting of the seventh sprint in a 12 sprint release, the stakeholders are surprised and upset that the product demonstrated is not what they expected and will incur additional costs that were not in their plan. What may have led to this?

- A. The Product Owner has not been managing the Development Team's tasks effectively.
- B. The stakeholders have not been attending the Sprint Reviews.
- C. The Product Owner has not made the stakeholders aware of the progress of the project.
- D. The Scrum Master has not been reporting on the progress of the Scrum Team.

Answer- B, C

Explanation

Scrum requires significant aspects of the process to be visible to those responsible for the outcome, including transparency with internal and external stakeholders.

The Product Owner is responsible for guiding the direction of the Product. He/she needs to continuously work with Stakeholders and the development team to ensure that the Product Backlog is transparent, visible, and understood.

On the other hand, the stakeholders should be regular in Scrum meetings. The question states that the Stakeholders are surprised to see the Product being built, which indicates that they were irregular in the previous Sprint review meeting.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.

Question 166:

In a large Agile project, multiple teams are working on the same product. Who should make sure that the outputs produced by each team can be integrated into one increment?

- A. Developers from each team.

- B. Project Manager.
- C. Product Owner.
- D. Scrum Master.

Answer – A.

Explanation

It's up to all development team members from each team to find their way and ensure that their outputs will create an integrated Increment that is potentially releasable.

Question 167:

Ryan is the newly appointed Project Manager of a software development project plagued with too many change requests. What is the most likely reason for a high amount of Change requests?

- A. Incomplete Stakeholder Register.
- B. Incomplete Project Scope statement
- C. Failure to implement risk responses.
- D. The Change Management plan did not authorize a Change Control Board.

Answer: A.

Explanation

Stakeholder Register contains identification information of stakeholders and their expectations and requirements, and at which phase their influence or impact will be most critical for the project. Though there can be many reasons for a high amount of change requests, option A seems to be the best choice. If project stakeholders are not identified in the early stage of the project, their project requirements will not be included in the project scope leading to an increased number of Change requests.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 514.

Question 168:

The development team prepares the Sprint Backlog in the Sprint planning session. What consists in the Sprint Backlog?

- A. Sprint Goal.
- B. Product Goal.
- C. The Product Backlog items selected for the Sprint
- D. Product Vision statement

Answer – A, C

Explanation

The Sprint Goal, the Product Backlog items selected for the Sprint, plus the plan for delivering them are together referred to as the Sprint Backlog.
scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page – 9.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

Question 169:

Ryan is the new CEO of the software development company. He is currently addressing the new organization's vision and mission statement to his leadership team in a meeting. The organization Vision and the Mission are examples for: -

- A. CPA
- B. Business Value.
- C. EEF.
- D. An assumption.

Answer – C

Explanation

The organization Vision and the Mission are examples for Enterprise Environment Factors Internal to the Organization.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 38.

Question 170:

During Closing, the project manager reviews a document that states the conditions to be met to close or cancel the project or phase. Which document is the Project Manager currently reviewing?

- A. Project Charter.
- B. Requirement Documentation.
- C. Project Management plan.
- D. Scope management plan.

Answer – A.

Explanation

Here, the Project Manager is reviewing the Project exit criteria defined in Project Charter.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) 81.

Question 171:

Ryan and his high-performance team are developing an Innovative IT product that will serve modern-day infrastructure needs. The product is 75% ready. Being overwhelmed with Ryan's new technology solution, the Senior Management wanted to demonstrate the same to a prospective customer. What should Ryan do?

- A. Reject the Senior Management proposal as the product is not yet ready.
- B. Share the Minimum marketable feature of the product to the customer.
- C. Have a meeting with the customer team to understand the problem statement.
- D. Set up a meeting with his team to make them ready for product demonstration.

Answer – C.

Explanation

Ryan should meet the customer and understand their needs. Giving a product demo or sharing the MMF may not be apt, as customer problems may differ. The best approach is meeting the customer, understanding their problem statement, and reviewing how his IT product can solve the customer's problem.

Question 172:

An IT project following Agile Scrum is not doing well. The Development team is struggling to achieve the Sprint goal due to variability in workload. The work arrives unpredictably; the team is finding it challenging to make predictable commitments. What went wrong in this situation?

- A. Product Backlog is not ordered properly.
- B. The Kanban method should have been adapted instead of the Scrum.
- C. The Development Team didn't apply Lean philosophy at work.
- D. The Product Owner has not participated in most Sprint Planning meetings.

Answer: B.

Explanation

The selection of the correct methodology or framework based on business needs is very important for an Agile project. As per the question, since there is variability in workload, Scrum or other timeboxed iterations may not work. The Kanban Method is well suited for the scenario stated in the question. The Kanban allows for a continuous flow of work and value to the customer. In Kanban, there are no fixed-length sprints. Instead, the team pulls tasks from a prioritized backlog of things that need to be done. Releases occur frequently or whenever a shippable product gets created.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 104.

Question 173:

Match the following

SL no	Process/Framework	Guiding Principles
1	DSDM	a. Apply Systems Thinking.
2	AgileUP	b. Focus on high-value activities.
3	SAFe®	c. Focus on Business need.

- A. 1-a, 2-b, 3-c.
- B. 1-b, 2-c, 3-a.
- C. 1-c, 2-b, 3-a.
- D. 1-b, 2-a, 3-c.

Answer - C.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 110-112.

Question 174:

Your company has hired four experts to handle specialized tasks that involve pure risk for a complex chemical engineering project. What should the Project Manager ensure before the task allocation of these experts? Select the best answer.

- A. Ensure experts clearly understand the overall project objectives.
- B. Ensure experts are involved in all project planning meetings.
- C. Ensure experts generate business value as quickly as possible.
- D. Confirm project compliance requirements and ensure safety training for all the new joiners.

Answer – D.

Explanation

The project manager has to manage Project compliance effectively. Since four newly hired experts will be working in activities that involve pure risk, safety training is mandatory before they start their project work. The Project Manager should confirm project compliance requirements, use methods to support compliance, and analyze the consequences of noncompliance before work gets allocated to these resources.

Here , Safety training is an example for Safety compliance.

PMI PMP Examination Content Outline – June 2019 – Domain III – Task 1.

Question 175:

Ryan is the newly joined Project Manager of a construction project who has joined the project near the end of planning. After reviewing the agreement and discussing the project scope with the team, he thinks that

the contract's timeline is unreasonable. What should he do to address this situation?

- A. Update the contingency reserve.
- B. Inform the sponsor to change the project charter with a new timeline.
- C. Schedule a meeting with the Subject Matter Expert and the sponsor to set realistic expectations.
- D. Review the resource management plan.

Answer – C.

Explanation

If the project timeline is not realistic, the project manager needs to provide proof to the sponsor to set realistic expectations. Such information can be provided by the SME (Subject Matter Expert), which may help the sponsor decide the course of action.

PMI PMP Examination Content Outline – June 2019 – Domain I – Task 8.

Question 176:

In Agile, Servant Leadership empowers the team. Which of the following characteristics of servant leadership enable project leaders to become more agile and facilitate the team's success?

- A. Helping people grow.
- B. Planning task execution.
- C. Negotiate project agreements.
- D. Promoting safety, respect trust

Answer - A, D

Explanation

The following characteristics of Servant Leadership enable project leaders to become more agile and facilitate the team's success: -

1. Promoting self-awareness.
2. Listening.
3. Serving those on the team.
4. Helping people grow.
5. Coaching Vs. controlling.

6. Promoting safety, respect, trust.

7. Promoting the energy and intelligence of others.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 34.

Question 177:

Match the following

SL no	Name	Description
1	Management.	a. Ability to be persuasive.
2	Leadership.	b. Reach consensus on project needs.
3	Influencing	c. Doing the things right.
4.	Negotiation.	d. Doing the right things.

A. 1-a, 2-c, 3-d, 4-b.

B. 1-c, 2-d, 3-a, 4-b

C. 1-d, 2-c, 3-a, 4-b.

D. 1-b, 2-a, 3-c, 4-d.

Answer – B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 64, 341, 350.

Question 178:

Arun is a Project Manager of an IT company. He is currently working on a middleware project that is following the traditional waterfall approach. A recent change in the software product version will change the project scope and timeline. The project needs to be completed in the next four months. What should the Project manager do?

A. Create a documented Change Request.

B. Close the project and request customer to start a new Agreement

C. Consult the issue with Sponsor and other stakeholders.

D. Add five more resources to the Project.

Answer – A.

Explanation

Formal, documented CR is the best course of action for the CR arising from change in Software version.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 117.

Question 179:

Retrospectives help the agile team learn from its previous work on the product and its process. Which of the following Agile principles covers this concept?

Select the best answer.

- A. Working Software is the primary measure of progress.
- B. Simplicity- the art of maximizing the amount of work not done is essential.
- C. At regular intervals; the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.
- D. Business people and developers must work together daily throughout the project.

Answer – C.

Explanation

Retrospective allows the team to learn, improve, and adapt its process. One of the principles behind the Agile Manifesto: “At regular intervals; the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.” Supports this concept.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9, 50, 98.

Question 180:

The Senior Leadership team of a manufacturing company has decided to provide product ownership training to all the domain experts of various departments. Whom should the senior management approach for such a service?

Select the best answer.

- A. Scrum Masters of various ongoing projects.
- B. Meet the domain experts directly and ask them to pursue Product Owner certification.
- C. Product Owners of successful Agile projects.
- D. Agile PMO.

Answer – D.

Explanation

An Agile PMO is multidisciplinary. They provide services like product owner training, coordination of Agile training courses, mentoring to help people transition to an Agile mindset and upgrade their skills.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 82.

Chapter 6: PMP Practice Test - 50 Questions-2021

Question 1:

Ryan is a Project Manager of a complex Cloud Infrastructure migration project of a leading manufacturing company. In the mid-stage, the company got merged with another manufacturing company. The company name and its policies got revised.

The project is ready to move to the Business realization phase, and the customer has formally accepted the project outcome.

What should the Project Manager do next to start the closure process?

- A. Refer to the Project Charter for the closing procedures.
- B. Examine the Scope baseline.
- C. Refer to the Organization Process Assets.
- D. There is no need to close the project as it's moving to the Business realization phase.

Answer – C.

Explanation

The question states that the project is ready to move to operations (Business Realization phase), which implies that the project manager should start the closure procedure. Since the company got merged in the recent past, there should be some changes in its OPA. On top of it, OPA is an input for the Close Project/phase process that contains project closure guidelines or requirements. Organization Process Assets include the corporate policies and procedures which the project manager should follow for formal project closure.

Therefore, among all options, option C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 126.

Question 2:

Ryan is a project manager of a fixed price project. The project team is currently engaged in a meeting with a few key stakeholders to finalize its

final scope approval. The Stakeholders discovered that some work packages are not required, and nor the tasks associated with these work packages are part of the contract. What went wrong in this scenario? Choose the best answer

- A. Requirement documentation was not properly created.
- B. Improper creation of WBS.
- C. The Project Scope statement was not prepared properly.
- D. The wrong development approach was chosen for the project

Answer- C.

Explanation

The Project Scope Statement includes the Project Scope, major deliverables, acceptance criteria and Project exclusions. As per the question, Project team is engaged in Scope baseline approval meeting, where stakeholders discovered that some of the out of scope requirements have been added in the Project scope. So, among all options given above, C is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 154.

Question 3:

The total estimated Project Cost is: -

- A. BAC.
- B. EAC.
- C. ETC.
- D. PMB.

Answer – B.

Explanation

BAC or PMB is the approved project budget that has been baselined during planning. So A and D are incorrect answers. ETC gives an approximate idea of how much money will be required to complete the remaining balance of work. So C is incorrect.

The estimated project budget is your latest cost estimate or forecast , represented by EAC (Estimate At Completion). So, B is the correct answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s)- 264,265.

Question 4:

In a project planning meeting, the project team decides to change the project management plan to get rid of a high probability risk.

What does this signify in relation to risk management?

- A. The Project Team adopted an Avoid Risk Response strategy.
- B. Project Team tried to enhance the success of the project by eliminating a known risk.
- C. Project team adopted a mitigation strategy to reduce the impact of a high probability risk.
- D. Project adopted a Transference strategy to transfer the risk.

Answer – A.

Explanation

The question implies that the project team is trying to get rid of a possible threat by changing the Project Management Plan, which is an example of Risk avoidance. People may get confused between option A and B. But option A looks like a better answer. Further, option B talks about a known risk, which is not correct. A risk can be Known-Unknown or Unknown-Unknown. So, among all options provided A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 443.

Question 5:

During planning, two team members disagree on adding one of the Solution requirements in the Project Scope Statement. According to one team member, if the need gets added, it will benefit the project later on. But, another team member says the addition of such a requirement would increase the ambiguity risks, so it's better to keep it in the project exclusion list. The Project Scope Statement needs to be finalized soon, so both the team members reach out to the Project Manager for an expert opinion. The Project manager said, "It's good to see that both of you have agreed on the majority of the contents of the Project Scope Statement, let's make it final as of now. In regard to this particular requirement, let's discuss with the sponsor post Phase 1".

Which of the following conflict-resolution techniques the Project

Manager applied over here?

- A. Smoothing.
- B. Compromising.
- C. Reconciling.
- D. Directing.

Answer – A.

Explanation

Smoothing, also known as accommodating, emphasizes areas of agreement rather than areas of difference. Here the Project Manager applied the smoothing technique.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 349.

Question 6:

A long-awaited Research and development project is on the verge of closure. Sponsor and Global Delivery Head of the buyer organization congratulate the Project Manager of the performing organization. The team asks the Project Manager to throw a glamorous party. Which of the following statements given below ensures that the contract can be considered complete?

- A. Formal acceptance of project deliverables.
- B. Final Acceptance of Project deliverables.
- C. A formal letter from the Customer stating the acceptance and completion of project work and the contract.
- D. Completion of Lessons learned.

Answer – C.

Explanation

This is a classic example of a tricky question where the hint is there in the last line of question. The question talks about Contract Closure, not Project Closure. So, Option B and D are not correct since those are performed during Closing. Option A talks about the Validate Scope process, where project deliverables are formally accepted, but that does not ensure closure of the contract. Contract closure is part of the Control Procurements process, where

the seller is provided with a formal notice that the Contract is complete. So, among all the provided options, option C can be the best answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 499.

Question 7:

Ryan is a project manager of a software development project which has gone behind schedule as 2 Testers have suddenly resigned, and they are unreachable. The Team decided to introduce Automated testing instead of manual testing for a set of tasks on a critical path. However, this decision would add two more servers in the project Infrastructure landscape, and the network may not have adequate capacity to handle the extra load of these two servers, which may impact the testing. What is the type of risk mentioned in the above scenario?

- A. Secondary Risk.
- B. Residual Risk.
- C. Ambiguity risk.
- D. Variability Risk.

Answer – A.

Explanation

A secondary risk arises as a direct result of implementing a risk response. Here as the question states, the Team has decided to implement a planned risk response of doing automated testing, and if that is done , it may create a new risk for the project. So, among all options , option A is the best answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 439.

Question 8:

The Project Manager of the Buyer organization is not convinced of the cost estimates provided by the prospective sellers for the company's upcoming construction project. After meeting with the sponsor, the project manager decides to award the contract to one of the preferred sellers with a clause that the seller has to provide all details of their legitimate direct costs incurred during the contract's life cycle for the final invoice payment. The agreement also states that the buyer organization will pay \$ 6,000 regardless of the invoiced cost.

What type of contract has been awarded in the above scenario?

- A. Time & Material Contract (T&M).
- B. Lump-Sum Contracts.
- C. Cost Plus Fixed Fee(CPFF).
- D. Cost Plus Incentive Fee(CPIF).

Answer - C.

Explanation

The question here talks about the CPFF contract, where the seller has to provide the actual invoices for payment. Also, they get a fixed fee as their profit amount.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 472.

Question 9:

Two developers are about to start implementing two different Approved change requests, as directed by the project manager.

What tool should assist the team members to know the list of authorized tasks they need to perform?

- A. Change log.
- B. Project Management Information System (PMIS).
- C. Work Performance Data.
- D. Work Breakdown Structure (WBS).

Answer – B.

Explanation

The question states that the project manager has asked two team members to perform the Direct and Manage Project Work process. Of the options provided, project management information system (PMIS) is the only tool or technique listed. The change log and WBS are inputs to the process, while work performance data is an output. The tool that actual team members are required to refer to here is the Work Authorization system which is actually a sub-system of PMIS.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 95.

Question 10:

A Project Manager who is new to an IT organization is managing a project that is running behind schedule. After talking with the team and reviewing all documents, he found that a critical resource working on a Critical path task had suddenly gone on vacation for personal reasons. A backup plan for onboarding a new resource is also available. What went wrong in the situation mentioned above?

- A. The risk response was not approved by the change control board.
- B. The Project Team failed to have a fallback plan.
- C. The issue has not been documented in the issue log.
- D. No action has been taken to implement the risk response.

Answer – D.

Explanation

The above situation suggests that the risk response was planned but never implemented. The project team already considered the above situation during planning and made the necessary resource replacement plan if such a situation triggers. But the new resource was never on-boarded. The common problem in risk management is that project teams may spend a lot of time and effort in identifying, analyzing, and documenting risks and risk responses, still then no action is taken to manage those risks.

So, here , among all options, option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 449-450.

Question 11:

The endeavor for keeping errors out of the process is also termed as: -

- A. Prevention.
- B. Inspection.
- C. Tolerances.
- D. Audit.

Answer – A.

Explanation

Prevention is keeping errors out of the process. Prevention over Inspection is

the biggest slogan of Modern Quality Management.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 274.

Question 12:

The Project Team has done a Root Cause Analysis of all defects detected during inspection. The Project Manager used a tool to organize and group the results of Root Cause Analysis that he will present in the next Senior Management review meeting.

Which of the following tools has been used by the Project Manager in the above scenario?

- A. Affinity Diagram.
- B. Cause and Effect Diagram.
- C. Mind Mapping.
- D. Histogram.

Answer- A.

Explanation

The scenario states that the Project Manager used an affinity diagram to organize and group the result of RCA.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 293.

Question 13:

A key stakeholder has submitted a Change Request, which will not impact the project schedule. What should the Project Manager do NEXT?

- A. Approve the Change Request and engage the Project Team to start the implementation procedure.
- B. Seek help from the Change Control Board.
- C. Ask Sponsor to authorize the Change Request.
- D. Evaluate the impacts of change on other project constraints.

Answer – D.

Explanation

The Project Manager should evaluate the other impacts of the project first. These include the impact on other project level constraints like scope, cost, quality, resource, etc. Once evaluated, CCB (if it exists) can be notified to take a final decision of the Change Request.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 117.

Question 14:

In a Go/No-Go decision meeting, the members of the senior management are evaluating two Projects i.e. Project X and Project Y. Project X has a net present value(NPV) of \$40,000 and Project Y has an NPV of \$25000. What is the opportunity cost If Project X is selected over Project Y?

- A. \$40,000.
- B. \$15,000.
- C. \$25,000.
- D. \$65,000.

Answer – C.

Explanation

The opportunity cost is the monetary value of the lost opportunity. In this case, since Project Y is not selected, the opportunity cost becomes \$25000. Refer :-PMP Exam Prep 9th edition 2018 – Rita Mulcahy.

Question 15:

An important stakeholder of a Health care project has raised a change request that needs to be implemented as early as possible.

Who has the ultimate authority to approve the change request?

- A. Functional Manager.
- B. Project Manager.
- C. CCB.
- D. Sponsor.

Answer – D.

Explanation

Every project has a sponsor. The projects have a CCB team when required,

not always. The approval authority for a Change Request is generally defined in the Change Management plan. Option C is a good option, but unless Sponsor is part of CCB, CCB decision cannot be marked as final. For all baseline level changes, Sponsor approval is mandatory even after CCB approval (if sponsor is not part of CCB). Furthermore, all projects do not have CCB.

Here in the question, ultimate is the keyword. All change requests raised in large projects require sponsor approval even if there is CCB in the project.so, among all options, option D is the best choice.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 115.

Question 16:

Ryan is a project manager of a construction project. The project team has identified some variability risks that need to be addressed to keep the project on track. Which of the following processes can be used to address variability risks?

- A. Perform Quantitative Risk Analysis.
- B. Perform Qualitative Risk Analysis.
- C. Perform Integrated Risk Review Analysis.
- D. Plan Non-event risks.

Answer – A.

Explanation

Variability risks can be addressed by using a simulation technique called Monte Carlo analysis which is performed as a part of the Perform Quantitative Risk Analysis process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 399, 433.

Question 17:

A new Project Manager has been appointed during the early stage of planning. In his first meeting with the sponsor, he understands the need for mandatory licensed software for the project. Which of the following documents should the Project Manager review to know whether the software has already been procured?

- A. Resource Management Plan.
- B. Project Charter.
- C. Assumption log.
- D. Physical resource assignments.

Answer – B.

Explanation

The question states that the Project has just entered the planning phase. So option D is incorrect as Physical resource assignments get created during execution.

The assumption log records the assumption and constraints available at the project level, so Option C is also wrong.

The Resource Management plan determines how resources would be acquired, and it does not contain information on available resources. Further, since the Project is at an early stage of planning, the Resource Management plan may not be drafted yet.

So, the answer is B by following the elimination technique. Project Charter contains Pre-approved financial resources.

Sometimes, Physical and human resources are determined in advance, even before the Resource Management plan gets prepared, also called Pre-assignment. Pre-assignment includes team members or physical resources already assigned during the Develop Project Charter process. So, among all the options provided, B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 81,333.

Question 18:

During execution in a Hybrid project, one key stakeholder requests the Project Manager for a change in a work package for which work has not started yet.

What should the Project Manager do NEXT?

- A. Ask the stakeholder to raise a formal change request.
- B. Update the work package and the Project Management Plan.
- C. Document the stakeholder request in the assumption log.
- D. Document the change request in the change log.

Answer – A.

Explanation

The scenario in the question states that the Project is in Execution, which means baselines are already created. Once baselines are created, formal change requests are required to establish any changes in the project control documents.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 115.

Question 19:

Ryan is a project manager of a Software Development project. The project is 60 percent complete, but there are still some open risks that need to be addressed.

Which tool would help the project manager to fetch the information about the reserves available in the current state?

- A. Alternative analysis.
- B. Technical performance analysis.
- C. Reserve analysis.
- D. Variance analysis.

Answer – C.

Explanation

Reserve analysis compares the amount of Contingency reserves remaining to the amount of risk remaining at any time in the project in order to determine if the remaining reserve is adequate.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 456.

Question 20:

Phase 1 of a manufacturing project is complete. During the planning stage of Phase 2, the Project Manager and the team, and two key stakeholders are trying to assess the type and quantities of raw materials necessary to perform and complete the project work. Which of the following tools and techniques cannot be used as a part of this process?

- A. Analogous estimating.

- B. Parametric estimating.
- C. Expert Judgment.
- D. Three-point estimation.

Answer- D.

Explanation

The question states that the project team is involved in the Estimate Activity Resources process. They are estimating the type and quantities of physical resources required to complete project work. Physical resources cannot be evaluated at the activity level, using three-point estimation.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 321.

Question 21:

During Execution, a newly joined team member wants to review the detailed description of the scope of the work assigned to him. Which document should team members review?

- A. Project Scope Statement
- B. WBS Dictionary.
- C. Requirement Documentation.
- D. Activity Lists.

Answer – B.

Explanation

Here in this scenario, the Team member wants to review the scope of work assigned to him, which means he has to review work package level information.

Some may get confused between options B and D. Activity Lists may relate to the work package, but they do not contain a detailed description of the work package.

The WBS dictionary is a document that provides detailed deliverables, activity, and scheduling information about each component in the WBS. It also contains the description of work at the work package level. Further, from a high-level perspective, the Activity List focuses on your scheduled activities (i.e., on the "tasks"). In contrast, the WBS Dictionary focuses on

describing your deliverables (the "scope").
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 162.

Question 22:

Arun is a Project Manager of a Telecom equipment manufacturing project. The project is risk-prone in some areas and running behind schedule. The project still has some unhandled open risks. The corresponding SPI and CPI values are 0.81 and 1.15. The Project Manager decides to bring the schedule back on track and complete the project in the next 45 days as one of the competitors has a plan to launch a similar product in the next two months.

What is the best course of action in the given scenario?

- A. The Project Manager should apply a crashing technique to bring the project back on track.
- B. The project manager should apply Fast tracking technique to bring the project back on track.
- C. The project manager should soon initiate risk response planning for all open risks.
- D. The Project Manager should reduce the project scope for early release.

Answer – A.

Explanation

The question states that the project is running behind schedule but within budget, and the project manager has to complete the project soon to remain competitive in the market. Since the project is already risk-prone, Fast-tracking is not advisable in the current situation. As per the case stated above, crashing is the best solution. Crashing is a schedule compression technique that shortens the project schedule without affecting the project scope.

Approving overtime, bringing in additional resources, and paying to expedite delivery activities are good examples of crashing. Further, as the project is still under budget, the crashing technique's application won't hamper the project budget to a great extent.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 215.

Question 23:

You are a project manager of a website development project. The project team has done an initial stakeholder analysis and grouped all stakeholders according to their level of authority and level of concern. Which of the following engagement strategies should be applicable for end-users who are responsible for handling day-to-day operational activities?

- A. Manage Closely.
- B. Keep Informed.
- C. Keep Satisfied.
- D. Monitor.

Explanation

Answer - B.

The question implies that the Project Team has categorized the identified stakeholders using the Power/Interest grid. Business operational end-users are those people who usually handle the day-to-day operational activities of a company. They may not be powerful but may have a high interest in the project, as they are the ones who will use the website when it gets developed. In the Power-Interest grid, they fall under - Low Power/High-Interest category. The project team's responsibility is to keep these people adequately informed and talk to them from time to time to ensure that no significant issues crop up. These people can often be beneficial, especially with the details of your intervention.

So, among all options provided, B is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 512.

Question 24:

The document that shows the organization's commitment to the project is ready. The Portfolio Governing body chairperson and other authorized representatives are having a meeting to discuss and decide the outcome of the document prepared. They are also reviewing the Cost-Benefit analysis, Feasibility study report, and other factors that justified the project's requirement.

Which of the following documents got prepared as per the scenario mentioned above?

- A. Program Charter.
- B. Business Case.
- C. Project Charter.
- D. Project Management Plan.

Answer - C.

Explanation

As per the scenario stated above, the Portfolio governing body is having a meeting in deciding the project charter. The Project Charter shows the organization's commitment to the project and provides a direct link between the project and the strategic objectives of the organization.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 75-78.

Question 25:

During Planning, the project management team decided not to change the project management plan for addressing one risk of low probability and high impact. Instead, the team plans additional reserves in case such a situation occurs. What response strategy has been planned in this scenario?

- A. Risk Avoidance.
- B. Passive risk Acceptance.
- C. Risk Mitigation.
- D. Active Risk Acceptance.

Answer – D.

Explanation

The scenario states that, Active Risk acceptance strategy has been adopted to handle the low probability and high impact risk. In this case, no response plan has been initiated, and a contingency reserve has been kept to handle the risk if it occurs.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443.

Question 26:

Ryan is the recently joined Project Manager of a complex software

development project. After receiving detailed knowledge transfer from the outgoing senior project manager, he has started conducting team meetings and actively got involved in maintaining documents like communication management plan, stakeholder register, and a few other necessary project documents. The outgoing project manager is closely observing Ryan's activities and, in time, helping him with suggestions. What Knowledge management technique has been followed in the above situation?

- A. Knowledge Transfer.
- B. Knowledge fairs and cafes.
- C. Reverse shadowing.
- D. Job shadowing.

Answer – C.

Explanation

In Job Shadowing, you follow an expert and gain knowledge from him or a group, whereas Reverse shadowing is a situation where an expert follows you and rectifies your way of work if required. The question states that the new Project Manager has started taking responsibilities under the close observation of the outgoing senior Project Manager, which is similar to the Reverse-KT situation. Knowledge fairs and cafés are a creative knowledge sharing session among a group of people of common interests, generally face-to-face.

So, among all options provided, C is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 103.

Question 27:

In a project, the Project performance appraisal gets carried out as a part of which of the following process groups?

- A. Closing.
- B. Executing.
- C. Monitoring and Controlling.
- D. Executing and Monitoring and Controlling.

Answer – B.

Explanation

In general, the Manage Team process of the Executing process group takes care of Project performance appraisal. In a Project Performance Appraisal team members get feedback from project work supervisors.

Question 28:

The customer has confirmed the formal acceptance of the delivered product. Before releasing the entire team, the Project Manager engaged the team to document the final lessons learned so that future projects can benefit from it.

What should the Project Manager do next?

- A. Review project documents for any open issues or risks.
- B. Organize a celebration party for the team.
- C. Archive all project information for future use by the organization.
- D. Ask customers for a formal contract closure letter.

Answer – C.

Explanation

The question states that the Project Manager is involved in the Close Project/Phase process.

Option D states about the Control Procurement process. So, it's not correct. The review of the project documents for open risks and issues happen before the Final lessons learned are prepared. Therefore, option A is also not correct. Option B may not be the best answer. The celebration party may happen at any time.

All project information is archived for its future uses. Further, the statement in previous examination content outline- "Archive project documents and materials using generally accepted practices in order to comply with statutory requirements and for potential use in future projects and audits" support the selection of option C as the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 123.

Question 29:

As a Project Manager of a mid-sized adaptive software development

project, you are using a tool to determine how much work is remaining to be done in the current iteration.

What tool has been referred over here?

- A. Burndown Chart.
- B. Burnup Chart.
- C. Why-why Diagram.
- D. Run Chart

Answer – A.

Explanation

The question states that the Project Manager is analyzing schedule control. Run Chart evaluates how a process is running, So D is not true. Why-why diagram is mainly a tool for RCA, so C is not applicable here. A burndown chart shows how much work is remaining to be done, whereas a burnup chart shows how much work has been completed, and the total amount of work. So, among all options provided, A is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 226.

Question 30:

Ryan is a Project Manager of a project that involves the renovation of a Chemical plant building. There is a lot of toxic and radioactive wastes inside the building. What is the best approach the Project Manager can take to transfer the risk?

- A. Take advice from the project sponsor.
- B. Ask customers to remove the toxic and radioactive wastes.
- C. Hire an outside organization to do the work.
- D. Ask Team to work in those areas of the building which don't have toxic wastes.

Answer – C.

Explanation

Transfer risk response strategy involves shifting ownership of a threat to a third party to manage the risk and bear the impact, if the threat occurs. The question states that the project manager has to find the best approach that

involves transference of pure risk that is currently part of the project. Among all the given options "Hiring an outside organization" to do the work that involves pure risk is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443.

Question 31:

Hierarchical charts act as an important data representation tool for planning resource management. Which of the following is not an example of Hierarchical charts?

- A. RACI Chart.
- B. Organization breakdown structure(OBS).
- C. Work breakdown structure (WBS).
- D. Resource breakdown structure.

Answer – A.

Explanation

B, C, D are examples of Hierarchical Charts where positions and relationships are shown in graphical, top-down format. RACI chart is an example RAM (Responsibility assignment matrix) and is not an example of Hierarchical charts.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 316,317.

Question 32:

As a Project Manager of the seller organization, you are working with the team to build an actionable plan to interact effectively with stakeholders.

What can help you to assess the political climate as well as stakeholder risk appetites of the buyer organization?

- A. Stakeholder Register.
- B. Enterprise environmental factors (EEF).
- C. Resource Management Plan.
- D. Risk management Plan.

Answer – B.

Explanation

The question implies that the Project Team is involved in the Plan Stakeholder Engagement process. All the provided options are input to this process. But, it's the EEF, that helps to determine the organizational culture, political climate, as well as stakeholder risk appetites.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 519.

Question 33:

Ryan is a Project Manager of a Software development project. During execution, a key stakeholder requests him for a scope change and asks about the correct change control process that will be followed throughout the project.

Which control document should the Project Manager refer to fetch this necessary information?

- A. Scope Management Plan.
- B. Configuration Management Plan.
- C. Change Management Plan.
- D. Quality Management Plan.

Answer – C.

Explanation

The Change Management plan provides the direction for managing the change control process and describes how the change requests throughout the project will be formally authorized and incorporated.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 88,116.

Question 34:

All of the following are input for Create WBS process except:

- A. Requirement Documentation.
- B. Project Scope Statement.
- C. Scope Management Plan.
- D. Scope baseline.

Answer – D.

Explanation

A, B, C are input to the Create WBS process, where D is the output of the Create WBS process. So, D is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 156.

Question 35:

Ryan is the Project Manager of a new software development project. He decides to meet another Project Manager who has handled a similar project in the past. But he finds out that the person he wants to meet has left the organization a month back.

What should Ryan do to gather the necessary information on the past project?

- A. Talk to the team members who worked in the past project.
- B. Refer to the Lessons learned document of the past project.
- C. Consult with sponsor and ask for necessary help.
- D. Talk to the customer of the past project

Answer – B.

Explanation

Lessons learned are collected throughout the Life Cycle of the project. During closing, the final Lessons learned is collected and transferred to the Lessons Learned repository so that future projects benefit out of it. So, here among all options provided, B is the best option in the current scenario.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 104, 128.

Question 36:

Arun is a directive PMO of an IT organization. A recent status report of various projects reveals that there are multiple projects where resources are over-allocated. Since over allocation is against company policy, he immediately advises all the related project managers to remove over allocation by applying the best-suited resource optimization technique even if the project schedule gets impacted.

What resource optimization technique is best suited in this scenario?

- A. Crashing.

- B. Fast tracking.
- C. Resource leveling.
- D. Resource smoothing.

Answer – C.

Explanation

Resource leveling is a resource optimization technique in which start and finish dates are adjusted based on resource constraints. This technique can change the critical path and acts as an effective aid to control over-allocation. Resource smoothing is also a resource optimization tool, but it doesn't change the project schedule, so D is not correct. Option A and B are examples of schedule compression techniques. So, among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 211.

Question 37:

Ryan is a newly joined Project Manager of a construction project which is running behind schedule. The project has some ongoing issues with a preferred vendor responsible for supplying cement for the project. He decides to facilitate a meeting between the Buyer and Seller management team to resolve all misunderstandings and create a win-win solution. What should he review to know the alternate dispute mechanisms applicable to the project?

- A. Communication Management Plan.
- B. Procurement Management Plan.
- C. Latest audit results of the procurement process.
- D. Agreements.

Answer – D.

Explanation

Alternative Dispute Resolution (ADR) is the procedure for settling disputes without litigation. At project level, ADR mechanisms are generally defined in the agreements in advance.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 489.

Question 38:

The Project team decides to change the project management plan to eliminate a high probability threat. Which process takes care of such actions?

- A. Monitor Risks.
- B. Plan Qualitative risk analysis.
- C. Plan Risk Responses.
- D. Perform Integrated Change Control.

Answer - C.

Explanation

The question talks about Risk avoidance strategy which can be applied for addressing high probability risks. Risk response planning is made during the Plan Risk Responses process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 437,443.

Question 39:

Statistical Sampling is a data gathering tool that involves choosing a part of a population of interest for inspection. The sample size and frequency of measurement of the tool are determined as a part of which of the following processes?

- A. Control Quality.
- B. Plan Quality Management
- C. Manage Quality.
- D. Plan Scope Management

Answer – B.

Explanation

Statistical Sampling is a tool that controls the quality process used to test a sample population for inspection. The size and the frequency measurements are determined during the Plan Quality Management Process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 303.

Question 40:

Ryan is a Project Manager of a research and development project which has a team in 4 different countries. The project uses a videoconferencing facility to address team meetings. Here videoconferencing is an example of:

- A. Interactive communication.
- B. Push communication.
- C. Pull Communication.
- D. Interpersonal communication.

Answer – A.

Explanation

Interactive communication happens between two or more parties for performing a multi-directional exchange of information in real time. Videoconferencing is a good example for Interactive communication. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 374.

Question 41:

Ryan is a Project Manager of a software development project that has four phases. Phase 1 of the project didn't go well. The customer did not accept 60% of the deliverables. In a retrospective meeting, the IT department director asks the project manager to furnish all the reasons that resulted in customer non-acceptance of the verified deliverables. What should the project manager review to know the reasons behind such non-acceptance?

- A. Verified deliverables of the control quality process.
- B. Work Performance Data of validate scope process.
- C. Requirements Traceability Matrix.
- D. Work Performance Information of the Validate Scope process.

Answer - D.

Explanation

The situation states that the customer did not accept 60% of the deliverables during the inspection. Work Performance Information, also an output for

Validate Scope process, includes information about project progress, such as which deliverables have been accepted and which have not been accepted, along with the reasons behind acceptance and non-acceptance.

So, among all options provided, D is the best answer.`

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 166.

Question 42:

Ryan is a Project Manager of a construction project. The Project Management team is currently engaged in a meeting to decide whether the work related to building two swimming pools in a posh residential complex area will be managed in-house or outsourced to a preferred seller.

In which of the following processes the project management team is currently engaged?

- A. Acquire Resources.
- B. Plan Procurement Management.
- C. Conduct Procurement.
- D. Plan Resource Management.

Answer – B.

Explanation

The situation states that the project management team is currently involved in the Plan Procurement Management process. The data analysis tool: - Make-or-buy analysis is being used in this scenario to come up with a Make-or-buy decision, one key output for the Plan Procurement Management process.

Make-or-buy analysis generally determines whether work will be accomplished by the team or would be outsourced.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 466, 473, 479.

Question 43:

You are a project manager of a software development project of a matrix organization. The project is currently in execution. The functional manager has decided to remove two essential resources from your project assigned to critical path tasks.

What should you do as a Project Manager?

- A. Escalate the situation to the sponsor.
- B. Overturn the decision of the Functional manager.
- C. Negotiate with the Functional manager and reach a consensus.
- D. Submit a change request to change the project schedule.

Answer - C.

Explanation

In Project Management, negotiation is a popular tool for reaching a consensus on project needs. Here the Project Manager needs to apply his interpersonal skill by negotiating with the Functional Manager and try to achieve a win-win result. Option B is incorrect because, in Matrix organization, the Functional Manager plays the resource manager's role. Option A or D can only be considered if option C fails.

So, among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 47, 341, 488, 527.

Question 44:

The CCB has to decide on a change request related to the scope change. The Change request evolved during the mid of the project due to an upcoming government regulation change. If the scope change gets implemented, 90% of the management reserve will get exhausted. What is the best decision the CCB can take in the given scenario?

- A. Approve the Change Request and advise the Project manager to add funds in Management reserve.
- B. Discuss the situation with the sponsor and ask him to make the final decision.
- C. Deny the Change Request since the project budget will be impacted.
- D. Defer the Change Request and wait until government regulation gets implemented.

Answer – B.

Explanation

Here in this situation, the scope change needs to be implemented as the CR evolved out of government regulation change. So, option C is wrong. The

Project Manager has no authority to change the Management reserve, so choice A is also not applicable. Choice D is also incorrect because it's never wise to wait for the government regulation to get implemented.

Project Sponsor is an individual with overall accountability of the project and is responsible for fund allocation and maintenance. Among all options provided, B is the best answer, as it's the Project Sponsor who can make the best decision here.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 120.

Question 45:

An Agile team works best in a Co-located environment. Which of the following are the goals for co-location? Choose the best two.

- A. Improved Team dynamics.
- B. Deliver finished value as an independent team.
- C. Better Communication.
- D. Small team, less than ten people.

Answer – A, C

Explanation

Colocation is one of the important attributes for successful Agile teams. The goals for colocation include: -

- Better Communication.
- Improved team dynamics.
- Knowledge sharing.
- Reduced cost of learning.
- Able to commit to working with each other.

Among all options, options A and C are the correct ones.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) –40.

Question 46:

Match the Following

SL no	Process Group	Process
1	Executing	a. Identify Stakeholders.
2	Initiating	b. Validate Scope.
3	Monitoring and Controlling	c. Develop Project Management Plan.
4	Planning	d. Develop Team.

- A. 1-b, 2-c, 3-a, 4-d.
- B. 1-d, 2-a, 3-b, 4-a.
- C. 1-c, 2-b, 3-a, 4-d.
- D. 1-d, 2-a, 3-b, 4-c.

Answer- D.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 25.

Question 47:

An Agile approach can reduce issues in an environment for a high degree of uncertainties. Which of the following are examples of pain points that Agile addresses under these circumstances? Select any two.

- A. Low defects.
- B. Technical Debt.
- C. Stakeholder engagement
- D. Unclear requirements.

Answer – B, D

Explanation

Project teams may experience many pain points when a project contains a high degree of uncertainty, change, and complexity. Agile approaches directly address these pain points by performing many troubleshooting

activities. Examples of pain points include unclear purpose, unclear requirements, technical debt, high defects, etc.
Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 58,59.

Question 48:

You are the new project manager for a non-profit organization project where most team members have mobility limitations or physical disabilities. What team model suits here?

- A. Co-located team model.
- B. Virtual Team model.
- C. Task Force model.
- D. Pre-assignment model.

Answer – B.

Explanation

Virtual teams are defined as groups of people with a shared goal who fulfill their roles with little or no time spent meeting face to face. The Virtual team model works well for people who work from home offices. The people with mobility limitations or disabilities who cannot travel can work from home and carry on their work.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 333.

Question 49:

Match the following

SL no	Type	Description/Example
1	OPA	a. Employee capability.
2	Hybrid	b. Organizational knowledge bases.
3	EEF	c. Organizational Structure Type.

- A. 1-a, 2-b, 3-c.
- B. 1-b, 2-c, 3-a.
- C. 1-c, 2-a, 3-b.
- D. 1-b, 2-a, 3-c.

Answer – B.

Explanation

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s)- 38,47.

Question 50:

Which of the following are characteristics of the Agile development team? Select any two.

- A. Responsible for guiding the direction of the product
- B. Are made up of T- shaped people.
- C. Accountable for creating the Sprint backlog.
- D. Are made up of I - shaped people.

Answer – B, C.

Explanation

The agile development team is cross-functional and self-organizing. The team comprises Generalizing Specialists or T-Shaped people and is accountable for preparing the iteration plan and iteration backlog. They are empowered to decide their tasks and have the best insight into task execution. No one tells the development team how to turn product backlog items into increments of potentially releasable functionality.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40-42.

Chapter 7: Brain Teasers - 25 Qs

Question 1:

Ryan is a Project Manager of a prestigious bridge construction project. Near the end of the execution, the project team realized a significant risk of completing the project in the available timeline. So a change request is raised to update the project schedule. Such an action is an example of which of the following risk response planning?

- A. Mitigation.
- B. Escalation.
- C. Acceptance.
- D. Avoidance.

Answer- D.

Explanation

Changing the Project Schedule means changing the Project Management Plan, an example of Risk Avoidance. Here as per the scenario stated in the question, there is a big threat of completing the project within the given timeline. The Project team tried to remove the effect of the threat by updating the schedule.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443,445.

Question 2:

In a Project that is following the hybrid approach, initially, all changes were going through by strictly following the integrated change control process. Near the end of execution, due to heavy workload, CCB is not able to clear all change requests in time. As a result of which team has started deciding on changes requested by stakeholders.

What should the Project Manager do over here to prevent the scope creep?

- A. Ensure that the team documents all Change Requests and takes sponsor approval before implementing such changes.
- B. Escalate against CCB to higher management

- C. Review the situation with the CCB.
- D. Review the situation with the PMO.

Answer – C.

Explanation

Here, the Project Manager needs to resolve the conflict. He has to work with the CCB team to know the reason for their heavy workload. A is not a good option, because giving full control to the team (developers, Testers, etc.) may not be a recommended option.

So, out of all options provided, reviewing the situations with CCB is the best answer.

Question 3:

Ryan is a Project Manager of a software development project which is in execution and running with many issues and threats. The Sponsor asks him to prepare the necessary documents to terminate the project. What should the project manager review to know the general terms and conditions and the termination clause?

(Choose the best answer)

- A. Business Case.
- B. Project Management Plan.
- C. Agreement.
- D. Project Charter.

Answer – C.

Explanation

The Termination clause and the General Terms and Conditions are available in the contractual agreement. The Project Manager has to review the agreement to proceed with the termination procedure.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 489.

Question 4:

Ryan is the CEO of an IT services company. Currently, the company has acquired another Software development company. In a steering committee meeting, Ryan asks his senior leadership team to plan and

**initiate the activities related to the merging of two organizations.
Merging two organizations is an example for:-**

- A. Business Development Plan.
- B. Project.
- C. Program.
- D. Business Realization plan.

Answer: B.

Explanation

The merging of two organizations is an example of a Project because the project is a temporary endeavor and has a start and end date.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 5.

Question 5:

In a Software development project with an adaptive life cycle, a vendor has completed the delivery that they are supposed to do for the current iteration. The Customer has been accepted for the deliverables with minor latency issues. The Company needs to close year-end books. What should the Project Manager do?

- A. Settle payment with written assurance from the vendor about resolving the latency issues.
- B. Take legal action against the vendor.
- C. Deduct amount equivalent to detect repair and pay that amount as service order.
- D. Keep the project open. delay the closure procedures.

Answer - A.

Explanation

Sometimes we need to think practically. Some may select option D thinking that the question talks about the Close Project / Phase process. But the question has nothing to do with the Closing phase, it just talks about "year-end book closure"; Minor latency issues can be addressed later too considering the fact year-end book closure is mandatory, and option A also states that written assurance has to be taken from the vendor to fix the latency

issues.

Among all options, the most practical answer is option A.

Question 6:

The Customer requests the project manager to share the latest work performance report in WhatsApp messenger as he is out on vacation. What should the project manager do next?

- A. Immediately share the report to the customer.
- B. Politely refuse to share the report in WhatsApp's messenger as it is unethical.
- C. Discuss the issue with team and senior management
- D. Review Organization Process Assets to know policies and procedures.

Answer – D.

Explanation

In this scenario, the project manager should first review the Organization Process Assets(OPA) to know the corporate policies and procedures for social media, ethics and security.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 383.

Question 7:

A Project manager of a large project is conducting a kick-off meeting to ensure that the stakeholders clearly understand project goals, objectives, benefits, and risks for the project, as well as their roles and responsibilities. The kick-off meeting in large projects can be held in support of which of the following processes?

- A. Develop Project Management Plan.
- B. Manage Team.
- C. Manage Stakeholder Engagement
- D. Monitor Communication.

Answer – C.

Explanation

In large projects, kick-off meetings take place with processes in the Executing Process Group. Kick-off meetings can be held in support of the Manage Stakeholder Engagement process to foster the appropriate level of stakeholder involvement in the project.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 86,528.

Question 8:

The Project team is involved in the preparation of a control chart and setting parameters for specification and control limits. The Control charts are established as a part of which of the following processes?

- A. Control Quality.
- B. Manage Quality.
- C. Plan Quality Management.
- D. Plan Scope Management

Answer - B.

Explanation

Many may get confused with A , but the Control charts are established in the Manage Quality process, and the parameters such as mean, specification limits, control limits are defined during the process.

During the Control Quality process , they are used to determine if the results of a process are within acceptable limits.

Rita Mulcahy 9th edition, RMC Publications, Inc.,2018, Page - 338.

Question 9:

The Project Charter can be approved by :

- A. Project Manager.
- B. A particular stakeholder who falls under the Outward bracket when classified based on direction of influence.
- C. A Particular stakeholder who falls under Upward bracket when classified based on direction of influence.
- D. CCB Team.

Answer - C.

Explanation

The Project Charter should be approved by the sponsor or an authorized representative who can take financial decisions as well as provide financial resources for the project. Since Project Sponsor falls under the Upward bracket when stakeholders are classified based on direction of influence, C seems to be the best answer among all options.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 77, 513

Question 10:

In a construction project, a thunderstorm damaged a lot of raw materials required for the project. The project manager instead of taking any preventive measures bought an insurance policy to handle this threat. Now, the Insurance company has refused to pay out the insured amount.

What could have been planned better in order to reduce the risk impact?

- A. Adoption of a mitigation plan to reduce the impact of risk.
- B. Escalation to the higher management when the risk was identified.
- C. Hiring another Insurance company.
- D. Keeping more funds in the project management reserve.

Answer - A.

Explanation

The project manager has to select the correct risk response planning. The Transfer strategy is used when the existing project team has no or little solution to tackle such risks. But in this scenario, the project manager should have planned a mitigation strategy to reduce the impact of the threat. Also, while buying insurance, it's the duty of the project manager to check its clauses; and on what grounds the insurance company may deny paying the insured amount.

Among all options, option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 443.

Question 11:

The Project Manager is currently involved in confirming the formal acceptance of the seller's work. Which process is getting executed in the

given scenario?

- A. Validate Scope.
- B. Close Project/Phase.
- C. Perform Integrated Change Control.
- D. Control Procurements.

Answer – B.

Explanation

The Confirmation of formal acceptance of sellers' work happens during closing (close Project/phase process).

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 123.

Question 12:

Ryan is the new project manager of a software development project who has joined during execution. The project is getting executed using an agile development approach where the scope of work gets progressively elaborated. He wants to review the remaining scope of work which is not decomposed to the lowest level but has known work content. What should he review to get this necessary information?

- A. Review all control accounts in WBS.
- B. Review the Project Management Plan.
- C. Review the Project Scope Statement
- D. Review the planning packages which are yet to be decomposed further.

Answer – D.

Explanation

A Planning package is a WBS component below the control account and above the work package with known work content, without detailed scheduled activities. In a project, where scope gets progressively elaborated, planning packages are decomposed later to work at package level when greater amounts of information and more accurate estimates become available.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Question 13:

A performing organization that was known for its on-time delivery is having problems in recent times. The team anticipated this as a risk and updated the risk in the risk register with a suitable response plan. As expected, the performing organization delivered 2 weeks late. What should be done first to handle this situation?

- A. Update risk register.
- B. Inform the stakeholders about the delay.
- C. Ask the sponsor for advice.
- D. Respond to the risk.

Answer – D.

Explanation

Here is in this situation, an identified risk has occurred, and the risk has become an issue. So, Implementation of previously agreed-upon risk response plan (Contingency Plan) is the best action required to be taken care of first. As per the question, since a risk response plan already exists, responding to the risk by implementing planned risk responses is the best choice. So, among all options, option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 449.

Question 14:

Ryan is a Project Manager of a large manufacturing project. Currently, the project is on the verge of completion of the initiation phase. What strategy in the current situation should be applied with the Unaware group of stakeholders of the project?

- A. Keep satisfied.
- B. Monitor.
- C. Keep informed.
- D. Manage Closely.

Answer – B.

Explanation

The question talks about the stakeholders, not the key stakeholders. The Unaware group of stakeholders falls under those who are unaware of the project and its potential impacts. Near the end initiation, the project team is not in a position to derive a complete picture of current versus desired level engagement of stakeholders as the planning phase is not initiated. The best solution is to keep these sets of stakeholders in Low Power –Low Interest in Power-Interest grid and monitor their movement and activities about the project. So, among all options, B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 512, 521.

Question 15:

The Project team is currently involved in the determination of the cost-effectiveness of the response strategy of a particular risk. Which of the following tools and techniques can help the project team to do the required analysis?

- A. Cost-benefit analysis.
- B. Alternatives analysis.
- C. Reserve analysis.
- D. What-if analysis.

Answer – A.

Explanation

As per the question, the project team is involved in the Plan risk responses process. So by adopting the elimination rule, option C and D can be ruled out. Cost-benefit analysis helps in determination of Cost effectiveness of the response strategy. The ratio of change in impact level divided by implementation cost gives cost effectiveness of the response strategy. Higher ratio indicates more effective response.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 446.

Question 16:

The CPI of your project has decreased to 0.87 due to constant deviation in exchange rates . What should the project manager do immediately?

- A. Qualify this as risk and document the concern in the risk register.
- B. Qualify this as risk and document the concern in the risk register.
- C. Request sponsor for additional fund.
- D. Raise a change request to change the contract type.

Answer – B.

Explanation

According to the question, in the current situation, the CPI of the project has decreased to 0.87 due to deviation in exchange rates, which means an unidentified risk has occurred over here and it's an issue now and the best practice for the Project Manager is to document the same in the issue log first then plan a workaround.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 458.

Question 17:

A severe storm has damaged a bridge that connects to Airport. The state government has decided to award the bridge reconstruction project to a renowned vendor to rebuild the bridge within a period of 6 months.

What contract type is most suitable in the given scenario?

- A. Fixed Price Incentive Fee (FPIF).
- B. Fixed Price Award Fee (FPAF).
- C. Time and Material (T&M).
- D. Cost Plus Incentive Fee (CPIF).

Answer – A.

Explanation

A Cost-Reimbursable contract is ideal when the scope of work is uncertain and expected to change. Here, as per the situation stated in the question, scope of work is pretty clear, so option D is not correct.

T&M is not generally suitable for construction projects; therefore, C is not the right choice.

Option B is a slightly better choice than C and D. But in the Fixed Price Award Fee contract, the buyer satisfaction is evaluated subjectively. Since, there is an urgency in completing the bridge construction work within a

period of 6 months, FPAF contract may not attract the vendor.

Fixed Price Incentive Fee (FPIF) contract would be most suitable in the given scenario, as it will give some amount of flexibility to both the buyer (state government in this case) and the seller. In the FPIF contract, financial incentives can be adjusted based on the seller meeting specified performance criteria, such as getting the work done faster (as in this case).

So, here among all choices, option A is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) –471.

Question 18:

Qnx company has delivered a business-critical project. The Customer has formally accepted the project outcome and praised the contribution of the Qnx company. Senior Management suggests the CEO of the company make a press release of the company's current project success story. The PMO Head asks the project manager to check the project level policies for information sharing.

What should the project manager refer to?

- A. Communication Management Plan.
- B. Lessons learned from a register of similar projects of the past.
- C. Project Charter
- D. Business Case.

Answer - A.

Explanation

The scenario in the question states that a business complex project has been formally accepted by the customer, which means the Validate scope process has been completed and the project has moved to the closing phase. Here, since the senior management has suggested making a press release of the current project success, the PMO head asks the project manager to check the methods or technologies that can be used to convey the project information. Here, Option D is ruled out as the business case defines the business need and cost benefit analysis that justify the project. Option C is also ruled out as the project charter states the project success criteria, signing authority for project closure. It doesn't reflect the communication methods to be followed in the project.

Option B is a good choice, but since every project is unique, it's not the best choice. It's the Communication Management Plan that provides communication guidelines for conveying project information. It also states the methods or technologies used to convey the project information, such as a press release.

So, among all options, option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 377.

Question 19:

Ryan is a project manager of a high priority Project of which budget is not a constraint. During planning, senior management advises him to come up with a detailed cost estimate for both planned and unplanned work and a report that will predict the project outcome based on the combined effect of risks. Which of the following is most essential for such a high priority project?

Choose the best answer

- A. Ensure risks are identified and qualified.
- B. Ensure risks are quantified.
- C. Ensure stakeholder register contains information for risk owners.
- D. Ensure risk responses are planned for all top risks.

Answer – B.

Explanation

Options A, C, D are applicable for projects of all priorities. They cannot be the best answers. One of the criteria for determining whether the Perform Quantitative Risk analysis needs to be done in a project is the priority of the project.

Question also states that the project manager has to furnish a report that will predict the project outcome based on combined effects on risks. This report is the Risk report which gets updated as a part of the Perform Quantitative Risk analysis process. Perform Quantitative Risk analysis process ensures risks are quantified and provides a numerical estimate of the overall effect of risk on the objectives of the project.

Among all options, option B is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Page(s) –429.

Question 20:

Ryan is the new project manager of a software development project who has joined during execution. After meeting and discussing with the team, he realizes that the project communication management plan is no more effective and requires a major update. Which of the following processes will issue a change request for the required update?

- A. Monitor Communication.
- B. Manage Communication.
- C. Manage Stakeholder Engagement
- D. Perform Integrated Change Control.

Answer – A

Explanation

As per the scenario stated in the question, the Monitor Communication process is being carried on. The Change request is a valid output for the process.

Option B is ruled out as the change request is not an output for the Manage Communication process. Though change request is an output for the Manage Stakeholder Engagement process, the change requests in the Manage Stakeholder Engagement process mainly results in changes to project or product scope. So, option C is also not the correct answer as per scenario stated in the question. The Perform Integrated Change Control process approves and manages a change request. So, option D is also ruled out.

Here in this case, the Monitor Communication will produce the change request as an output that will get approved as part of the Perform Integrated Change Control process. Then the project communication management plan process will make the necessary update in order to implement the approved change request.

So, among all options , Option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 389.

Question 21:

The project manager announced a grand celebration party for the team after the successful completion of a business-complex software development project. The Team also received a special appreciation note from the customer. Still, several team members are worried as they don't have clarity about their next assignment.

What type organization structure is mentioned in the scenario stated above?

- A. Hybrid.
- B. Functional.
- C. Projectized.
- D. Strong Matrix.

Answer – C.

Explanation

Here, as per the scenario stated in the question, the project has successfully moved to the Closing phase.

Option C is the best answer here, as in a projectized organization, the teams and resources are demobilized once the project is completed. Since such organizations run their revenue through execution of projects, employees may feel that they can be laid off once the project gets over.

Question 22:

A project risk for which the mitigation plan was done during planning seems to be outdated. The project team gathered in a risk reassessment meeting and modified the risk response plan for which additional funds will be required if the response plan needs to be implemented.

What reserve should be used to access this additional fund?

- A. Contingency reserve.
- B. Management reserve.
- C. Reserve from cost baseline.
- D. Reserve from CPA

Answer – B

Explanation

As per scenario stated in the question, an outdated risk response plan has

been reassessed, and the project is in execution. Risk Reassessment meetings happen during the Monitor Risks process. The implementation of the modified response plan will add more cost to the project budget. It means that, cost baseline will be impacted.

Option C and D are outright rejected.

Since Contingency reserve is a part of the Cost baseline, and it gets created before the baseline gets finalized, it is also not the correct option.

Here, in this situation, a new response plan has been created because the old one was outdated, and for implementing it, extra funds are required. If additional funds are required for implementing a risk response plan created post planning, management reserve needs to be used. For accessing management reserves, a formal change request should be in place.

So, among all options, option B is the best answer.

Question 23:

Ryan is the new project manager who joins the software development project during the early stage of planning. He wants to review the responsibilities and the authority level of him as a project manager.

What should he refer to?

- A. Resource Management Plan.
- B. Business Case.
- C. CPA
- D. Project Charter.

Answer D.

Explanation

As per the scenario stated in the question, the project has moved to the first iteration of planning.

Since the project is in the early stage of planning, the Resource Management Plan may not be created, therefore option A is ruled out. Option B is outright rejected, as the Business case states the Business justification of the project. It doesn't state the roles, responsibilities and authority level of the project manager.

Option C- OPA can be referred for deciding roles, responsibilities, authority level of the project manager, but it's not the best answer, as every project is unique.

Among all options, option D – Project Charter is the best answer. The project charter formally authorizes the existence of a project, and provides the project manager with the authority to apply organizational resources to project activities. The project manager is first assigned in the project charter. The project charter also defines the roles, responsibilities and the authority level of the project manager in a project.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 81.

Question 24:

During the execution of a construction project, the client requests to add a clause in the contract due to sudden regulatory change concerning the state-level wage policies.

What should the project manager do?

- A. Update the project management plan.
- B. Review the risk policies and existing risk response plan.
- C. Ask clients to follow the formal Change Request procedure.
- D. Ignore the client request as the contract has already been signed.

Answer - C

Explanation

Since compliance is mandatory, a change request needs to be submitted, and the contract has to be modified by following the Contract Change Control System.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) - 117,470

Question 25:

A Project Manager wants to review the technical requirements the seller is required to satisfy before he goes for inspection of the deliverables produced by the seller. What should Ryan review?

- A. Work Performance Data.
- B. Requirements Management Plan.
- C. Requirements Documentation.
- D. Quality reports.

Answer – C

Explanation

The situation in the question states that the project manager would perform inspection of the seller's work, which happens as a part of the Control procurements process.

All four options are input for the Control Procurements process.

The Work Performance Data reflects the project status such as activities in progress, completed, invoices that have been paid, etc. So option A is not correct.

The Requirements Management Plan describes how contractor requirements will be analyzed, documented, and managed. Therefore, option B is also not correct here.

The Quality reports help to identify seller processes, procedures that are out of compliance. So, option D is also incorrect.

It's the Requirements documentation, that include the intellectual property rights as well as the technical requirements the seller is required to satisfy.

So, among all options, option C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 495.

Chapter 8: Agile Questions - 50

Questions

Question 1:

In an Agile project, who is accountable for the end product that is built?

- A. Project Manager.
- B. Servant Leader.
- C. Product Owner.
- D. Development Team.

Answer – C.

Explanation

In an Agile project, the product owner defines the product vision and is responsible for maximizing the value of the product. Product owners are also ultimately responsible and accountable for the end product that is built.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41,153

Question 2:

In an agile project, the entire team is engaged in the retrospective meeting to analyze the areas of improvement of future releases of the current project. Retrospectives happen as a part of which of the following process groups?

- A. Executing.
- B. Monitoring and Controlling.
- C. Planning.
- D. Closing.

Answer – B

Explanation

Retrospectives help the team to learn from its previous work on the product and its processes and make improvements. It has following objectives: -
What went well?

What could have been better?

What's new that can be tried in future Sprints?

What do we need to escalate?

In typical Agile projects, retrospectives are usually the last ceremony done in a phase or iteration event and are generally part of the Monitoring and Controlling process group. Lots may get confused with option D, i.e. closing as lessons learned meetings which are slightly similar to Retrospective meetings happen during closing. But in closing, the objective of the lessons learned meeting is to finalize the lessons learned register and transfer them to lessons learned repository so that future projects benefit out of lessons learned in the current project.

Here the question states that the team is engaged in a retrospective meeting “to analyze the areas of improvement for future releases of the current project” which indicates such meetings are held as a part of the Monitoring and Controlling process group. In Agile projects, Retrospectives are also often conducted to check the effectiveness of the quality processes.

As per PMBOK, for agile projects, Retrospectives are conducted during Control Schedule, Control Quality, Monitor Stakeholder Engagement processes, and all are part of the Monitoring and Controlling process group. So, among all provided options, option B seems to be the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 25, 224, 276, 305, 535.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 50,51.

Question 3:

Ryan is a product owner of a software development project that is running using Agile-Scrum. Currently, he is engaged in a quick meeting with the team where team members are updating their task completion status and current risks and problems that are blocking their work. Such types of meeting are popularly known as: -

- A. Status Update meeting.
- B. Risk review meeting.
- C. Lessons Learned meeting.
- D. Daily stand-up meeting.

Answer – D.

Explanation

As per the question, the project is running using an agile approach. Meetings, or "ceremonies" are an important part of agile projects, and daily stand-ups are one of the examples to it.

The Daily Stand-up meeting happens every weekday and is held at the same time. This is not a Status Update Meeting. Ideally it should not continue more than 15 minutes.

In Daily stand-up meeting each team member should answer 3 questions: -

- a. What I was able to accomplish since last meeting.
- b. What I will try to accomplish by the next meeting.
- c. What is blocking me (risks and problems).

So among all options, Option D is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 364.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 53.

Question 4:

In a software development project, an agile team is currently demonstrating some added features of the Working Software prepared in the current iteration to key project stakeholders for formal acceptance of the increment. What agile ceremony is being conducted in the given scenario?

- A. Iteration review meeting.
- B. Daily Stand-up meeting.
- C. Status review meeting.
- D. Iteration Retrospective meeting.

Answer – A.

Explanation

In an Agile project, Working software is used to measure the progress of the completed work. After completion of work included in the current iteration, the potentially shippable product increment of the current iteration is demonstrated to the customer for approval and formal feedback of the delivered product. All these happen as a part of the Iteration review meeting. So, among all options, option A is the best answer.

Question 5:

In a software development project that is using the Agile development approach, the project team and the product owner are currently engaged in the determination of the number of iterations required to produce a finished product and the duration of each iteration based on the business goal, dependencies, and impediments. What process is being carried out in the given scenario?

- A. Control Schedule.
- B. Develop Schedule.
- C. Control Quality.
- D. Define Scope.

Answer – B

Explanation

As per the question, the Product owner and team are involved in Agile Release planning that happens as a part of the Develop Schedule process. Agile release planning provides a high-level summary timeline of the release schedule (typically 3 to 6 months) based on the product roadmap and the product vision for the product's evolution.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 216.

Question 6:

In a typical Agile project , who has the best insight into task execution?

- A. Project Manager.
- B. Product Owner.
- C. Team members.
- D. Scrum Master

Answer – C

Explanation

Agile projects are composed of self-organized teams, whereas the Project Manager or the Scrum Master acts as a facilitator. So, options A and D are

incorrect. The product owner ensures product vision and is responsible for managing the product backlog. Therefore, option B is also not correct.

So, based on the elimination rule, option C is the correct answer. Further, Agile follows the principle “The best architecture, requirements, and designs emerge from self-organizing teams.” In agile, as team members are closest to work, they have the best insight into task execution. So, among all options provided, option C looks to be the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 39,41.

Question 7:

Ryan is a Project Manager of a manufacturing company. The company has decided to adapt On-demand scheduling practices for one of their upcoming projects.

Which of the following Agile methodologies or frameworks use the on-demand scheduling approach?

- A. Scrum.
- B. DSDM.
- C. Large Scrum Scrum (LeSS).
- D. Kanban.

Answer – D.

Explanation

The On-Demand scheduling approach is typically used in the Kanban system. Kanban is a scheduling system for Lean manufacturing and Just In Time (JIT) manufacturing that mainly focuses on flexibility and continuous delivery. On-demand scheduling ensures that the rate of demand controls the rate of production, and Kanban is the best solution for this practice.

Among all options provided, option D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 177.

Question 8:

A sprint review meeting is occurring for a project that is following Agile-Scrum methodology. After being satisfied with the demo, the product owner requests an additional feature.

What should the Project team do next?

- A. Ask the Product owner to raise a formal change request.
- B. Add the request in the product backlog.
- C. Reject the change as the final release will happen in the next 2 weeks.
- D. Raise the concern with the sponsor.

Answer – B.

Explanation

The scenario states that a sprint review meeting is occurring for an Agile project. Option D is not relevant here. Option C is also incorrect because one of the fundamental Agile principles says: - “Welcome changing requirements, even late in development.” So, the Agile team welcomes change at any point of time in the project. Option A is applicable for projects using the Predictive approach; here, the project follows the Agile-scrum methodology. Therefore, option A is also not correct here. Based on the elimination rule, B is the right answer.

Further, the product backlog is the prioritized list of all the work items needed to build the product. The product backlog is a living and dynamic document that includes features to be made, functions, Functional and non-functional requirements, enhancements, defects, and bug fixes. As per the given situation stated in the question, a new feature request needs to add to the product backlog. The product owner himself is accountable for maintaining the product backlog. So, here, among all options, B is the right choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 131.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9,52,85,153.

PMI-ACP Exam prep – 2nd Edition, RMC Publications, 2015, Page(s) -42,43.

Question 9:

Your agile project has completed 2 releases. The sponsor asks you what new features would be introduced in Release - 4 and the timeline of the release. What needs to be referred to here?

- A. Working Software.
- B. Current Sprint backlog.
- C. Product roadmap.
- D. Product backlog.

Answer – C.

Explanation

The Product roadmap drives release plans. It shows the release dates and high level contents of the releases, so it will be the best deliverable for answering the question asked by the sponsor. Working software may be good for showing the sponsor what has already been built, but they are not targeted at communicating the release schedule for upcoming features, and neither are the product or sprint backlog. So, among all options provided, option C is the best choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 216.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 52.

Question 10:

The Project Team has decided to adopt the Agile development approach during early stages of planning. What is the primary goal for doing project work using agile?

- A. Speed.
- B. Correctness of the solution.
- C. Manage cost
- D. Providing Customer value via frequent deliveries and feedback.

Answer: D.

Explanation

Agile emphasizes on four golden values: - “Individuals and Interactions Over Process and tools”, “Working software over comprehensive documentation”, “Customer collaboration over contract negotiation”, Responding to change over following a plan”, and the primary goal behind adoption such life cycle approach is to generate customer value via frequent deliveries and feedback. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 18.

Question 11:

In a typical Agile project following scrum framework , who plays the role of a servant leader?

- A. Scrum Master.
- B. Product Owner.
- C. Team Lead.
- D. Sponsor.

Answer – A.

Explanation

Servant Leaders play the role of a facilitator in an Agile project. Their job is to ensure the process not the project. The main focus of a Servant Leader is to understand and address the needs of the development team in order to enable the highest possible team performance.

In an Agile project following the scrum framework, Scrum Master plays the role of a Servant leader whereas the Product owner is accountable for the product to be built.

Here, among all options , option A is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101,154.

Question 12:

Naveen is a directive PMO of an IT organization. As an Agile expert, he decides to promote osmotic communication across projects under his portfolio. Osmotic communication is best suited in which of the following team environments?

- A. Virtual teams.
- B. Co-located teams.
- C. Project Team in forming stage of Tuckman ladder.
- D. Matrix Team.

Answer – B.

Explanation

Osmotic communication means that information flows into the background hearing of members of the team, so that they pick up relevant information as

though by osmosis. This is normally accomplished in a co-located environment, where team members can accommodate themselves in the same room.

Osmotic communication requires teams who work in close proximity and are mostly involved in Face-to-face communication. So, option A is ruled out. Option D talks about a team which has 2 bosses, so option D is also incorrect. Option C talks of the first stage of team development, where the Project Leader adopts a directive leadership style and manages the team. At this stage, team cohesion is not yet developed and communication needs are also not clear to team members. So option C is also not the best choice. As co-located teams work in close proximity, which is also the key requirement for osmotic communication, option B is the best choice among all provided options.

The term has been coined by Alistair Cockburn, co-author of Agile Manifesto.

Cockburn, A. (2006). Agile Software Development: The Cooperative Game. (Page 111, 112).

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 340.

Question 13:

An Agile project using scrum methodology with relatively fixed scope is running behind schedule for the last 3 sprints. What may be the best possible reason for such a delay?

- A. Project is plagued with too many Change requests.
- B. The Project Manager didn't follow the RACI chart while doing task allocation.
- C. A few team members are not 100% allocated to the project
- D. Development team is in the Forming stage of the Tuckman ladder.

Answer – C.

Explanation

Option A and B are applicable for projects running with a predictive development approach, so both the options are ruled out. Option D may be one of the reasons, but the question has not produced enough information that the Team is in the Forming stage. Further, the question talks about the last 3

sprints, that doesn't indicate that the team is at the forming stage. Option C seems the best choice among all options. The key requirement for any agile projects are 100 percent dedicated team members for increased focus and productivity. Partial allocation may lead to situations like multitask and task switching. Multitasking slows the team progress and thereby reduces the throughput. People experience productivity losses during task switching, as it consumes working memory and people are less likely to remember the context when they multitask. When team members are 100% dedicated to the team, the team has the fastest possible throughput.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 44-45.

Question 14:

In a typical agile project, Product backlog is the repository for product scope. Who owns the product backlog?

- A. Project Manager.
- B. Sponsor.
- C. Development Team.
- D. Product owner.

Answer – D.

Explanation

Product owner ensures the product vision and is accountable for managing the product backlog.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40,41.

Question 15:

Team members are not happy as the product owner often sets up informal meetings for project status updates, which is impacting their scheduled plan to meet the iteration goal.

What can be a Servant Leader's best course action to manage the given situation?

- A. Advise the product owner to take part in daily stand-up meetings.
- B. Update team charter.
- C. Give coaching to the team to handle such unplanned necessary

meetings.

D. Escalate the issue to the higher management.

Answer – A.

Explanation

Fourth Agile Principle states that “Business people and the developers work together daily throughout the project” and daily stand-up is one of ceremonies for Agile projects across methodologies that happens on a daily basis. Here as per the scenario stated in the question, the Servant leader should advise the Product owner to attend daily stand-up meetings which will help him to remain updated of the current state of the project on a daily basis. On the other hand, the team will also be happy as it will save their time and they can concentrate on their current iteration goal.

Among all options, option A seems to be the best collaborative approach that the Servant Leader can take for win-win consensus.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 53,54.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 535.

Question 16:

Scrum is a popular agile framework. Which of the following represents a scrum artifact?

- A. Sprint.
- B. Sprint backlog.
- C. Scrum backlog.
- D. Sprint planning.

Answer – B.

Explanation

Options A and D are examples for Scrum events, so both the options are incorrect. Option C, i.e. “Scrum backlog” is a made -up term, so C is also ruled out. Option B – Sprint backlog is an example for one of the three scrum artifacts. Other two Scrum artifacts include – product backlog, increments. Therefore, among all options, B is the right answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) –

101.

Question 17:

An agile project team is working to manage quality for their project. What should the team primarily focus on considering quality management?

- A. Concentrating on small batch systems.
- B. Doing Root Cause Analysis periodically.
- C. Determining Quality metrics.
- D. Conducting frequent benefit analysis.

Answer – A.

Explanation

Agile methods involve frequent quality review steps that are built into the project. They focus on small batch systems in order to uncover the inconsistencies and the quality issues as early in the project as possible when the overall life cycle costs are low. Recurring retrospectives are conducted to check the effectiveness of the quality processes, so that improvements can be made when they are needed.

Here , among all options, Option A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 276.

Question 18:

Which of the following Agile principles helps to promote elimination of wastes there by minimizing scope creep and encouraging to do only those categories of work that generate business value?

- A. Working Software is the measure of progress.
- B. Continuous to technical excellence and good design enhances agility.
- C. Simplicity- the art of maximizing the amount of work not done — is essential.
- D. Business people and developers must work together daily throughout the project.

Answer – C.

Explanation

Maximizing the amount of work not done means maximizing the amount of unnecessary work not done. Agile methods stress the importance of prioritizing all work, including changes, based on business value. The expectation of the entire team is that unless there is business value, the work will not get done. Therefore, work that on many traditional projects was completed without question will not be completed on an Agile project, because it doesn't make business sense to do so. Maximizing the work not done helps minimize scope creep.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9
Refer:- <https://www.pmi.org/learning/library/determine-organization-agile-scrum-ready-6129>

Question 19:

In a Sprint review meeting, the product owner dismisses the sprint and recommends the newly appointed Scrum Master to arrange Agile coaching for the Team. The Team worked hard to meet the sprint goal and came up with a detailed PowerPoint presentation for the sprint review meeting. What went wrong in the situation mentioned above?

- A. Nothing. So-urn master should apply his influencing skill to convince the product owner to reconsider his decision.
- B. Presentation made by the team doesn't not satisfy the Sprint goal.
- C. Team should have included test cases and defect logs in their presentation.
- D. Team should have presented working software to gain formal acceptance of the increment.

Answer – D.

Explanation

One of the core Agile principles states “Working Software is the primary measure of progress”. In any Iteration review meeting, the team presents the working software or working system to gain customer (in this scenario – the product owner) acceptance. By adopting this practice, focus is shifted to working results rather than documentation and design. In agile, we measure progress based on the emerging product or service we are creating. In this scenario, the team didn't follow the mandatory Agile practice and the

product owner rightly dismissed the sprint and advised the Scrum master to plan coaching for the team.

Therefore, as per the situation stated in the question, option D, seems to be the best choice among all options.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9.
PMI-ACP Exam prep – 2nd Edition, RMC Publications, 2015, Page(s) -34.

Question 20:

Ryan is a Project Manager of a software development project. While preparing the Project Management Plan, he decides to use the Agile development approach instead of the traditional and commonly used waterfall approach.

What has influenced him to decide agile as a development approach?

- A. Organization Process Assets (OPA).
- B. Enterprise Environmental Factors. (EEF).
- C. Life Cycle and Development approach.
- D. Change Management Plan.

Answer – B.

Explanation

As per the scenario stated in the question, the project manager is involved in the Develop Project Management Plan process for preparing an Integrated Project Management Plan.

Option C and D are ruled out as they are additional components of Project Management Plan, which are generated as output for the Develop Project Management Plan process.

Both Organization Process Assets and Enterprise Environmental Factors are valid inputs to the process and help in the preparation of the Project Management Plan.

While OPAs influence the process with tailoring and closure guidelines for the project along with policies, processes, procedures (e.g. Change Control procedure) that needs to be taken care while developing the plan, it's the Enterprise Environmental Factors. (EEF) which can help in the selection process of the most suitable development approach. Project Management Body of Knowledge for the focus area helps in the determination of Agile software development.

Further , as per the scenario stated in the question, Ryan's organization has predominantly followed the traditional waterfall approach, so chances for having historical data for other development approaches are minimal. So, referring to OPA, as in this case for selection of Agile as a Development approach may not add much value.

So, as per the above question, among all options, option B is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 84.

Question 21:

In an Agile Software development project, a tester is writing a test before work related to coding has begun. Which of the following technical practices is mentioned in the given scenario?

- A. Test-Driven Development.
- B. Sprint Review.
- C. Feature-Driven Development.
- D. Kanban.

Answer – A.

Explanation

Here as per scenario stated in the question, option B is ruled out as it's an example for a scrum event. Option C and D are examples for Agile methodology. So, by applying elimination technique, A is the correct answer. A Test-Driven Development is a technique where tests are defined before work starts, so that work in progress is validated continuously.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 56,154.

Question 22:

Arun is a Servant leader of a complex software development project that is running using an Agile development approach. Two developers are having mixed opinions on a user story.

What Arun needs to do here?

- A. Arun should assess the level of conflict and intervene appropriately.
- B. Consult the product owner.

- C. Entire team should gather to discuss the issue and come up with a collective solution.
- D. Arun should ask for further clarifications from both the developers.

Answer - C.

Explanation

On an empowered agile team, it's up to the team members to resolve their technical disputes collectively. The scrum master, or the product owner is unlikely to have the knowledge required to make such technical decisions. So, among all options, option C is the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41-43, 61.

Question 23:

The project team is using an Iteration burn-down chart to predict the remaining work in the current iteration backlog. Iteration Burn-down chart is an example of which of the following tools and techniques?

- A. Data gathering.
- B. Data analysis.
- C. Data representation.
- D. Decision making.

Answer – B.

Explanation

The burn-down chart is an example of a data analysis tool that shows the total effort against the amount of work for each iteration.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 226,687.

Question 24:

Which of the following is not an example of the four values of the Agile Manifesto?

- A. Individuals and interactions over processes and tools.
- B. Working software is the primary measure of progress.
- C. Responding to change over following a plan.

D. Customer collaboration over contract negotiation.

Answer – B.

Explanation

The options - A, C, D fall under four values of the Agile manifesto. Option B is an example for one of the twelve Agile principles behind the Agile manifesto. So, among all options, option B is the right answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 8,9.

Question 25:

Ryan is the newly appointed Project Manager of a software development project which has a high degree of uncertainties and scope is not yet fully defined. Considering all the factors, he decides to adopt an Agile development approach for running the project. Now, he has to come up with an initial cost estimate before the project work begins.

What can be the best approach for the preparation of an initial cost estimate as per the stated scenario?

- A. Using Bottom-up estimation technique to come up with a detailed cost estimate.
- B. Using Analogous estimating technique to come up with a high level cost estimate.
- C. Doing reserve and technical performance analysis to come up with an initial cost estimate.
- D. Using light-weight estimation methods to generate a high-level cost forecast.

Answer – D.

Explanation

Projects with a high degree of uncertainty or if the scope is not fully defined may not benefit from detailed cost calculations because there are frequent changes. In an agile environment, lightweight estimates can be used to generate a high-level cost of project labor costs, which can be adjusted when changes are made. A detailed estimate is then used for a short-term planning horizon.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,

Question 26:

Ryan is a project manager of a large software product development project that needs to be executed from five different cities. Senior management has decided to run the project using an agile approach by following the philosophies of the scrum as much as possible.

What framework is most suited as per the given situation?

- A. Scrum.
- B. Kanban.
- C. LeSS
- D. Feature-Driven Development

Answer – C.

Explanation

The situation in the question states about a very large project that needs to be executed using an agile approach by following philosophies of scrum.

Considering the fact that a very large project needs a large team too, the project manager has to break the large team into multiple smaller teams and use program management to synchronize and coordinate.

Option A and B can be ruled out as such frameworks are applicable for relatively smaller teams. The situations required for Feature-Driven development is not stated in the question. It's generally used to meet the specific needs of software development projects. So, option D is also ruled out.

LeSS (Large Scale Scrum) is a framework that extends scrum with scaling guidelines while preserving the original purposes of scrum. It maintains many of the practices and ideas of one-team Scrum like a single product owner and one product backlog, one Definition of Done for all teams, one Potentially Shippable Product Increment at the end of each Sprint. It's most applicable for many teams (cross functional) working together on one product.

So, among all options, option C is the best choice.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 113,121,152

Question 27:

The ongoing sprint for a software development project that is being executed by following Agile –Scrum needs to be canceled as the sprint goal has become obsolete due to a sudden change in law associated with IT security. Who has authority to cancel the sprint?

- A. Project Manager.
- B. Sponsor.
- C. Product Owner.
- D. Scrum Master.

Answer – C

Explanation

The situation in the question states that an ongoing sprint of a software development project needs to be canceled due to changes in security law. Here, options A, and B are ruled out as the Scrum team consists of Product owner, Scrum master, the development team.

Option D is also incorrect as the scrum master ensures the process, not the project.

The product owner is responsible for maximizing the value of the product, and only he has authority to cancel the sprint, although he or she may do under the influence from key stakeholders, sponsor, project manager or any other stakeholders A sprint gets canceled when sprint goal becomes obsolete due to sudden change associated to organizational policies, external environmental factors or due to any adverse situations impacting the project. Here, among all options, option C is the best one.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf

Question 28:

In a typical business complex and mid-sized agile project, what does the team attempt to develop every Sprint?

- A. Product backlog.
- B. Source code ready to be handed off to QC for testing.
- C. Final working software

D. A Potentially Releasable Product Increment

Answer - D

Explanation

The output of every Sprint is called a Potentially Releasable Product Increment. It is the sum of the Product Backlog Items completed during a Sprint. Delivering Potentially Releasable Product each Sprint is fundamental to the Scrum, since in agile as work is divided into simple pieces, it can be finished in a short iteration.

The product backlog is a living document that has anything and everything in relation to project work. It is the ordered list of all the work presented in story form for a team. It is the single source of requirements for any changes to be made to the product. The Product owner is responsible for the product backlog. Therefore, option A is not correct.

Agile development team is composed of a cross-functional team (where developers, testers, QA, QC, etc.) work together to deliver an increment. So, option B makes no sense.

In Agile, "Working software is the primary measure of progress". Final working software is produced in the last sprint, not in every sprint, as asked in the question. Therefore, option C is also ruled out.

As per situation stated in the question, among all options, option D is the correct choice.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

www.scrumguides.org/docs/scrumguide/v2017/2017-Scrum-Guide-US.pdf

Question 29:

During the last sprint of a software development project that is being executed by following the Scrum framework, a key stakeholder requests a change that may improve the performance of the final product but may also affect the planned schedule. What should the product owner do?

- A. Reject the change as it's out of project scope.
- B. Analyze the new requirement with the team and check options for refactoring.

- C. Issue a formal change request.
- D. Escalate the situation to the senior management

Answer – B.

Explanation

Option C is not correct as Change requests are applicable for projects following waterfall or predictive approach, and not applicable for Agile projects. Option A is not recommended, as the Agile projects “welcome changing requirements, even late in development”. Since, the product owner is the sole person responsible for maintaining the product backlog, option D is generally not required.

As per the scenario stated in the question, the stakeholder has requested a change (non – functional requirement) which may cause performance improvement. Analyzing the request with the team and identifying options for refactoring is the best choice among all options.

So, among all choices, option B is the best fit.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9, 153.

<https://Agilemanifesto.org>.

Question 30:

You are a project manager of a software development project that is following the Agile – Scrum framework. The product owner and the development team have finalized a clear sprint goal for the upcoming sprint. Who determines how the scrum team will accomplish the goal?

- A. Product owner.
- B. Scrum Master.
- C. The development team.
- D. Project Manager.

Answer – C.

Explanation

In agile projects, the development teams are self-organized and cross-functional. They are empowered to organize and manage their own work. No one tells the development team how to turn product backlog into increments of potentially releasable functionality. Cross-functional teams work together

to accomplish the sprint goal.

As per situation stated in the question, among all options, option C is the correct choice.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40-41.

www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf

Question 31:

Match the following

SL no	Scrum Team	Role
1	Scrum Master	a. Accountable for maximizing the value of the product.
2	Product Owner	b. Responsible for delivering potential releasable product on a regular cadence.
3	Developers	c. Accountable for Scrum team's effectiveness.

- A. 1-c, 2-b, 3-a.
- B. 1-c, 2-a, 3-b.
- C. 1-b, 2-c, 3-b.
- D. 1-a, 2-c, 3-b.

Answer - B

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 151.

2020 –Scrum Guide – Page 5-6

Question 32:

An Agile team is currently engaged in a Sprint planning meeting. What is the outcome of a Sprint Planning meeting?

- A. Scrum backlog.
- B. Sprint backlog.
- C. Product backlog.
- D. User story.

Answer – B

Explanation

Sprint planning is an event in a scrum that kicks off the Sprint. Sprint planning aims to define what the Sprint can deliver and how the team will achieve the work. The sprint backlog is a Scrum artifact and the outcome of the Sprint planning. The Sprint Backlog is composed of: -

- a. The Sprint Goal (why)
- b. The set of Product Backlog items selected for the Sprint (what).
- c. The actionable plan for delivering the Increment (how).

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

Question 33:

Ryan is a Project Manager of a Software development project which is using the Agile-Scrum framework. Which of the following are examples of Scrum events? Choose the best two.

- A. Sprint Planning.
- B. Increments.
- C. Sprint review.
- D. Product backlog.

Answer - A and C.

Explanation

Increments and Product backlog are examples of Scrum Artifacts. Therefore, options B and D are ruled out.

Sprint planning and the Sprint review are examples of Scrum events. Other examples include-: Sprint, Daily Scrum, Sprint retrospective.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 101.

Question 34:

You are a Project Manager of a project where the customer expects a high frequency of delivery in short intervals. The project scope will be emerging gradually, and there is a high degree of complexity in the project. What Life Cycle approach is best suited here?

- A. Iterative.
- B. Predictive with phased approach.
- C. Agile.
- D. Incremental

Answer – C

Explanation

A Project with a high degree of complexity in the project may have a lot of requirements change. Agile is suitable for a project that has BOTH a high degree of change AND a high frequency of delivery to the customer.

Option A -Incorrect- Iterative - Iterative approach is best for a project that has a high degree of change (but a low frequency of delivery to the customer). So, option A is not the best choice.

Option - B -Incorrect- Predictive with Phased approach - Since the project has a high degree of complexity, this is not the best choice.

Option C - Correct - Agile - Agile projects are both iterative and incremental. This is the best choice here as the project requires a high frequency of delivery and a high degree of change.

Option D - Incorrect - Incremental - Incremental approach is best for a project that has a high frequency of delivery with a low degree of change. So, option D is not the best choice here.

So, here, among all options, option C is the best choice.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 19.

Question 35:

An Agile team is currently estimating the tasks for the current sprint. Which of the following Scrum Ceremonies teams is currently engaged?

- A. Sprint Review.
- B. Sprint Planning.
- C. Sprint Retrospective.
- D. Backlog grooming.

Answer – B.

Explanation

The purpose of the Sprint Planning is to estimate and forecast the work that can be accomplished by the team in the given Sprint. It generally happens on

the first day of the Sprint. In Sprint planning, the team estimates the size and complexity of the PBIs. Typically, the team would estimate only a few top PBIs for meeting the Sprint Goal.

Question 36:

The Project team has committed to deliver 14 story points. Only 10 story points can be covered in the current iteration. What should the team do here?

- A. Extend the Iteration.
- B. Apply crashing technique.
- C. Complete 10 story points, move rest to the product backlog.
- D. Consult the situation with the CCB.

Answer - C

Explanation

Since all iterations are timeboxed, the duration won't change. Generally, the iteration plan is not changed, and the resources are not added. Instead, all incomplete works are returned to the backlog.

Therefore, among all options, Option C is the best choice.

Question 37:

A day before a Sprint review meeting, the Project Manager found that the team didn't test user stories properly. What could be the reason behind it?

- A. Incomplete Definition of Done.
- B. Ambiguous Product Scope.
- C. Poor Project Management
- D. Lack of coordination between Team and the Project Manager.

Answer – A.

Explanation

Definition of Done (DOD) - Build the thing right (Deliverables).

The Definition of Done consists of the tasks, types, levels of testing, and all other criteria that the team thinks are needed for the completion of a user story. The user story should be tested against all these criteria before

announcing it finished and delivering it to the customer. The insufficient testing of the user stories indicates the incomplete definition of done. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 151.

Question 38:

The Agile team is currently engaged in a retrospective meeting, where they are trying to find out the root cause of a problem. What technique can the team use here?

- A. Kano analysis.
- B. Expert Judgment.
- C. Voting.
- D. Five Whys.

Answer – D.

Explanation

The Five Whys is a team exercise used to get to the root cause of a given problem. Here, among all options provided, option D is the best answer.

Question 39:

Which one of the following are true about the Definition of Done? Choose any two.

- A. It reflects the entire Product Scope.
- B. An increment is born when a Product Backlog item meets the Definition of Done.
- C. The Definition of Done is the product acceptance criteria.
- D. The Definition of Done is a formal description of the state of the increment when it meets the quality measures required for the product

Answer – B, D

Explanation

Definition of Done (DOD) - Build the thing right (Deliverables).

Option A is outright rejected. Product scope refers to the features or functions that characterize a product. Definition of Done is a formal description of the state of the increment when it meets the quality measures required for the

product.

Option C is also not correct. Acceptance Criteria means - building the right thing. It reflects functionality. The Definition of Done is building the thing right. It reflects the deliverables.

Among all choices, options B and D are the best answers.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 151.

www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page 12.

Question 40:

Your company has hired a vendor to work on an Agile project with six user stories for which the scope is well defined. What type of contractual agreement is most suitable here?

- A. FP-EPA.
- B. Have a Fixed Price contract with change for free clause.
- C. Have a separate Fixed price contract for each of the User Stories.
- D. Have a T&M Contract for a period of one year.

Answer – C.

Explanation

A collaborative approach needs to be adopted while deciding on an Agile contract where a shared risk-reward relationship exists between the buyer and the seller. Rather than having one Single agreement for the entire project scope, six fixed-price contracts for each User Stories can be in place. This strategy limits financial risk for the seller, and for the customer, it gives more control over money flow.

Among all options, option C seems to be the best answer.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 77.

Question 41:

A day before the Iteration review meeting, a defect was discovered. The Project Manager engaged the entire team to resolve the issue. Such a technique in which multiple team members focus on resolving a specific impediment is popularly called: -

- A. Task Switching.
- B. Grouping.
- C. Anti-pattern.
- D. Swarming.

Answer – D.

Explanation

Here the team adopted the Swarming technique. It's a technique in which multiple team members focus collectively in resolving a specific impediment. Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 154.

Question 42:

A Senior Manager instructs the development team to add a work item in the current sprint. What should the development team do in this situation?

- A. Initiate another Sprint Planning meeting to add the piece of work in the current Sprint Goal.
- B. Ask the Senior Manager to raise a formal change request
- C. Inform the Product Owner so that the Product Owner can negotiate with the Senior Manager.
- D. Add the work item in the Sprint Backlog so that this work can be done in the next Sprint

Answer – C.

Explanation

Option B is applicable for projects following a predictive Life Cycle approach. So, option B is ruled out. Sprint goals are generally not changed during the middle of the Sprint. So, option A is also not correct. Sprint backlog is prepared for the current Sprint, not for a future sprint. Therefore, option D is also not a valid choice.

The Product Owner is responsible for maximizing the value of the product resulting from the work of the Development Team. The Product Owner is the sole person responsible for managing the Product Backlog. No one can force the Development Team to work in a different set of requirements, as the team is empowered to decide their tasks. So, informing the Product Owner so that

he/she can work with the Senior Manager is the best choice among all options.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40,41.

Question 43:

In a development project, the team could not achieve the Sprint Goal as two key developers were working on multiple projects. Which of the categories of waste does this fall?

- A. Task Switching.
- B. Mobbing.
- C. Follow-the-Sun development process.
- D. Fractional allocation.

Answer – A.

Explanation

Being assigned to different projects is an example of task switching, which creates waste.

In general, task switching productivity loses somewhere between 20% and 40%.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 44,45.

Question 44:

In the middle of execution, a new Scrum Master joins a software development project following Agile Scrum. He wants to know the scope of the project, and the work completed so far. What should he do first?

- A. Review the Project Scope Statement
- B. Review the Burndown chart.
- C. Review the Burnup chart
- D. Schedule a meeting with the team to understand the Project Scope and work completed.

Answer – C.

Explanation

Burnup charts show the work completed.

Option A – Incorrect – In general, the Project Scope statements are not created for Agile-Scrum projects. The contents of a typical Project Scope Statement reside in the Product Backlog.

Option B – Incorrect – Burndown chart is the graphical representation of the work remaining versus the time left in the timebox.

Option C – Correct - A Burnup Chart is a tool used to track how much work has been completed, and show the total amount of work for a project or iteration.

Option D – Incorrect – Meeting with the team is also a good option. But it's better for the Scrum Master to review the Burnup chart first before meeting the team to have a clear understanding of project work completion.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 63,150.

Question 45:

A Product Owner is not happy because the Scrum team's velocity is comparatively lower than the other Scrum project teams in the organization. How should the Scrum Master react?

- A. Schedule a meeting with the team and find a way to improve the velocity.
- B. Review the Burndown chart to find reasons for low velocity.
- C. Advise the Product Owner and the team to onboard more resources to improve the velocity.
- D. Meet the Product Owner and explain him comparing other teams velocity is not correct

Answer – D.

Explanation

Velocity can't be compared across teams. Every project is unique with varied scope. Velocity is the average number of Story points delivered in the last 3 to 5 sprints. And it is an estimate specific to a project. The Scrum Master is responsible for facilitating team collaboration and removing barriers, if any. He should help the Product Owner to understand comparing Velocity with other teams is not a correct approach.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 64.

Question 46:

In a Sprint Planning meeting, the development team is slicing a user story. What are they doing?

- A. Decomposing a planning package to a work package.
- B. Collecting and validating requirements.
- C. Dividing the story into pieces that can be completed in the current iteration.
- D. Estimating the User Story.

Answer – C.

Explanation

Slicing a user story is the process of dividing a User Story into pieces that can be completed in one iteration. The other options are incorrect.

Question 47:

The Scrum Master is accountable for establishing scrum as defined in the Scrum Guide. Which of the following are responsibilities of a typical Scrum Master?

Choose the best two.

- A. Coaching the team members in self-management and cross-functionality.
- B. Ensuring that the Product Backlog is transparent, visible and understood.
- C. Creating the Sprint Backlog.
- D. Removing barriers between stakeholders and Scrum Teams.

Answer – A, D

Explanation

Development team is primarily responsible for creating the Sprint Backlog. So, option C is not correct. The Product owner manages the Product Backlog; the Scrum Master has little role here. Therefore, option B is also not correct. The Options A and D are typical Scrum Master roles. So, among all options, the options A and D are the best answers.

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 40,41.

Question 48:

An Agile team committed to delivering 50 story points in the Sprint Planning meeting of the third Sprint for a 10 Sprint release project. During the middle of the Sprint, a team member who is supposed to deliver ten story points suddenly goes for a vacation due to personal emergencies. What is the most appropriate action here?

- A. The Scrum Master and the Product Owner should request the team to work overtime and complete the work.
- B. Deliver 40 story points for the Sprint.
- C. Schedule a meeting with the Functional Manager for a replacement resource to work on the current Sprint.
- D. The Team member should be asked to complete the work remotely.

Answer: - B.

Explanation

In such situations the Team should deliver what they can for the current Sprint.

Option – A – Incorrect – First of all, the team is accountable for task execution. Overtime is counterproductive in the long term view and doesn't adhere to the Agile Principle of maintaining a sustainable pace.

Option – B – Correct – In this case, the team should deliver what is possible within the iteration. The best approach should be to deliver 40 Story points and explain the variance in the Sprint Review meeting.

Option – C – Incorrect – New resource onboarding may be time consuming. It's not a logical approach.

Option –D – Incorrect – Asking the team member to work remotely doesn't adhere to the Agile principles. Further, there is no guarantee whether the team member is in a position to work remotely.

Question 49:

Agile teams are cross-functional. What does the word cross-functional team mean?

- A. Team internally decides what needs to be done.

- B. Team members are skilled in software technologies.
- C. Team members are collaborative.
- D. Team members have all the skills necessary to create value each Sprint.

Answer: - D.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 41.
www.scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf – Page 5

Question 50:

Match the following

SL no	Agile meaning	Terminology
1	Working Software is the Primary measure of progress	a. Story Point.
2	A unit-less measure used in relative user story estimation techniques.	b. User Story.
3	A brief description of deliverable value for a specific user.	c. Agile Principle.

- A. 1-a, 2-c, 3-b.
- B. 1-c, 2-a, 3-b.
- C. 1-c, 2-b, 3-a.
- D. 1-b, 2-a, 3-c.

Answer – B.

Explanation

Agile Practice Guide - Project Management Institute Inc., 2017, Page(s) – 9,154,155.

Chapter 9: PMP Practice Test - 50 ITTO Questions

Question 1:

The Control Procurement process receives a very important input from the Direct and Manage Project Work process. Which one of the following acts as the valid input for the process?

- A. Approved change requests.
- B. Work performance data.
- C. Procurement documents.
- D. Agreements.

Answer – B.

Explanation

Work performance data, generated by the Direct and Managed Project Work process contains information related to activity duration, delivery status, start and end dates of scheduled activities which may act as an important input for the Control Procurements process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 493, 95.

Question 2:

The Control chart acts as an important tool in determining the stability of a process. It's an example of which of the following Tools and techniques?

- A. Data Gathering.
- B. Data Analysis.
- C. Data Representation.
- D. Decision Making.

Answer: C.

Explanation

The Control chart is an example of a Data Representation technique used to

Control Quality process to determine its stability.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 304.

Question 3:

Simulation is an important technique used in the Perform Quantitative Risk Analysis process. It's an example of which of the following tools and techniques?

- A. Decision Making.
- B. Data Representation.
- C. Data Analysis.
- D. Decision Analysis.

Answer - C.

Explanation

Simulation is a Data Analysis technique that explores the impact of variances on deliverables or outcomes during a given project. Variances constitute a wide variety of unforeseen and unexpected events that may delay or impact the timeliness of deliverables. In the Perform Quantitative Risk Analysis process, Simulations are typically performed using a Monte Carlo analysis. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 433.

Question 4:

The Project Manager is making sure that the work performed to deliver a product has been completed as defined in the Project Management Plan. What part of the project management process group is getting carried out in the given situation?

- A. Planning.
- B. Executing.
- C. Monitoring and Controlling.
- D. Closing.

Answer – D.

Explanation

During closing, the project manager reviews the Project Management Plan to ensure all project work is completed and the project has met its objectives. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 123.

Question 5:

The Control schedule process gives an important output that acts as an input for the Monitor and Control Project Work process. What output of the Control Schedule process is referred to in this scenario?

- A. Work Performance Data.
- B. Work Performance Information.
- C. Schedule baseline.
- D. Work Performance Reports.

Answer – B.

Explanation

Work performance information is a summarized or analyzed version of work performance data. Most of the processes in the Monitoring and Controlling Process Group have Work Performance Information as outputs, and it becomes inputs to the Monitor and Control Project Work process. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 222-223, 228.

Question 6:

As a Project Manager, you are holding team building activities for your project team. Which process is getting performed in this scenario?

- A. Develop Team.
- B. Manage Team.
- C. Monitor Communication.
- D. Control Resources.

Answer – A.

Explanation

Organizing Team building activities is an effective aid for building a high performance team, which happens during the Develop team process. In the

Develop Team process, the project manager tries to improve the overall team environment to enhance project performance. So, among all options provided, A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 337.

Question 7:

The Project team is currently engaged in Quality planning. They are using several tools to come up with a proper Quality Management Plan. Which of the following is not a tool or technique of the Plan Quality Management process?

- A. Interviews.
- B. Cost-Benefit analysis.
- C. Quality Metrics.
- D. Mind mapping.

Answer - C.

Explanation

A, B, D are not correct because they are defined tools and techniques used for Quality Planning. A quality metric specifically describes a project or product attribute and how the Control Quality process will verify compliance to it.

Quality metrics are outputs from the Plan Quality Management process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 281 - 284.

Question 8:

As a Project Manager, you are estimating the type and quantities of materials, and suppliers necessary to perform project work. What enterprise environmental factors can influence your estimation process?

- A. Duration estimating databases.
- B. Resource location, availability.
- C. Productivity metrics.
- D. Policies and procedures relating to supplies and equipment.

Answer – B.

Explanation

The scenario states that the Project Manager is performing the Estimate Activity Resources process and EEF is a valid input to the process. D is an example of OPA, so D is not correct. A, C are examples of EEF but they are used as input for the Estimate Activity Durations process, not the Estimate Activity Resources process. So, A, C are also incorrect. B is an example of EEF (enterprise environmental factors) which is used as input for the Estimate Activity Resources process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 323.

Question 9:

Which of the following techniques can be used during closing to improve the metrics of the organization by comparing the end result with what was initially planned?

- A. Trend Analysis.
- B. Document Analysis.
- C. Regression Analysis.
- D. Variance Analysis.

Answer - D.

Explanation

Here the question states about the data analysis technique called Variance Analysis which helps to compare plan Vs. Actual results. Among all options provided, D is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 127.

Question 10:

Ryan is a project manager of an IT development project. Currently he is involved in the preparation of the Cost Management Plan. Which of the following techniques can help in the preparation of the Cost Management Plan?

- A. Cost of Quality.
- B. Alternative Analysis.
- C. Earned Value Analysis.

D. Analogous estimating.

Answer – B.

Explanation

The situation states that the Project Manager is involved in the Plan Cost Management process. A and D are not correct as both the techniques are used in the Estimate Costs process, not Plan Cost Management process. C is wrong, because it's used as a technique for the Control Costs process. B is a valid technique used in the Plan Cost management process to review strategic funding options and consideration of ways to acquire project resources such as making, purchasing, renting, or leasing. So, among all options provided B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 238.

Question 11:

One of your Project deliverables is ready for customer inspection. Which of the following acts as a key input for the process in relation to Customer inspection?

- A. Accepted Deliverable.
- B. Work Performance Information.
- C. Verified Deliverable.
- D. Change Request.

Answer – C.

Explanation

Customer inspection of Project deliverables is a part of the Validate scope process.

In this very process, Customer inspects the deliverable(s), that are verified by QC team in the Control Quality process. In most frequent situations, QC team checks the correctness of the deliverable implemented in the recent Direct and Manage Project work process, and marks them as Verified deliverable(s) if all conditions are satisfied. This Verified deliverable(s) acts as an input for the Validate Scope process for customer inspection, followed by formal acceptance by the customer. Options A and D are not correct because they are output to the Validate Scope process. B is not a valid input for the validate

scope process. So among all options provided, C is the right answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 163, 165.

Question 12:

During planning, the Project Manager and the team are involved in the process of identifying and documenting relationships among the project activities. With the help of a software they are connecting each activity, other than the first and the last one to at least one predecessor and successor activity with an appropriate logical relationship. Which process is getting performed in this scenario?

- A. Sequence Activities.
- B. Define Activities.
- C. Create WBS.
- D. Plan Schedule Management.

Answer – A.

Explanation

As per the given scenario, the project team is involved in the Sequence Activities process where activities are sequenced and proper logical relationships among activities are established to define the logical sequence of work in order to obtain greatest efficiency.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 187, 188.

Question 13:

As a Project Manager, Ryan is engaged in the development of the Risk Management Plan. What tools and techniques can help him to determine the risk appetite of Project Stakeholders?

- A. Assumption and Constraint Analysis.
- B. Stakeholder Analysis.
- C. SWOT Analysis.
- D. Stakeholder Register.

Answer – B.

Explanation

The scenario states that the Project Manager is currently involved in the Plan Risk Management process. Options A and C are not correct, because they are used as Tools and Techniques for Identify Risks process. Stakeholder Register is an example for a project document, it's not a Tools and Techniques. So, option D is also not correct. Stakeholder Analysis is a valid Data Analysis technique used in Plan Risk Management Process in order to determine the risk appetite of the project stakeholders. So, among all options provided, Option B is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 401,404.

Question 14:

Which of the following techniques can be used during closing to evaluate the satisfaction of the stakeholders , gather lessons learned, and to celebrate success?

- A. Trend Analysis.
- B. Document Analysis.
- C. Regression Analysis.
- D. Meetings.

Answer - D.

Explanation

During Closing, Meetings can be used to confirm that the deliverables have been accepted, to validate that the exit criteria is met, to evaluate the satisfaction of the stakeholders, gather lessons learned, and to celebrate success. Typically, the types of meetings that happen during Closing are: - Close-Out reporting meetings, Customer wrap-up meetings, Lessons learned meetings, and celebration meetings. So , among all options provided, D is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 127.

Question 15:

In an Internal project initiated by a car manufacturing company, several Subject Matter Experts have gathered to decide the outcome of a deliverable in regard to its formal acceptance. Ultimately a decision

making technique that is also used in requirement gathering as well as making decisions on Change Request has been applied to reach a consensus.

What decision making technique has been used in the given scenario?

- A. Voting.
- B. Inspection.
- C. Meetings.
- D. Autocratic decision making.

Answer – A.

Explanation

The question here talks about the Validate Scope process and several experts have to decide the Formal acceptance of a deliverable. Voting is a decision making technique which is widely used in this process to reach a decision or judgment.

B is a technique used in the validate scope process but it's not a Decision Making technique. So, B is not correct. C is again not a decision making technique, so C is also not correct. D is an example decision making technique but generally not used in the Validate Scope process. Further in "Autocratic decision making", one individual takes a decision, but here the question clearly states about reaching consensus or agreement by applying a decision making technique. So, D is also correct.

Voting is a decision making technique that is applied, to reach a consensus. Other than Validate scope process, it's also used in Collect Requirement and Perform Integrated Change Control processes as stated in the question.

Therefore, among all options provided, A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 163, 166, 113,138.

Question 16:

Your project is near completion and the entire Project Team is involved in a process that involves the confirmation of the formal acceptance of the final product. Which of the following is not an input of the process mentioned in the above scenario?

- A. Project Charter.

- B. Business Case.
- C. Enterprise Environmental Factors (EEF).
- D. Organizational Process Assets (OPA).

Answer – C.

Explanation

The question here talks about Close projects or phase processes. A, B, D are valid input to the process. EEF is not an input to the Close Project or Phase process. So, among all options provided C is the correct answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 121.

Question 17:

Arun is the sponsor of a software development project. He has involved a few experts in the preparation of Project Charter. Which of the following is an example of tools and techniques that can be used in the process of developing the Project Charter?

- A. Business Case.
- B. Need Assessment.
- C. Project Management Information System (PMIS).
- D. Conflict Management

Answer – D.

Explanation

The scenario states that Arun, the sponsor, has involved few experts in the Develop Project Charter process. The Develop Project Charter is the process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities. This process has four tools and techniques: Expert Judgment, Data Gathering, Interpersonal and Team skills (Conflict management, facilitation, etc.), Meetings.

A and B are not examples of Tools and Techniques, so they are not correct. C is a tool and technique but not used in the Develop Project Charter process. Conflict management is an example of interpersonal and team skills technique used in the Develop Project Charter process. Conflict management can be helpful in bringing stakeholders into alignment on various elements of

the charter. So , among all options provided D is the right answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 75,80.

Question 18:

You are looking for information on personnel rates, travel costs, and other costs that are needed to estimate the overall project budget. Where is the best place to find this information?

- A. Basis of Estimates.
- B. Resource Calendars.
- C. Resource Management Plan.
- D. Cost Management Plan.

Answer – C.

Explanation

The scenario suggests you are carrying out the Determine Budget process. Basis of Estimates is an input to the process but it contains supporting details of cost estimates. It generally specifies any basic assumptions dealing with inclusion or exclusion of indirect or other costs in the project budget, so A is not correct. B is not an input for the Determine Budget process, so B is not correct. D is an input for the Determine Budget process, but the Cost Management Plan describes how the project cost will be structured in the project budget. So, D is also wrong. Based on the elimination rule, C is the right answer. Further, the Resource Management Plan which is a valid input to the process, provides information on rates (personnel and other resources), estimation of travel costs, and other foreseen costs that are necessary to estimate the overall project budget. Therefore among all options provided, C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 250.

Question 19:

The Checklists are important tools used in Quality Management. Checklists are examples of which of the following tools and techniques?

- A. Data Gathering.
- B. Data Analysis.

- C. Data Representation.
- D. Decision Making.

Answer – A.

Explanation

Checklists are examples of data gathering techniques that are generally used to check if a list of requirements has been satisfied.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 288, 292.

Question 20:

The technology and functional experts of an IT organization have gathered to come up with a procurement strategy and bid documents for the company's Next Gen IT innovation project. Procurement strategy and Bid documents are created as a part of which of the following processes?

- A. Conduct Procurements.
- B. Develop Project Charter.
- C. Plan Procurement Management
- D. Control Procurements.

Answer – C.

Explanation

Procurement Strategy and Bid documents are output for the Plan Procurement Management process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 466, 476, 477.

Question 21:

Expert Judgment is an example of Important Tools and Techniques used in Project Management. Expert judgment is a tool and technique used in all of the following processes except:

- A. Implement Risk Responses.
- B. Conduct Procurements.
- C. Identify Stakeholders.

D. Manage Communications.

Answer - D.

Explanation

A, B, C are not correct. In the Implement Risk responses process, experts with specialized knowledge are sometimes required to validate or modify risk responses if needed. Expert Judgment is used in Conduct Procurements to evaluate proposals. In the Identify Stakeholder process, it is used to ensure all stakeholders have been identified, listed, and assessed appropriately. By following the elimination rule, D is the right answer. Further, Manage Communications is part of the Executing Process Group. As per PMBOK6, expert judgment is not required while managing communications. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 378.

Question 22:

The Control Quality process uses several tools and techniques to ensure the project outputs are complete, correct, and meet customer expectations. Which of the following is not an example of tools and techniques used in the control quality process?

- A. Histogram.
- B. Design for X.
- C. Questionnaires and surveys.
- D. Scatter diagrams.

Answer – B.

Explanation

A , C, D are tools and techniques used in the Control quality process. B is an example of a technique used in the Manage Quality process. Among all options provided, B is the right answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 295, 298.

Question 23:

The Organization Process Assets play a huge role in influencing the management of a project. Organization Process Assets may get updated

as an output in which of the following processes?

- A. Manage Team.
- B. Manage Quality.
- C. Implement Risk Responses.
- D. Manage Project Knowledge.

Answer – D.

Explanation

In most of the processes, Organization Process Assets(OPA) acts as an input. But there are few processes where OPA gets updated as an output, Manage Project Knowledge is one of them. Other processes where OPA gets updated as an output are: - Manage Communication, Acquire Resources, Develop Team, Monitor Risks, Plan Procurement Management, Control Procurements, Close project/phase.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 98.

Question 24:

During inspection, the customer asks the Project Manager if the software that the project has developed so far conforms to all the requirements. What should be used to measure completion of the product scope?

- A. Requirements Management Plan.
- B. Requirement Documentation.
- C. Project Management Plan.
- D. Requirements Traceability Matrix

Answer – B.

Explanation

The given scenario indicates that the Validate Scope process is getting conducted to measure completion of the product scope.

The completion of the project scope is measured against the components of the project management plan, i.e. scope baseline. Completion of the product scope is measured against the product requirements documentation, which also serves as an input to the validate scope process. So, among all options provided B is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 131,165.

Question 25:

Meetings are used as a tool and technique in many project management processes. Which of the processes below do not use meetings as a tool or technique?

- A. Identify Stakeholders.
- B. Monitor Risks.
- C. Plan Communications Management
- D. Define Scope.

Answer – D.

Explanation

Options A, B, C use meeting as a technique. Meetings are not among the tools and techniques explicitly listed for the Define Scope process. So, among all options, option D is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 150,692.

Question 26:

The Project Team is currently engaged in using an iterative planning technique in which they are doing detailed planning for work to be accomplished in near term, but for future work they are only doing high level planning. What process is getting executed in the current scenario?

- A. Define Activities.
- B. Define Scope.
- C. Estimate Activity Duration.
- D. Determine Budget.

Answer – A.

Explanation

The scenario states that the project team is currently engaged in applying Rolling Wave planning technique where work to be done in near term is planned in detail, whereas the work to be done in future is planned at higher

level. Further this technique is effectively used in the Define Activities process for planning the work to be accomplished. So, among all options available, option A is the right choice.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 185.

Question 27:

You are a new Project Manager who has joined during the middle of execution. You want to review the planned training strategies for all your team members. What should you refer to fetch this information?

- A. Project Team assignments.
- B. Resource Management Plan.
- C. Team performance assessments.
- D. Communication Management Plan.

Answer – B.

Explanation

Here in this scenario, Resource Management Plans should be referred to as Training strategies for team members that are recorded in the Resource Management Plan.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 319.

Question 28:

As a Project Manager, you are engaged in the process of communicating and working with stakeholders to meet their needs, expectations, and foster appropriate stakeholder involvement.

What should you refer to determine the methods, format, and technologies to be used for stakeholder communication?

- A. Stakeholder Engagement Plan.
- B. Lessons Learned Register.
- C. Communication Management Plan.
- D. Stakeholder Register.

Answer – C.

Explanation

The scenario states that the Project Manager is involved in the Manage Stakeholder Engagement Process. All 4 given options are input to the process. But it's the communications management plan that provides all the communications details for stakeholder engagement that includes methods, formats, and technologies to be used for stakeholder communication. So, among all options provided C is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 523, 525.

Question 29:

Which of the following components of the project management plan is an input to the Plan Risk Management process?

- A. Scope Management Plan.
- B. Cost Management Plan.
- C. Stakeholder Engagement Plan.
- D. All Components of the Project Management Plan.

Answer – D.

Explanation

All Components of the Project Management Plan are inputs to the Plan Risk Management Process. Information in the various project management plan components might influence the Plan Risk Management process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 401-403.

Question 30:

The following process ensures that the project stays aligned with the business needs and also helps in comparison of the actual project performance against the project management plan. What process is referred to in the given scenario?

- A. Monitor and Control Project Work
- B. Perform Integrated Change Control.
- C. Control Scope.
- D. Define Scope.

Answer – A.

Explanation

B is not correct, because the Perform Integrated Change Control process manages the change requests raised across the life cycle of the project. C is incorrect because the control scope process only manages the changes to the scope baseline, it doesn't compare project performance against the project management plan. D is incorrect because Define Scope is a process of Planning process group which mainly deals with project inclusions and exclusions. So, by applying elimination technique, A is the right answer. Further, Monitor and Control Project Work is the process of tracking, reviewing, the overall project progress to meet the performance objectives defined in the Project Management Plan. This is accomplished by comparing actual project performance against the Project Management Plan on a regular basis. So, among all options provided, option A is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 222-223, 228.

Question 31:

As a Project Manager, you are in the process of developing the approved version of the time-phased project budget. Which of the following acts as an important output to the process?

- A. Basis of Estimates.
- B. Funding Limit Reconciliation.
- C. Project Funding Requirements.
- D. Cost Management plan updates.

Answer – C.

Explanation

The scenario states that the Project Manager is performing the Determine Budget process. The major output for the process is cost baseline, which is not there in option. Other than cost baseline, project funding requirements, project document updates (cost estimates, project schedule, risk register) are other outputs.

Option A is an input to the Determine Budget process, so A is not correct. Option B is a technique used in the Determine Budget process, and it's not an output. The Cost Management Plan is not updated as a part of the Determine

budget process, so D is also not correct. Project funding requirements include total funding requirements and periodic funding requirements (e.g., Quarterly, annually) and acts as a valid output for the Determine Budget process. So, among all options, C is the best answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 248, 256.

Question 32:

Your project has successfully moved to the Closing phase. You want to know who will sign off the project. What needs to be reviewed here?

- A. Project Management Plan.
- B. Project Charter.
- C. Agreement.
- D. Communication Management Plan.

Answer – B.

Explanation

In this scenario, the Project Charter should be reviewed to know the approval requirements, and who will sign off the project. Some may get confused with agreements i.e. option C, but Agreements take care of contract closure, not project closure.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 124.

Question 33:

Networking is an example of interpersonal and team skills that creates the conditions for project teams to share knowledge among one another. Which of the following processes uses networking as tools and technique?

- A. Develop Team.
- B. Manage Team.
- C. Manage Project Knowledge.
- D. Create WBS.

Answer – C.

Explanation

Networking is an example of a tool and technique used in processes like Manage Project Knowledge, Manage Communications, Monitor Stakeholder Engagement, that creates the condition to share tacit and explicit knowledge for stakeholders in an informal fashion.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 98, 104.

Question 34:

Project KPIs are measurable indicators that help to track the performance of a project. Which of the following processes generates KPIs for the first time?

- A. Direct and Manage Project Work.
- B. Monitor and Control Project Work.
- C. Close Project / Phase.
- D. Control Procurements.

Answer – A.

Explanation

KPIs are examples for Work Performance Data(WPD) which is generated as an output for Direct and Managed Project work processes.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 90, 95.

Question 35:

An unknown-unknown risk has occurred. The Project Team has planned a workaround. Workaround gets implemented as a part of which of the following processes?

- A. Monitor Risks.
- B. Direct and Manage Project Work.
- C. Implement Risk Responses.
- D. Control Quality.

Answer – B.

Explanation

Some may get confused with C, but Implement Risk responses are for Known-Unknown risks. Workaround is planned when an unknown risk occurs in the project. They are identified as a part of Monitor Risks process, where the project team, post identification raises a change request (also a valid output for the process) that gets processed for review and disposition through Perform Integrated Change Control process and ultimately when approved gets implemented as a part of the Direct and Manage Project Work process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 457.

**Workaround is not part of PMBOK6, but we must remember the PMP exam is beyond PMBOK. It is available in Rita and other Project Management books.

Question 36:

Quality Function Deployment (QFD) is a technique popularly used in manufacturing industries to determine critical characteristics for new product development. QFD is an example of which of the following types of tools and techniques?

- A. Data Analysis.
- B. Decision Making.
- C. Interpersonal and Team Skills.
- D. Data Gathering.

Answer- C.

Explanation

Quality Function Deployment (QFD) is a Facilitation technique widely used in the Manufacturing industry that helps to determine critical characteristics of a new product. QFD starts by collecting customer needs, also known as Voice of the Customer. These needs are then objectively sorted and prioritized, and goals are set for achieving them.

QFD, which falls under Facilitation technique category, is an example of Interpersonal and Team Skills technique used during the Collect Requirements process.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 145.

Question 37:

Which of the following Data Representation tools and techniques is used in both Plan Stakeholder Engagement as well as Monitor Stakeholder Engagement process?

- A. Mind Mapping.
- B. Multi-criteria Decision Analysis.
- C. Presentations.
- D. Stakeholder Engagement Assessment Matrix.

Answer – D.

Explanation

Mind Mapping is an example of Data Representation tools and Technique used in the Plan Stakeholder Engagement process, but not in the Monitor Stakeholder Engagement process, so A is not correct. Multi-criteria Decision Analysis is an example decision analysis tool and techniques which is used in Monitor Stakeholder Engagement process, but not Plan Stakeholder Engagement process, so B is also not correct. Presentations are example communication skills used in the Monitor Stakeholder Engagement process, but not the Plan Stakeholder Engagement process, so C is not correct.

Stakeholder Engagement Assessment Matrix is an example of Data Representation technique used in both Plan Stakeholder Engagement as well as Monitor Stakeholder Engagement process.

So, among all options, Option D is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 516, 530.

Question 38:

Performance Measurement Baseline, the additional component for Project Management Plan is an integrated scope-schedule-cost plan for the project work against which project execution is compared to measure and manage performance. Performance Measurement Baseline acts as an input to which of the following processes?

- A. Control Scope.
- B. Develop Project Management Plan.
- C. Control Resources.

D. Control Quality.

Answer – A.

Explanation

Performance Measurement Baseline is an additional component for the Project Management Plan. It was returned as an output for the Develop Project Management Plan process, So B is not the right answer. C and D doesn't have a Performance Measurement Baseline as input or output, so C and D are incorrect. The Performance Measurement Baseline acts as a valid input for the Control Scope process. In the Control Scope process, when Earned Value analysis is used, the Performance Measurement Baseline is compared to actual results to determine if a change, corrective action or preventive action is necessary. So, among all options provided A is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 169.

Question 39:

Which of the following is NOT an example for tools or techniques used in the Conduct Procurements process?

- A. Expert judgment
- B. Proposal evaluation.
- C. Source selection criteria.
- D. Negotiations.

Answer – C.

Explanation

A, B, D are valid examples of Tools and Techniques in the procurement process. Source Selection Criteria (Procurement Documentation) describe how bidder proposals will be evaluated. It's an input to the Conduct Procurement process, and does not belong to the Tools and Techniques category.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 482,485.

Question 40:

A Change Request for modification of Procurement Statement of Work has been approved by following a defined Change Control process. The approved Change Request generated by the Perform Integrated Change Control process in this situation becomes input to which of the following processes?

- A. Validate Scope.
- B. Control Quality.
- C. Monitor and Control Project Work.
- D. Control Procurements.

Answer – D.

Explanation

Approved Change Requests generated by Perform Integrated Change Control process becomes input to Direct and Manage Project Work, Control Quality and Control Procurements processes. Option A and C are not correct because these processes don't have Approved Change Request as input. Though the Control Quality process has an Approved Change request as input but Approved Change Requests for Control Quality process only gets affected by actions like defect repair, preventive or corrective actions, and it never includes any requests in relation to modification of contracts or related contractual documents. Approved change requests such as modifications to the contract or the procurement statement of work would affect the Control Procurements process and it also acts as valid input to the process. So, among all options provided D is the best answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 492,496.

Question 41:

The Entire Project Management Team is attending a workshop conducted by a trained moderator to rank most useful ideas for further brainstorming or for prioritization. What tools and techniques have been referred to in this scenario?

- A. Voting.
- B. Nominal group technique.
- C. Facilitation.

D. Focus groups.

Answer – B.

Explanation

Nominal group technique is part of the Interpersonal and team skills category of technique. It is a technique that enhances brainstorming with a voting process used to rank the most useful ideas for further brainstorming or for prioritization. It is a structured form of brainstorming that involves the entire group of stakeholders participating in the requirement gathering (Collect Requirements) process .

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 144.

Question 42:

CCB has approved 4, deferred 2, and rejected 3 change requests. As a result of this, which of the following gets updated in the next step to record all these CCB decisions?

- A. Project Document.
- B. The Project Management Plan.
- C. The Change Management Plan.
- D. Issue log.

Answer – A.

Explanation

The disposition of all Change Requests are recorded in the Change log as a part of project document update. Change Log is also an example for a project document that maintains the status of all change requests. So, among all options, A is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 120.

Question 43:

Decomposition is a popular technique used for dividing and subdividing the project scope and project deliverables into smaller, and more manageable parts. Decomposition is used as a tool and technique in which of the two processes?

- A. Create WBS and Estimate Activity Resources.
- B. Define Activities and Define Scope.
- C. Create WBS and Define Activities.
- D. Create WBS and Define Scope.

Answer – C.

Explanation

Decomposition is explicitly listed as a tool and technique for both the Create WBS and the Define Activities processes. So, among all options C is the right answer.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 156,185.

Question 44:

The following technique is used in both Plan Communication Management as well as Monitor Stakeholder Engagement processes to understand the strategies of the organization, know who wields power and influence, and to develop a communication strategy.

What tool and technique is mentioned in the given scenario?

- A. Networking.
- B. Political awareness.
- C. Communication styles assessment
- D. Expert Judgment.

Answer – B.

Explanation

A is not correct, because Networking is used as a technique in the Monitor Stakeholder Engagement process but not used in the Plan Communication Management process. Communication styles assessment and Expert judgment are used in the Plan Communications Management process but not in the Monitor Stakeholder Engagement process, so both C and D are also not correct. By following elimination rules, PMBOK 6; B is the right answer as Political awareness is used in both Plan Communication Management as well as Monitor Stakeholder Engagement processes. Political awareness helps the project manager to develop communication strategies to connect with stakeholders by understanding who wields power and influence in the

required field.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 366,530, 534.

Question 45:

A Project Manager of a large project goes for a long sabbatical leave just a week before Project kick-off meeting. Ryan is the new Project Manager who has joined as his replacement. As a newly joined Project Manager, he wants to know what development approach has been decided to follow in the project. What should Ryan refer to fetch this necessary information?

- A. Project Charter.
- B. Communication Management Plan.
- C. Project Management Plan.
- D. Stakeholder Register.

Answer – C.

Explanation

Here in the scenario, the newly joined Project Manager should review the Project Management Plan. Development approach is the additional component for Project Management Plan that defines whether predictive, Iterative, agile, or a hybrid model or development approach or model should be used.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017,
Page(s) – 88.

Question 46:

Ground rules set the expected behaviors of the project team members, with regard to stakeholder engagement. Which of the following processes define Ground rules?

- A. Manage Stakeholder Engagement
- B. Develop Project Charter.
- C. Plan Quality Management.
- D. Plan Resource Management.

Answer – D.

Explanation

The Ground rules are defined in the Team Charter which is an output for Plan Resource Management Process. So, among all options, D is the right answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 312, 320, 528.

Question 47:

Ryan is a Project Manager of a complex engineering project. He along with all risk owners have just completed an important risk management process and updated the Risk register with information on agreed-upon risk response strategies, triggers, warning signs of risk occurrence, residual risks that are expected to remain because they are deliberately accepted. What risk management process has been concluded in the given scenario?

- A. Plan Risk Responses.
- B. Perform Qualitative Risk Analysis.
- C. Identify Risks.
- D. Monitor Risks.

Answer – A.

Explanation

The situation states that the project manager as well as the risk owners has just concluded the Plan Risk Responses process. Risk Register gets updated as a valid output for Plan Risk Response process with information like agreed-upon response strategies, triggers, warning signs of risk occurrence, residual risks that are expected to remain because they are deliberately accepted, etc.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 448.

Question 48:

Ryan is the Project Manager of a construction company. Currently he is engaged in the first iteration of the Identify Stakeholder process for a new road construction project of his company.

What can help him with the initial list of stakeholders who may get positively or negatively affected by the project?

- A. Project Charter.
- B. Business Case.
- C. Project Management Plan.
- D. Benefit Management Plan.

Answer – B.

Explanation

In the first iteration of the Identify Stakeholder process, Business case and the Benefit Management Plan are sources of information about project's stakeholders. A is not correct because the project charter may help in the identification of key stakeholder list, this information to project charter receives from Business case. To be more precise, the question asked about stakeholders not key stakeholders. C is incorrect, because Project Management Plan is not yet created at this stage. Some may get confused between B and D, as both the options are close. The Benefit Management Plan emphasizes those stakeholders who will benefit from the delivery of the outcomes of the project. Business case helps with an initial list of stakeholders affected by the project (both positively and negatively, in most possible scenarios). So, among all options B is the best answer. (PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 509.

Question 49:

One of the outputs from the Control Quality process gives an important input to the Manage Quality process that represents documented results of Control Quality activities. What has been referred to in the given scenario?

- A. Quality Report.
- B. Work Performance Information.
- C. Quality Control Measurements.
- D. Quality Metrics.

Answer - C.

Explanation

A is the output for the Manage Quality process, it's not the input. So A is not correct. Work Performance Information is neither input or output to the

Manage Quality process, though it's a valid output for the Control Quality process. So, B is also not correct. D is an input for the Manage Quality process, but not an output for the Control Quality process. Therefore, D is also not correct. Based on the elimination rule, C is the right answer. Further Quality Control Measurements is a valid output for the Control Quality process that represents documented results of Control Quality activities. It acts as a valid input for the Manage Quality process to analyze and evaluate the quality of the processes and deliverables of the project against the standards of the performing organization or the requirements specified. So, considering all options, option C is the right answer.
(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 288, 291, 298, 305.

Question 50:

During closing what would help to measure whether the benefits of the project were achieved as planned?

- A. Project Management Plan.
- B. Benefit Management Plan.
- C. Business Case.
- D. Lessons Learned Register.

Answer – B.

Explanation

A Benefit Management Plan is a type of Business document that describes how and when the benefits of the project will be delivered, and describes the mechanism that should be in place to measure those benefits. During Closing, it is reviewed to measure whether the benefits of the project were achieved as planned.

(PMBOK Guide) – Sixth Edition, Project Management Institute Inc., 2017, Page(s) – 125.

Chapter 10: Additional 200 Mock Questions

Question 1:

There are three types of project management charts to help display schedules and timelines in more visual ways. If you have a stakeholder who only wants to see the start and end dates of major milestones, which type of chart is best-suited to her needs?

- A. Project Schedule Network Diagrams
- B. Milestone Charts
- C. Pivot Tables
- D. Bar Charts (aka Gantt Charts)

Answer: B.

Explanation

A Bar Chart is a graphical display of project elements, dates and durations, typically used by project teams. A Milestone Chart is a graphical display of significant events in the project, usually completion of major deliverables. This type of display is often used for Executive Management. Project Schedule Network Diagrams are a graphical way to view tasks, dependencies, and the critical path of a project. These often resemble Visio diagrams (flowcharts) and provide a clear "sequence of events" that has to take place from now through project completion. Pivot Tables are an Excel feature that hasn't much to do with displaying a project schedule. A Milestone Chart will give your stakeholder what she's looking for.

Question 2:

There are five process groups in the project management process. Which number is Executing?

- A. 4
- B. 3
- C. 2

D. 1

Answer: B.

Explanation

The five processes, in order: Initiating, Planning, Executing, Monitoring & Controlling, Closing. Executing is third.

Question 3:

Your organization may have processes or procedures specific to the Closing process group, including contract closure, deliverable acceptance and resource reassignment. Which of the following could also be required by your organization with a certain process or procedure?

- A. Time Reporting
- B. Project evaluation and project audit
- C. Project Audit
- D. Project Evaluation

Answer: B.

Explanation

Responses A and C are correct. Though your organization may also have processes or procedures surrounding time reporting, that is not relevant to the Closing process group.

Question 4:

Martin is about to change his project's scope baseline for the fourth time, as his team continues discovering missed business requirements as they work through evaluating different die-casting vendors. So far the first baseline change was to the scope, which also resulted in a schedule change adding 2 months to the timeline. The second baseline change was to the project's cost, as was the third. Their most recent discovery has resulted in yet another schedule change and the change request to add six more weeks to the project schedule was just approved by the project's steering committee. When Martin re-baselines the schedule, what is the "starting point" he should use to calculate the new dates?

- A. The original project schedule/estimated completion date

- B. The first baseline
- C. The second baseline
- D. The third baseline

Answer: D.

Explanation

Changes to baselines are only made from the last baseline forward. Past performance has not changed. This protects the integrity of the baselines and the historical data of past performance.

Question 5:

As Charlene's project team is working through the Plan Procurement Management process, they realize that the supplier of nanotechnology they need to execute their product to current specifications, is so backlogged on orders that they won't be able to make the date the team had outlined in their project schedule. What input to the Plan Procurement Management Process is this an example of?

- A. Enterprise Environmental Factors (EEFs)
- B. Organizational Process Assets (OPAs)
- C. Business documents
- D. Project documents

Answer: A.

Explanation

The first three EEFs (one of the inputs, together with OPAs (Organizational Process Assets) given for most processes within the PMBOK) listed for the Plan Procurement Management process are 1) Marketplace conditions, 2) Products, services, and results that are available in the marketplace and 3) Sellers, including their past performance or reputation. Since Charlene's project is now impacted by an EEF - something they cannot control - she has to undergo whatever her planned mitigation strategies were if the risk of a supplier not having what they needed arose.

Question 6:

Brad has been decomposing work packages to develop the Work Breakdown Structure (WBS) for his new underground railroad

(subway/tube) station project. When he shows the first draft to his project owner, he advises that it's missing such deliverables as Project Scope Statement, Assumptions Log and Risk Register. Which role's work package did Brad neglect to include in his WBS?

- A. Project Sponsor
- B. Business Analyst
- C. Business Process Owner
- D. Project Manager

Answer: D.

Explanation

The WBS represents all product and project work, including the project management work. The total of the work at the lowest levels should roll up to the higher levels so that nothing is left out and no extra work is performed. This is sometimes called the 100 percent rule.

Question 7:

Eva's just taken a position in the engineering department of her local phone company. Part of her job description includes the fact that she may be called upon to participate in various engineering work groups while in the role and eventually she may even wind up leading some of those projects herself but that if she does she'll have little to no authority and her manager would be managing the budget rather than her. What type of organizational structure does Eva's new employer have?

- A. Functional (centralized)
- B. Project-oriented (composite, hybrid)
- C. Virtual
- D. Matrix (balanced)

Answer: A.

Explanation

Page 47 of the PMBOK contains Table 2-1. Influences of Organizational Structures on Projects and this is where you will find and must learn the various characteristics that describe all of these types of organizations. In Eva's case, while the work group described could have placed her

organization into either Matrix (balanced) or Functional (centralized), the deciding factor for Functional (centralized) was the little to no authority for the project manager and the fact that the functional manager is the one who manages the budget.

Question 8:

Which of the following is not a baseline that should be included in the project management plan?

- A. Cost
- B. Scope
- C. Management Reviews
- D. Schedule

Answer: C.

Explanation

While a management review could theoretically be baselined by comparing subsequent reviews to the first one done, the three baselines that should be included in every project management plan are the scope baseline, the schedule baseline and the cost baseline.

Question 9:

What or who is the natural liaison between the organization's portfolios, programs, projects, and the organizational measurement systems?

- A. Project Manager
- B. Portfolio Manager
- C. Project Management Office
- D. Program Manager

Answer: C.

Explanation

The PMO integrates data and information from organizational strategic projects and evaluates how higher-level strategic objectives are being fulfilled. The PMO is the natural liaison between the organization's portfolios, programs, projects, and the organizational measurement systems (e.g., balanced scorecard). The projects supported or administered by the

PMO may not be related other than by being managed together. The specific form, function, and structure of a PMO are dependent upon the needs of the organization that it supports.

Question 10:

Your organization contracts with a vendor to deliver their online banking software to you as part of your project to transform your customers' digital experience. If the vendor doesn't deliver the software with some functionality that you were expecting to receive, or the functionality works but not in the way you had wanted it to, what may you have to do as part of your duties during Monitoring and Controlling?

- A. Adjust the Communication Plan
- B. Create a Change Request
- C. Update the Project Management Plan
- D. Analyze Data

Answer: B.

Explanation

During the Validate Scope process, you can use the tool & technique of inspection to determine if a deliverable has met your requirements. During that time, or perhaps later in a group meeting, it may be decided that a change request is needed if a deliverable cannot be accepted or the scope needs to change to accommodate what actually was delivered, if the project team deems it acceptable. If the scope is validated, the deliverable is accepted. If not, a change request may be required.

Question 11:

Danielle is managing a project to build a new passenger-bearing rocket that a private company is planning to launch into space in five years. She has been reviewing her activity list and adding information as she receives it, and tying together predecessors and their successors, leads and lags, resource requirements and the locations where all of the different components are being manufactured. What are all of these things Danielle is adding to the project's activities called?

- A. Project Management Plan

- B. Activity Attributes
- C. Resource Requirements
- D. Project Schedule

Answer: B.

Explanation

Activity attributes. These extend the description of the activity by identifying multiple components associated with each activity. The components for each activity evolve over time. During the initial stages of the project, they include the unique activity identifier (ID), WBS ID, and activity label or name. When completed, they may include activity descriptions, predecessor activities, successor activities, logical relationships, leads and lags, resource requirements, imposed dates, constraints, and assumptions. Activity attributes can be used to identify the place where the work has to be performed, the project calendar the activity is assigned to, and the type of effort involved. Activity attributes are used for schedule development and for selecting, ordering, and sorting the planned schedule activities in various ways within reports.

Question 12:

How many Data Gathering techniques are there?

- A. 15
- B. 5
- C. 9
- D. 3

Answer: C.

Explanation

There are 9. Benchmarking, Brainstorming, check sheets, checklists, focus groups, interviews, market research, questionnaires & surveys and statistical sampling.

Question 13:

How many communication skills are there?

- A. 10

- B. 5
- C. 1
- D. 2

Answer: D.

Explanation

There are 2. Feedback and presentations.

Question 14:

Which of the following is considered an input to Directing & Managing project work?

- A. Project Management Plan
- B. Meetings
- C. Deliverables
- D. Issues Log

Answer: A.

Explanation

Meetings are part of a Project Manager's Tools & Techniques for managing the project through the Executing phase. An Issues Log and Deliverables are both Outputs of the Executing phase. The inputs to Executing are: Project Management Plan, Project Documents, Approved Change Requests, Enterprise Environmental Factors (EEFs) and Organizational Process Assets (OPAs)

Question 15:

Kalani works for an airline and has been asked to update the Project Sponsor, and the sponsor's peer managers, on how the project team is going to ensure the new passenger jet he's introducing into the fleet meets the airline's documented quality standards. What Input to the Project Management Plan could Kalani create to both answer their questions and help him manage quality throughout the project life cycle?

- A. Stakeholder Engagement Plan
- B. Quality Management Plan
- C. Assumption Log

D. Scope Management Plan

Answer: B.

Explanation

One of the outputs to the Plan Quality Management process group is the Quality Management Plan. The quality management plan is a component of the project management plan that describes how applicable policies, procedures, and guidelines will be implemented to achieve the quality objectives. It describes the activities and resources necessary for the project management team to achieve the quality objectives set for the project.

Question 16:

How many processes are there mapped to the Process Groups and Knowledge Areas?

- A. 40
- B. 45
- C. 49
- D. 50

Answer: C.

Explanation

Refer to the matrix displayed in this module for a complete review of the processes and how they are mapped to all Process Groups and Knowledge Areas.

Question 17:

Which is the overarching project management process that must be done at all times no matter which part of the project lifecycle you're in?

- A. Executing
- B. Planning
- C. Controlling & Monitoring
- D. Initiating

Answer: C.

Explanation

No matter whether you're just starting a project, deep in the planning phase, executing on your solution or closing up the project, a project manager is expected to control and monitor what's happening at every step of the way, and is therefore the only process that overarches all the others.

Question 18:

Tammy is so focused on managing the 98 members of her project team that when Accounting advises that she left the resource cost estimate out of her submitted project budget, she realizes that she never created this type of plan, within which all of those estimated resource hours and amounts would have been documented.

- A. Resource Management
- B. Schedule Management
- C. Budget Management
- D. Scope Management

Answer: A.

Explanation

The resource management plan provides information on rates (personnel and other resources), estimation of travel costs, and other foreseen costs that are necessary to estimate the overall project budget.

Question 19:

One of the two major influences on projects is known as EEFs. What does "EEF" stand for?

- A. Enterprise Environmental Factors
- B. Empirical Evaluation Failures
- C. Evaluation Elemental Factors
- D. Enterprise Estimation Field

Answer: A.

Explanation

The correct answer is Enterprise Environmental Factors. EEFs originate from the environment outside of the project and often outside of the enterprise, and may have an impact at the organizational, portfolio, program, or project level.

Question 20:

There are 7 individuals or groups that comprise a project team. What is the group who will be in charge of providing project artifacts or completing specific tasks toward a certain end called?

- A. Key Stakeholders
- B. Deliverable Resources
- C. Deliverable Owners
- D. Executive Sponsor

Answer: C.

Explanation

Artifacts such as the project charter, business requirements document and Training Plan are all deliverables. So, too, are outputs of the project itself such as new process flows if you're improving processes, new policies and procedures and a new customer experience. Those responsible for delivering these items throughout the life of the project are the deliverable owners and should be called out in a responsibility matrix (more on that to come).

Question 21:

Fill in the blank: Plan Stakeholder _____ is the process of developing approaches to involve project stakeholders based on their needs, expectations, interests, and potential impact on the project.

- A. Engagement
- B. Input
- C. Meetings
- D. Communication

Answer: A.

Explanation

Plan Stakeholder Engagement is the process of developing approaches to involve project stakeholders based on their needs, expectations, interests, and potential impact on the project. The key benefit is that it provides an actionable plan to interact with stakeholders effectively. This process is performed periodically throughout the project as needed. Having meetings with stakeholders, communicating with them and getting their input are all

things that need to be done as part of engagement.

Question 22:

Fill in the blank: Portfolio management aligns portfolios with organizational strategies by selecting the right programs or projects, prioritizing the work, and providing the needed _____.

- A. Resources
- B. Assets
- C. Project Manager
- D. Funds

Answer: A.

Explanation

While money is necessary to fund any project - even if it's to pay the salaries of internal employees as the project's only cost - providing that is simply one of several ways that needed resources can be supplied (e.g. via buying needed materials or hiring consultants). Organizational assets of many kinds also fall under the broader label of resources but simply stating "needed assets" is too limiting for the broader needs of portfolio management. And finally, while many may argue that having the right project manager could make or break a project, a project manager is once again just one of many different resources a portfolio may require to execute its objectives.

Question 23:

You've engaged a contractor to deliver a specific number of plexiglass display cases for a merchandise display in the new store you're building as Construction Project Manager for a retail business. They are on a CPFF contract and you as the Project Manager are responsible for monitoring payments to them as well as their adherence to the schedule and quality. What two kinds of payments are you making to this vendor?

- A. All allowable costs plus a fixed fee that was calculated as a percentage of the initial estimated project costs
- B. A fixed price with additional agreed-upon incentives if they exceed performance expectations
- C. The actual time they work and the materials they purchase in order to complete the work

- D. All legitimate costs, but mostly coming from an assessment of how well they met the pre-agreed-upon criteria for what constitutes satisfactory deliverables

Answer: A.

Explanation

The definition contained within Response A is T&M or Time and Materials. The definition contained within Response C is for Cost plus award fee (CPAF). The definition contained within Response D is Fixed price incentive fee (FPIF). The correct definition to CPFF is Response B, because CPFF = Cost plus fixed fee.

Question 24:

Saheed's General Contractor has just finished laying the foundation for the first of seven houses being built within a new subdivision project he's managing. He's about to meet with the person leading the GC's team so they can perform a walkthrough of the site together to ensure a mutual understanding of the work in progress. What is this walkthrough an example of, within the context of monitoring and controlling vendor work?

- A. Claims Administration
- B. Inspection
- C. Data Analysis
- D. Audit

Answer: B.

Explanation

An inspection is a structured review of the work being performed by the contractor. This may involve a simple review of the deliverables or an actual physical review of the work itself. On a construction/engineering/infrastructure project, inspections involve walkthroughs of the site by both the buyer and the contractor to ensure a mutual understanding of the work in progress.

Question 25:

Once a project management plan has been baselined, what is the only

way through which any changes may be made to it (which process group)?

- A. Close Project or Phase
- B. Validate Scope
- C. Monitor and Control Project Work
- D. Perform Integrated Change Control

Answer: D.

Explanation

Before the baselines are defined, the project management plan may be updated as many times as necessary. No formal process is required at that time. But, once it is baselined, it may only be changed through the Perform Integrated Change Control process. Consequently, change requests will be generated and decided upon whenever a change is requested.

Question 26:

The lessons learned register is created as an output of which process early in the project?

- A. Monitor and Control Project Work
- B. Develop Project Management Plan
- C. Manage Project Knowledge
- D. Direct and Manage Project Work

Answer: C.

Explanation

The lessons learned register is created as an output of this process early in the project. Thereafter it is used as an input and updated as an output in many processes throughout the project. The persons or teams involved in the work are also involved in capturing the lessons learned. Knowledge can be documented using videos, pictures, audios, or other suitable means that ensure the efficiency of the lessons captured. At the end of a project or phase, the information is transferred to an organizational process asset called a lessons learned repository.

Question 27:

Your project sponsor asks why you've included a 15% contingency in your cost estimate. In which subsidiary plan would he have seen this?

- A. Cost Management Plan
- B. Requirements Traceability Matrix
- C. Risk Management Plan
- D. Project Charter

Answer: A.

Explanation

The Cost Management Plan contains cost estimates, which generally includes consideration of risk response planning, such as contingency plans. A Risk Management Plan would have defined the risk and even the mitigation strategy, but it would not have explicitly defined an associated contingency cost. The Requirements Traceability Matrix documents business requirements, while the project charter, completed in the Initiating phase, will generally not do much more with costs than provide a very high-level dollar amount, if any at all.

Question 28:

Andrei has gained a reputation during his decade as a project manager, of being the one who can negotiate with even the toughest of vendors and project team members. He's also highly skilled at problem solving and is known for his interpersonal skills. No matter the conflicts that arise, Andrei is always given consistently high praise from his project team members, stakeholders and external customers and vendors alike. Which of the three project manager competencies from the Talent Triangle does Andrei demonstrate?

- A. Strategic and Business Management
- B. Technical Project Management
- C. Dealing With Difficult People
- D. Leadership

Answer: D.

Explanation

Leadership skills involve the ability to guide, motivate, and direct a team.

These skills may include demonstrating essential capabilities such as negotiation, resilience, communication, problem solving, critical thinking, and interpersonal skills. Projects are becoming increasingly more complicated with more and more businesses executing their strategy through projects. Project management is more than just working with numbers, templates, charts, graphs, and computing systems. A common denominator in all projects is people. People can be counted, but they are not numbers.

Question 29:

The definition of project management is the application of knowledge, skills, tools, and techniques to project activities to meet the what?

- A. Project requirements
- B. Deadlines set forth by the Board of Directors
- C. Step by step outline of project management process found in the PMBOK
- D. Sponsors and stakeholder's expectations

Answer: A.

Explanation

Though expectations and deadlines are important to a project manager, first and foremost their duty is to meet project requirements, in whatever form those requirements take.

Question 30:

For which project management process can the Requirements Traceability Matrix (RTM) be both an Input and an Output?

- A. Identify Stakeholders
- B. Plan Procurement Management
- C. Create WBS
- D. Determine Budget

Answer: B.

Explanation

As an input, you are potentially using the RTM to inform your Procurement Management Plan, whereas the results of decisions that are made within the

Plan Procurement Management process may require you to update your RTM with new information. The RTM is not typically created yet at the time you're identifying stakeholders, and while it may help inform the processes required to create a WBS and determine a budget, the RTM is not an output of either of those processes.

Question 31:

In iterative life cycle projects, a high-level vision will be developed for the overall project, but the detailed scope is determined one iteration at a time, and the detailed _____ for the next iteration is carried out as work progresses on the current project scope and deliverables.

- A. budget
- B. requirements gathering
- C. analysis
- D. planning

Answer: D.

Explanation

The preparation of a detailed project scope statement builds upon the major deliverables, assumptions, and constraints that are documented during project initiation. During project planning, the project scope is defined and described with greater specificity as more information about the project is known. Existing risks, assumptions, and constraints are analyzed for completeness and added or updated as necessary. The Define Scope process can be highly iterative. In iterative life cycle projects, a high-level vision will be developed for the overall project, but the detailed scope is determined one iteration at a time, and the detailed planning for the next iteration is carried out as work progresses on the current project scope and deliverables.

Question 32:

While Sophie is updating the cost and schedule forecasts for the project she's managing, she realizes that as they proceed with executing the project management plan there are three other project documents she needs to keep updated. Which of the following is NOT one of those she should be updating while monitoring and controlling her project?

- A. Risk Register

- B. Issues Log
- C. Project Charter
- D. Lessons Learned Register

Answer: C.

Explanation

Project documents that may be updated as a result of this process include but are not limited to: -Cost forecasts, -Issue log, -Lessons learned register, -Risk register, and -Schedule forecasts. A project charter is a one-and-done point-in-time document that is not ever updated after it is signed off.

Question 33:

Scope, Time and Cost are three sides of what is sometimes called the Iron Triangle, Talent Triangle or Triple Constraints of a project. What's important about these three things?

- A. Time and Cost provide you with what's in the scope of the project.
- B. You have to track them separately without any connection among them.
- C. Changing one of them will change both of the others in some way.
- D. They are the responsibility of the project's Executive Sponsor to monitor on a daily basis.

Answer: C.

Explanation

While the "Triple Constraint" model referred more to the the quality management of a project originally, it's come to be used also as a tool for project managers to define how a change in schedule will affect the scope and/or the budget, how a change in scope will affect the schedule and or budget and how a change in budget will impact the schedule and/or scope.

Question 34:

Which of the following measures is considered the most critical EVA (Earned Value Analysis) metric and measures the cost efficiency for the work completed?

- A. VAC (Variance at Completion)

- B. AC (Actual Cost)
- C. CV (Cost Variance)
- D. CPI (Cost Performance Index)

Answer: D.

Explanation

The cost performance index (CPI) is a measure of the cost efficiency of budgeted resources, expressed as a ratio of earned value to actual cost. It is considered the most critical EVA metric and measures the cost efficiency for the work completed. Equation: $CPI = EV \text{ (Earned Value)} / AC \text{ (Actual Cost)}$.

Question 35:

Harry's been tasked with being the project manager on one country's side of a commercial venture. His counterpart at the company his firm has entered into a manufacturing agreement with, a man called Qammar, has asked for the two of them to meet soon for a discussion about the politics between their countries and the differences in culture within their companies, both of which may pose risks to the project. Within the Plan Communications Management process group, which interpersonal and team skills are both Harry and Qammar demonstrating they understand the importance of here?

- A. Political awareness
- B. Cultural awareness
- C. Both Political awareness and Cultural awareness
- D. Communication style assessment

Answer: C.

Explanation

Operating from two different countries, especially if those countries have fundamental differences in culture and a contentious political relationship, is a challenge for any project manager and is a risk to any project. Qammar and Harry hope to mitigate both political and cultural risks by discussing openly and honestly those things that may need mitigation strategies or require different ways of communicating outward and upward, as a result.

Question 36:

What does the emerging trend acronym PMIS stand for, keeping in mind it's something referred to in the Monitoring & Controlling process group?

- A. Project Management Information System
- B. Portfolio Management Information System
- C. Process Management Information System
- D. Program Management Information System

Answer: A.

Explanation

A PMIS or project management information system can provide continuous monitoring and insight into the health of projects and identify areas needing attention and focus. The PMIS is one of the tools and techniques of the Monitor and Control Project Work process.

Question 37:

Ediz has been talking to a few other project managers about various communication tools and techniques they use, at his PMI chapter's annual celebration dinner. While discussing the different PMISs they're all using, Ediz shares that he prefers providing status reports via a combination of web conferencing, since his stakeholders are scattered across the United States, and publishing the reports and a project monitoring dashboard on a password-protected website only they can access, which also serves as an online project management tool they use to provide statuses to each other. Which PMIS Tool or Technique are Ediz's preferred methods an example of?

- A. Electronic project management tools
- B. Electronic communications management
- C. All three of these answers are correct
- D. Social media management

Answer: C.

Explanation

The following definitions/examples from the PMBOK demonstrate why Ediz's preferences demonstrate all three of the PMIS tools and techniques

under Project Management Information System (PMIS): -Electronic project management tools: Project management software, meeting and virtual office support software, web interfaces and work management tools -Electronic communications management: Email, fax, and voicemail; audio, video and web conferencing; and websites and web publishing -Social media management: Websites and web publishing; and blogs and applications, which offer the opportunity to engage with stakeholders and form online communities

Question 38:

Benjamin is an IT project manager in the aerospace industry, where technology and functionality of equipment are changing daily. He knows that the project manager stays informed about current industry trends. He takes this information and sees how it may impact or apply to the current projects. How many trends does he need to keep on top of in his industry in order to meet these demands upon his expertise?

- A. 7
- B. 12
- C. 10
- D. 6

Answer: A.

Explanation

The correct answer is 7. The project manager stays informed about current industry trends. The project manager takes this information and sees how it may impact or apply to the current projects. These trends include but are not limited to: Product and technology development; New and changing market niches; Standards (e.g., project management, quality management, information security management); Technical support tools; Economic forces that impact the immediate project; Influences affecting the project management discipline; and Process improvement and sustainability strategies.

Question 39:

The Code of Ethics and Professional Conduct includes both aspirational standards and mandatory standards for project managers. The Code was created to instill confidence in the project management profession

and to help an individual in making wise decisions, particularly when faced with difficult situations where the individual may be asked to compromise his or her _____ or values.

- A. PMP Designation
- B. Knowledge
- C. Beliefs
- D. Integrity

Answer: D.

Explanation

PMI publishes the Code of Ethics and Professional Conduct [5] to instill confidence in the project management profession and to help an individual in making wise decisions, particularly when faced with difficult situations where the individual may be asked to compromise his or her integrity or values. The values that the global project management community defined as most important were responsibility, respect, fairness, and honesty. The Code of Ethics and Professional Conduct affirms these four values as its foundation.

Question 40:

How many Knowledge Areas are encompassed within the Closing process group?

- A. 10
- B. 1
- C. 6
- D. 4

Answer: B.

Explanation

Only Knowledge Area 4, Project Integration Management, has a component within the Closing process group, and that is: Close Project or Phase.

Question 41:

Of the four categories of life cycles, which has the goal "Correctness of Solution?"

- A. Incremental
- B. Agile
- C. Iterative
- D. Predictive

Answer: C.

Explanation

Agile Practice Guide Section 3.1 p.18 Table 3-1. Characteristics of Four Categories of Life Cycles states that the goal of Predictive is Manage Cost, the goal of incremental is speed, the goal of iterative is correctness of solution and the goal of agile is customer value via frequent deliveries and feedback.

Question 42:

There are four generic phases of a project life cycle and though not all five of the process groups match those four phases exactly, the Closing process group - while also being able to apply to individual project phases - most closely aligns itself to which generic project phase?

- A. Starting the project
- B. Ending the project
- C. Carrying out the work
- D. Organizing and Preparing

Answer: B.

Explanation

Ending the project is where you will be able to apply all of the Closing process groups though you can also use some of them if you are closing out contracts or project phases independent of the full and complete end of a project.

Question 43:

After some discussion with project team members, you discover that you know a lot more about details for near-term deliverables than you do about the more nebulous details further out. What kind of planning would work well for your project?

- A. Six Sigma

- B. Critical Path
- C. Waterfall
- D. Rolling Wave

Answer: D.

Explanation

Six Sigma focuses on business process improvements. Critical path determines the duration of the longest path through the project. The waterfall methodology, also called predictive, is a plan-driven approach to project management and within the waterfall and agile methodologies both, can exist the Rolling Wave for scenarios such as this. Rolling Wave is an iterative which allows more focus upon what is known near-term and less upon what happens further out in the project.

Question 44:

How many interpersonal and team skills techniques are there?

- A. 17
- B. 16
- C. 10
- D. 15

Answer: A.

Explanation

There are 17. Active listening, communication styles assessment, conflict management, cultural awareness, decision making, emotional intelligence, facilitation, influencing, leadership, meeting management, motivation, negotiation, networking, nominal group technique, observation/conversation, political awareness and team building.

Question 45:

Which of the following processes is NOT part of Project Schedule Management?

- A. Develop schedule
- B. Estimate costs
- C. Estimate Activity Durations

D. Define Activities Sequence

Answer: B.

Explanation

Project schedule management contains the following processes within the Planning phase: Plan Schedule Management, Define Activities, Sequence Activities, Estimate Activity Durations and Develop Schedule. Estimate costs is its own separate and distinct knowledge area with its own inputs, tools & techniques and outputs.

Question 46:

If you work in a hospital's Information Technology department, and you are in charge of managing only network and server changes because you're IT, while the neuroscience managers are responsible for implementing a remote stroke assessment network, the billing department manager is responsible for implementing new billing codes hospital-wide and the Facilities manager is responsible for the project to modernize all hospital elevators, what structural type of organization is your employer?

- A. Balanced matrix organization
- B. Functional organization
- C. Weak matrix organization
- D. Projectized organization

Answer: B.

Explanation

Each department in a functional organization will do its project work independently of other departments. That's not to say that other departments don't get called upon for tasks, such as the neuroscience manager requiring IT to open up network ports or firewalls for their software to work, but the actual project management happens within each functional silo rather than from a centralized Project Management Office.

Question 47:

How many decision-making techniques are there?

- A. 3
- B. 1
- C. 5
- D. 2

Answer: D.

Explanation

There are 2. Multicriteria decision analysis and voting.

Question 48:

Methodology is a system of practices, techniques, procedures, and rules used by those who work in a discipline which project managers apply in their management. (PMBOK is NOT a methodology, but a recommended reference for what practice?

- A. program management
- B. tailoring
- C. project management
- D. product management

Answer: B.

Explanation

Methodology is a system of practices, techniques, procedures, and rules used by those who work in a discipline which project managers apply in their management. (PMBOK is NOT a methodology, but a recommended reference for tailoring). Tailoring is necessary because each project is unique.

Question 49:

All of the following are Tools and Techniques of the Close Phase, Project or Contract process group EXCEPT:

- A. Data Analysis
- B. Expert Judgments
- C. Meetings
- D. Stakeholder Analysis

Answer: D.

Explanation

Stakeholder Analysis is a Tool and Technique from the Initiating process group. Responses A, B and C are the three Tools and Techniques from the Closing process group.

Question 50:

A project is contracted on a Cost-Plus-Incentive-Fee (CPIF) basis. The contract states that if the final costs are less than expected costs, the sharing formula for cost savings is 80:20. The targeted cost is US\$750,000 with a 10 percent fee. If the project comes in at US\$720,000, what would be the total cost of the contract?

- A. US \$756,000
- B. US \$720,000
- C. US \$750,000
- D. US \$736,000

Answer: A.

Explanation

In a Cost-Plus-Incentive-Fee (CPIF) type of contract, the seller is reimbursed for allowable costs for performing the contract work and receives a predetermined fee. In some cases, if the final costs are less than the expected costs, then both the buyer and the seller benefit from the cost saving based on a pre-negotiated sharing formula. For this question: the predetermined fee is 10% of US\$750,000 = US\$75,000. Since the project came in at US\$720,000, the cost saving is $750,000 - 720,000 = 30,000$. The sharing formula is 80:20, so the additional payout to the seller = $(20/100) * 30,000 = 6,000$. Therefore, the total cost of the contract = $720,000 + 30,000 + 6,000 = \text{US\$}756,000$. Clearly this is to the disadvantage of the buyer, whose original estimate was less than this amount.

Question 51:

Dana is in the midst of facilitating a workshop that her project stakeholders are attending, in which she has asked them to write down one project risk per Post-It Note and identify where on a four-quadrant diagram the risk should be placed by judging whether it's high or low impact and high or low probability. What type of analysis is Dana facilitating?

- A. Quantitative Risk Analysis
- B. Identify Risks
- C. Qualitative Risk Analysis
- D. Plan Risk Responses

Answer: C.

Explanation

Perform Qualitative Risk Analysis is the process of prioritizing individual project risks for further analysis or action by assessing their probability of occurrence and impact as well as other characteristics. The key benefit of this process is that it focuses efforts on high-priority risks. 1. Identify Risks. 2. Perform Qualitative Risk Analysis. 3. Perform Quantitative Risk Analysis. 4. Plan Risk Responses.

Question 52:

The time has come for Devon to decide which tools he's going to use to manage configuration change control on his software development project. Which of the following is NOT a potential tool he could choose for this specific purpose?

- A. Requirements Traceability Matrix
- B. Identify configuration item
- C. Perform configuration item verification and audit
- D. Record and report configuration item status

Answer: A.

Explanation

Of the options presented, the only one that is not a tool for configuration change control is the requirements traceability matrix, which is a grid that links product requirements from their origin to the deliverables that satisfy them. In order to facilitate configuration and change management, manual or automated tools may be used. Configuration control is focused on the specification of both the deliverables and the processes, while change control is focused on identifying, documenting, and approving or rejecting changes to the project documents, deliverables, or baselines. Tool selection should be based on the needs of the project stakeholders including organizational and environmental considerations and/or constraints. Tools should support the

following configuration management activities: -Identify configuration item. Identification and selection of a configuration item to provide the basis for which the product configuration is defined and verified, products and documents are labeled, changes are managed, and accountability is maintained. -Record and report configuration item status. Information recording and reporting about each configuration item. -Perform configuration item verification and audit. Configuration verification and configuration audits ensure that the composition of a project's configuration items is correct and that corresponding changes are registered, assessed, approved, tracked, and correctly implemented. This ensures that the functional requirements defined in the configuration documentation are met.

Question 53:

During which project phase does Monitoring and Controlling project work really ramp up?

- A. Initiating
- B. Planning
- C. Closing
- D. Executing

Answer: D.

Explanation

Monitoring and Controlling project work ramps up once the project is in execution.

Question 54:

Which of the following is not an Interpersonal and Team Skill used to Develop a Project Charter?

- A. Facilitation
- B. Meeting Management
- C. Focus Groups
- D. Conflict Management

Answer: C.

Explanation

Facilitation: the ability to effectively guide a group event to a successful decision, solution, or conclusion. Conflict Management can be used to help bring stakeholders into alignment on the objectives, success criteria, high-level requirements, project description, summary milestones, and other elements of the charter. Meeting Management: includes preparing the agenda, ensuring that a representative for each key stakeholder group is invited, and preparing and sending the follow-up minutes and actions. On the other hand, Focus Groups are a way to gather data for the project charter, during which you may use any of the other three - the interpersonal and team skills - to elicit what you need from the focus group.

Question 55:

Project management is divided into how many knowledge areas?

- A. 7
- B. 5
- C. 6
- D. 10

Answer: D.

Explanation

Project Management Divided into 10 Knowledge Areas (Integration, Scope, Schedule, Cost, Quality, Resource, Communication, Risk, Procurement and Stakeholder Management)

Question 56:

It's time for Emma to do her weekly status report again. It's a very important tool she's found is effective at letting all of her stakeholders know where things stand at the end of each week, regardless of what level of engagement the stakeholder is at. What input to "Manage Communications" is this status report an example of?

- A. Project Documents
- B. Project Management Plan
- C. EEFs and OPAs
- D. Work Performance Reports

Answer: D.

Explanation

Work Performance Reports -Reports are circulated to the project stakeholders through this process -Can include status and progress reports -Work performance reports can contain earned value graphs and information, trend lines and forecasts, defect histograms, contract performance information, and risk summaries -They can be presented as dashboards, heat reports, stop light charts, or other representations useful for creating awareness and generating decisions and actions

Question 57:

George is tasked with standing up a new retail store for a medical marijuana business in a state where medicinal use has been legalized. However, the state requires any organization who's opening up a medical marijuana dispensary to adhere to some very strict guidelines, many of which must occur in a very specific order, such as obtaining certain licenses and passing several inspections before the company can begin construction on their new building. Which input to "Sequencing Activities" do all of these government regulations he must abide by fall under?

- A. Organizational Process Assets
- B. Enterprise Environmental Factors
- C. Activity List
- D. Project Management Plan

Answer: B.

Explanation

The enterprise environmental factors that can influence the Sequence Activities process include but are not limited to: -Government or industry standards, -Project management information system (PMIS), -Scheduling tools, and -Organization work authorization systems.

Question 58:

Which of these Tools & Techniques is NOT used to Manage Stakeholder Engagement?

- A. Advertising
- B. Management Skills

- C. Interpersonal Skills
- D. Communication Methods

Answer: A.

Explanation

Advertising is one of the Tools & Techniques used in Conducting Procurements. Different stakeholders may require different communication methods, or specific interpersonal or management skills from us as the project managers, so our toolkit needs to be full of ways to do this.

Question 59:

Susan is putting together three agile teams to address the project she's just been assigned to. Of the following, which fits into the ideal number of people that she should have on each team?

- A. 11
- B. 5
- C. 2
- D. 10

Answer: B.

Explanation

Agile teams focus on rapid product development so they can obtain feedback. In practice, the most effective agile teams tend to range in size from three to nine members. Ideally, agile teams are colocated in a team space. Team members are 100% dedicated to the teams. Agile encourages self-managing teams, where team members decide who will perform the work within the next period's defined scope. Agile teams thrive with servant leadership. The leaders support the teams' approach to their work.

Question 60:

$tE = (tO + tM + tP) / 3$ is the formula for Three-Point Estimating in which you estimate project duration with very little historical data to go on. In the equation, E represents the Estimate, O the Optimistic duration based on the best-case scenario and M is the Most Likely activity duration given a number of analyzed factors. What does the P represent?

- A. Pie-in-the-sky
- B. Pessimistic
- C. Prescient
- D. Purposeful

Answer: B.

Explanation

Pessimistic is correct. The purpose of three-point estimation in order to determine your best activity duration is to identify the best-case (optimistic), worst-case (pessimistic) and most likely scenarios, and then essentially take the average of them by dividing the three numbers by 3. So if your best-case is 10 hours, your worst-case is 24 hours and your most likely duration is 17 hours, you add $10 + 24 + 17$, which equals 51. And then 51 divided by 3 is 17 (and if you use the Excel AVERAGE formula, you'll find you get the same answer). This method is beneficial if you are basing your activity durations on peoples' judgments rather than known data. For example, if you know it takes 8 hours from start to finish to stand up a working server, then you don't need three-point estimation because you already know that activity's duration for a fact.

Question 61:

Fill in the blank: _____ analysis is an analytical technique used to determine the basic underlying reason that causes a variance, defect, or risk.

- A. Process Improvement
- B. Documented Data
- C. Alternative Process
- D. Root Cause

Answer: D.

Explanation

Root cause analysis is an analytical technique used to determine the basic underlying reason that causes a variance, defect, or risk. A root cause may underlie more than one variance, defect, or risk. It may also be used as a technique for identifying root causes of a problem and solving them. When all root causes for a problem are removed, the problem does not recur. The

other three options as shown are not valid processes within the Project Management Book of Knowledge.

Question 62:

What are the three types of bid documents that can be used when seeking bids for goods or services for external companies?

- A. SOW, RFI and RFP
- B. RFP, SOW and TOR
- C. RFI, RFQ and SOW
- D. RFI, RFQ and REP

Answer: D.

Explanation

A procurement Statement of Work (SOW) is one component of an RFI, RFQ or RFP. It is not, on its own, used as a bid document. A TOR, or Terms of Reference, is similar to an SOW, often the two terms being used interchangeably. Like an SOW, a TOR is merely a component of bid documents, but is not used on its own for such. An RFI (Request for Information), RFQ (Request for Quote) and an RFP (Request for Proposal) are the standard bid documents utilized when an organization wants to solicit external vendors for goods or services.

Question 63:

While writing his Risk Management Plan, Steve's been interviewing project stakeholders and recording various of their responses to his questions regarding how much variance from what's planned they'll accept. For example, the project sponsor is okay if the budget is never more exact than plus or minus 25% of its current forecast, while an Information Systems stakeholder will not accept any deviation from the technical requirements that she provided to the RFP process the project undertook. In the meantime, the portfolio steering committee has made it clear they will be tolerant of a 4% change in budget from what's forecasted. Which output of the Risk Management Plan is being described here?

- A. Reporting Formats
- B. Definitions of risk probability and impacts

- C. Stakeholder risk appetite
- D. Risk Management Plan

Answer: C.

Explanation

Stakeholder risk appetite. The risk appetites of key stakeholders on the project are recorded in the risk management plan, as they inform the details of the Plan Risk Management process. In particular, stakeholder risk appetite should be expressed as measurable risk thresholds around each project objective. These thresholds will determine the acceptable level of overall project risk exposure, and they are also used to inform the definitions of probability and impacts to be used when assessing and prioritizing individual project risks.

Question 64:

Magnus has inherited a project that was handled from inception as a predictive life cycle project. However, he quickly discovers through trial and error that the stakeholders don't exactly know what they want until they see it, and decides it would be better to provide them with the first part of the new toy car they're building, then add the next bit of functionality (remote control) to it, followed by the ability for it to climb over obstacles in its path, then to spread hidden wings out from its sides and finally to launch itself into the air and fly. The team agrees that each milestone can be met in 3 months. What type of project life cycle is Magnus switching to?

- A. Adaptive
- B. Incremental
- C. Hybrid
- D. Iterative

Answer: B.

Explanation

In a predictive life cycle, the project scope, time, and cost are determined in the early phases of the life cycle. Any changes to the scope are carefully managed. Predictive life cycles may also be referred to as waterfall life

cycles. -In an iterative life cycle, the project scope is generally determined early in the project life cycle, but time and cost estimates are routinely modified as the project team's understanding of the product increases. - Iterations develop the product through a series of repeated cycles, while increments successively add to the functionality of the product. -In an incremental life cycle, the deliverable is produced through a series of iterations that successively add functionality within a predetermined time frame. The deliverable contains the necessary and sufficient capability to be considered complete only after the final iteration. -Adaptive life cycles are agile, iterative, or incremental. The detailed scope is defined and approved before the start of an iteration. Adaptive life cycles are also referred to as agile or change-driven life cycles. -A hybrid life cycle is a combination of a predictive and an adaptive life cycle. Those elements of the project that are well known or have fixed requirements follow a predictive development life cycle, and those elements that are still evolving follow an adaptive development life cycle.

Question 65:

Which project roles are responsible for identifying both individual and overall project risks?

- A. Project Manager
- B. Business Analyst
- C. Project Sponsor
- D. All project roles

Answer: D.

Explanation

The Project Management Book of Knowledge (PMBOK) makes it clear that "all project stakeholders should be encouraged to identify individual project risks. It is particularly important to involve the project team so they can develop and maintain a sense of ownership and responsibility for identified individual project risks, the level of overall project risk, and associated risk response actions."

Question 66:

After closing a project out, the deliverables go to the end users. What happens to the project records?

- A. They're taken by an auditing team
- B. You hand them off to the next project manager
- C. They become part of the Organizational Process Assets (OPAs)
- D. They become part of the Enterprise Environmental Factors (EEFs)

Answer: C.

Explanation

They become part of the Organizational Process Assets (OPAs). Things like Lessons Learned and Close Reports may provide valuable insights to everything from forensic audits of the project to other project managers. But you don't ever just give the project assets to other teams or departments; you ensure they become part of the organization's core knowledge so that anyone in future may access and benefit from it.

Question 67:

Tabitha's halfway finished with her project initiation tasks and is contemplating when she should hold her project kick-off meeting. The project is small, involving only two departments and their mutual processes and the same team is handling both the initiation and the planning processes. By this criteria, when should Tabitha schedule her kick-off meeting to occur?

- A. shortly after Initiation, in the Planning process group
- B. at the beginning of each project phase
- C. at the end of the Planning process group, at the start of the development / implementation in the Executing process group
- D. she should have conducted it before she started the Initiating processes

Answer: A.

Explanation

For small projects, there is usually only one team that performs the planning and the execution. In this case, the kick-off occurs shortly after initiation, in the Planning Process Group, because the team is involved in planning. In large projects, a project management team normally does the majority of the planning, and the remainder of the project team is brought on when the initial planning is complete, at the start of the development/implementation. In this

instance, the kick-off meeting takes place with processes in the Executing Process Group. Only multiphase projects will typically include a kick-off meeting at the beginning of each phase.

Question 68:

Which of the following does not describe a Project Manager's role in Managing Project Resources?

- A. The project manager should ensure that all procurements meet the specific needs of the project, while working with procurement specialists to ensure organizational policies are followed
- B. The project manager is responsible for proactively developing team skills and competences while retaining and improving team satisfaction and motivation
- C. The project manager should be both the leader and the manager of the project team

Answer: A.

Explanation

While procurements include project resource management, the key to knowing that it was NOT one of the truths of Project resource management is that this is explicitly called out as part of the project procurement process. That procurement might be one of the ways that the project manager accomplishes their need to acquire team members or obtain equipment required to complete the project.

Question 69:

A project charter is not considered to be a _____ because there is no consideration or money promised or exchanged in its creation.

- A. Project Output
- B. Deliverable
- C. Contract
- D. Project Artifact

Answer: C.

Explanation

An approved project charter formally initiates a project, but contractual agreements are handled separately, often with the Procurement Management arena.

Question 70:

Your project has undergone two change orders and you need to gain an understanding of how those together have impacted the original baselined project budget solidified during Planning. Which of the following variance analysis formulas would you use?

- A. $SV = EV - PV$
- B. $CPI = EV/AC$
- C. $CV = EV - AC$
- D. $VAC = BAC - EAC$

Answer: C.

Explanation

$SV = EV - PV$ Schedule Variance = Earned Value - Planned Value
 $VAC = BAC - EAC$ Variance at Completion = Budget at Completion - Estimate at Completion
 $CPI = EV/AC$ Cost Performance Index = Earned Value - Actual Cost
 $CV = EV - AC$ Cost Variance = Earned Value - Actual Cost

Question 71:

Which of the following pairs are Tools & Techniques for Managing Quality?

- A. Decision-Making and Audits
- B. Problem-Solving and Quality Improvement Methods
- C. Data Gathering and Data Analysis
- D. All of these

Answer: D.

Explanation

External risks are: Regulatory, Competitors, Suppliers, Customer and Acts of God. Technology is one of the sub-categories under the Technical type of risk.

Question 72:

Which type of Project Management Office provides a consultative role to projects by supplying templates, best practices, training, access to information, and lessons learned from other projects?

- A. Directive
- B. External Consultant
- C. Supportive
- D. Controlling

Answer: C.

Explanation

Supportive PMOs provide a consultative role to projects by supplying templates, best practices, training, access to information, and lessons learned from other projects. This type of PMO serves as a project repository. The degree of control provided by the PMO is low. In contrast, Controlling PMOs provide support and require compliance through various means. The degree of control provided by the PMO is moderate. Compliance may include (Adoption of project management frameworks or methodologies, use of specific templates, forms, and tools, conformance to governance frameworks). And finally, Directive PMOs take control of the projects by directly managing the projects. Project managers are assigned by and report to the PMO. The degree of control provided by the PMO is high. Whether or not the PMI is an external consultant is irrelevant to this question.

Question 73:

Trevor is working to determine whether or not the next three deliverables of his marketing project will be as late as the three that have already put him behind schedule. Since they're all being delivered by the same vendor, Trevor devises a calculation that becomes a mathematical model that he can input deliverable type, estimated duration and his new calculated factor of "lag time" into, and obtain what he feels is a good projection for how late - or not - the next three deliverables are likely to be. What type of analysis has Trevor just used to make this determination?

- A. Cost-benefit analysis
- B. Variance analysis

- C. Trend Analysis
- D. Earned value analysis

Answer: C.

Explanation

Correct answer: Trend analysis is used to forecast future performance based on past results. It looks ahead in the project for expected slippages and warns the project manager ahead of time that there may be problems later in the schedule if established trends persist. This information is made available early enough in the project timeline to give the project team time to analyze and correct any anomalies. The results of trend analysis can be used to recommend preventive actions if necessary. Earned value provides an integrated perspective on scope, schedule, and cost performance. Variance analysis reviews the differences (or variance) between planned and actual performance. This can include duration estimates, cost estimates, resources utilization, resources rates, technical performance, and other metrics. And finally, Cost-benefit analysis helps to determine the best corrective action in terms of cost in case of project deviations.

Question 74:

Which of the following risk parameters is defined by the degree to which a risk is perceived to matter by one or more stakeholders?

- A. Dormancy
- B. Controllability
- C. Propinquity
- D. Detectability

Answer: C.

Explanation

In addition to assessing the likelihood and impact of a risk upon the project, there are nine other characteristics that the PMBOK advises may help you evaluate and develop other risks and risk mitigation strategies, including all of the answers here. But the one defined in the question is propinquity, which means that if a risk is perceived as being high then it's high even if the reality of the risk may not be as impactful as perception indicates.

Question 75:

Information such as objectives, success criteria, high level requirements and pre approved financial resources are to be found in which process's outputs?

- A. Develop Project Charter
- B. Define Activities
- C. Create WBS
- D. Develop Project Management Plan

Answer: A.

Explanation

The project charter is the document issued by the project initiator or sponsor that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities. It documents the high-level information on the project and on the product, service, or result the project is intended to satisfy, such as: Project purpose; Measurable project objectives and related success criteria; High-level requirements; High-level project description, boundaries, and key deliverables; Overall project risk; Summary milestone schedule; uu Pre Approved financial resources; Key stakeholder list; Project approval requirements (i.e., what constitutes project success, who decides the project is successful, and who signs off on the project); Project exit criteria (i.e., what are the conditions to be met in order to close or to cancel the project or phase); Assigned project manager, responsibility, and authority level; and Name and authority of the sponsor or other person(s) authorizing the project charter.

Question 76:

Kirstie's project team is having fun designing a new coffee maker intended to easily mimic the fancy drinks made at major coffee houses around the world. As the team brainstorms what features and functions they want the coffee maker to have, Project Manager Kirstie and assigned Business Analyst Michelle are asking the Subject Matter Experts various questions to help them refine what they want the product to do and how they want it to work. This team is involved in which aspect of Define Scope?

- A. Data Analysis
- B. Decision Making
- C. Expert Judgment
- D. Product Analysis

Answer: D.

Explanation

While it may take bits and pieces of all four of these categories to make the best coffee maker possible, the clear winner here is Product Analysis because, as the PMBOK 6th Edition shares with us on p. 153: Product analysis can be used to define products and services. It includes asking questions about a product or service and forming answers to describe the use, characteristics, and other relevant aspects of what is going to be delivered. Each application area has one or more generally accepted methods for translating high-level product or service descriptions into meaningful deliverables. Requirements are captured at a high level and decomposed to the level of detail needed to design the final product. Examples of product analysis techniques include but are not limited to: J175Product breakdown, Requirements analysis, Systems analysis, Systems engineering, Value analysis, and Value engineering.

Question 77:

Katsu really wants the risk workshop he's scheduled to be successful, because he knows there are a lot of risks inherent in the project he's managing. Add to that the fact that the project's two most impacted stakeholders have historically not gotten along well and he's understandably nervous about the outcome. What is the one factor mentioned in the Perform Qualitative Risk Analysis process section in the PMBOK, that Katsu could add to his risk workshop to increase effectiveness of the meeting?

- A. Risk Owner
- B. Business Analyst
- C. Skilled Facilitator
- D. Project Sponsor

Answer: C.

Explanation

To undertake qualitative risk analysis, the project team may conduct a specialized meeting (often called a risk workshop) dedicated to the discussion of identified individual project risks. The goals of this meeting include the review of previously identified risks, assessment of probability and impacts (and possibly other risk parameters), categorization, and prioritization. A risk owner, who will be responsible for planning an appropriate risk response and for reporting progress on managing the risk, will be allocated to each individual project risk as part of the Perform Qualitative Risk Analysis process. The meeting may start by reviewing and confirming the probability and impact scales to be used for the analysis. The meeting may also identify additional risks during the discussion, and these should be recorded for analysis. Use of a skilled facilitator will increase the effectiveness of the meeting.

Question 78:

Mickey is the project manager for a team largely composed of individuals who work remotely from their own home offices. There are several tasks that would be completed more efficiently if the team could be gathered into a single room together for at least two weeks, which means Mickey needs to factor into his Project Management Plan an estimation of travel costs and lodging for these team members. In which Project Management Plan component should Mickey document that information?

- A. Project Management Plan
- B. Scope Baseline
- C. Resource Management Plan
- D. Cost Management Plan

Answer: C.

Explanation

The Resource Management Plan, described in Section 9.1.3.1 of the PMBOK, is the correct answer. The resource management plan provides information on rates (personnel and other resources), estimation of travel costs, and other foreseen costs that are necessary to estimate the overall project budget. It is one of the components of the Project Management Plan.

Question 79:

Your project is humming along through Execution without too many difficulties and then suddenly a key stakeholder leaves the company and someone who hasn't been involved with the project takes over the role for the foreseeable future. What plan may you need to update - and then act upon - as a result?

- A. Procurement Plan
- B. Scope Management Plan
- C. Communications Plan
- D. Quality Management Plan

Answer: C.

Explanation

While monitoring communications on your project, you may find that the project communication plan may need to be adjusted from time to time if the project stakeholders change or if the engagement level calls for different strategies to keep communication open and working on the project.

Question 80:

As Kaya is speaking with the people who will be performing various tasks on her project, one of her resources, a newer employee named Annika, reveals that it will take longer than the estimated 2 days to provision a new physical server for the software they have to install. Annika estimates it'll take closer to three full days until she becomes more familiar with the company's server image. Which process that Kaya is performing is going to be affected by Annika's information?

- A. Sequence Activities
- B. Estimate Activity Durations
- C. Plan Schedule Management
- D. Define Activities

Answer: B.

Explanation

Estimating activity durations uses information from the scope of work, required resource types or skill levels, estimated resource quantities, and

resource calendars. Other factors that may influence the duration estimates include constraints imposed on the duration, effort involved, or type of resources (e.g., fixed duration, fixed effort or work, fixed number of resources), as well as the schedule network analysis technique used.

Question 81:

There are four types of risk, and within each of those types are sub-categories. All of the following are examples, or sub-categories, of external risks EXCEPT:

- A. Acts of God
- B. Competitors
- C. Technology
- D. Suppliers

Answer: C.

Explanation

External risks are: Regulatory, Competitors, Suppliers, Customer and Acts of God. Technology is one of the sub-categories under the Technical type of risk.

Question 82:

Which of the following areas are least likely to fall under the Initiating process group?

- A. Obtaining approval to proceed to the next phase of the project
- B. Developing a business case that can be used to inform the creation of a project charter.
- C. Understanding a project's purpose and objectives
- D. Engaging project stakeholders

Answer: B.

Explanation

A project manager must engage stakeholders to obtain the information necessary to complete a stakeholder register. The project manager must also understand a project's purpose and objectives well enough to articulate them in a project charter, and may be the person engaging decision-makers after

the charter's written, in order to obtain approval to proceed. The development of a business case does not happen within the project management life cycle at all, but is a precursor to something becoming a project, and thus while a project manager may have involvement at that stage, it occurs before the Initiating process begins.

Question 83:

When you work to Acquire Resources, which of the following are you NOT working to acquire?

- A. Materials
- B. Project Team Members
- C. Money
- D. Facilities

Answer: C.

Explanation

Project Resources are considered to be: team members, facilities, equipment, materials, supplies and other resources necessary to complete project work. Money falls under the definition of a project budget, and in most cases this is not something a project manager tries to acquire this far into the project.

Question 84:

Which output from a previous process group should the Project Manager review to ensure that all project work is completed and that the project has met its objectives in order to close a project (or phase)?

- A. Requirements Traceability Matrix (RTM)
- B. Project Charter
- C. Project Scope
- D. Project Management Plan

Answer: D.

Explanation

The Project Management Plan integrates the components of the other plans developed during the Planning processes. It is therefore the Project Management Plan which contains all of the information needed for a Project

Manager to assess whether or not it is valid to close a project or phase when the time comes. All of the other answers listed can be inputs to the Project Management Plan but do not contain enough information for this evaluation.

Question 85:

How do you manage a project's core objectives during the Executing phase?

- A. Managing Budget
- B. Managing time
- C. Managing Quality
- D. Managing Expectations

Answer: C.

Explanation

Though managing expectations, the budget and time are all important qualities of an effective Project Manager, a project's core objectives are specifically managed by ensuring what's being produced during the Executing phase is of acceptable quality. Specifically, Quality Management is the process of translating the Quality Management Plan into executable quality activities that incorporate the organization's quality policies into the project.

Question 86:

Victoria schedules a workshop with her project team members together to work on a team charter. She begins by explaining to them that the goal is to establish team values, agreements and operating guidelines they will all abide by in order to decrease misunderstandings and increase productivity. Which of the following items might the team charter they develop include?

- A. Project Organization Chart
- B. Training Strategies
- C. Meeting Guidelines
- D. Recognition Plan

Answer: C.

Explanation

The team charter is a document that establishes the team values, agreements, and operating guidelines for the team. The team charter may include but is not limited to: Team values, Communication guidelines, Decision-making criteria and process, Conflict resolution process, Meeting guidelines, and Team agreements. The team charter establishes clear expectations regarding acceptable behavior by project team members. Early commitment to clear guidelines decreases misunderstandings and increases productivity.

Question 87:

The Initiating process group contains within it components from which two knowledge areas?

- A. Integration Management and Communications Management
- B. Integration Management and Stakeholder Management
- C. Communications Management and Quality Management
- D. Scope Management and Schedule Management

Answer: B.

Explanation

There are two deliverables produced within the Initiating phase. The project charter is the component that falls under Integration Management. The Stakeholder Register falls under the Stakeholder Management knowledge area. The other knowledge areas listed do not have components in the Initiating phase.

Question 88:

Which of the following project processes are not repeated at the beginning of each iterative phase of a project using an agile (or adaptive) life cycle?

- A. Manage Project Knowledge
- B. Collect Requirements
- C. Create Work Breakdown Structure
- D. Define Scope

Answer: A.

Explanation

Managing project knowledge is something a project manager should be doing throughout the entire life cycle of a project regardless of what phase or iteration a project is in. Three processes (Collect Requirements, Define Scope, and Create WBS) are repeated for each iteration. On the contrary, in a predictive project, these processes are performed toward the beginning of the project and updated as necessary, using the integrated change control process.

Question 89:

If you engage any external vendors or contractors for project work, what do you have to ensure is closed out during the Closing process group?

- A. Contracts
- B. Formal communication between them and you
- C. REP
- D. Open risks

Answer: A.

Explanation

The contracts made with any external entities for project work must be closed out, which includes final payments and sign-off on completion. Though you may formally end communication between you and them, that is not a requirement of the Closing phase. Similarly, if there are open risks then the project is either not ended or those risks are being transferred to Operations as part of the Closing process. And finally, an RFP is a point-in-time document used to solicit bids from external vendors for the work you need performed or the goods you need to procure. An RFP is not a living-breathing document that will need to be closed by the time you've reached the end of the project.

Question 90:

Which of the following is not one of the three bidding documents used to solicit proposals from prospective sellers during the Plan Procurement Management process?

- A. Request for Proposal (REP)
- B. Request for Quotation (REQ)
- C. Request for Information (REI)

D. Statement of Work (SOW)

Answer: D.

Explanation

The RFI, RFQ and RFP are bid documents. The statement of work (SOW) for each procurement is developed from the project scope baseline and defines only that portion of the project scope that is to be included within the related contract. The SOW describes the procurement item in sufficient detail to allow prospective sellers to determine if they are capable of providing the products, services, or results. It is a component of the three bid documents.

Question 91:

Philippe is asked to complete a Stakeholder Engagement Assessment Matrix. He discovers that all of the identified stakeholders he can label as Neutral, as they are neither supportive or unsupportive of the project. However, there is one stakeholder who, when he sits down to meet with them, isn't even aware of the project or its impacts. How should Philippe label that particular stakeholder in his matrix?

- A. supportive
- B. unaware
- C. leading
- D. resistant

Answer: B.

Explanation

A stakeholder engagement assessment matrix supports comparison between the current engagement levels of stakeholders and the desired engagement levels required for successful project delivery. The engagement level of stakeholders can be classified as follows: -Unaware. Unaware of the project and potential impacts. -Resistant. Aware of the project and potential impacts but resistant to any changes that may occur as a result of the work or outcomes of the project. These stakeholders will be unsupportive of the work or outcomes of the project. -Neutral. Aware of the project, but neither supportive or unsupportive. -Supportive. Aware of the project and potential impacts and supportive of the work and its outcomes. -Leading. Aware of the project and potential impacts and actively engaged in ensuring that the

project is a success.

Question 92:

How many potential financial measures that a project needs to meet, could be documented in the business case?

- A. 20
- B. 10
- C. 5
- D. 15

Answer: D.

Explanation

One of the project success criteria is meeting the agreed-upon financial measures documented in the business case. These financial measures may include but are not limited to: Net present value (NPV), Return on investment (ROI), Internal rate of return (IRR), Payback period (PBP), and Benefit-cost ratio (BCR). Meeting business case nonfinancial objectives; Completing movement of an organization from its current state to the desired future state; Fulfilling contract terms and conditions; Meeting organizational strategy, goals, and objectives; Achieving stakeholder satisfaction; Acceptable customer/end-user adoption; Integration of deliverables into the organization's operating environment; Achieving agreed-upon quality of delivery; Meeting governance criteria; and Achieving other agreed-upon success measures or criteria (e.g., process throughput).

Question 93:

Cara works for a company that sells multi-functional furniture and she's tasked with a project to produce one new ten-page "how to" pamphlet every two weeks for sixteen weeks, in order to provide customers with quick and easy reference guides for their new multi-functional line of beds. Cara's team gained a little ground during the recent holiday season because none of them took vacations. At next week's steering committee meeting she is going to use a new EVM technique to explain to project stakeholders how far ahead of schedule the project is based on what they've accomplished versus when they originally meant to have accomplished that goal. What emerging Project Cost Management technique is Cara going to use?

- A. Earned Schedule (ES)
- B. Planned Value (PV)
- C. Actual Time (AT)
- D. Schedule Performance Index (SPI)

Answer: A.

Explanation

ES (Earned Schedule) is an extension to the theory and practice of EVM (Earned Value Management). Earned schedule theory replaces the schedule variance measures used in traditional EVM (earned value – planned value) with ES and actual time (AT). Using the alternate equation for calculating schedule variance $ES - AT$, if the amount of earned schedule is greater than 0, then the project is considered ahead of schedule.

Question 94:

Project governance refers to the framework, functions, and _____ that guide project management activities in order to create a unique product.

- A. Structure
- B. Phases
- C. Processes
- D. Templates

Answer: C.

Explanation

The correct answer is processes, as project governance requires them in order to provide benchmarks against which performance can be measured. Phases are how projects are structured and templates can be provided to create documents, but even if required by governance they are all capable of being referenced by processes as the overarching mechanism under the framework, to carry out the functions.

Question 95:

Lucius is writing a benefits management plan and one of the biggest reasons his company wants to undertake the project to redesign their customer-facing website is the first target benefit he includes: to provide

their customers with an enhanced user experience that results in a 20% increase in positive customer feedback from their next customer satisfaction survey. This is an example of which type of target benefit?

- A. ROI
- B. Cost-Based
- C. Tangible
- D. Intangible

Answer: D.

Explanation

Target benefits are the expected tangible and intangible business value to be gained by the implementation of the product, service, or result. ROI the calculation of Return On Investment and Cost-Based is not a valid measure of a target benefit.

Question 96:

In which of the project management processes will you find stakeholders engaging with the project for the first time?

- A. Plan Scope Management
- B. Identify Risks
- C. Develop Project Management Plan
- D. Develop Project Charter

Answer: D.

Explanation

While stakeholders are engaged throughout every phase of the project, and can be involved in all of the project management processes, inputs and outputs, the first time a stakeholder engages with the project is during the Initiation process of Develop Project Charter.

Question 97:

The PMBOK clearly defines differences between documents produced as part of, and for, the project itself versus documents created during the life of a project which are business documents. How many total project documents is it possible to create during the life of your project?

- A. 33
- B. 15
- C. 27
- D. 20

Answer: A.

Explanation

Table 1-2. Project Management Plan and Project Documents on page 559 of PMBOK 6th Edition lists 33 possible project documents that are created by, from and for the project itself. These are not to be confused with business documents such as Organizational Process Assets (OPAs).

Question 98:

Nadia works for a construction firm in a Canadian company. The construction season is so short due to inclement weather during the winter months that much of the outdoor construction has to be done during June, July and August. As she and her project team evaluate proposals from brick vendors for material that will be used in building a remotely located school, they realize that each of their three top choices come with their own set of potential problems that will have to be monitored closely if they're chosen. Once the team selects a vendor, where should Nadia record these things which may put their project's scope, schedule or budget in jeopardy?

- A. Requirements Traceability Matrix
- B. Assumption Log
- C. Issue Log
- D. Risk Register

Answer: D.

Explanation

Each approved seller comes with its own unique set of risks, depending on the seller's organization, the duration of the contract, the external environment, the project delivery method, the type of contracting vehicle chosen, and the final agreed-upon price. A Risk Register will help Nadia and her team ensure that their chosen vendor's known risks are monitored, and that mitigation strategies are identified and agreed upon with the vendor.

Question 99:

If your boss asks you to take on a collection of projects, programs, sub-portfolios, and operations managed as a group to achieve strategic benefits, which of the following titles would you be acting as?

- A. Operations Manager
- B. Portfolio Manager
- C. Program Manager
- D. Project Manager

Answer: B.

Explanation

It would be a portfolio manager who would oversee all these. A Project Manager manages a project. A Program Manager manages a group of related projects in a coordinated way. An Operations Manager might be a Stakeholder in your project, program or portfolio, but not the actual portfolio manager.

Question 100:

As you monitor project progress, you begin to realize that some of the assumptions the team made about resource availability at the time the Assumption Log was created may have been incorrect. Which of the following project documents might you need to update now, as an output to Monitoring and Controlling Project Work?

- A. Issues log and risk register
- B. Milestone List
- C. Risk Register
- D. Issue Log

Answer: A.

Explanation

Both the Issue Log and Risk Register are project documents that are outputs of the Monitoring and Controlling Project Work process, along with Cost Forecasts, Lessons Learned Register and Schedule Forecasts. The Milestone List is an Input only, while the others are both Inputs AND Outputs.

Question 101:

Directing and Managing Project Work and Managing Project Knowledge are the two components of which part of which of the following:

- A. Core Objectives Management
- B. Facilitation Management
- C. Executing Process
- D. Project Integration Management

Answer: D.

Explanation

Directing and managing project work is the process of leading and performing the work defined in the project management plan and implementing approved changes to achieve the project's objectives. Managing project knowledge is the process of using existing knowledge and creating new knowledge to achieve the project's objectives and contribute to organizational learning. These two components define Project Integration Management.

Question 102:

What must be resolved prior to formally closing a project, phase or contract?

- A. Issues
- B. Group Conflicts
- C. Risks
- D. Requirements

Answer: A.

Explanation

If risks still needed to be resolved, you wouldn't not be ready to close the project. If requirements are determined to not have been met, the project team can still agree to close the project with those caveats or other mitigation strategies in place. Although it's nice to want to remove group conflicts prior to disbanding the project team, that's not always possible nor is it required by the process. The only items which must be resolved - or assigned for

resolution post-completion - are issues which are still outstanding post-implementation.

Question 103:

Elena's project to implement a new cost accounting system within a sub-division of her company's Finance department has to be implemented three months before the end of the organization's fiscal year so the old system can be decommissioned prior to the new system taking on all the information required for year-end reporting. Which of the four PDM logical relationships exists between the old cost accounting system and the new one Elena's implementing?

- A. Finish-to-start (FS)
- B. Start-to-start (SS)
- C. Start-to-finish (SF)
- D. Finish-to-finish (FF)

Answer: C.

Explanation

The correct answer is Start-to-finish (SF). PDM includes four types of dependencies or logical relationships. A predecessor activity is an activity that logically comes before a dependent activity in a schedule. A successor activity is a dependent activity that logically comes after another activity in a schedule. -Finish-to-start (FS). A logical relationship in which a successor activity cannot start until a predecessor activity has finished. -Finish-to-finish (FF). A logical relationship in which a successor activity cannot finish until a predecessor activity has finished. -Start-to-start (SS). A logical relationship in which a successor activity cannot start until a predecessor activity has started. -Start-to-finish (SF). A logical relationship in which a successor activity cannot finish until a predecessor activity has started.

Question 104:

Which of the following does a cost management plan NOT establish?

- A. Level of Precision
- B. Level of Accuracy
- C. Units of Measure
- D. Contract Costs

Answer: D.

Explanation

The cost management plan is a component of the project management plan and describes how the project costs will be planned, structured, and controlled. The cost management processes and their associated tools and techniques are documented in the cost management plan. For example, the cost management plan can establish the following: -Units of measure. Each unit used in measurements (such as staff hours, staff days, or weeks for time measures; meters, liters, tons, kilometers, or cubic yards for quantity measures; or lump sum in currency form) is defined for each of the resources. -Level of precision. This is the degree to which cost estimates will be rounded up or down (e.g., US\$995.59 to US\$1,000), based on the scope of the activities and magnitude of the project. -Level of accuracy. The acceptable range (e.g., $\pm 10\%$) used in determining realistic cost estimates is specified, and may include an amount for contingencies.

Question 105:

As Walter reviews the Requirements Traceability Matrix (RTM) for his project, he realizes that all of them describe the higher-level needs of the organization as a whole, such as the business issues or opportunities, and reasons why a project has been undertaken. What category of requirements have been gathered?

- A. Functional requirements
- B. Stakeholder requirements
- C. Technical requirements
- D. Business requirements

Answer: D.

Explanation

Though entirely possible that all the requirements are stakeholder and/or functional requirements, the definition provided in the question leads to Business requirements being the correct answer, as the question did not indicate where the requirements came from or what precisely they were for.

Question 106:

Which of the following project documents used to help determine the

budget for a project, indicates how the various estimates were derived and can be used to make a decision on how to respond to variances?

- A. Cost estimates
- B. Risk register
- C. Project schedule
- D. Basis of estimates

Answer: D.

Explanation

The amount and type of additional details supporting the duration estimate vary by application area. Regardless of the level of detail, the supporting documentation should provide a clear and complete understanding of how the duration estimate was derived. Supporting detail for duration estimates may include: Documentation of the basis of the estimate (i.e., how it was developed), Documentation of all assumptions made, Documentation of any known constraints, Indication of the range of possible estimates (e.g., $\pm 10\%$) to indicate that the duration is estimated between a range of values), Indication of the confidence level of the final estimate, and Documentation of individual project risks influencing this estimate.

Question 107:

Estuardo's project team is evaluating risks during a facilitated workshop. As the Project Manager, he's concerned that everyone is calling out the risks they're naming as high impact but they don't seem to be able to determine how likely it is that each will occur. He speaks with the facilitator and they agree that the stakeholders need a visual representation of this risk probability and impact assessment to get everyone on the same page regarding the risks they've called out. What Project Management Process are Estuardo and his team engaged in?

- A. Implement Risk Responses
- B. Perform Quantitative Risk Analysis
- C. Plan Risk Responses
- D. Perform Qualitative Risk Analysis

Answer: D.

Explanation

Perform Qualitative Risk Analysis is the process of prioritizing individual project risks for further analysis or action by assessing their probability of occurrence and impact as well as other characteristics. In contrast, Perform Quantitative Risk Analysis is the process of numerically analyzing the combined effect of identified individual project risks and other sources of uncertainty on overall project objectives. And finally, Estuardo's team cannot plan or implement responses to risks when they have not yet agreed upon what those are or the likelihood and impact of them.

Question 108:

How many data representation techniques are there?

- A. 10
- B. 8
- C. 5
- D. 15

Answer: D.

Explanation

There are 15. Affinity diagrams, cause-and-effect diagrams, control charts, flowcharts, hierarchical charts, histograms, logical data model, matrix diagrams, matrix-based charts, mind mapping, probability and impact matrix, scatter diagrams, stakeholder engagement assessment matrix, stakeholder mapping/representation and text-oriented formats.

Question 109:

A few of the managers who were identified as Interested Stakeholders, have come to Lily, the project manager for a project to revamp a company's Marketing website, and asked her to give them an updated project budget. Lily tells them it will take a little time, as she'll have to evaluate the relationship between the consumption of project funds and the work being accomplished by those expenditures. The managers protest that she doesn't need to do all that to control project costs; that all she needs to do is tell them how much of the budget has been spent. Lily stands her ground, arguing that only by monitoring work completed vs. money spent can she give them an accurate picture of where things stand. Which Project Cost Management process is Lily pulling these

principles from?

- A. Control Costs
- B. Determine Budget
- C. Plan Cost Management
- D. Estimate Costs

Answer: A.

Explanation

Updating the budget requires knowledge of the actual costs spent to date. Any increase to the authorized budget can only be approved through the Perform Integrated Change Control process. Monitoring the expenditure of funds without regard to the value of work being accomplished for such expenditures has little value to the project, other than to track the outflow of funds. Much of the effort of cost control involves analyzing the relationship between the consumption of project funds and the work being accomplished for such expenditures. The key to effective cost control is the management of the approved cost baseline

Question 110:

Nigel has just taken over as the third project manager assigned to a project that has seen considerable delays and cost overruns, caused by difficulties in getting the vendor to deliver what's in the contract. He gathers vendor representatives and his company's stakeholders together for a face-to-face meeting to bring everyone back to the same page. While they're taking a break he overhears a conversation in which someone who works for the vendor admits that the only reason they got the contract is because their president is an old college buddy of Nigel's company's Vice President of Finance, not because they could actually produce what their customer needed. What should Nigel do?

- A. Notify his project's stakeholders immediately
- B. Report an ethics violation to the Project Management Institute immediately
- C. Confront the vendor immediately
- D. Nothing. It doesn't matter how they got the contract. The fact is that Nigel's there to manage what exists no matter how it

happened to come about

Answer: A.

Explanation

Nigel has to notify his project's stakeholders immediately, as it is up to them to decide whether or not to take action based on this new information. They may confront the vendor openly, they may confront their own VP of Finance or they may decide to just let it go and try to force the vendor to perform according to the contract terms. But that's their decision to make, not Nigel's. However, if Nigel had chosen NOT to report this new knowledge then he would've been in violation of the PMI Code of Ethics and Professional Conduct and someone could have reported HIM to the PMI for that lapse in professional responsibility.

Question 111:

A directive PMO (Project Management Office) differs from a supportive PMO in that a directive PMO is responsible for providing which critical role/resource to a project, while the supportive PMO acts in consultative roles?

- A. Business Process Owner
- B. Risk Manager
- C. Project Manager
- D. Software Systems Analyst

Answer: C.

Explanation

Directive PMOs take control of the projects by directly managing the projects. Project managers are assigned by and report to the PMO. The degree of control provided by the PMO is high.

Question 112:

An input to identifying project stakeholders is:

- A. Enterprise Environmental Factors (EEF)
- B. Project Management Plan
- C. Business Case

D. Statement of Work (SOW)

Answer: A.

Explanation

The only one of these four that is a true input to the Stakeholder Register is the EEF. Both the Business Case and SOW are prerequisites to the project Initiating phase, and the Project Management Plan is a deliverable that comes out of the next phase, Planning.

Question 113:

Which of the following is not a key benefit of the Initiating process "Develop Project Charter?"

- A. it provides a direct link between the project and the strategic objectives of the organization
- B. shows the organizational commitment to the project
- C. creates a formal record of the project
- D. promotes the engagement of team members as local domain experts in integration management

Answer: D.

Explanation

Develop Project Charter is the process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities. Key benefits of this process are that it provides a direct link between the project and the strategic objectives of the organization, creates a formal record of the project, and shows the organizational commitment to the project. The one that doesn't belong is "promotes the engagement of team members as local domain experts in integration management" because this describes how iterative and agile approaches benefit project integration management.

Question 114:

Which of the following is a common type of responsibility assignment matrix that uses responsible, accountable, consult, and inform statuses to define the involvement of stakeholders in project activities?

- A. Role-Responsibility-Authority Form
- B. RACI Chart
- C. Work Breakdown Structure (WBS)
- D. Resource Breakdown Structure (RBS)

Answer: B.

Explanation

The key is in the definition here: responsible, accountable, consult, and inform - these provide the acronym RACI, so RACI Chart is the correct answer. A Role-Responsibility-Authority Form can achieve the same result, but it's done in a text format, usually looking somewhat like an outline. Both the WBS and the RBS can be used to show positions and relationships in a graphical, top-down format, but these do not explicitly state the responsibilities of various roles to the level of detail that a RACI Chart does.

Question 115:

What is the most fundamental characteristic of a project team in a projectized organization?

- A. They work together in a business or operations unit as well as being on the same project team together.
- B. They don't need to follow all of the steps in the project management process.
- C. They move from one project to the next as a whole unit.
- D. They work together to manage the project rather than having a single project manager.

Answer: C.

Explanation

The project manager and the staff work together as a unit, and are co-located from one project to the next. Project managers have a great deal of independence and authority. They functionally own the project team members like a manager has staff.

Question 116:

Billy and his team are embroiled in a five-year project that is both complex and at times, incredibly frustrating. Their efforts are all the

more difficult by the fact that the 52 team members work in eight different locations around the globe. In an effort to help individual team members work together effectively by coming together face-to-face, he organizes a four-day retreat and starts the second day off by splitting everyone into teams of four for an all-day scavenger hunt meant to build relationships and camaraderie. What is this type of development activity an example of?

- A. Influencing
- B. Motivation
- C. Negotiation
- D. Team Building

Answer: D.

Explanation

Team building is conducting activities that enhance the team's social relations and build a collaborative and cooperative working environment. Team building activities can vary from a 5-minute agenda item in a status review meeting to an offsite, professionally facilitated event designed to improve interpersonal relationships. The objective of team-building activities is to help individual team members work together effectively. Team-building strategies are particularly valuable when team members operate from remote locations without the benefit of face-to-face contact. Informal communication and activities can help in building trust and establishing good working relationships. While team building is essential during the initial stages of a project, it should be a continuous process.

Question 117:

One of the tools and techniques for developing the project schedule involves identifying which series of tasks will take the longest to complete. What is this technique called?

- A. Data Analysis
- B. Resource Optimization
- C. Critical Path Method
- D. Schedule Compression

Answer: C.

Explanation

The correct answer is the Critical Path Method, in which you identify which path (series of tasks) within the project is the longest from start to finish, which includes taking into consideration all predecessors and successors. Data Analysis methods include "what-if" scenarios and simulations. Schedule Compression refers to fast tracking a project, which is high risk, or crashing a project, which is high cost. And finally, resource optimization involves scheduling projects around resource availability versus other factors.

Question 118:

Herman's project team is completely virtual. Which of the following is not an example of communication technology he should consider to address team development issues that can arise from a project team not working in the same physical location together?

- A. Fax Machines
- B. Video Conferencing
- C. Audio Conferencing
- D. Email

Answer: A.

Explanation

Communication technology is important in addressing the team development issues in colocated and virtual teams. It helps build a harmonious environment for the colocated team and a better understanding for the virtual team, especially those working in different time zones. Examples of communication technology that may be used are: -Shared portal. A shared repository for information sharing (e.g., website, collaboration software or intranet) is effective for virtual project teams. -Video conferencing. Video conferencing is an important technique for effective communication with virtual teams. -Audio conferencing. Communication within a team using audio conferencing is another technique to build rapport and confidence within virtual teams. -Email/chat. Regular communications using email and chat is also an effective technique.

Question 119:

In agile, also called adaptive, project environments, the expectations of a project manager do not change. However, their focus changes to building a collaborative decision-making environment while control of detailed product planning and delivery is assigned to whom?

- A. Project Sponsor
- B. Project Team
- C. Program Manager
- D. Steering Committee

Answer: B.

Explanation

Iterative and agile approaches promote the engagement of team members as local domain experts in integration management. The team members determine how plans and components should integrate. The expectations of the project manager do not change in an adaptive environment, but control of the detailed product planning and delivery is delegated to the team.

Question 120:

Holding's Holdings is a land development firm currently negotiating a contract with an electrical contractor to do all of the cabling work for all of the houses in a new subdivision it's building. In the past, Holding's Holdings has experienced problems with vendors they contract with farming the work out to other companies, and not getting the results they had expected. So this time around, they make it clear to the new electrical contractor that they are to perform all the cabling work that's in the contract themselves. Which professional services delivery method is this an example of?

- A. buyer/ services provider acts as the representative
- B. buyer/ services provider with subcontracting allowed
- C. buyer/ services provider with no subcontracting
- D. joint venture between buyer and services provider

Answer: C.

Explanation

Once the make-or-buy analysis is complete and the decision is made to

acquire from outside the project, a procurement strategy should be identified. The objective of the procurement strategy is to determine the project delivery method, the type of legally binding agreement(s), and how the procurement will advance through the procurement phases. For professional services, delivery methods include: buyer/services provider with no subcontracting, buyer/services provider with subcontracting allowed, joint venture between buyer and services provider, and buyer/services provider acts as the representative. In the case of this question, the correct answer is buyer/services provided with no subcontracting, because Holding's Holdings doesn't want any of the work "farmed out."

Question 121:

The project life cycle is organized into four phases that are typically sequential. How do the costs and staffing levels of a project change from the beginning to the middle to the end of the project throughout the life cycle?

- A. The staffing is higher at the very end of a project because of what had to be done to complete it, so costs are higher, too.
- B. Both costs and staffing levels start low, increase rapidly, and decrease.
- C. Costs and staffing are both higher at the beginning of a project because of unknowns.
- D. Both costs and staffing levels start low and stay low until the end of Executing a project and then remain high through completion.

Answer: B.

Explanation

Project trends show that the highest output of cost and staffing occur in Executing, though the ramp-up to get there is slow. In addition, changing the project scope or project outcome costs a lot more over time.

Question 122:

The Communication Model for Cross-Cultural Communication begins with the Sender being the one to Encode a message. He or she then transmits it, and it is received by the Receiver and decoded. The Receiver then returns some form of acknowledgement to the sender confirming receipt. But what is the thing called which can and does disrupt both the

sender transmitting their message and the receiver transmitting their feedback.

- A. Noise
- B. Feedback Message
- C. Not having a Communication Plan
- D. Culture

Answer: A.

Explanation

In and of itself, not having a communication plan will not be a contributing factor in disrupting the communication between a sender and a receiver. A cultural difference may cause - but is not in and of itself the thing disrupting) the correct answer, which is Noise. And finally, the Feedback Message - if any is provided by the Receiver - can also be disrupted by Noise. Noise can be anything from personal biases, to cultural differences to actual noise such as loud music or language barriers inhibiting clarity.

Question 123:

Which of the following is NOT part of the PMI Talent Triangle?

- A. Technical Project Management
- B. Strategic and Business Management
- C. Leadership
- D. Continuing Education

Answer: D.

Explanation

While continuing education is indeed one of the ways a project manager can increase his or her knowledge and/or effectiveness in any of the three project manager competencies, the competencies themselves are the other three answers shown here. The talent triangle focuses on three key skill sets: 1)Technical project management. The knowledge, skills, and behaviors related to specific domains of project, program, and portfolio management. The technical aspects of performing one's role. 2)Leadership. The knowledge, skills, and behaviors needed to guide, motivate, and direct a team, to help an organization achieve its business goals. 3)Strategic and

business management. The knowledge of and expertise in the industry and organization that enhanced performance and better delivers business outcomes.

Question 124:

Which of the following is not considered a data gathering technique for developing a project charter?

- A. Interviews
- B. Responsibility Assignment Matrix
- C. Brainstorming
- D. Focus Groups

Answer: B.

Explanation

While the Responsibility Assignment Matrix, or RAM, may help you identify who is responsible for helping with various parts of the charter, the three main techniques for gathering data to put into a project charter are conducting focus groups, interviews or brainstorming sessions with Subject Matter Experts (SMEs).

Question 125:

Joni's about to attend a meeting in which a very important decision on which part to use for the widget her project team is producing, will be made. She believes from what most of her team's told her that Part A is the best choice but has no vote as it is a business and manufacturing decision. The project sponsor has agreed that he will abide by whatever the Majority Vote is at today's meeting. If there are 18 people in attendance including Joni, how many must vote for Part A in order for it to be chosen?

- A. 18
- B. 9
- C. 10
- D. 17

Answer: B.

Explanation

Though there are 18 people at the meeting, one of them is Joni. But Joni is the project manager and as the problem tells us, she does not have a vote in which part is chosen since it's a business and manufacturing decision. This leaves 17 people eligible to vote. The definition of a Majority Vote is that more than 50% of the members of the group have to vote a certain way for that to be the decision. $50\% \times 17 = 8.5$. So if 9 people vote for Part A, then Part A will be chosen. (Of note: Had Joni also been allowed a vote, then Majority would have had to be 10 or more votes for Part A in order for it to win, as $50\% \times 18 = 9$, and more than 50% automatically takes us to 10 votes.)

Question 126:

This two-word item is held at the end of a phase and is alternatively known as phase review, stage gate, kill point, and phase entrance or phase exit.

- A. process phase
- B. phase conclusion
- C. process end
- D. phase gate

Answer: D.

Explanation

A phase gate is held at the end of a phase. The project's performance and progress are compared to project and business documents including but not limited to project business case, project charter, project management plan and benefits management plan.

Question 127:

Garrett's in charge of a robotics project where a prototype robot designed to enter buildings, provide live video feed and navigate using its own Artificial Intelligence has to be built so it can then be tested for how it can maneuver in a hostage negotiation situation. Which of the four dependency attributes is this an example of?

- A. External
- B. Internal
- C. Mandatory

D. Dependency

Answer: C.

Explanation

Mandatory dependencies often involve physical limitations, such as on a construction project, where it is impossible to erect the superstructure until after the foundation has been built, or on an electronics project, where a prototype has to be built before it can be tested. Discretionary dependencies are established based on knowledge of best practices within a particular application area or some unusual aspect of the project where a specific sequence is desired, even though there may be other acceptable sequences. External dependencies involve a relationship between project activities and non-project activities, and internal dependencies involve a precedence relationship between project activities and are generally inside the project team's control. Because the prototype is being built from several different contractors' components, they are not all inside the project team's control and that's why this isn't a "Mandatory Internal" dependency.

Question 128:

One of the techniques for managing quality is Data Representation. There are many ways this can be accomplished, including flowcharts, affinity diagrams, histograms, matrix diagrams and cause-and-effect diagrams. What is another technique that can be used in this area?

- A. Gantt Charts
- B. Project Schedules
- C. Data Tables
- D. Scatter Diagrams

Answer: D.

Explanation

Data representation is meant to provide a look at data in a more visual graphical way than listing the data in columns, rows, tables or cells. Project schedules are word and number based data and while Gantt Charts do help give a more visual view into the schedule, the key in this question was that the listed techniques occur in conjunction with managing quality on a project, and a Gantt Chart is unrelated to Project Quality Management.

Question 129:

Organizational process assets (OPAs) and Enterprise environmental factors (EEFs) can both have influences on your organization's project management. What are other possible influences?

- A. Organizational cultures and styles
- B. Organizational structures
- C. All of these
- D. Organizational communications

Answer: C.

Explanation

Any organization, while not necessarily always to the negative, is influenced by all five of these factors to some degree, both operationally and in the management and execution of projects.

Question 130:

Jayanthi worked for a large manufacturing company and has just been assigned to a project to upgrade each manufacturing facility's equipment. This is a project that is undertaken every ten years, and the company keeps good Organizational Process Assets and project artifacts on file. What project artifact or place should she look at first to learn the pitfalls, successes and challenges the project team encountered the last time this upgrade project was done?

- A. Change Management Documentation
- B. Lessons Learned Repository
- C. Project Management Plan
- D. Customer Acceptance Documentation

Answer: B.

Explanation

While it's true that Jayanthi may find the other resources useful, the first place to look is the Lessons Learned Repository. Lessons learned and knowledge gained throughout the previous project may save Jayanthi and her team a lot of time, money and trouble if accounted for - or at least identified as potential risks - from the moment she receives the assignment.

Question 131:

Hilda is a project manager for a local farm that's in the midst of adding new vegetable crops to their crop rotation. To be certain that these new crops meet their quality guidelines for taste and appearance, Hilda's team is going to randomly select 5 out of every 50 beets, radishes and onions to be tested, analyzed and inspected from the very first harvest. What form of sampling is this method an example of?

- A. Population Parameter
- B. Non-Probability
- C. Statistical
- D. Stratified

Answer: C.

Explanation

Statistical sampling involves choosing part of a population of interest for inspection. The sample is taken to measure controls and verify quality.

Question 132:

Dorian's managing a project to refresh all servers for a global organization with more than 60 offices worldwide. It's had a lot of ups, downs and unexpected changes to both scope and schedule, and as the project team completes server refreshes for the sixteenth office, the project steering committee asks to understand the project's cost performance. What formula should Dorian apply in order to provide them with this information?

- A. $CV = EV - AC$
- B. $SV = EV - PV$
- C. $EAC = AC + New\ ETC$
- D. $SPI = EV/PV$

Answer: A.

Explanation

CV, which stands for Cost Variance, is the amount of budget deficit or surplus at a given point in time. It is a measure of cost performance on a project and is achieved by subtracting the actual costs (AC) of the project

from the earned value (EV). However, in order to use this equation you have to first calculate Earned Value, which shows how much of the budget and time should have been spent, considering the amount of work done. So if you have spent 3/4 of your budget but you're only supposed to have spent 1/4 of it, then your EV is not at all good. Conversely, if you're halfway through the project and have only spent 1/4 of your estimated spend, then you are ahead of the game. $EV = \text{Total Project Budget} \times (\text{times}) \text{Percent (\%)} \text{ of project completion}$. The EAC formula is to calculate the estimate at completion, which is not what the steering committee asked for. SV is the schedule variance, which has nothing to do with the cost variance equation. And finally the SPI is the schedule performance index which also does not help in determining the cost variance of a project at a given point in time.

Question 133:

Why is it important to identify and document stakeholders as soon as possible after the project charter has been signed and a project manager has been assigned to a project?

- A. To provide visibility into aspects of the project that may not have been considered had stakeholders been misidentified or excluded.
- B. Because a stakeholder can negatively influence the project, it's best to know about them as soon as possible to help mitigate the risk they pose.
- C. All of these.
- D. Doing so increases a project's chances of success by ensuring those who need to be involved are involved from the beginning.

Answer: C.

Explanation

Starting a project's Planning phase without knowing a) who all the impacted people or companies are, b) understanding why they are impacted, positively or negatively, by the project's outcome, and c) having a good handle on the environment within which the project is operating, sets a project up for failure rather than success. If the sales manager is not engaged on a project trying to implement new sales goal calculations then he may block the project from implementing even if the project team's spent 6 months trying to find the best solution, if for no other reason than that he has not lent his expertise

to the solution identification process which means something may have been missed that's important to him or the company, or there may be state laws he knows but the project team didn't consider.

Question 134:

Esther's project is tasked with changing the material her company uses to manufacture outdoor welcome mats, so that each mat costs \$1.85 apiece to manufacture. In addition, the project sponsor wants the new manufacturing techniques to start in February of the new year, which means all of the research has to be completed during the winter holiday season when many employees take time off. What are these two parameters (cost and timing) Esther's project has to manage within, called?

- A. requirements
- B. deadlines
- C. scope
- D. constraints

Answer: D.

Explanation

Tailoring is necessary because each project is unique; not every process, tool, technique, input, or output identified in the PMBOK® Guide is required on every project. Tailoring should address the competing constraints of scope, schedule, cost, resources, quality, and risk. The importance of each constraint is different for each project, and the project manager tailors the approach for managing these constraints based on the project environment, organizational culture, stakeholder needs, and other variables. The cost limitation and the schedule limitation are both constraints for Esther's project.

Question 135:

As Ginger prepares her construction project team for an in-depth analysis of the feasibility of using the existing concrete slab and embedded plumbing for the new retail store they're going to build in the site of a previous owner whose company went out of business, a report comes in from an on-site inspector indicating that while the concrete slab is in good enough shape to support another structure, the sewage and water pipes embedded in the concrete are in the early stages of

disintegrating. Which of the following types of internal EEFs is now going to result in Ginger's project having to replace all of the pipes before they can begin building the new store?

- A. Resource availability
- B. Geographic distribution of facilities and resources
- C. Infrastructure
- D. Employee capability

Answer: C.

Explanation

The Enterprise Environmental Factor Ginger's dealing with is Infrastructure. Examples include existing facilities, equipment, organizational telecommunications channels, information technology hardware, availability, and capacity.

Question 136:

When first looking at writing a project charter, where should you be drawing critical success factors for your project, including financial success factors?

- A. Business Case
- B. Project Management Plan
- C. Basis of Estimates
- D. Benefits Management Plan

Answer: A.

Explanation

Described in Section 1.2.6 of the PMBOK, the business documents that can be considered as inputs for the Determine Budget process include but are not limited to: Business case: The business case identifies the critical success factors for the project, including financial success factors; and Benefits management plan, which includes the target benefits, such as net present value calculations, timeframe for realizing benefits, and the metrics associated with the benefits. Theoretically, the project would not have proceeded to the initiating processes unless it had been approved to proceed as a project, and one of the factors leading to that decision should have been

critical success factors.

Question 137:

When David finishes aggregating all of his cost estimates for the project he's managing, he adds a contingency of 15% as per his project sponsor's direction. What is another name for the contingency that David's adding to what will become the overall project budget?

- A. Data Analysis
- B. Management Reserves
- C. Expert Judgment
- D. Cost Aggregation

Answer: B.

Explanation

A data analysis technique that can be used in the Determine Budget process includes but is not limited to reserve analysis, which can establish the management reserves for the project. Management reserves are an amount of the project budget withheld for management control purposes and are reserved for unforeseen work that is within scope of the project. Management reserves are intended to address the unknown unknowns that can affect a project. The management reserve is not included in the cost baseline but is part of the overall project budget and funding requirements. When an amount of management reserves is used to fund unforeseen work, the amount of management reserve used is added to the cost baseline, thus requiring an approved change to the cost baseline.

Question 138:

The PMBOK lists many different components that can comprise a project management plan including a Quality Management Plan, a Scope Baseline and a Development Approach. How many total components does the PMBOK state is possible to have in your project management plan?

- A. 10
- B. 15
- C. 16
- D. 18

Answer: D.

Explanation

Table 1-2. Project Management Plan and Project Documents on page 559 of PMBOK 6th Edition lists 18 possible components to the Project Management Plan.

Question 139:

What are the two deliverables expected to be completed by the end of the Initiating phase of a project?

- A. Risk Management Plan and Procurement Plan
- B. Stakeholder Register and Project Charter
- C. Stakeholder Register and Communication Plan
- D. Business Requirements Document and Project Charter

Answer: B.

Explanation

When you're initiating a project, you are making identifications and gathering data at a high level in order to obtain approval to proceed to the Planning phase. All of the other documents listed within the multiple choice answers besides Stakeholder Register and Project Charter, are deliverables developed during the Planning phase.

Question 140:

When you are planning Quality Management for your project, there are different techniques for data representation. Which of the four following techniques includes a model known as a SIPOC, an acronym which stands for: suppliers, inputs, process, outputs, and customers?

- A. Flowcharts
- B. Logical Data Mapping
- C. Mind Mapping
- D. Matrix Diagrams

Answer: A.

Explanation

Flowcharts are also referred to as process maps because they display the

sequence of steps and the branching possibilities that exist for a process that transforms one or more inputs into one or more outputs. Flowcharts show the activities, decision points, branching loops, parallel paths, and the overall order of processing by mapping the operational details of procedures that exist within a horizontal value chain. One version of a value chain is the SIPOC. You can see a sample of this model in Figure 8-6 on page 285 of the PMBOK.

Question 141:

One possible input that a project manager may use to help them write a project charter are Organizational Process Assets, or OPAs. Which of the following are NOT OPAs?

- A. Your company's phase gating process by which the organization approves a project to proceed to the next phase)
- B. Laws that the local government has just passed regarding the manner in which your company is allowed to do business
- C. Your organization's Project Management templates
- D. Your IT department's change control process for software implementations

Answer: B.

Explanation

If laws are passed which impact how your organization does business, those fall under the category of Enterprise Environmental Factors or EEFs. The other three examples are indeed OPAs: templates are often enforced or desired by an organization, for everything from forms that're filled out to project artifacts such as the charter. Change control processes may need to be factored into a project's schedule simply because it's an internal control standard which must be met regardless of why. And finally, if your company controls how a project moves from one phase to the next then that is yet another internal hurdle you must sail over in order to proceed with the project. None of those are caused by forces external to your organization, but instead are caused by internal mechanisms and decisions.

Question 142:

The "Develop the project management plan" process achieves which of these goals?

- A. Copying and pasting what's in the charter (output of Initiating) into a Planning phase document.
- B. Defines, prepares and coordinates all subsidiary plans and integrates them into a comprehensive plan.
- C. Defines an iterative and agile approach to the project to promote engagement of team members.
- D. It formally sets in stone an entire set of detailed business requirements, technical requirements and vendor requirements.

Answer: B.

Explanation

The Project Management Plan integrates - sometimes at a high level - the components of the other plans developed during the Planning phase. From the Risk Management Plan to the Project Schedule to the Quality Management Plan and the project budget, the Project Management Plan becomes the "one-stop shop" outlining how the project will proceed to meet its objectives. It doesn't contain details down to the nth degree; it contains enough information to be the roadmap, and the subsidiary documents contain further detail for those times when it's required.

Question 143:

Kelly is identifying which of her assigned project team members has the most experience with her organization's financial software, as she's been tasked with a project to replace it with a newer product. Her company holds a training on the software every three months and keeps note of who's taken the training in personnel files. Which type of OPA can Kelly use to identify the resource she needs?

- A. Formal knowledge-sharing and information-sharing procedures
- B. Organizational standards policies, processes, and procedures
- C. Organizational communication requirements
- D. Personnel administration

Answer: D.

Explanation

One of the Organizational Process Assets (OPAs) available to Kelly that undertakes the management of project knowledge is Personnel

administration. These types of OPAs include, for example, employee development and training records, and competency frameworks that refer to knowledge-sharing behaviors.

Question 144:

When the project team brainstormed all of the possible risks that could occur during the lifetime of the project, they also came up with mitigation plans - what to do to keep the risk from happening, and what to do if it did occur. But what do you now do with all of that during the Executing phase?

- A. Wait until the risk is realized and then try to remember what the mitigation strategies were.
- B. Make sure it's saved in a Risk Register and/or Risk Report.
- C. Give the required level of effort to implement the agreed upon responses to manage the risks proactively.
- D. Forget about them and focus on getting the Executing tasks completed on time.

Answer: C.

Explanation

A common problem with Project Risk Management is that project teams spend effort in identifying and analyzing risks and developing risk responses, then risk responses are agreed upon and documented in the risk register and risk report, but no action is taken to manage the risk. Only if risk owners give the required level of effort to implement the agreed upon responses will the overall risk exposure of the project and individual threats and opportunities be managed proactively.

Question 145:

The way a RACI chart is matrixed typically assumes that how many people will be accountable for a work package?

- A. None
- B. 1
- C. 3 or more
- D. 2

Answer: B.

Explanation

The RACI matrix format shows all activities associated with one person and all people associated with one activity. This also ensures that there is only one person accountable for any one task to avoid confusion about who is ultimately in charge or has authority for the work.

Question 146:

During a contract life cycle, which of the following items can a seller NOT be considered as?

- A. Selected Source
- B. Contractor
- C. Supplier or Vendor
- D. Bidder

Answer: B.

Explanation

The seller can be viewed during the contract life cycle first as a bidder, then as the selected source, and then as the contracted supplier or vendor. The seller may very well be a contractor, but that does not apply across the board to all sellers.

Question 147:

A project is underway for a team of toy makers to create by hand as many toy reindeer as possible. Their noses have to glow and their antlers have to be covered with glitter. In addition, all four of their legs have to move and they have to be painted pale brown. The team builds up a good rhythm and over time they begin to average completing 20 reindeer toys per hour. By the time 3 weeks have passed their boss wants to know how many reindeer are left to be made and how that compares to the amount of time left to complete them. What is a graphical way that Adam can show this information to the team's boss?

- A. Milestone Chart
- B. Burnup Chart Only
- C. Burndown Chart or Burnup Chart

D. Burndown Chart Only

Answer: C.

Explanation

Some iteration-based projects use burndown charts to see where the project is going over time. Some project teams prefer burnup charts. Burnup charts show the work completed. The charts are based on the same data but displayed in two different ways. Teams may prefer how to see their data.

Question 148:

How many Data Analysis techniques are there?

- A. 27
- B. 30
- C. 21
- D. 25

Answer: A.

Explanation

There are 27. Alternatives analysis, Assessment of other risk parameters, Assumption and constraint analysis, Cost of quality, Cost-benefit analysis, Decision tree analysis, Document analysis, Earned value analysis, Influence diagrams, Iteration burndown chart, Make-or-buy analysis, Performance reviews, Process analysis, Proposal evaluation, Regression analysis, Reserve analysis, Risk data quality assessment, Root cause analysis, Sensitivity analysis, Simulation, Stakeholder analysis, SWOT analysis, Technical performance analysis, Trend analysis, Variance analysis and What-if scenario analysis.

Question 149:

Fill in the blank: A portfolio is a collection of projects, programs, subsidiary portfolios, and _____ managed as a group to achieve strategic objectives.

- A. Requirements
- B. Operations
- C. Budgets

D. Resources

Answer: B.

Explanation

This definition of a portfolio comes straight from the Project Management Book of Knowledge. It is Operations which rounds out that definition, as Operations within the context of organizational structure support a company's strategic objectives at the same level as projects do...the hierarchy is: Strategic Objectives --> Portfolio --> Programs --> Projects & Operations --> Shared Resources and Stakeholders.

Question 150:

Which of the following is not a Project Procurement Management process?

- A. Monitor Risks
- B. Plan Procurement Management
- C. Conduct Procurements
- D. Control Procurements

Answer: A.

Explanation

As with all processes contained within the PMBOK Guide, you first plan and then do (conduct) and then monitor or in this case, control. Monitoring risks is an overarching process a Project Manager must do throughout the project life cycle, not specifically tied to the project procurement processes.

Question 151:

How many processes are there in Project Integration Management?

- A. 11
- B. 9
- C. 5
- D. 7

Answer: D.

Explanation

The correct answer is 7, and the Project Integration Management processes are: Develop Project Charter—The process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities. Develop Project Management Plan—The process of defining, preparing, and coordinating all plan components and consolidating them into an integrated project management plan. Direct and Manage Project Work—The process of leading and performing the work defined in the project management plan and implementing approved changes to achieve the project's objectives. Manage Project Knowledge—The process of using existing knowledge and creating new knowledge to achieve the project's objectives and contribute to organizational learning. Monitor and Control Project Work—The process of tracking, reviewing, and reporting overall progress to meet the performance objectives defined in the project management plan. Perform Integrated Change Control—The process of reviewing all change requests; approving changes and managing changes to deliverables, organizational process assets, project documents, and the project management plan; and communicating the decisions. Close Project or Phase—The process of finalizing all activities for the project, phase, or contract.

Question 152:

What is a project stakeholder?

- A. A person or group of people who perceive themselves to be affected by the project.
- B. An individual, group, or organization, who may affect, be affected by, or perceive itself to be affected by a decision, activity, or outcome of a project.
- C. An individual, group or organization who believes very much in the project objectives and does everything they can to promote it to the rest of the involved parties.
- D. Someone who runs the project

Answer: B.

Explanation

Stakeholders can be individuals, groups or entire companies, and what makes them stakeholders is the most important part of this definition, which is "who

may affect, be affected by or perceive itself to be affected by." Remember that for many, perception is reality and so even if their belief is only a perception, they are no less a stakeholder than someone probably affected by the project. Stakeholders do not run the project and those who perceive themselves as affected are not the only possible types of stakeholders. The final incorrect response describes project champions - these are the proponents of the project...the cheerleaders, if you will...to other individuals, groups and organizations. While champions are usually stakeholders, not all stakeholders are champions.

Question 153:

As the director of her organization's Project Management Office, Lisa is heavily involved in working with department managers as they work out their project budgets for upcoming fiscal years. She also works closely with the management team at her organization as they hammer out their rolling three- and five-year business plans annually. In this way Lisa is proactively ensuring that the organization's strategy, mission, goals and objectives align with the projects they're choosing to execute in the coming months and years. Which of the three project manager competencies from the Talent Triangle does she demonstrate?

- A. Leadership
- B. Strategic and Project Management
- C. Technical Project Management
- D. Strategic and Business Management

Answer: D.

Explanation

In order to make the best decisions regarding the successful delivery of their projects, project managers should seek out and consider the expertise of the operational managers who run the business in their organization. These managers should know the work performed in their organization and how project plans will affect that work. The more the project manager is able to know about the project's subject matter, the better. The project manager should apply the following knowledge and information about the organization to the project to ensure alignment: Strategy, Mission, Goals and objectives, Priority, Tactics, and Products or services (e.g., deliverables).

Question 154:

What is the purpose of the initiating process?

- A. To create a project charter and stakeholder register.
- B. To kick off the planning phase of a project.
- C. To develop a business case for the purpose of obtaining authorization to embark upon the project management process.
- D. To define a new project or a new phase of an existing project by obtaining authorization to start the project or phase.

Answer: D.

Explanation

Though the project charter and stakeholder register are outputs of the initiating phase, they are not the reason the phase is done. The purpose of this phase is to obtain someone's (or a group of someones') authorization to start a project. The business case doesn't work here because that's a precursor to the initiating phase, allowing a company to decide whether there's enough of a case to even begin writing a charter, for example. And finally, the planning phase isn't kicked off by the initiating phase. The planning phase simply begins once signoff on the initiating phase is obtained.

Question 155:

When looking for someone to provide goods or services for a project, least cost, qualifications only, fixed budget and sole source are just four possibilities for which Plan Procurement Management tool or technique?

- A. Expert Judgment
- B. Source Selection Analysis
- C. Data Gathering
- D. Data Analysis

Answer: B.

Explanation

It is necessary to review the prioritization of the competing demands for the project before deciding on the selection method. Since competitive selection methods may require sellers to invest a large amount of time and resources upfront, it is a good practice to include the evaluation method in the

procurement documents so bidders know how they will be evaluated. Commonly used selection methods include the following: least cost, qualifications only, quality-based/highest technical proposal score, quality and cost-based, sole source and fixed budget.

Question 156:

How many project characteristics are there that you can assess to determine the organizational structure of the company you're working with or for?

- A. 4
- B. 6
- C. 8
- D. 10

Answer: B.

Explanation

There are six, as demonstrated in page 47 of PMBOK, Table 2-1. Influences of Organizational Structures on Projects. The six are: how work groups are arranged, a project manager's authority, a project manager's role, resource availability, who's managing the project budget and how many (and what type) of project management administrative staff an organization has.

Question 157:

How many personality characteristics does the PMBOK indicate an effective project needs to have some level of ability within, in order to be successful?

- A. 8
- B. 11
- C. 10
- D. 15

Answer: B.

Explanation

The correct answer is 11. Personality refers to the individual differences in characteristic patterns of thinking, feeling, and behaving. Personality

characteristics or traits include but are not limited to: Authentic (e.g., accepts others for what and who they are, show open concern); Courteous (e.g., ability to apply appropriate behavior and etiquette); Creative (e.g., ability to think abstractly, to see things differently, to innovate); Cultural (e.g., measure of sensitivity to other cultures including values, norms, and beliefs); Emotional (e.g., ability to perceive emotions and the information they present and to manage them; measure of interpersonal skills); Intellectual (e.g., measure of human intelligence over multiple aptitudes); Managerial (e.g., measure of management practice and potential); Political (e.g., measure of political intelligence and making things happen); Service-oriented (e.g., evidence of willingness to serve other people); Social (e.g., ability to understand and manage people); and Systemic (e.g., drive to understand and build systems). An effective project manager will have some level of ability with each of these characteristics in order to be successful. Each project, organization, and situation requires that the project manager emphasize different aspects of personality.

Question 158:

When she was assigned to a project to add one more flight between Honolulu and Tokyo to her airline's daily scheduled flights, Xia was told by the portfolio steering committee exactly how many passengers they expected to be traveling this route by 3 months, 6 months, 9 months and 12 months after implementation. Into which project document did Xia record this information?

- A. Marketing Plan
- B. Benefits Management Plan
- C. Business Case
- D. Project Charter

Answer: B.

Explanation

The benefits management plan is used to measure whether the benefits of the project were achieved as planned. And while Xia is a project manager, she was actually assigned the project prior to its official initiation, by helming the effort to do the research that led up to the formation of a business case and benefits management plan, which is how she and her team were able to get

the steering committee to agree to initiate the project.

Question 159:

Who does the Project Management Book of Knowledge (PMBOK) say is responsible for what the project team produces?

- A. Project Manager
- B. Project Business Owner
- C. Executive Sponsor
- D. Project Sponsor

Answer: A.

Explanation

On p. 56 of PMBOK 6th Edition, the Project Management Institute explains the role of a project manager as being similar to that of an orchestra's conductor, stating, "The project manager and conductor are both responsible for what their teams produce—the project outcome or the orchestra concert, respectively. The two leaders need to take a holistic view of their team's products in order to plan, coordinate, and complete them."

Question 160:

There are three main goals for a project manager with respect to their responsibilities during Project Resource Management. The PM has to both acquire and manage project team resources. But what is the third responsibility a project manager has to their project's team members?

- A. Meet With Team Regularly
- B. Develop Team
- C. Do Personality Assessments of Project Team Members
- D. Celebrate Victories with Team

Answer: B.

Explanation

Though celebrating victories with the project team is strongly encouraged, meeting with the project team regularly is considered a best practice and doing personality assessments of project team members may foster better communication within the project as a whole, the third responsibility of a

project manager to his or her team is to develop them. This will help both them and the project by improving competencies, team member interaction, and overall team environment to enhance project performance

Question 161:

What is the Closing process used to close out?

- A. One project phase
- B. The whole project
- C. Contract
- D. All of these

Answer: D.

Explanation

The closing process group consists of the processes performed to formally complete or close a project, phase, or contract. This process group verifies that the defined processes are completed within all of the Process Groups to close the project or phase, as appropriate, and formally establishes that the project or project phase is complete.

Question 162:

The PMBOK defines the project life cycle as having four phases, the first of which is "Starting the Project." Which of the following is not one of the other three phases?

- A. Organizing and Preparing
- B. Carrying out the work
- C. Ending the Project
- D. Initiating

Answer: D.

Explanation

Project life cycle: The series of phases that a project passes through from its start to its completion. Initiating is a Project Management Process Group. It is not one of the phases of the project life cycle. It is important to note that Project Management Process Groups are not project phases, either.

Question 163:

The Tuckman Ladder (1965) is one of the models used to describe team development, with Forming being the first stage. How many stages of development does Tuckman outline that a team goes through?

- A. 6
- B. 8
- C. 5
- D. 4

Answer: C.

Explanation

One of the models used to describe team development is the Tuckman ladder (Tuckman 1965) which includes 5 stages of development that teams go through. Tuckman maintained that these phases are all necessary and inevitable in order for the team to grow, to face up to challenges, to tackle problems, to find solutions, to plan work, and to deliver results. The five stages are: Forming, Storming, Norming, Performing and Adjourning.

Question 164:

What two things have to be defined before you can accurately estimate resources?

- A. The project schedule and all possible risk responses.
- B. The project scope and the business requirements.
- C. The project scope and the Work Breakdown Structure (WBS).
- D. The Work Breakdown Structure (WBS) and data analysis.

Answer: C.

Explanation

Though the project charter and stakeholder register are outputs of the initiating phase, they are not the reason the phase is done. The purpose of this phase is to obtain someone's (or a group of someones') authorization to start a project. The business case doesn't work here because that's a precursor to the initiating phase, allowing a company to decide whether there's enough of a case to even begin writing a charter, for example. And finally, the planning phase isn't kicked off by the initiating phase. The planning phase simply begins once signoff on the initiating phase is obtained.

Question 165:

After lessons learned are gathered from all those involved in the project, what can you then do with the project resources to help complete the closing processes?

- A. Release them to be reassigned
- B. Say good-bye
- C. Celebrate with them
- D. Put them back into storage (if they are 'things such as computers, calculators or extra materials)

Answer: A.

Explanation

Those Responses A and C are kind things to do, and Response D is very practical, releasing the resources so they can be reassigned elsewhere - whether those resources are people, facilities or materials - are the most beneficial to the organization, because this act ensures that if those resources are needed elsewhere, you've freed them for that needed reassignment.

Question 166:

The key benefits of the Manage Team process are all of the following EXCEPT:

- A. Keeps Stakeholders Engaged
- B. Resolves Issues
- C. Influences Team Behavior
- D. Manages Conflict

Answer: A.

Explanation

The key benefit of this process is that it influences team behavior, manages conflict, and resolves issues. As a result of managing the project team, change requests are submitted, the human resources management plan is updated, issues are resolved, input is provided for performance appraisals, and lessons learned are added to the organization's database. Managing the project team is not directly beneficial to how, why or the results of keeping project stakeholders engaged.

Question 167:

Jake's just received word from two project stakeholders that there are changes needed to the project scope or the widget they're making won't meet customers' needs. Jake now must follow the integrated change control process to handle the scope change requests. Which project document is he definitely going to need to update as part of this process?

- A. Milestone List
- B. Requirements Traceability Matrix (RTM)
- C. Quality Metrics
- D. Change Log

Answer: D.

Explanation

Any formally controlled project document may be changed as a result of this process. A project document that is normally updated as a result of this process is the change log. The change log is used to document changes that occur during a project. While it's true that if the scope changes are approved that Jake may need to come up with different use cases for his RTM, or different quality metrics to account for the new scope or new milestones to add to his existing milestone list, none of those documents are to be updated as part of performing integrated change control processes. The purpose of these processes is simply to shepherd the requested change through to its conclusion.

Question 168:

The project manager may serve as an informal _____ by educating the organization as to the advantages of project management with regard to timeliness, quality, innovation, and resource management.

- A. trainer
- B. ambassador
- C. teacher
- D. liaison

Answer: B.

Explanation

A professional project manager may choose to orient and educate other professionals regarding the value of a project management approach to the organization. The project manager may serve as an informal ambassador by educating the organization as to the advantages of project management with regard to timeliness, quality, innovation and resource management.

Question 169:

When are the business case and the benefits management plan developed?

- A. Prior to project initiation
- B. During the initiating processes
- C. During the closing processes
- D. While the project is executing its project management plan

Answer: A.

Explanation

Both the business case and the benefits management plan are developed prior to the project being initiated. Additionally, both documents are referenced after the project has been completed. Therefore, they are considered business documents rather than project documents or components of the project management plan.

Question 170:

When the stakeholders on Quinton's software development project became upset by seemingly unforeseen delays in the development of the user interface portion of the project, the project sponsor looked to Quinton, as the Project Manager, for understanding as to why the team had not realized that the one person needed to create that user interface would be unavailable due to his daily operational tasks. Quinton could look at the resource assumptions recorded on the assumption log that's been kept since the creation of the project charter, but he suspects the overallocation of this particular resource is a systemic, larger issue the organization needs to address. Where might he investigate outside of his own project documents, to see if his hunch is correct?

- A. Project Charter
- B. Emails

- C. Issues Log
- D. Business Case

Answer: D.

Explanation

The key wording in this scenario involved the project resource's operational obligations keeping him from completing the project work. High - level strategic and operational assumptions and constraints are normally identified in the business case before the project is initiated and will flow into the project charter. Chances are - in this case, at least - that the assumption was made during the business case that all resources required would be available as needed. That assumption would have then carried on through the project charter and if no one sought to validate it, would explain why the project suddenly finds itself without the one resource it needs to keep the project moving. If you assume a resource of any kind is available, and you schedule the project and make promises based on that information, then you may find yourself in Quinton's unfortunate position where he now has to explain why he didn't check to be certain during the Planning processes that the original business case assumptions were accurate.

Question 171:

Augustin attends a steering committee one day where he, as the project manager of a multi-million dollar construction project, is celebrated and praised by all the steering committee members for finding a flaw in the original architect's blueprints that could have cost the city nearly one million dollars if it had been discovered after the building was already under construction. However, Augustin wasn't the one who actually discovered the error; he was merely the one who brought it to the architect's attention, and he tells the committee members as much. In being honest and forthright about who the credit should really go to, which leadership quality has Augustin displayed?

- A. Giving credit to others where due
- B. Accepting feedback gracefully
- C. Giving feedback constructively
- D. Managing expectations

Answer: A.

Explanation

Augustin demonstrated several leadership qualities, and all of the options presented are ones all project managers, as leaders, should display. However, for the particular scenario presented, "Giving credit to others where due" is the correct answer. A leader proves him or herself a leader by not taking credit for something that wasn't entirely their doing. Even if the Project Manager did have a part in what occurred, they should still always give credit to all involved parties. This behavior goes a long way toward solidifying loyalty and operating with integrity and in an ethical manner.

Question 172:

This document is issued to prospective vendors when more information is needed on how they would satisfy the requirements and/or how much it will cost.

- A. RFQ
- B. RFI
- C. RFP
- D. SOW

Answer: A.

Explanation

Depending on the goods or services needed, the bidding documents can include a request for information, request for quotation, request for proposal, or other appropriate procurement documents. The conditions involving their use are presented below: 1) Request for information (RFI). An RFI is used when more information on the goods and services to be acquired is needed from the sellers. It will typically be followed by an RFQ or RFP. 2) Request for quotation (RFQ). An RFQ is commonly used when more information is needed on how vendors would satisfy the requirements and/or how much it will cost. 3) Request for proposal (RFP). An RFP is used when there is a problem in the project and the solution is not easy to determine. This is the most formal of the "request for" documents and has strict procurement rules for content, timeline, and seller responses.

Question 173:

There are eight inputs to the Close Project or Phase process. Of those listed here, which - when the project is audited - will factor into determining how well the project did what the requirements and scope set out to do?

- A. Accepted deliverables
- B. Project documents
- C. Project charter
- D. Business documents

Answer: A.

Explanation

Of the answers listed, project documents, business documents and the project charter cannot provide you with the information needed to determine whether or not the project's scope baseline and requirements were met, other than by conveying to you what those were. It's going to be the accepted deliverables that provide that insight - did those deliverables meet expectations? Meet quality metrics? Meet requirements? Fulfill the scope? Satisfy the stakeholders?

Question 174:

Project management is divided into how many process groups?

- A. 8
- B. 5
- C. 4
- D. 10

Answer: B.

Explanation

Project Management is divided into 5 Process Groups (Initiating, Planning, Executing, Monitoring & Controlling and Closing Process Group)

Question 175:

Rosa is very good at tailoring both traditional and agile tools, techniques and methods for each project she's engaged to manage, and she takes the time needed to plan and prioritize diligently even if it sometimes puts her

at odds with teammates or sponsors who want to do work first and ask questions later. Which of the three project manager competencies from the Talent Triangle does Rosa demonstrate?

- A. Technical Project Management
- B. Strategic and Business Management
- C. Leadership
- D. Agile Project Management

Answer: A.

Explanation

According to research. The top project managers consistently demonstrated several key technical project management skills including, but not limited to, the ability to: Focus on the critical technical project management elements for each project they manage. This focus is as simple as having the right artifacts readily available. At the top of the list were the following: Critical success factors for the project, Schedule, Selected financial reports, and Issue log. Tailor both traditional and agile tools, techniques, and methods for each project. Make time to plan thoroughly and prioritize diligently. Manage project elements, including, but not limited to, schedule, cost, resources, and risks.

Question 176:

The Plan Quality Management process exists within the Planning phase of a project. In the Monitoring & Controlling phase of a project, you Control Quality. What is the missing related process you undertake during the Executing phase of a project?

- A. Manage Quality
- B. Categorize Quality
- C. Define Quality
- D. Prioritize Quality

Answer: A.

Explanation

Manage Quality is the missing process, and it occurs during the Executing phase of a project, when your job is to manage the quality of whatever it is

that's being executed. The other three options are not PMBOK processes.

Question 177:

Which subsidiary plan developed in the Planning phase describes how goods or services will be obtained?

- A. Process Improvement Plan
- B. Resource Management Plan
- C. Procurement Management Plan
- D. Stakeholder Engagement Plan

Answer: C.

Explanation

The Procurement Management Plan includes the processes necessary to purchase or acquire products, services, or results needed from outside the project team. The Resource Management Plan focuses on the process of defining how to estimate, acquire, manage, and use team and physical resources. The Stakeholder Engagement plan contains detailed plans on how effective stakeholder engagement can be realized. And finally, the Process Improvement Plan supports continuous process improvement activities as undertaken on behalf of the performing organization and falls under Project Quality Management.

Question 178:

The very first of a Project Manager's 7 responsibilities is to develop the project charter. What is the next major milestone that you as a Project Manager are expected to deliver?

- A. Work Breakdown Structure
- B. Project Change Request
- C. Project Management Plan
- D. Risk Management Plan

Answer: C.

Explanation

While all of these are indeed project deliverables required at one point or another in the project life cycle, they are not all authored by the project

manager, nor are they all major milestones. The Project Management Plan is both, as well as being the very next focus point of the Project Manager, whose Work Breakdown Structure, project schedule and other related artifacts will all become necessary components of said plan.

Question 179:

When you are tasked with "project knowledge management," what type of project knowledge is it that you are managing?

- A. Explicit Knowledge
- B. Tacit Knowledge
- C. Knowledge of the project's requirements
- D. Both Tacit and Explicit Knowledge

Answer: D.

Explanation

Project knowledge management through the lens of the Project Management Institute, involves the management of both objective information (explicit) and subjective information (tacit). While explicit knowledge is structured and fixed (e.g. "the facts"), tacit knowledge is personal and context specific (e.g. feeling or judgment). Knowledge of the project's requirements is handled via the documentation of requirements, which is one example of explicit knowledge, but isn't the only knowledge you are tasked with managing on a project.

Question 180:

All of the following statements are true of a project charter EXCEPT:

- A. It documents high-level requirements to satisfy stakeholder needs.
- B. It formally authorizes the project or phase and authorizes the project manager.
- C. A document that contains a detailed plan and guidance for training required as part of the project
- D. It documents business needs, assumptions and constraints.

Answer: C.

Explanation

This definition is for a training plan. At the point at which you're writing a project charter - before the project's even begun - you don't have a firm enough grasp on the solution you'll be implementing to project what kind of training will be required.

Question 181:

There's a set of technical guidelines known as Dxf (Design for X) which can be used to manage quality. Using the DfX that's decided upon within your project may result in cost reduction, quality improvement, better performance, and what?

- A. Improved Engagement
- B. Better Communication
- C. Customer Satisfaction
- D. Shortened Project Schedule

Answer: C.

Explanation

Design for X (DfX) is a set of technical guidelines that may be applied during the design of a product for the optimization of a specific aspect of the design. DfX can control or even improve the product's final characteristics. The X in DfX can be different aspects of product development, such as reliability, deployment, assembly, manufacturing, cost, service, usability, safety, and quality. Using the DfX may result in cost reduction, quality improvement, better performance, and customer satisfaction. The other three choices here have nothing to do with product design.

Question 182:

Donna is a project manager for a large SaaS company who works from her home office. On her team is a business analyst who works out of the company's headquarters in New York, a developer who works from his home in Southern California and a web designer who works in one of the company's satellite offices in Seattle, Washington. What type of (physical) project team is Donna a part of?

- A. Dispersed
- B. Lolocatea
- C. Cohesive

D. Distributed

Answer: A.

Explanation

Some organizations have been able to create colocated, cross-functional teams; others have a different situation. Instead of having all team members colocated, some organizations have distributed or dispersed teams.

Distributed teams have cross-functional teams in different locations.

Dispersed teams may have each team member working in a completely different location, either in an office or from home. A colocated team is an agile team working together in a single space regardless of their function or role on the team, and cohesive describes how a team operates, not their physical location in relation to one another.

Question 183:

Lindsey's project involves a lot of people who work in shifts, rather than a "9 to 5" day. It's going to be difficult for her to remember when each of the project team members is available, and how that availability affects their assigned tasks, without something to help manage that.

What's one output of the Acquire Resources process that will provide her with a tool she can use throughout the rest of the project to keep the resources organized?

- A. Project Team Assignments
- B. Resource Calendar
- C. Physical Resource Assignments
- D. Change Requests

Answer: B.

Explanation

Of the choices presented, only the Resource Calendar meets Lindsey's needs. A resource calendar identifies the working days, shifts, start and end of normal business hours, weekends, and public holidays when each specific resource is available. Information on which resources (such as team resource, equipment, and material) are potentially available during a planned activity period is used for estimating resource utilization. Resource calendars also specify when and for how long identified team and physical resources will be

available during the project. This information may be at the activity or project level. This includes consideration of attributes such as resource experience and/or skill level, as well as various geographical locations.

Question 184:

Which of the following is NOT a possible output to the Validate Scope process?

- A. Project Management Plan
- B. Work Performance Information
- C. Change Requests
- D. Accepted Deliverables

Answer: A.

Explanation

The Project Management Plan is one of the inputs to the Validate Scope process. The other three are possible outputs. Deliverables that meet the acceptance criteria are formally signed off and approved by the customer or sponsor. Formal documentation received from the customer or sponsor acknowledging formal stakeholder acceptance of the project's deliverables is forwarded to the Close Project or Phase process. Work performance information includes information about project progress, such as which deliverables have been accepted and which have not been accepted and the reasons why. And finally, the completed deliverables that have not been formally accepted are documented, along with the reasons for non-acceptance of those deliverables. Those deliverables may require a change request for defect repair. The change requests are processed for review and disposition through the Perform Integrated Change Control process.

Question 185:

Power can originate with traits exhibited by the individual or the organization. Power is often supported by other people's perception of the leader. It is essential for project managers to be aware of their relationships with other people. Relationships enable project managers to get things done on the project. There are numerous forms of power at the disposal of project managers. How many are there?

- A. 10

- B. 9
- C. 14
- D. 12

Answer: C.

Explanation

Power and its use can be complex given its nature and the various factors at play in a project. Various forms of power include but are not limited to: Positional (sometimes called formal, authoritative, legitimate) (e.g., formal position granted in the organization or team); Informational (e.g., control of gathering or distribution); Referent (e.g., respect or admiration others hold for the individual, credibility gained); Situational (e.g., gained due to unique situation such as a specific crisis); Personal or charismatic (e.g., charm, attraction); Relational (e.g., participates in networking, connections, and alliances); Expert (e.g., skill, information possessed; experience, training, education, certification); Reward-oriented (e.g., ability to give praise, monetary or other desired items); Punitive or coercive (e.g., ability to invoke discipline or negative consequences); Ingratiating (e.g., application of flattery or other common ground to win favor or cooperation); Pressure-based (e.g., limit freedom of choice or movement for the purpose of gaining compliance to desired action); Guilt-based (e.g., imposition of obligation or sense of duty); uu Persuasive (e.g., ability to provide arguments that move people to a desired course of action); and Avoiding (e.g., refusing to participate). Top project managers are proactive and intentional when it comes to power. These project managers will work to acquire the power and authority they need within the boundaries of organizational policies, protocols, and procedures rather than wait for it to be granted.

Question 186:

As you and your team assess project risks, you learn that there are two vulnerabilities in the current version of a piece of software you have already selected as your chosen solution. Because the software vendor is already aware of these vulnerabilities, they agree to manage the risk and to bear the impact if your company is affected by these risks before they release security patches to address them. What is this strategy for handling threats called?

- A. escalate
- B. avoid
- C. transfer
- D. accept

Answer: C.

Explanation

Transfer involves shifting ownership of a threat to a third party to manage the risk and to bear the impact if the threat occurs. Risk transfer often involves payment of a risk premium to the party taking on the threat. Transfer can be achieved by a range of actions, which include but are not limited to the use of insurance, performance bonds, warranties, guarantees, etc. Agreements may be used to transfer ownership and liability for specified risks to another party.

Question 187:

Which of the following is NOT one of the three dimensions of complexity within projects?

- A. Ambiguity
- B. System Leflavlör
- C. Human behavior
- D. Processes

Answer: D.

Explanation

Processes is the correct answer, as it is NOT one of the three dimensions. Complexity within projects is a result of the organization's system behavior, human behavior, and the uncertainty at work in the organization or its environment. These three dimensions of complexity are defined as: System behavior. The interdependencies of components and systems. Human behavior. The interplay between diverse individuals and groups. Ambiguity. Uncertainty of emerging issues and lack of understanding or confusion.

Question 188:

Bonita's just been assigned as the project manager for a project in which she'll be responsible for replacing the entire HVAC system in one of her healthcare network's medical buildings. Bonita doesn't really know what

an HVAC system is, so she does some research and finds an online forum where she finds out not only what the acronym stands for, but is able to post questions about it that experts in the field are happy to answer. This Networking Bonita is doing is a Tool and Technique of which process?

- A. Direct and Manage Project Work
- B. Develop Project Management Plan
- C. Manage Project Knowledge
- D. Develop Project Charter

Answer: C.

Explanation

Manage Project Knowledge is the process of using existing knowledge and creating new knowledge to achieve the project's objectives and contribute to organizational learning. Knowledge management tools and techniques connect people so they can work together to create new knowledge, share tacit knowledge, and integrate the knowledge of diverse team members. The tools and techniques appropriate in a project depend on the nature of the project, especially the degree of innovation involved, the project complexity, and the level of diversity (including diversity of disciplines) among team members.

Question 189:

What document should be completed before a project is considered, in order to assess the viability of continuing with an initiative into the project management processes?

- A. Project Charter
- B. Project Management Plan
- C. Business requirements document
- D. Business Case

Answer: D.

Explanation

A Business Case is a documented economic feasibility study used to establish the validity of the benefits of a selected component lacking sufficient definition and that is used as a basis for the authorization of further project

management activities; it lists the objectives and reasons for project initiation.

Question 190:

There are three potential solutions for Travel Trucks Unlimited's project which aims to refresh their fleet of delivery trucks to ensure they're getting the best gas mileage and are as safe as possible for their drivers. They could replace the entire fleet of 20 trucks with brand-new vehicles, they could simply do an overhaul on each of their existing trucks or they could choose the best 10 and overhaul those, while completely replacing the worst 10 trucks. What form of analysis should they do now to support selection of the best of several alternative courses of action?

- A. Influence Diagram
- B. Simulation
- C. Sensitivity Analysis
- D. Decision Tree Analysis

Answer: D.

Explanation

Decision trees are used to support selection of the best of several alternative courses of action. Alternative paths through the project are shown in the decision tree using branches representing different decisions or events, each of which can have associated costs and related individual project risks (including both threats and opportunities). The end-points of branches in the decision tree represent the outcome from following that particular path, which can be negative or positive. The decision tree is evaluated by calculating the expected monetary value of each branch, allowing the optimal path to be selected.

Question 191:

Joan is working through the various activities required to build a virtual server environment onsite for the software development project she's managing. Very quickly she realizes that the tasks related to the IT Network resource assigned to the project, need more time than what was originally allocated. She thinks there may be some other conflicts with resources who are already working on other projects, that she may need for tasks on hers. What is a visual method Joan can use to show potential resource conflicts or issues, that is considered one potential output to the

Develop Schedule process?

- A. Project Calendar
- B. Resource Histogram
- C. Applied Schedule Reserve
- D. Alternative Schedule

Answer: B.

Explanation

The schedule data for the project schedule model is the collection of information for describing and controlling the schedule. The schedule data includes, at a minimum, the schedule milestones, schedule activities, activity attributes, and documentation of all identified assumptions and constraints. The amount of additional data varies by application area. Information frequently supplied as supporting detail includes but is not limited to: Resource requirements by time period, often in the form of a resource histogram; Alternative schedules, such as best-case or worst-case, not resource-leveled or resource-leveled, or with or without imposed dates; and uu Applied schedule reserves. Schedule data could also include such items as resource histograms, cash-flow projections, order and delivery schedules, or other relevant information.

Question 192:

Electric Light Bulbery has just entered into a contract with Brite Lite Chandeliers and Lamps to supply them with 5,000 specially made conical light bulbs per week for twelve weeks, totalling 60,000 light bulbs, for \$350,000. However, if Electric Light Bulbery can deliver all 60,000 bulbs in only 10 weeks, they'll get a \$20,000 bonus and whatever it might cost to do so is on them. What type of contract did these two companies sign?

- A. Fixed price with economic price adjustments (FPEPA)
- B. Cost plus incentive fee (CPIF)
- C. Fixed price incentive fee (FPIF)
- D. Firm fixed price (FFP)

Answer: C.

Explanation

Electric Light Bulbery has a fixed price contract, in that they'll be paid a certain amount for delivering all the light bulbs within the specific timeframe, but they'll be rewarded if they can deliver the entire amount two weeks early. This is a fixed price (\$350,000) with an incentive fee (\$20,000). A firm fixed price would not have included the bonus (incentive). There is no mention of any economic price adjustments, so it's not an FPEPA and the only reason the contract is not a CPIF is because Electric Light Bulbery has set a fixed price regardless of cost. If meeting the shorter 10-week timeframe costs them additional funds, that's not something they're going to get reimbursed for in this particular scenario.

Question 193:

When you think about how best to tailor project integration management to your project's needs, which of the following items is NOT a question you will want to ask?

- A. What are the project status reporting requirements?
- B. How important is quality management going to be on this project?
- C. What is an appropriate project life cycle?
- D. What is the change management process going to be on this project?

Answer: B.

Explanation

Even if your project is very small, or delivering something seemingly simple, there are still going to be acceptance criteria that will have to be met. The question is not about the importance, as quality management is always important. Responses B, C and D are good examples of questions that you can ask yourself and the team in order to help tailor the management of a project to more realistic parameters for the depth and breadth of each individual project. Other questions you can ask include how you're going to manage knowledge, which management process will be most effective and how best can you collect lessons learned.

Question 194:

All of the following are outputs of the Control Quality process EXCEPT:

- A. Change requests

- B. Work performance information
- C. Quality control measurements
- D. Organizational process assets updates

Answer: D.

Explanation

Organizational process assets updates are an output of controlling risks. The outputs of controlling quality are: Quality control measurements, Verified deliverables, Work performance information, Change requests, Project management plan updates and Project documents updates.

Question 195:

Amber and Aurora are both performing software development tasks on a project to develop a new medical software to help doctors tune hearing aids more accurately for an individual patient. As Amber continues her work, Aurora sees her code degrading right in front of her eyes and realizes that one of the risks she had voiced earlier on in the project is coming true: the more Aurora tells the software to fine-tune the higher decibel ranges, the less capable it becomes of accurately identifying the lower decibels. She brings it to Aurora's attention and after a short discussion they decide they need to report it to the Project Manager, Nishan, who pulls out the Risk Management Plan and sees that the team had developed a risk response should this occur. What should Nishan do now that the risk has now come to pass?

- A. Tell Amber and Aurora not to worry about it
- B. Cencic with the project sponsor
- C. Execute the previously defined risk response
- D. Update the Risk Management Plan to say the risk happened

Answer: C.

Explanation

Proper attention to the Implement Risk Responses process will ensure that agreed-upon risk responses are actually executed. A common problem with Project Risk Management is that project teams spend effort in identifying and analyzing risks and developing risk responses, then risk responses are agreed upon and documented in the risk register and risk report, but no action is

taken to manage the risk. Only if risk owners give the required level of effort to implement the agreed-upon responses will the overall risk exposure of the project and individual threats and opportunities be managed proactively.

Question 196:

Integration is a critical skill for project managers. Which of the following is NOT one of the levels that performing integration takes place at?

- A. Integration at the System Level
- B. Integration at the Context Level
- C. Integration at the Process Level
- D. Integration at the Cognitive Level

Answer: A.

Explanation

The correct answer is "Integration at the System Level" because "System Level" is made up. Project management may be seen as a set of processes and activities that are undertaken to achieve the project objectives. Although there is no stated definition on how to integrate the project processes, it is clear that a project has a small chance of meeting its objective when the project manager fails to integrate the project processes where they interact. The project manager should strive to become proficient in all of the Project Management Knowledge Areas. In concert with proficiency in these Knowledge Areas, the project manager applies experience, insight, leadership, and technical and business management skills to the project. Finally, it is through the project manager's ability to integrate the processes in these Knowledge Areas that makes it possible to achieve the desired project results. Project managers need to be cognizant of the project context and these new aspects when managing the integration. Then project managers can decide how to best use these new elements of the environment in their projects to achieve success.

Question 197:

It's Ruth's next task to perform quantitative risk analysis on the various critical tasks and activities within the project she's managing, and she feels the best way for her to do so is to provide and explain a tornado diagram at next week's steering committee meeting. What type of

analysis is Ruth planning to undertake?

- A. Simulation
- B. Decision tree
- C. Monte Carlo
- D. Sensitivity

Answer: D.

Explanation

The Monte Carlo analysis is one type of a Simulation analysis and the decision tree involves alternate project paths drawn like tree branches. One typical display of sensitivity analysis is the tornado diagram, which presents the calculated correlation coefficient for each element of the quantitative risk analysis model that can influence the project outcome. This can include individual project risks, project activities with high degrees of variability, or specific sources of ambiguity. Items are ordered by descending strength of correlation, giving the typical tornado appearance.

Question 198:

Which of the following is not considered a decision-making technique when performing integrated change control?

- A. Autocratic Decision-Making
- B. Voting
- C. Influencing Outcomes
- D. Multicriteria Decision Analysis

Answer: C.

Explanation

Influencing outcomes is not a decision-making technique. Rather, it's a technique that anyone can use in an attempt to influence the outcome of a decision being made using one of the other three techniques listed here.

Question 199:

Your project to overhaul your office's temperature regulation system had an original, approved budget of \$575,200. By the time you were halfway through the project you had estimated the budget at completion

would be \$582,760. After all invoices had been paid, as you were performing the final checks in preparation for writing your project closure report, your calculations indicated that the actual expenditure was \$596,231. What is the Variance at Completion of this project?

- A. \$13,471
- B. (\$ 13.47)
- C. \$21,031
- D. (\$21,031)

Answer: A.

Explanation

To calculate the Variance at Completion (VAC) you use the following formula: $VAC = BAC \text{ (Budget At Completion)} - EAC \text{ (Estimate At Completion)}$. Using the amounts provided in the question, then, $BAC (\$596,231) \text{ minus } EAC (\$582,760) = VAC (\$13,471)$.

Question 200:

Helga is everyone's favorite project manager because she allows any project team she leads to make their own decisions and establish their own goals. What is Helga's leadership style?

- A. Laissez-faire
- B. Servant leader
- C. Interactional
- D. Transactional

Answer: A.

Explanation

Research describes numerous leadership styles that a project manager can adopt. Some of the most common examples of these styles include but are not limited to: -Laissez-faire (e.g., allowing the team to make their own decisions and establish their own goals, also referred to as taking a hands-off style); -Transactional (e.g., focus on goals, feedback, and accomplishment to determine rewards; management by exception); -Servant leader (e.g., demonstrates commitment to serve and put other people first; focuses on other people's growth, learning, development, autonomy, and well-being;

concentrates on relationships, community and collaboration; leadership is secondary and emerges after service); -Transformational (e.g., empowering followers through idealized attributes and behaviors, inspirational motivation, encouragement for innovation and creativity, and individual consideration); -Charismatic (e.g., able to inspire; is high-energy, enthusiastic, self-confident; holds strong convictions); and -Interactional (e.g., a combination of transactional, transformational, and charismatic).